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Multi-Currency Portfolio Optimization with General Risk Measures

Master's Thesis

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Acknowledgments

Abstract

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Chapter 1

Introduction

1.1 Motivation: International Diversification, Currency Risk, and Tail Risk

1.2 Contributions

a systematic comaprison across different risk measures, universe sizes, which the exsititng literature covers only partially.

the mechanism analysis of the tail measures

the universe size results. The joint advantage depends on the currency share of the decision variables.. dimensionality effect.

1.3 Thesis Structure

Chapter 2

Literature Review

International diversification exposes an investor to currency risk, and early studies ask whether this risk should be hedged at all. Solnik [58] shows in an equilibrium model of the international capital market that, under specific conditions, it is optimal for investors to fully hedge their currency exposure. Perold and Schulman [52] argue that US investors should fully hedge their foreign holdings, since hedging substantially reduces risk while leaving the long-run expected return unchanged. Full hedging, however, treats the currency exposure only as a risk to be removed and ignores that currency positions can also diversify the risk of the assets.

The literature on currency overlays therefore studies how to choose the currency positions so that they improve the risk-return profile of a given asset portfolio. Glen and Jorion [20] find that adding forward contracts significantly improves the performance of internationally diversified portfolios that contain bonds, and Campbell et al. [10] show that the US dollar, the euro, and the Swiss franc tend to move against world equity markets, which makes them attractive to risk-minimizing equity investors despite their low average returns. More recent studies hedge dynamically. Boudoukh et al. [5] split the currency positions into a hedge component, which reduces the volatility of the equity portfolio, and an alpha component based on value, momentum, and carry signals, and Opie and Riddiough [50] exploit a forecastable component of currency factor returns. All of these strategies, however, take the asset portfolio as given. Van Binsbergen et al. [63] show that delegating the investment decisions for different asset classes to separate managers can be costly for the investor, and the same argument applies to the separation of the asset and the currency decisions in a currency overlay.

Jorion [27] shows that a currency overlay adds value to global equity and bond portfolios but improves them less than an integrated approach that chooses the asset and the currency positions together. Subsequent studies develop such integrated models further. Suzuki et al. [61] include forward currency contracts in an international portfolio optimization model, and Konno and Li [30, 31] select the individual stocks and bonds of several countries in a single mean-variance or mean absolute deviation program instead of the usual two-stage allocation, and hedge the currency risk with forward contracts. Topaloglou et al. [62] choose the asset weights and the currency hedge of each market together by minimizing Expected Shortfall over simulated scenarios, and Mhiri and Prigent [46] consider the higher moments of the returns in international portfolio optimization. Burkhardt and Ulrych [8] point out that the evidence for joint optimization was mostly established in sample, where the joint program can only improve on the overlay program, whereas out of sample the advantage can be lost to overfitting. They add regularization to joint and sequential mean-variance optimization and find that the regularized joint programs achieve on average higher out-of-sample Sharpe ratios than the overlay programs. Lucescu and Ulrych [37] apply L^1 and L^2 penalties to Expected Shortfall optimization and find that joint optimization performs better in the smaller of their

two investable universes.

Whatever the risk measure, the optimized portfolio depends on estimates from a finite sample. Merton [45] shows that the precision of an estimated expected return depends only on the length of the sample period, so that expected returns are much harder to estimate than variances. Michaud [47] calls mean-variance optimizers estimation-error maximizers, since they overweight the assets with large estimated expected returns and small estimated variances, which are the most likely to be estimated with large errors, and DeMiguel et al. [15] find that none of the optimized portfolios they consider consistently outperforms the equally weighted portfolio out of sample. Minimizing risk alone, as Lucescu and Ulrych do in their backtests, avoids the estimation of expected returns but it does not avoid the estimation error in the risk measure itself. This error grows with the number of assets relative to the number of observations, and Kondor et al. [28] show that close to a critical value of this ratio the optimized weights fluctuate so strongly from sample to sample that the optimization loses its meaning. The joint program uses the dependence between the individual assets and the currencies, whereas the overlay program takes the currencies into account only through their dependence on the return of the fixed asset portfolio, which Burkhardt and Ulrych regard as a form of dimensionality reduction. The size of the investable universe may therefore affect the comparison of the two approaches.

The estimation error also depends on the risk measure one chooses to optimize. Markowitz [39] measures the risk of a portfolio by the variance of its return, which penalizes returns above the mean as much as returns below it, and in his later book [40] he proposes the semi-variance as a downside alternative, whose efficient sets Markowitz et al. [41] compute. Konno and Yamazaki [32] propose the mean absolute deviation, which turns portfolio optimization into a linear program. Artzner et al. [3] define coherent risk measures, a class to which Expected Shortfall belongs [1], and Rockafellar and Uryasev [54, 55] show that Expected Shortfall can be minimized by a linear program. Krokhmal [33] extends Expected Shortfall to higher-moment coherent risk measures, which put more weight on the largest losses, and Rockafellar et al. [56] introduce deviation measures, which quantify the dispersion of a return rather than its losses. Since these measures depend on different parts of the return distribution, they are affected differently by estimation error, and the measures of tail risk, which depend only on the largest losses, are particularly affected. Kondor et al. [28] show that portfolios optimized under Expected Shortfall are more sensitive to estimation error than those optimized under the variance, and Ciliberti et al. [12] show that, without bounds on the weights, the optimization has no finite solution in some samples, and that for large portfolios a solution ceases to exist once the ratio of the number of assets to the number of observations exceeds a critical value. Kondor and Varga-Haszonits [29] show that this instability affects all coherent risk measures, since no optimal portfolio exists in a sample in which one portfolio dominates another. Each of the multi-currency studies above relies on a single risk measure, which leaves open how much the comparison of the two approaches depends on that choice.

One approach to reducing the estimation error is to shrink the estimates. James and Stein [24] show that, for three or more independent normal means, shrinking the sample means toward zero lowers their total expected squared error, and Ledoit and Wolf [34, 35] shrink the sample covariance matrix toward a structured target. Jagannathan and Ma [23] show that a no-short-sale constraint is equivalent to shrinking the large entries of the sample covariance matrix and reduces the risk of the optimized portfolio. Another approach is to regularize the weights directly. DeMiguel et al. [14] bound the norm of the weights of minimum-variance portfolios, Brodie et al. [7] add an L^1 penalty, which yields sparse and stable portfolios, and Li [36] combines the L^1 and the L^2 penalty. For coherent risk measures, Still and Kondor [60] relate regularized Expected Shortfall optimization to support vector regression and show

that an L^2 penalty acts as a diversification pressure that stabilizes the solution, and Caccioli et al. [9] show that any L^p penalty with $p > 1$ removes the instability, whereas the L^1 norm removes it only as a hard constraint, such as a ban on short sales. Papp et al. [51] analyze the effect of an L^1 constraint on Expected Shortfall analytically. These methods are developed for portfolios of assets in a single currency. Burkhardt and Ulrych [8] carry them over to the mean-variance optimization of assets and currencies, and Lucescu and Ulrych [37] to Expected Shortfall.

In this thesis we extend this literature in four directions. First, we compare joint and sequential optimization under five risk measures, the variance, the semi-variance, the mean absolute deviation, Expected Shortfall, and second-order Expected Shortfall, in the same out-of-sample framework with and without regularization. Second, we study how the comparison depends on the size of the investable universe. Third, we add a mean term to the programs of all five risk measures, with expected asset returns that are pooled by economy and shrunk and expected currency returns that are derived from the forward premia. Finally, we distinguish the advantage that joint optimization has in sample by construction from the advantage that it realizes out of sample.

Chapter 3

Optimization Framework

In this chapter, we formulate the portfolio optimization programs that we evaluate in Chapters 5 and 6. In Section 3.1 we state the risk-based portfolio optimization problem for a single currency and introduce the five risk measures that we consider in the thesis. In Section 3.2 we derive the return of an international portfolio in the domestic currency and decompose it into the return of the fully hedged asset portfolio and the excess return of the currency overlay. In Section 3.3 we formulate the regularized multi-currency programs of the sequential and the joint optimization approaches. Finally, in Section 3.4 we add a mean term to these programs and specify the expected asset returns and the expected excess currency returns estimators.

3.1 Risk-Based Portfolio Optimization

Portfolio optimization theory studies how an investor should allocate their wealth across a set of different risky assets. In one of the most influential works in finance, Markowitz [39] proposes to choose among the assets by trading off the expected portfolio return against its variance. This requires estimates of the expected asset returns, which are less precise than estimates of variances and covariances. Merton [45] shows that the precision of an estimated expected return depends only on the length of the sample period and not on how often the returns are observed within it, whereas the precision of an estimated variance increases with the observation frequency. Michaud [47] describes mean-variance optimizers as estimation-error maximizers, since they overweight the assets with large estimated expected returns and small estimated variances, which are the assets most likely to have large estimation errors. DeMiguel et al. [15] find that, out of sample, none of the optimized portfolios they consider consistently outperforms the equally weighted portfolio, because the gain from optimal diversification is more than offset by estimation error. Following Lucescu and Ulrych [37], we therefore first consider portfolios that minimize a measure of risk alone and require no explicit estimates of expected returns. We consider the expected portfolio return later in this chapter.

We initially formulate the portfolio optimization problem for an investor who holds N assets denominated in a single currency and extend the setting to multiple currencies in the next section. Let $\boldsymbol{x}_t \in \mathbb{R}^N$ denote the portfolio weights chosen at time t and held until $t+1$, and $\boldsymbol{R}_{t+1} \in \mathbb{R}^N$ the vector of simple asset returns over the period from t to $t+1$. The portfolio return at time $t+1$ is then $\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}$, and the weights satisfy the budget constraint $\boldsymbol{x}_t^\top \mathbf{1}_N = 1$, where $\mathbf{1}_N \in \mathbb{R}^N$ is the vector of ones. A risk measure ρ assigns to the random portfolio return a real number that quantifies its risk. We write ρ as a function of the portfolio return $\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}$, so that the risk-based portfolio optimization problem is

$$\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \rho(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}), \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1. \tag{3.1}$$

The risk measure is evaluated under the distribution of $\boldsymbol{R}_{t+1}$ conditional on the information available at time t.

3.1.1 Variance

In the portfolio theory of Markowitz [39], the risk of a portfolio is measured by the variance of its return. Let $\mathbb{E}_t[\cdot]$ denote the expectation conditional on the information available at time t, and let $\boldsymbol{\mu}_t = \mathbb{E}_t[\boldsymbol{R}_{t+1}] \in \mathbb{R}^N$ and $\boldsymbol{\Sigma}_t = \mathbb{E}_t\big[(\boldsymbol{R}_{t+1} - \boldsymbol{\mu}_t)(\boldsymbol{R}_{t+1} - \boldsymbol{\mu}_t)^\top\big] \in \mathbb{R}^{N\times N}$ denote the expected asset return and the conditional covariance matrix of the asset returns. The variance of the portfolio return is

$$\mathrm{Var}_t(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}) = \mathbb{E}_t\big[(\boldsymbol{x}_t^\top \boldsymbol{\mu}_t - \boldsymbol{x}_t^\top \boldsymbol{R}_{t+1})^2\big] = \boldsymbol{x}_t^\top \boldsymbol{\Sigma}_t \boldsymbol{x}_t.$$

The variance thus depends on the distribution of the asset returns only through their covariance matrix.

With the variance as the risk measure, the objective of problem (3.1) is the quadratic form $\boldsymbol{x}_t^\top \boldsymbol{\Sigma}_t \boldsymbol{x}_t$. Since $\boldsymbol{\Sigma}_t$ is positive semi-definite, the objective is convex in the weights, and the problem is a convex quadratic program, which standard solvers handle efficiently [6]. If $\boldsymbol{\Sigma}_t$ is positive definite, the first-order conditions of the Lagrangian further give the unique solution in closed form

$$\boldsymbol{x}_t^* = \frac{\boldsymbol{\Sigma}_t^{-1}\mathbf{1}_N}{\mathbf{1}_N^\top \boldsymbol{\Sigma}_t^{-1}\mathbf{1}_N},$$

which is known as the global minimum-variance portfolio. The closed form is a particularity of the variance, since the problems under the other risk measures we consider have no closed-form solution in general and must be solved numerically. It no longer applies once we add the leverage constraint and the penalties later in this chapter, although the problem remains a convex quadratic program. The closed form also shows that the weights depend on the covariance matrix through its inverse, so that errors in an estimate of $\boldsymbol{\Sigma}_t$ carry over directly to the portfolio.

Since the conditional distribution of $\boldsymbol{R}_{t+1}$ is unknown, $\boldsymbol{\Sigma}_t$ has to be estimated. For the variance and for the other risk measures later in the chapter, we use q historical realizations $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_q \in \mathbb{R}^N$ of $\boldsymbol{R}_{t+1}$. In the empirical analysis, these are the daily asset returns over a lookback window of the ten years before time t, so that $q \approx 2{,}500$. We denote their sample mean by $\bar{\boldsymbol{y}} = \frac{1}{q}\sum_{k=1}^{q} \boldsymbol{y}_k$.

The natural estimate of $\boldsymbol{\Sigma}_t$ is the sample covariance matrix of the realizations

$$\bar{\boldsymbol{\Sigma}}_t = \frac{1}{q}\sum_{k=1}^{q} (\boldsymbol{y}_k - \bar{\boldsymbol{y}})(\boldsymbol{y}_k - \bar{\boldsymbol{y}})^\top.$$

It has $N(N+1)/2$ distinct entries, all estimated from the same q observations, and when N is large relative to q, its eigenvalues are more dispersed than those of $\boldsymbol{\Sigma}_t$, with the smallest eigenvalues biased downward [35]. Through the inverse in the closed form, the minimum-variance portfolio then concentrates on the combinations of assets whose risk is most underestimated, and its in-sample variance understates its out-of-sample variance [17]. When $N \geq q$, the sample covariance matrix is singular, and the closed form does not exist.

We therefore estimate $\boldsymbol{\Sigma}_t$ with the shrinkage estimator of Ledoit and Wolf [34], which combines the sample covariance matrix with a structured target. The sample covariance matrix imposes no structure but has a large estimation error when N is large relative to q, while a structured target has few parameters and therefore a small estimation error but is biased whenever its structure does not hold. A weighted average of the two trades off the estimation error of the

one against the bias of the other. Ledoit and Wolf use the constant-correlation target, which keeps the sample variances of the assets and replaces every pairwise correlation by the average sample correlation.

To define the estimator, let $y_{k,i}$ denote the i-th entry of $\boldsymbol{y}_k$, $\bar{y}_i$ the i-th entry of $\bar{\boldsymbol{y}}$, and $\mathring{y}_{k,i} = y_{k,i} - \bar{y}_i$ the demeaned return of asset i in realization k. We suppress the dependence on t of the scalar quantities that follow. The sample covariance matrix $\bar{\boldsymbol{\Sigma}}_t$ has the entries

$$\bar{\sigma}_{ij} = \frac{1}{q} \sum_{k=1}^{q} \mathring{y}_{k,i}\, \mathring{y}_{k,j},$$

from which we obtain the sample correlations $r_{ij} = \bar{\sigma}_{ij} / \sqrt{\bar{\sigma}_{ii}\bar{\sigma}_{jj}}$ and their average $\bar{r} = \frac{2}{N(N-1)} \sum_{i<j} r_{ij}$. The constant-correlation target $\boldsymbol{T}_t$ has the diagonal entries $\bar{\sigma}_{ii}$ and the off-diagonal entries $\bar{r}\sqrt{\bar{\sigma}_{ii}\bar{\sigma}_{jj}}$, and the Ledoit–Wolf estimate is the convex combination

$$\hat{\boldsymbol{\Sigma}}_t = \hat{\delta}\, \boldsymbol{T}_t + (1 - \hat{\delta})\, \bar{\boldsymbol{\Sigma}}_t$$

with the shrinkage intensity $\hat{\delta} \in [0, 1]$.

Ledoit and Wolf choose the intensity that minimizes the expected squared Frobenius distance between $\hat{\boldsymbol{\Sigma}}_t$ and $\boldsymbol{\Sigma}_t$, that is, the expected sum of the squared differences between their entries. The optimal intensity is asymptotically proportional to $1/q$, and its estimate is

$$\hat{\delta} = \max\Big\{0, \min\Big\{\frac{\hat{\pi} - \hat{\nu}}{q\, \hat{\omega}}, 1\Big\}\Big\},$$

where

$$\begin{aligned}
\hat{\pi} &= \sum_{i,j} \hat{\pi}_{ij}, \qquad \hat{\pi}_{ij} = \frac{1}{q} \sum_{k=1}^{q} \big(\mathring{y}_{k,i}\, \mathring{y}_{k,j} - \bar{\sigma}_{ij}\big)^2, \\
\hat{\nu} &= \sum_{i} \hat{\pi}_{ii} + \bar{r} \sum_{i \neq j} \sqrt{\bar{\sigma}_{jj} / \bar{\sigma}_{ii}}\; \hat{\vartheta}_{ij}, \\
\hat{\vartheta}_{ij} &= \frac{1}{q} \sum_{k=1}^{q} \big(\mathring{y}_{k,i}^2 - \bar{\sigma}_{ii}\big)\big(\mathring{y}_{k,i}\, \mathring{y}_{k,j} - \bar{\sigma}_{ij}\big), \\
\hat{\omega} &= \sum_{i \neq j} \big(\bar{r}\sqrt{\bar{\sigma}_{ii}\bar{\sigma}_{jj}} - \bar{\sigma}_{ij}\big)^2.
\end{aligned}$$

The quantity $\hat{\pi}$ estimates the sum of the asymptotic variances of the entries of $\sqrt{q}\, \bar{\boldsymbol{\Sigma}}_t$, and $\hat{\nu}$ the sum of the asymptotic covariances between the entries of $\sqrt{q}\, \boldsymbol{T}_t$ and those of $\sqrt{q}\, \bar{\boldsymbol{\Sigma}}_t$, which arise because both matrices are computed from the same realizations. The quantity $\hat{\omega}$ is the squared Frobenius distance between $\boldsymbol{T}_t$ and $\bar{\boldsymbol{\Sigma}}_t$, which measures how far the structure of the target is from the data. The intensity therefore increases with the estimation error of the sample covariance matrix, decreases with the misspecification of the target, and tends to zero as q grows, so that $\hat{\boldsymbol{\Sigma}}_t$ approaches the sample covariance matrix when many observations are available. For the positive average correlations typical of stock returns, the target is positive definite, and $\hat{\boldsymbol{\Sigma}}_t$ is then positive definite whenever $\hat{\delta} > 0$, even when $N \geq q$.

The variance program replaces $\boldsymbol{\Sigma}_t$ by the Ledoit–Wolf estimate

$$\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \boldsymbol{x}_t^\top \hat{\boldsymbol{\Sigma}}_t \boldsymbol{x}_t, \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1,$$

and we write it in the standard form of a quadratic program in Appendix B.2.1.

One of the common criticisms of the variance is that it penalizes returns above the expected portfolio return as much as returns below it, although an investor is typically concerned with underperformance rather than outperformance [41]. This motivates the semi-variance, which we consider next.

3.1.2 Semi-Variance

The semi-variance only considers the deviations of the portfolio return below a threshold, so that returns above the threshold do not add to the risk of the portfolio. Markowitz et al. [41] consider two thresholds, the expected portfolio return and a benchmark return, which may also be a constant. Measured against a benchmark, the semi-variance captures the risk of underperforming the benchmark, while measured against the expected return, it captures the dispersion of the portfolio return below its mean. We use the expected portfolio return as the threshold. With $z^+ = \max\{z, 0\}$, the semi-variance of the portfolio return is

$$\mathrm{SVar}_t(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}) = \mathbb{E}_t\Big[\big((\boldsymbol{x}_t^\top \boldsymbol{\mu}_t - \boldsymbol{x}_t^\top \boldsymbol{R}_{t+1})^+\big)^2\Big].$$

As for the variance, the conditional expectation has to be estimated. With the historical realizations $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_q$ and their sample mean $\bar{\boldsymbol{y}}$ of Section 3.1.1, we replace $\boldsymbol{\mu}_t$ by $\bar{\boldsymbol{y}}$ and the conditional expectation by the empirical average over the realizations, which gives the sample semi-variance

$$\frac{1}{q}\sum_{k=1}^{q}\big((\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k)^+\big)^2 = \boldsymbol{x}_t^\top \boldsymbol{\Sigma}^-(\boldsymbol{x}_t)\, \boldsymbol{x}_t, \tag{3.2}$$

where

$$\boldsymbol{\Sigma}^-(\boldsymbol{x}_t) = \frac{1}{q}\sum_{k\,:\,\boldsymbol{x}_t^\top(\boldsymbol{y}_k - \bar{\boldsymbol{y}})<0} (\boldsymbol{y}_k - \bar{\boldsymbol{y}})(\boldsymbol{y}_k - \bar{\boldsymbol{y}})^\top$$

is the sample semi-covariance matrix of the realizations in which the portfolio return falls below its sample mean.

The sample semi-covariance matrix has the same form as the sample covariance matrix of Section 3.1.1, but it only contains the realizations in which the portfolio underperforms its mean, and which realizations these are depends on the weights. The semi-covariance matrix is therefore endogenous [18], so unlike for the variance, there is no fixed matrix that could be estimated before the optimization. For the same reason, the problem has no closed-form solution, and the Ledoit–Wolf estimator of Section 3.1.1 cannot be used. Estrada [18] proposes a heuristic semi-covariance matrix that does not depend on the weights, but it only approximates the semi-variance of the portfolio. We therefore minimize the sample semi-variance directly.

Each summand of the sample semi-variance (3.2) is only piecewise quadratic in the weights. Following Markowitz et al. [41], we introduce for each realization k an auxiliary variable $u_k \geq 0$ subject to $u_k \geq \boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k$, and collect them in $\boldsymbol{u} = [u_1, \ldots, u_q]^\top$. Since the objective $\frac{1}{q}\sum_{k=1}^{q} u_k^2$ increases in every $u_k \geq 0$, each u_k equals $(\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k)^+$ at the optimum, and the semi-variance program is the quadratic program

$$\begin{aligned}
\boldsymbol{x}_t^*, \boldsymbol{u}^* = \underset{\boldsymbol{x}_t,\, \boldsymbol{u}}{\arg\min} \quad & \frac{1}{q}\sum_{k=1}^{q} u_k^2, \\
\text{s.t.} \quad & \boldsymbol{x}_t^\top \mathbf{1}_N = 1, \\
& \boldsymbol{x}_t^\top (\boldsymbol{y}_k - \bar{\boldsymbol{y}}) + u_k \geq 0, \quad u_k \geq 0, \quad k = 1, \ldots, q.
\end{aligned}$$

Each realization adds one auxiliary variable and one linear constraint, so that the size of the program grows with the number of realizations, whereas the variance program works with a covariance matrix whose size grows quadratically with the number of assets. We write the program in the standard form of a quadratic program in Appendix B.2.2.

Like the variance, the semi-variance squares the deviations, so that a few large deviations below the mean can dominate its estimate, and minimizing it requires a quadratic program. The mean absolute deviation, which we consider next, measures the deviations linearly instead.

3.1.3 Mean Absolute Deviation

Konno and Yamazaki [32] propose the mean absolute deviation as an alternative to the variance. It measures the deviations of the portfolio return from its expected return by their absolute values instead of their squares. Large deviations therefore have less influence on the measure. The mean absolute deviation of the portfolio return is

$$\mathrm{MAD}_t(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}) = \mathbb{E}_t\big[|\boldsymbol{x}_t^\top \boldsymbol{\mu}_t - \boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}|\big].$$

The main motivation of Konno and Yamazaki is computational. Since the absolute value is piecewise linear, minimizing the sample mean absolute deviation can be formulated as a linear program rather than a quadratic program, which makes it more efficient for larger problems, and setting up the problem requires no estimate of the covariance matrix.

As for the semi-variance, we replace $\boldsymbol{\mu}_t$ by the sample mean $\bar{\boldsymbol{y}}$ and the conditional expectation by the empirical average over the realizations $\boldsymbol{y}_1, \dots, \boldsymbol{y}_q$ of Section 3.1.1, which gives the sample mean absolute deviation $\frac{1}{q}\sum_{k=1}^{q}|\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k|$. Since the deviations $\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k$ sum to zero over the sample, the sum of their positive parts equals the sum of their negative parts, and the absolute value of each deviation is the sum of its two parts. The sample mean absolute deviation is therefore twice the average shortfall of the portfolio return below its sample mean [59, 49]

$$\frac{1}{q}\sum_{k=1}^{q}|\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k| = \frac{2}{q}\sum_{k=1}^{q}(\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k)^+.$$

To write the problem as a linear program, we introduce, as for the semi-variance, an auxiliary variable $u_k \geq 0$ for each realization k subject to $u_k \geq \boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k$. Since the objective $\frac{2}{q}\sum_{k=1}^{q} u_k$ increases in every u_k, each u_k equals $(\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k)^+$ at the optimum, and the mean absolute deviation program is the linear program

$$\begin{aligned} \boldsymbol{x}_t^*, \boldsymbol{u}^* = \underset{\boldsymbol{x}_t, \boldsymbol{u}}{\arg\min} \quad & \frac{2}{q}\sum_{k=1}^{q} u_k, \\ \text{s.t.} \quad & \boldsymbol{x}_t^\top \mathbf{1}_N = 1, \\ & \boldsymbol{x}_t^\top (\boldsymbol{y}_k - \bar{\boldsymbol{y}}) + u_k \geq 0, \quad u_k \geq 0, \quad k = 1, \dots, q. \end{aligned}$$

It has the same constraints as the semi-variance program, but its objective is linear in the auxiliary variables. We write it in the standard form of a linear program in Appendix C.1.

The linear program is computationally efficient, but its solution is sensitive to estimation error. Kondor et al. [28] show that portfolios optimized under the mean absolute deviation are even more sensitive to estimation error than those optimized under the variance, and that their weights fluctuate more strongly from sample to sample.

The three risk measures considered so far measure the dispersion of the portfolio return around its mean. Expected Shortfall, which we consider next, instead measures the size of the largest losses.

3.1.4 Expected Shortfall

Expected Shortfall measures the risk of a portfolio by the average of its largest losses. Let $L(\boldsymbol{x}_t) = -\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}$ denote the loss of the portfolio at time $t+1$ and $\mathbb{P}_t(\cdot)$ the probability conditional on the information available at time t. The Value-at-Risk (VaR) at confidence level $\beta \in (0,1)$ is the β-quantile of the loss distribution

$$\mathrm{VaR}_\beta(\boldsymbol{x}_t) = \min\{\alpha \mid \mathbb{P}_t(L(\boldsymbol{x}_t) \leq \alpha) \geq \beta\}.$$

VaR does not take into account how large the losses beyond it are. Expected Shortfall is the expected loss given that the loss exceeds VaR

$$\mathrm{ES}_\beta(\boldsymbol{x}_t) = \mathbb{E}_t\big[L(\boldsymbol{x}_t) \mid L(\boldsymbol{x}_t) \geq \mathrm{VaR}_\beta(\boldsymbol{x}_t)\big],$$

so that it accounts for both the probability and the size of the losses in the tail. For loss distributions with atoms, such as the empirical distribution used below, Expected Shortfall is defined as the average of the quantiles of the loss distribution above β, which is equivalent to the conditional expectation for continuous distributions [1]. In its capital requirements for market risk, the Basel Committee on Banking Supervision has replaced VaR at the 99% level by Expected Shortfall at the 97.5% level [4], indicating that it is a more appropriate measure of risk than VaR.

Since $\mathrm{VaR}_\beta(\boldsymbol{x}_t)$ itself depends on the portfolio weights, the conditional expectation in the definition is not in a suitable form for optimization. Rockafellar and Uryasev [54, 55] show that Expected Shortfall is the minimum over $\alpha \in \mathbb{R}$ of the auxiliary function

$$F_\beta(\boldsymbol{x}_t, \alpha) = \alpha + \frac{1}{1-\beta}\,\mathbb{E}_t\big[(L(\boldsymbol{x}_t) - \alpha)^+\big],$$

and that the minimizing α is $\mathrm{VaR}_\beta(\boldsymbol{x}_t)$. Minimizing Expected Shortfall over the portfolio weights $\boldsymbol{x}_t$ is therefore equivalent to minimizing F_β jointly over $\boldsymbol{x}_t$ and α

$$\min_{\boldsymbol{x}_t} \mathrm{ES}_\beta(\boldsymbol{x}_t) = \min_{\boldsymbol{x}_t, \alpha} F_\beta(\boldsymbol{x}_t, \alpha),$$

where F_β is jointly convex in $(\boldsymbol{x}_t, \alpha)$. The derivation of this representation is given in Appendix A.1.1.

With the historical realizations $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_q$ of Section 3.1.1, we replace the conditional expectation in F_β by the empirical average, which gives the objective $\alpha + \frac{1}{q(1-\beta)} \sum_{k=1}^{q} (-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+$. As for the semi-variance and the mean absolute deviation, we introduce for each realization k an auxiliary variable $u_k \geq 0$ subject to $u_k \geq -\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha$. Since the objective increases in every u_k, each u_k equals $(-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+$ at the optimum, and the Expected Shortfall program is the linear program

$$\begin{aligned} \boldsymbol{x}_t^*, \alpha^*, \boldsymbol{u}^* = \underset{\boldsymbol{x}_t,\, \alpha,\, \boldsymbol{u}}{\arg\min} \quad & \alpha + \frac{1}{q(1-\beta)} \sum_{k=1}^{q} u_k, \\ \text{s.t.} \quad & \boldsymbol{x}_t^\top \mathbf{1}_N = 1, \\ & \boldsymbol{x}_t^\top \boldsymbol{y}_k + \alpha + u_k \geq 0, \quad u_k \geq 0, \quad k = 1, \ldots, q. \end{aligned}$$

Since Expected Shortfall measures losses rather than deviations from the mean, the constraints contain the realizations $\boldsymbol{y}_k$ themselves instead of their deviations from the sample mean. We write the program in the standard form of a linear program in Appendix A.2.1.

At the optimum, only the realizations whose loss exceeds α^*, which are the $(1-\beta)q$ realizations with the largest losses, contribute to the objective. The estimate of Expected Shortfall is therefore based on a fraction of the sample, and Kondor et al. [28] show that portfolios optimized under Expected Shortfall are more sensitive to estimation error than those optimized under the variance and the mean absolute deviation. Goldberg et al. [21] note that estimation error matters more for Expected Shortfall than for the variance, since most realizations only serve to determine which losses are the largest. The confidence level thus trades off the focus on the tail against the number of realizations that determine the estimate. We set $\beta = 0.8$, at which about 500 of the $q \approx 2{,}500$ daily realizations contribute to the objective. Testing

confidence levels from 60% to 99%, Goldberg et al. [21] also find that minimum Expected Shortfall portfolios outperform minimum-variance portfolios most at relatively low confidence levels, at which Expected Shortfall captures the asymmetry of the return distribution rather than its extreme losses.

Furthermore, Ciliberti et al. [12] show that, without bounds on the portfolio weights, the program is unbounded in some samples. For large portfolios, it has no finite solution once N/q exceeds a critical value, which is about 0.5 for high confidence levels, and the weights fluctuate from sample to sample as N/q approaches this value. In the large universe of the empirical analysis, which contains up to 1,400 stocks, N/q can exceed this value. The leverage constraint that we add later in this chapter bounds the positions and thereby ensures that the program has a solution.

Expected Shortfall weighs all losses beyond VaR equally, so that the largest losses in the tail count no more than the losses just beyond the quantile. Second-order Expected Shortfall, which we consider next, puts more weight on the largest losses.

3.1.5 Second-Order Expected Shortfall

Second-order Expected Shortfall belongs to the family of higher-moment coherent risk measures of Krokhmal [33], which measure the risk in the tail of the loss distribution through its higher moments and also contain Expected Shortfall. The second-order formulation replaces the expectation of the excess loss in the auxiliary function F_β of Section 3.1.4 by the L^2 norm $\|Z\|_{L^2} = (\mathbb{E}_t[Z^2])^{1/2}$ of the excess loss, which gives the auxiliary function

$$G_\beta(\boldsymbol{x}_t, \alpha) = \alpha + \frac{1}{1-\beta} \left\| (L(\boldsymbol{x}_t) - \alpha)^+ \right\|_{L^2},$$

and second-order Expected Shortfall at confidence level β is defined as its minimum over α

$$\mathrm{SOES}_\beta(\boldsymbol{x}_t) = \min_{\alpha \in \mathbb{R}} G_\beta(\boldsymbol{x}_t, \alpha).$$

Since the L^2 norm squares the excess losses, large losses beyond α receive more weight than in Expected Shortfall. Krokhmal notes that second-order Expected Shortfall measures the risk in the tail through the second moment of the losses, which relates it to the mean-variance framework. In a case study on fifty S&P 500 stocks, he finds that portfolios minimizing the higher-moment measures at the 90% level outperform those minimizing Expected Shortfall or the variance out of sample [33].

By definition, minimizing second-order Expected Shortfall is

$$\min_{\boldsymbol{x}_t} \mathrm{SOES}_\beta(\boldsymbol{x}_t) = \min_{\boldsymbol{x}_t, \alpha} G_\beta(\boldsymbol{x}_t, \alpha),$$

where G_β is jointly convex in $(\boldsymbol{x}_t, \alpha)$, which we show in Appendix A.1.2.

With the historical realizations $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_q$ of Section 3.1.1, we replace the L^2 norm in G_β by its empirical estimate $\big(\frac{1}{q}\sum_{k=1}^{q}\big((-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+\big)^2\big)^{1/2}$. As for Expected Shortfall, we introduce for each realization k an auxiliary variable $u_k \geq 0$ subject to $u_k \geq -\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha$, so that the empirical norm becomes $\frac{1}{\sqrt{q}}\|\boldsymbol{u}\|_2$. Since the objective is still not linear, we introduce one more auxiliary variable s subject to $\|\boldsymbol{u}\|_2 \leq s$, which equals $\|\boldsymbol{u}\|_2$ at the optimum, and the

second-order Expected Shortfall program is

$$\begin{aligned} \boldsymbol{x}_t^*, \alpha^*, \boldsymbol{u}^*, s^* = \underset{\boldsymbol{x}_t, \alpha, \boldsymbol{u}, s}{\arg\min} \quad & \alpha + \frac{1}{(1-\beta)\sqrt{q}}\, s, \\ \text{s.t.} \quad & \boldsymbol{x}_t^\top \mathbf{1}_N = 1, \\ & \boldsymbol{x}_t^\top \boldsymbol{y}_k + \alpha + u_k \geq 0, \quad u_k \geq 0, \quad k = 1, \ldots, q, \\ & \|\boldsymbol{u}\|_2 \leq s. \end{aligned}$$

The constraint $\|\boldsymbol{u}\|_2 \leq s$ requires the vector $[\boldsymbol{u}^\top, s]^\top$ to lie in the second-order cone, the set of vectors whose last entry is at least the Euclidean norm of the remaining entries [6, Section 2.2.3], so that the program is a second-order cone program [6, Section 4.4.2]. We write it in standard form in Appendix A.2.2.

The cone constraint makes the program harder to solve than the linear and quadratic programs of the other risk measures, and the second-order Expected Shortfall programs are computationally the most expensive programs in our backtests (see Appendix E.3). We use the same confidence level $\beta = 0.8$ as for Expected Shortfall, so that the two tail measures can be compared, and as for Expected Shortfall, the leverage constraint that we add later in this chapter bounds the positions and thereby ensures that the program has a solution.

3.1.6 Coherent Risk Measures and Deviation Measures

Expected Shortfall and second-order Expected Shortfall measure the size of the largest losses, whereas the variance, the semi-variance, and the mean absolute deviation measure deviations from the mean. The five risk measures can be divided into two classes, which differ in how they treat the return distribution.

Artzner et al. [3] call a risk measure $\mathcal{R}$ coherent if it satisfies four properties. Adding a riskless amount to the return reduces the risk by that amount, $\mathcal{R}(X + C) = \mathcal{R}(X) - C$, scaling the position by a positive factor scales the risk by the same factor, the risk of a combined position is at most the sum of the risks of its parts, so that diversification does not increase risk, and a return that is higher in every state of the world carries no more risk. VaR violates the third property, subadditivity, in general, whereas Expected Shortfall [1] and second-order Expected Shortfall [33] are coherent. Additionally, they are both strictly expectation bounded, that is, their risk exceeds the expected loss of every nonconstant return.

Rockafellar et al. [56] define deviation measures $\mathcal{D}$, which quantify the uncertainty of a return rather than its losses. They share positive homogeneity and subadditivity with coherent risk measures, are non-negative and zero only for constant returns, and replace the first property by translation invariance, $\mathcal{D}(X + C) = \mathcal{D}(X)$. The standard deviation, the semi-deviation, and the mean absolute deviation are deviation measures. The variance and the semi-variance are the squares of the standard deviation and the semi-deviation, so they are not positively homogeneous and therefore not deviation measures themselves. They are, however, translation invariant, and they have the same minimizers as the standard deviation and the semi-deviation.

The two translation properties capture the difference between the classes. A deviation measure depends only on the dispersion of the return around its mean and is unaffected by a shift of the whole distribution, whereas a coherent risk measure decreases when the distribution shifts upward. This fundamental difference between the two classes of risk measures is reflected in our empirical results in Chapters 5 and 6. More technical details on coherent risk measures and deviation measures are given in Appendix D. For brevity, we refer in the remainder of the thesis to the variance, the semi-variance, and the mean absolute deviation as the deviation measures, and to Expected Shortfall and second-order Expected Shortfall as the tail measures.

So far, we have treated the asset returns as given and denominated in the investor's domestic currency. An international investor, however, holds assets denominated in several currencies, whose returns in the domestic currency also depend on the exchange rates and on the currency positions that the investor takes to hedge them. All five risk measures are evaluated on the portfolio return in the domestic currency, so before we can formulate the multi-currency programs, we have to express this return in terms of the asset weights and the currency positions. In the next section we derive this representation, without assuming any underlying model for the joint distribution of asset and currency returns.

3.2 Model-Free Multi-Currency Portfolio Returns

Burkhardt and Ulrych [8] express the return of an international portfolio in the domestic currency in terms of its asset weights and currency positions, and use this representation for regularized mean-variance optimization, with the currency positions chosen either jointly with the asset weights or after them. Lucescu and Ulrych [37] apply the same representation to Expected Shortfall portfolios, and we follow their derivation and notation.

3.2.1 Unhedged Portfolio Returns

We consider an international investor who allocates their wealth across N assets denominated in up to $M+1$ currencies. Let the investor's domestic currency be indexed by 0 and let the remaining M foreign currencies be indexed by $\mathcal{C} = \{1, 2, \ldots, M\}$.

Let $P_{i,t} > 0$ be the price of asset i at time t in its local currency $c_i \in \mathcal{C} \cup \{0\}$. Define the corresponding simple local-currency asset return over the period from t to $t+1$ as

$$R_{i,t+1} = \frac{P_{i,t+1} - P_{i,t}}{P_{i,t}}.$$

Thus $P_{i,t+1} = P_{i,t}(1 + R_{i,t+1})$. For a domestic asset, $R_{i,t+1}$ is also the return in the domestic currency, as in the previous section.

For each foreign currency $c \in \mathcal{C}$, let $S_{c,t} > 0$ denote the exchange rate at time t, quoted as units of domestic currency per unit of foreign currency c. The associated simple currency return over the period from t to $t+1$ is

$$e_{c,t+1} = \frac{S_{c,t+1} - S_{c,t}}{S_{c,t}}.$$

Equivalently, $S_{c,t+1} = S_{c,t}(1+e_{c,t+1})$. For the domestic currency, we set $S_{0,t} \equiv 1$ and $e_{0,t+1} \equiv 0$.

The domestic-currency value of holding one unit of asset i at time t is $P_{i,t}\, S_{c_i,t}$. Therefore, the unhedged return of asset i in the domestic currency is

$$\begin{aligned} R^u_{i,t+1} &= \frac{P_{i,t+1}S_{c_i,t+1} - P_{i,t}S_{c_i,t}}{P_{i,t}S_{c_i,t}} \\ &= \frac{P_{i,t}(1 + R_{i,t+1})\, S_{c_i,t}(1 + e_{c_i,t+1}) - P_{i,t}S_{c_i,t}}{P_{i,t}S_{c_i,t}} \\ &= R_{i,t+1} + e_{c_i,t+1} + R_{i,t+1}\, e_{c_i,t+1}. \end{aligned} \tag{3.3}$$

As in the previous section, we denote the portfolio weights at time t by $\boldsymbol{x}_t = [x_{1,t}, \ldots, x_{N,t}]^\top$, with $x_{i,t}$ the fraction of total portfolio value invested in asset i, and impose the budget constraint $\boldsymbol{x}_t^\top \mathbf{1}_N = 1$.

Assuming no rebalancing between times t and $t+1$, the unhedged portfolio return in the domestic currency is given by

$$R^u_{\mathcal{P},t+1} = \sum_{i=1}^{N} x_{i,t}\, R^u_{i,t+1}.$$

3.2.2 Hedged Portfolio Returns

To manage currency risk, the investor can take positions in one-period currency forward contracts. For each foreign currency $c \in \mathcal{C}$, let $F_{c,t}$ be the forward rate quoted at time t for settlement at $t+1$, in units of domestic currency per unit of foreign currency c. We define the forward premium by

$$f_{c,t} = \frac{F_{c,t} - S_{c,t}}{S_{c,t}}.$$

The investor hedges by selling foreign currency forward. Let $\phi_{c,t}$ denote the relative notional of the short forward position in currency c at time t, expressed as a fraction of total portfolio value, with the convention that $\phi_{c,t} > 0$ corresponds to hedging. Per unit of notional, the short position pays $f_{c,t} - e_{c,t+1}$ in domestic currency over the period. Adding these forward payoffs to the unhedged portfolio return gives the hedged portfolio return

$$R^h_{\mathcal{P},t+1} = \underbrace{\sum_{i=1}^{N} x_{i,t}\, R^u_{i,t+1}}_{R^u_{\mathcal{P},t+1}} + \sum_{c=1}^{M} \phi_{c,t}\,(f_{c,t} - e_{c,t+1}). \tag{3.4}$$

3.2.3 Matrix Representation and Decomposition

To express the returns in a compact form suitable for optimization, we define the currency incidence matrix $\boldsymbol{C} \in \{0,1\}^{N\times M}$ by

$$C_{i,c} = \begin{cases} 1, & \text{if asset } i \text{ is denominated in foreign currency } c, \\ 0, & \text{otherwise.} \end{cases}$$

Each row of $\boldsymbol{C}$ has at most one nonzero entry, and the rows corresponding to domestic assets are identically zero. This allows us to define the portfolio's implicit foreign-currency exposure as

$$\boldsymbol{w}_t = \boldsymbol{C}^\top \boldsymbol{x}_t \in \mathbb{R}^M, \qquad w_{c,t} = \sum_{i:\, c_i = c} x_{i,t},$$

so that $w_{c,t}$ is the total portfolio weight invested in assets denominated in currency c. The implicit domestic exposure follows residually as $w_{0,t} = 1 - \mathbf{1}_M^\top \boldsymbol{w}_t$, since the asset weights sum to one.

We express returns in vector form $\boldsymbol{R}_{t+1} \in \mathbb{R}^N$, $\boldsymbol{e}_{t+1} \in \mathbb{R}^M$, and the forward premia as $\boldsymbol{f}_t \in \mathbb{R}^M$. The product $\boldsymbol{C}\boldsymbol{e}_{t+1} \in \mathbb{R}^N$ aligns currency returns to assets, and its i-th component is $e_{c_i,t+1}$ for foreign assets and 0 for domestic ones. Using (3.3), the vector of unhedged asset returns in the domestic currency can be written as

$$\boldsymbol{R}^u_{t+1} = \boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}),$$

where $\odot$ denotes the Hadamard product.

With the forward positions collected in $\boldsymbol{\phi}_t \in \mathbb{R}^M$, the hedged portfolio return (3.4) can be written as

$$R^h_{\mathcal{P},t+1} = \boldsymbol{x}_t^\top \boldsymbol{R}^u_{t+1} + \boldsymbol{\phi}_t^\top (\boldsymbol{f}_t - \boldsymbol{e}_{t+1}). \tag{3.5}$$

Setting $\boldsymbol{\phi}_t = \boldsymbol{w}_t = \boldsymbol{C}^\top \boldsymbol{x}_t$ neutralizes the implicit currency exposure. Substituting this and the unhedged return vector into (3.5) gives the fully hedged portfolio return

$$\begin{aligned} R^{fh}_{\mathcal{P},t+1} &= \boldsymbol{x}_t^\top \boldsymbol{R}^u_{t+1} + (\boldsymbol{C}^\top \boldsymbol{x}_t)^\top (\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\top \Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}) \Big] + \boldsymbol{x}_t^\top \boldsymbol{C}(\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\top \Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{e}_{t+1} + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}) + \boldsymbol{C}\boldsymbol{f}_t - \boldsymbol{C}\boldsymbol{e}_{t+1} \Big] \\ &= \boldsymbol{x}_t^\top \Big[\boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{f}_t + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}) \Big] \\ &= \boldsymbol{x}_t^\top \boldsymbol{R}^{fh}_{t+1}. \end{aligned}$$

The first-order currency terms $\pm\, \boldsymbol{x}_t^\top \boldsymbol{C}\boldsymbol{e}_{t+1}$ cancel, leaving the fully hedged asset return vector

$$\boldsymbol{R}^{fh}_{t+1} = \boldsymbol{R}_{t+1} + \boldsymbol{C}\boldsymbol{f}_t + (\boldsymbol{R}_{t+1} \odot \boldsymbol{C}\boldsymbol{e}_{t+1}).$$

More generally, define the net currency exposure $\boldsymbol{\psi}_t = \boldsymbol{w}_t - \boldsymbol{\phi}_t$. Since a forward position in the domestic currency has no payoff, we can set the domestic position to $\phi_{0,t} = 1 - \mathbf{1}_M^\top \boldsymbol{\phi}_t$. The domestic net exposure is then $\psi_{0,t} = w_{0,t} - \phi_{0,t} = -\mathbf{1}_M^\top \boldsymbol{\psi}_t$, so the net currency exposures sum to zero and form a zero-investment portfolio, the currency overlay. Substituting $\boldsymbol{\phi}_t = \boldsymbol{w}_t - \boldsymbol{\psi}_t$ into (3.5) expresses the hedged return in terms of net exposures

$$\begin{aligned} R^{h}_{\mathcal{P},t+1} &= \boldsymbol{x}_t^\top \boldsymbol{R}^u_{t+1} + (\boldsymbol{w}_t - \boldsymbol{\psi}_t)^\top (\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) \\ &= \boldsymbol{x}_t^\top \boldsymbol{R}^u_{t+1} + \boldsymbol{w}_t^\top (\boldsymbol{f}_t - \boldsymbol{e}_{t+1}) + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t) \\ &= \boldsymbol{x}_t^\top \boldsymbol{R}^{fh}_{t+1} + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t), \end{aligned} \tag{3.6}$$

where $\boldsymbol{e}_{t+1} - \boldsymbol{f}_t$ is the vector of excess currency returns. We adopt this representation in all programs that follow, as it makes the asset weights $\boldsymbol{x}_t$ and the net currency exposures $\boldsymbol{\psi}_t$ the decision variables.

3.3 Regularized Multi-Currency Portfolio Optimization

In practice, international investors often manage currency risk with a currency overlay. The asset portfolio is constructed first, without regard to its currency exposure, and a portfolio of currency positions is then laid over it, often by a specialized overlay manager [8]. Jorion [27] argues that this structure is inherently suboptimal, since it ignores the interactions between the assets in the underlying portfolio and the exchange rates. In his simulations, a currency overlay adds value to global equity and bond portfolios, but improves their performance less than an integrated approach to currency management. Burkhardt and Ulrych [8] find that regularized joint mean-variance optimization of assets and currencies also outperforms its currency overlay counterpart out of sample. For Expected Shortfall, Lucescu and Ulrych [37] report that joint optimization performs better in the smaller of their two investable universes.

We formulate both approaches for the five risk measures introduced earlier in this chapter, applied to the hedged portfolio return of the previous section. In sequential optimization, the asset weights $\boldsymbol{x}_t$ are chosen first by minimizing the risk of the fully hedged asset portfolio, and the currency overlay $\boldsymbol{\psi}_t$ is then chosen by minimizing the risk of the hedged portfolio return with the asset weights held fixed. In joint optimization, $\boldsymbol{x}_t$ and $\boldsymbol{\psi}_t$ are chosen together in a single program. The joint program can therefore exploit the dependence between the individual assets and the currencies, whereas the overlay program takes the currencies into account only through their dependence on the return of the fixed asset portfolio, which Burkhardt and Ulrych [8] regard as a form of dimensionality reduction.

3.3.1 Regularization

Due to estimation error, the optimal weights of these programs fluctuate from one sample to the next. Kondor et al. [28] find that the weights are even more sensitive to estimation error than the risk of the optimized portfolio, and that close to a critical value of the ratio of the number of assets to the number of observations, their fluctuations become so strong that the optimization loses its meaning. Bounding the weights, for example with a long-only constraint, prevents them from diverging, but Ciliberti et al. [12] note that large fluctuations remain and that many weights stick to the bounds.

Regularization reduces these fluctuations at the cost of introducing bias in the optimization. A penalty on the size of the weights pulls the solution toward the portfolios that the penalty favors. Empirically it tends to improve out-of-sample performance. DeMiguel et al. [14] find that minimum-variance portfolios with a constraint on the norm of the weights often have higher out-of-sample Sharpe ratios than portfolios based on shrinkage estimators. Brodie et al. [7] find that mean-variance portfolios with an L^1 penalty consistently outperform the equally weighted portfolio out of sample, and Li [36] finds that portfolios with constraints that induce sparsity and stability have better out-of-sample performance and lower turnover than many other strategies. In the multi-currency setting, Burkhardt and Ulrych [8] find that their regularized joint mean-variance programs outperform unregularized benchmarks, and Lucescu and Ulrych [37] find that regularization lowers the realized risk and raises the risk-adjusted returns of both the sequential and the joint optimization approach under Expected Shortfall.

Following Lucescu and Ulrych [37], we regularize the programs with an L^1 penalty, an L^2 penalty, and a leverage constraint. The L^1 penalty $\lambda_1\|\boldsymbol{x}_t\|_1 = \lambda_1 \sum_{i=1}^{N} |x_{i,t}|$, with $\lambda_1 \geq 0$, penalizes the absolute size of the weights. Brodie et al. [7] show that the penalty is effective in portfolio selection, where it yields portfolios with few active positions. Under the budget constraint, the long positions exceed the short positions by one, so that

$$\|\boldsymbol{x}_t\|_1 = 1 + 2\sum_{i=1}^{N}(-x_{i,t})^+.$$

The L^1 penalty therefore acts only on the short positions. Once λ_1 is large enough to remove them, the penalty becomes constant, and the solution is the long-only minimizer of the risk measure [7]. For Expected Shortfall, Papp et al. [51] show that removing short positions tends to remove the most volatile assets from the portfolio.

The L^2 penalty $\lambda_2\|\boldsymbol{x}_t\|_2^2 = \lambda_2 \sum_{i=1}^{N} x_{i,t}^2$, with $\lambda_2 \geq 0$, penalizes large weights more than small ones and spreads the portfolio more evenly over many assets. Under the budget constraint, $\|\boldsymbol{x}_t\|_2^2$ is smallest for the equally weighted portfolio, so the penalty shrinks the weights toward it.

The leverage constraint $\|\boldsymbol{x}_t\|_1 \leq \ell^a$, with $\ell^a \geq 1$, bounds the gross exposure of the asset portfolio. Setting $\ell^a = 1$ rules out short positions, and the value $\ell^a = 2$ that we use in the empirical analysis limits the total short position to half of the portfolio value. Ammann et al. [2] find that a leverage constraint improves the implementation of portfolios, since it lowers their transaction costs, the number of negative weights, and their volatility, and makes them less sensitive to the choice of their parameters. Together with the budget constraint, the leverage constraint defines the set of feasible asset weights

$$\mathcal{X} = \{\boldsymbol{x} \in \mathbb{R}^N : \boldsymbol{x}^\top \mathbf{1}_N = 1,\ \|\boldsymbol{x}\|_1 \leq \ell^a\}.$$

In the multi-currency programs that follow, the asset weights and the currency exposures have separate penalty weights, λ_1^a and λ_2^a for the assets and λ_1^c and λ_2^c for the currencies, since the

two differ in their constraints and transaction costs [37]. The currency exposures carry neither a budget constraint nor a leverage constraint, so both penalties shrink them toward zero, that is, toward the fully hedged portfolio, and the L^1 penalty sets the exposures to some currencies exactly to zero. Lucescu and Ulrych [37] note that the sparsity that the L^1 penalty promotes may be desirable for the selection of assets but less so for the currencies. The penalty weights are selected at every rebalancing date by the grid search described in Section 4.2.2.

3.3.2 Sequential Optimization

In sequential optimization, the asset weights and the currency exposures are chosen in two steps. The asset step chooses the asset weights by minimizing the risk of the fully hedged asset portfolio

$$\boldsymbol{x}_t^* = \underset{\boldsymbol{x}_t \in \mathcal{X}}{\arg\min} \Big\{ \rho(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}^{fh}) + \lambda_1^a \|\boldsymbol{x}_t\|_1 + \lambda_2^a \|\boldsymbol{x}_t\|_2^2 \Big\}. \tag{3.7}$$

This is the regularized version of the single-currency problem (3.1), with the fully hedged return vector $\boldsymbol{R}_{t+1}^{fh}$ in place of $\boldsymbol{R}_{t+1}$. The currency step then chooses the currency overlay by minimizing the risk of the hedged portfolio return (3.6), with the asset weights held fixed

$$\boldsymbol{\psi}_t^* \mid \boldsymbol{x}_t^* = \underset{\boldsymbol{\psi}_t \in \mathbb{R}^M}{\arg\min} \Big\{ \rho(\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)) + \lambda_1^c \|\boldsymbol{\psi}_t\|_1 + \lambda_2^c \|\boldsymbol{\psi}_t\|_2^2 \Big\}. \tag{3.8}$$

We derive the programs of both steps for the five risk measures in Appendices A.4, B.4, and C.3.

With $\boldsymbol{\psi}_t = \mathbf{0}_M$, the implicit currency exposure of the asset portfolio is fully hedged, and with $\boldsymbol{\psi}_t = \boldsymbol{w}_t = \boldsymbol{C}^\top \boldsymbol{x}_t^*$, the portfolio holds no forward positions and is unhedged. The two resulting asset-only portfolios serve as the benchmarks Full and Zero in Section 4.2.2.

3.3.3 Joint Optimization

The joint program chooses the asset weights and the currency exposures together. We stack them into the combined decision vector $\boldsymbol{\theta}_t = [\boldsymbol{x}_t^\top, \boldsymbol{\psi}_t^\top]^\top \in \mathbb{R}^{N+M}$ and the fully hedged and excess currency returns into the combined return vector $\boldsymbol{R}_{t+1}^{\text{all}} = [\boldsymbol{R}_{t+1}^{fh\top}, (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)^\top]^\top \in \mathbb{R}^{N+M}$, so that $\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}}$ is the hedged portfolio return (3.6). The joint program is then written as

$$\begin{aligned} \boldsymbol{\theta}_t^* = \underset{\boldsymbol{\theta}_t}{\arg\min} \Big\{ & \rho(\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}}) + \|\boldsymbol{S}_1 \boldsymbol{\theta}_t\|_1 + \|\boldsymbol{S}_2 \boldsymbol{\theta}_t\|_2^2 \Big\} \\ & \text{s.t. } \boldsymbol{\theta}_t^\top \boldsymbol{q}_{N,M} = 1, \quad \|\boldsymbol{L}^a \boldsymbol{\theta}_t\|_1 \le \ell^a, \end{aligned} \tag{3.9}$$

where the vector $\boldsymbol{q}_{N,M} \in \mathbb{R}^{N+M}$ has ones in the N asset positions and zeros in the M currency positions, and the matrix $\boldsymbol{L}^a \in \mathbb{R}^{(N+M)\times(N+M)}$ is diagonal with ones in the asset positions and zeros in the currency positions. The budget and the leverage constraint thus apply to the asset weights only, and the currency exposures remain unconstrained. The regularization matrix $\boldsymbol{S}_1 \in \mathbb{R}^{(N+M)\times(N+M)}$ is diagonal with λ_1^a in the asset positions and λ_1^c in the currency positions, and $\boldsymbol{S}_2 \in \mathbb{R}^{(N+M)\times(N+M)}$ is diagonal with $\sqrt{\lambda_2^a}$ and $\sqrt{\lambda_2^c}$ in the same positions, so that $\|\boldsymbol{S}_1\boldsymbol{\theta}_t\|_1 = \lambda_1^a\|\boldsymbol{x}_t\|_1 + \lambda_1^c\|\boldsymbol{\psi}_t\|_1$ and $\|\boldsymbol{S}_2\boldsymbol{\theta}_t\|_2^2 = \lambda_2^a\|\boldsymbol{x}_t\|_2^2 + \lambda_2^c\|\boldsymbol{\psi}_t\|_2^2$. This formulation allows us to treat the asset and the currency penalties of Section 3.3.1 separately within a single objective. We give the explicit matrices and derive the joint programs for the five risk measures in Appendices A.5, B.5, and C.4.

All programs of this section are convex. Since the L^1 penalties and the leverage constraint can be written with linear constraints, the regularized programs are quadratic programs under the

variance, the semi-variance, the mean absolute deviation, and Expected Shortfall, and second-order cone programs with a convex quadratic objective under second-order Expected Shortfall. Without the L^2 penalties, the programs under the mean absolute deviation and Expected Shortfall are linear programs. Standard solvers handle all of these programs efficiently [6].

3.4 Mean-Risk Portfolio Optimization

The programs of the previous section minimize a measure of risk without an explicit estimate of the expected returns, since such estimates are imprecise. The resulting portfolios, however, may give up excess returns that a moderately riskier allocation would earn. A mean term in the objective lets the programs trade off expected return against risk explicitly. With it, the joint program can also weigh the expected returns of the assets and the currencies against each other, whereas the overlay program considers the expected currency returns only once the asset weights are fixed.

Mean-risk optimization goes back to Markowitz [39], and the works on which our programs for the other four risk measures are based also trade off risk against expected return. Markowitz et al. [41] compute the portfolios that are efficient with respect to the expected return and the semi-variance, and Konno and Yamazaki [32], Rockafellar and Uryasev [54], and Krokhmal [33] minimize the mean absolute deviation, Expected Shortfall, and the higher-moment coherent risk measures, respectively, subject to a required expected return. In the multi-currency setting, Burkhardt and Ulrych [8] use a mean-variance objective with a risk aversion coefficient of 3 and estimate the expected returns with the Bayes–Stein estimator of Jorion [26]. Lucescu and Ulrych [37] formulate their programs with an objective that trades off the expected return against Expected Shortfall but evaluate them out of sample under infinite risk aversion, which removes the expected return from the objective, since estimates of expected returns are highly sensitive to estimation error [45, 15]. We add the mean term back to the programs of the previous section and specify the expected returns so that they are less affected by estimation error.

3.4.1 Mean-Risk Programs

To add a mean term to the sequential and joint programs, let $\boldsymbol{\mu}_t^{fh} = \mathbb{E}_t[\boldsymbol{R}_{t+1}^{fh}] \in \mathbb{R}^N$ and $\boldsymbol{\mu}_t^c = \mathbb{E}_t[\boldsymbol{e}_{t+1} - \boldsymbol{f}_t] \in \mathbb{R}^M$ denote the expected fully hedged asset returns and the expected excess currency returns, and let $\boldsymbol{\mu}_t^{\text{all}} = [(\boldsymbol{\mu}_t^{fh})^\top, (\boldsymbol{\mu}_t^c)^\top]^\top \in \mathbb{R}^{N+M}$, so that $\boldsymbol{\theta}_t^\top \boldsymbol{\mu}_t^{\text{all}}$ is the expected hedged portfolio return. Each program subtracts the expected return from the objective. We scale the expected return by $1/\gamma$, where $\gamma > 0$ is the risk aversion coefficient. The asset step (3.7) becomes

$$\boldsymbol{x}_t^* = \operatorname*{arg\,min}_{\boldsymbol{x}_t \in \mathcal{X}} \Big\{ \rho(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}^{fh}) - \frac{1}{\gamma} \boldsymbol{x}_t^\top \boldsymbol{\mu}_t^{fh} + \lambda_1^a \|\boldsymbol{x}_t\|_1 + \lambda_2^a \|\boldsymbol{x}_t\|_2^2 \Big\}, \tag{3.10}$$

and the currency step (3.8) becomes

$$\begin{aligned} \boldsymbol{\psi}_t^* \mid \boldsymbol{x}_t^* = \operatorname*{arg\,min}_{\boldsymbol{\psi}_t \in \mathbb{R}^M} \Big\{ & \rho(\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)) - \frac{1}{\gamma} \boldsymbol{\psi}_t^\top \boldsymbol{\mu}_t^c \\ & + \lambda_1^c \|\boldsymbol{\psi}_t\|_1 + \lambda_2^c \|\boldsymbol{\psi}_t\|_2^2 \Big\} \\ & \text{s.t. } \|\boldsymbol{\psi}_t\|_1 \le \ell^c, \end{aligned} \tag{3.11}$$

where the expected return of the fixed asset portfolio is a constant and is removed from the objective. The joint program (3.9) becomes

$$\begin{aligned} \boldsymbol{\theta}_t^* = \arg\min_{\boldsymbol{\theta}_t} \Big\{ \rho(\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}}) - \frac{1}{\gamma}\, \boldsymbol{\theta}_t^\top \boldsymbol{\mu}_t^{\text{all}} + \|\boldsymbol{S}_1 \boldsymbol{\theta}_t\|_1 + \|\boldsymbol{S}_2 \boldsymbol{\theta}_t\|_2^2 \Big\} \\ \text{s.t. } \boldsymbol{\theta}_t^\top \boldsymbol{q}_{N,M} = 1, \quad \|\boldsymbol{L}^a \boldsymbol{\theta}_t\|_1 \le \ell^a, \quad \|\boldsymbol{L}^c \boldsymbol{\theta}_t\|_1 \le \ell^c, \end{aligned} \tag{3.12}$$

where the matrix $\boldsymbol{L}^c \in \mathbb{R}^{(N+M)\times(N+M)}$ is diagonal with zeros in the asset positions and ones in the currency positions. Both the currency step and the joint program thus contain a currency leverage constraint, which the programs of Section 3.3 do not have. We give the mean-risk programs for the five risk measures in Appendices A.6, B.6, and C.5.

We use the same risk aversion coefficient in the asset step, the currency step, and the joint program. It determines the weight of the expected return relative to the risk measure and the penalties. As $\gamma \to \infty$, the mean term vanishes, and the programs reduce to the risk-minimization programs (3.7)–(3.9) with the additional currency leverage constraint. As $\gamma \to 0$, the mean term dominates, and the programs maximize the expected portfolio return subject to the constraints. More details on the calibration of the risk aversion coefficient are given in Appendix G.3.

The currency leverage constraint bounds the currency exposures, which otherwise carry neither a budget constraint nor a leverage constraint. Without it, the mean term can push the currency exposures toward very large positions. The asset weights are already bounded by the leverage constraint of Section 3.3.1, and for the currency exposures we set $\ell^c = 2$ as well.

3.4.2 Expected Asset Returns

The natural estimate of $\boldsymbol{\mu}_t^{fh}$ is the vector of sample means of the historical realizations of $\boldsymbol{R}_{t+1}^{fh}$. As discussed at the beginning of this chapter, however, the precision of an estimated expected return depends only on the length of the sample period [45]. For example, if the returns of a stock with an annual volatility of 30% are independent and normally distributed, the standard error of its sample mean over ten years is $30\%/\sqrt{10} \approx 9.5\%$ per year, and a 95% confidence interval for its expected return is about 37 percentage points wide.

To reduce the noise in the expected returns, we pool the stocks of each economy and estimate one expected return per economy. Each of the 14 economies of the empirical analysis has its own currency, so we index the economies by their currencies $c \in \mathcal{C} \cup \{0\}$, and asset i belongs to the economy of its local currency c_i. As in Section 3.1.1, we suppress the dependence on t of the scalar quantities that follow. Let $y_{k,i}^{fh}$ denote the fully hedged return of asset i in the k-th historical realization. The return of economy c in realization k is the equally weighted return of its N_c assets

$$z_{k,c} = \frac{1}{N_c} \sum_{i\,:\,c_i = c} y_{k,i}^{fh},$$

with the sample mean $\bar{z}_c = \frac{1}{q}\sum_{k=1}^{q} z_{k,c}$. Pooling reduces the number of estimated means from N to the number of economies $M+1$, while the number of realizations stays the same. The equally weighted economy return also diversifies part of the volatility of its assets, so its mean is estimated more precisely than the means of the individual assets. Every asset then uses the expected return of its economy, so the mean term depends on the asset weights only through the total weight of each economy.

The economy means are still imprecise, since the length of the sample period is unchanged, so we further shrink them with the estimator of James and Stein [24]. Let $\bar{\boldsymbol{z}} = [\bar{z}_0, \bar{z}_1, \ldots, \bar{z}_M]^\top \in$

$\mathbb{R}^{M+1}$ be the vector of the sample means, and let $\bar{s}^2$ denote the average of the sample variances $s_c^2 = \frac{1}{q-1}\sum_{k=1}^{q}(z_{k,c} - \bar{z}_c)^2$ of the economy returns. The shrunk economy means are

$$\hat{m}_c = (1 - \hat{\eta})\,\bar{z}_c, \qquad \hat{\eta} = \min\Big\{1, \frac{(M-1)\,\bar{s}^2}{q\,\|\bar{\boldsymbol{z}}\|_2^2}\Big\},$$

and the estimate $\hat{\boldsymbol{\mu}}_t^{fh}$, whose i-th entry is $\hat{m}_{c_i}$, replaces $\boldsymbol{\mu}_t^{fh}$ in the programs. James and Stein show that, for three or more independent normal means with a common variance, shrinking the sample means toward zero by such a factor lowers their total expected squared error below that of the sample means, whatever the true means are.

3.4.3 Expected Currency Returns

Unlike the expected asset returns, the expected excess currency returns are not estimated from the realizations but derived from the forward premia. By covered interest parity, the forward rate of currency c is $F_{c,t} = S_{c,t}(1 + r_{0,t})/(1 + r_{c,t})$, where $r_{0,t}$ and $r_{c,t}$ denote the domestic and the foreign interest rate from t to $t + 1$, so that the forward premium

$$f_{c,t} = \frac{r_{0,t} - r_{c,t}}{1 + r_{c,t}} \approx r_{0,t} - r_{c,t}$$

is approximately the difference between the domestic and the foreign interest rate. A net exposure to currency c therefore earns the spot return $e_{c,t+1}$ plus approximately the interest rate differential $r_{c,t} - r_{0,t}$, which is known as the carry. Since $f_{c,t}$ is known at time t, the expected excess return of currency c is

$$\mathbb{E}_t[e_{c,t+1} - f_{c,t}] = \mathbb{E}_t[e_{c,t+1}] - f_{c,t}.$$

Under uncovered interest parity, the expected change of the spot rate offsets the interest rate differential, $\mathbb{E}_t[e_{c,t+1}] = f_{c,t}$, and the expected excess return is zero, in which case the currency exposures would not enter the mean term. We instead set the expected excess currency returns to the negative forward premia

$$\boldsymbol{\mu}_t^c = -\boldsymbol{f}_t,$$

which approximate the carry, so that the mean term rewards long exposures to currencies with a higher interest rate than the domestic currency and short exposures to currencies with a lower one. Since the forward premia are observed at time t, nothing has to be estimated, and the expected currency returns change over time with the interest rate differentials.

Chapter 4

Data and Implementation

In this chapter, we describe the data and the backtest implementation with which we evaluate the optimization programs of Chapter 3 in Chapters 5 and 6. In Section 4.1 we describe the data sources and the construction of three investable universes of different sizes from the largest stocks of 14 global economies, and analyze the fully hedged asset returns, the excess currency returns, and the correlations between them. In Section 4.2 we describe the implementation of the backtest. We first present in Section 4.2.1 the modular structure of the backtesting framework. In Section 4.2.2 we present the backtested strategies and their benchmarks, and specify the rebalancing frequency, the lookback window, the selection of the hyperparameters, and the modeling of transaction costs. Finally, in Section 4.2.3 we define the performance metrics on which the strategies are evaluated in Chapter 5.

4.1 Data

The empirical evaluation of all portfolio optimization strategies in this thesis relies on a dataset of international equities, foreign exchange spot rates, and forward exchange rates sourced from Refinitiv Datastream. The one-month U.S. Treasury bill rate used in the return and performance metric calculations is obtained from Kenneth R. French's online data library.[1] The datasets span from June 1990 to June 2023, covering several different market regimes, including the dot-com crash, the global financial crisis, the European sovereign debt crisis, and the COVID-19 pandemic.

We construct the investable universe from 14 economies: Australia, Canada, Denmark, Norway, Sweden, Hong Kong, Japan, New Zealand, Singapore, South Africa, Switzerland, the Eurozone, the United Kingdom, and the United States. The universe is rebuilt at the beginning of each year from the largest stocks by market capitalization at the time (a rolling-window approach that mitigates survivorship bias). At the start of each year we rank stocks within each economy and restrict the following year's investments to that set. To study how the universe size affects the optimization approaches considered, we form three cross-sections, comprising the top 10, top 50, and top 100 stocks per economy. We refer to these as the small, medium, and large universes. Because the constituents change each year, each universe spans more distinct stocks over the full sample than are held on any single date, 407 for the top 10, 1,705 for the top 50, and 3,189 for the top 100, respectively.

The US dollar serves as the base currency throughout our analysis. For each stock we compute fully hedged returns, and for each currency the excess currency return, following the model-free return representation in Chapter 3.

[1] https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Economies	Assets (%)						Currencies (%)	
	Top 10		Top 50		Top 100			
	Return	Vol.	Return	Vol.	Return	Vol.	Excess Return	Vol.
Australia	4.57	29.94	3.10	37.72	-1.88	58.00	1.31	11.59
Canada	9.10	32.89	6.58	40.24	1.86	59.24	-0.03	7.63
Denmark	8.35	34.05	3.04	39.00	-3.47	48.52	-0.08	9.68
Eurozone	5.09	32.24	5.05	34.01	3.25	35.46	-0.40	9.58
Hong Kong	4.42	37.59	-0.15	45.77	-3.48	52.59	-0.16	0.56
Japan	5.37	33.69	4.41	34.01	4.09	35.38	-2.07	10.47
New Zealand	2.88	29.93	-1.13	37.16	-4.56	41.49	2.48	11.36
Norway	3.17	43.30	-1.59	50.29	-2.61	53.26	-0.36	11.73
Singapore	3.32	35.43	-1.99	54.57	-4.84	62.12	0.18	5.35
South Africa	3.53	37.72	1.01	43.84	-4.39	53.94	1.22	15.18
Sweden	9.54	34.50	6.92	42.50	5.37	47.65	-1.16	11.50
Switzerland	6.38	29.04	5.67	30.32	3.97	33.65	-0.12	10.99
United Kingdom	2.76	31.50	2.63	34.71	1.29	36.22	0.12	9.42
United States	12.59	33.58	7.99	39.07	6.86	42.84	0.00	0.00

Table 4.1: Summary statistics of fully hedged equity returns and excess currency returns by economy. Asset statistics are reported for three investable universes (the top 10, 50, and 100 stocks per economy). Excess currency returns are defined as the currency return net of the forward premium. All numbers are reported as annualized percentages, from the perspective of a USD-based investor.

Table 4.1 reports annualized average returns and volatilities for the fully hedged equity returns and excess currency returns, by economy. For each economy, the asset figures are the cross-sectional average of the daily returns of its individual stocks.

We can see that as the investable universe becomes larger, the average returns tend to fall. In the small universe every economy has a positive average return, ranging from 2.76% (the United Kingdom) to 12.59% (the United States). In the medium universe, the range drops from -1.99% (Singapore) to 7.99% (the United States), and in the large universe the range goes from -4.84% (Singapore) to 6.86% (the United States), with many economies now having negative average returns. The trend reflects the firm size, the largest stocks post the strongest returns, and the smaller firms added pull the average down. Volatilities exhibit the opposite pattern, rising with universe size, as smaller, more volatile stocks are added to the investable universe. Switzerland exhibits the lowest volatility in every case. Its volatility rises from 29.04% to 33.65%, and the upper end rises from 43.30% (Norway) in the small universe to 62.12% (Singapore) in the large universe.

Excess currency returns range between -2.07% for the Japanese yen and 2.48% for the New Zealand dollar, and excess currency volatilities range from 0.56% for the Hong Kong dollar, whose peg to the US dollar keeps it stable, up to 15.18% for the South African rand.

To quantify the dependence between asset and currency returns, we compute Pearson correlations between economy-level fully hedged equity returns and excess currency returns. For each economy we form an equally weighted daily portfolio of the fully hedged returns of its constituent stocks, resetting the constituents at each annual universe reconstitution. Correlating each economy's series with each currency's excess return over the full sample gives a 14×13 correlation matrix. The US dollar is dropped from the currency axis, since its excess return is zero by construction. We repeat this for the three universe sizes. The resulting heatmaps are shown in Figures 4.1 to 4.3.

Across all three universes the correlations range from -0.2 to 0.4 and remain mostly positive. The results are intuitive and agree with known empirical findings. The strongest positive

correlations belong to commodity and risk-sensitive currencies such as the AUD, CAD, NZD, and ZAR, whose equity markets and currencies rise together in good times and fall together in bad [11, 38, 44]. The JPY and CHF are the only negatively correlated currencies with most other economies (ranging from -0.2 to 0.03), in line with their role as safe havens that strengthen when equities fall [53]. The EUR, DKK, and HKD are minimally correlated, reflecting the HKD's peg to the US dollar and the DKK's peg to the EUR. This split between risky and safe-haven currencies is well documented [10, 53].

This is the dependence structure the overlay program discards when it optimizes the asset weights. It sets the asset weights first, using only equity information, and adds the currency hedge afterwards, whereas the joint program can act on it directly. The structure barely changes across universe sizes, and this matters for the results that follow. A shift in the relative performance of the joint and overlay programs across universe sizes in Chapters 5 and 6 is therefore not due to a change in the underlying dependence structure, and we analyze its causes in Chapter 6.

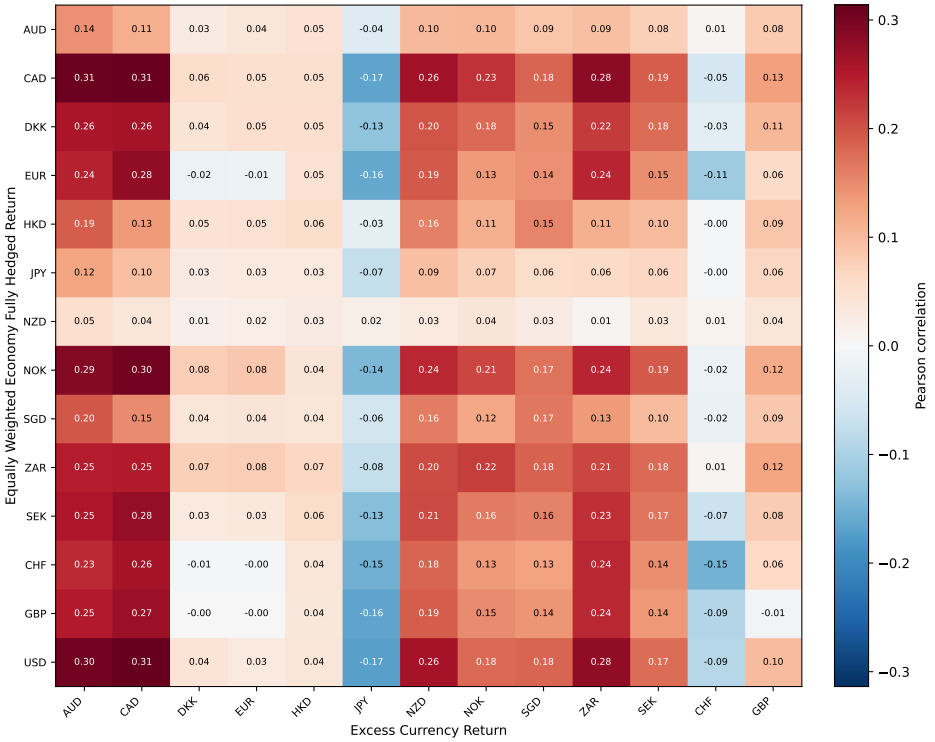


Figure 4.1: Pearson correlation between the fully hedged returns of the equally weighted economy portfolios and the excess currency returns in the small investable universe (top 10 stocks per economy). The US dollar is excluded from the currency axis, since it serves as the base currency and therefore its excess return is zero.

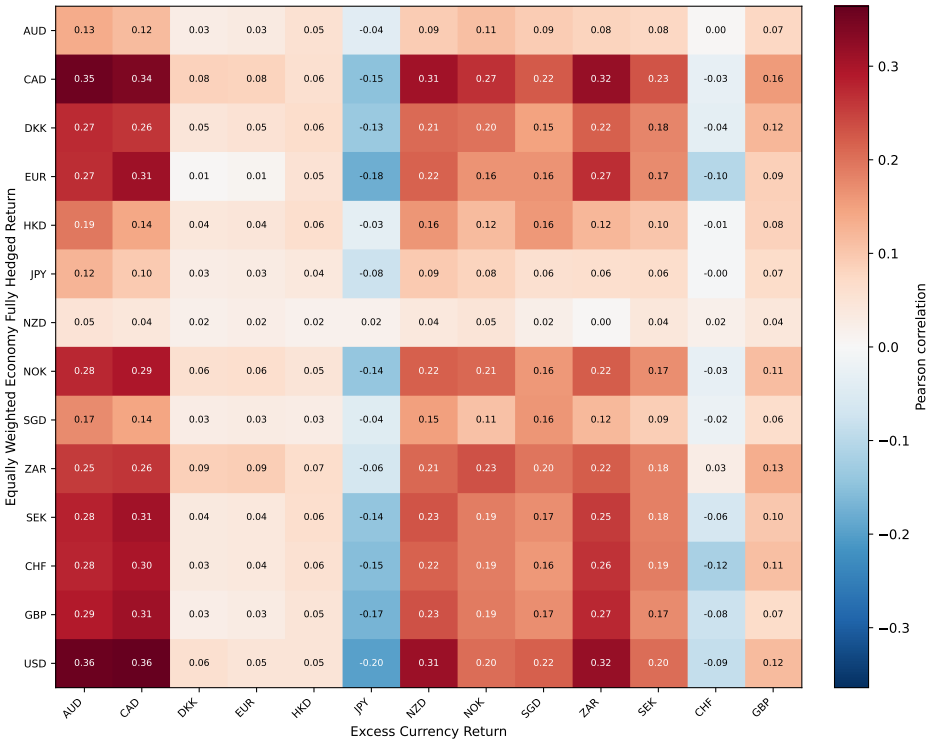


Figure 4.2: Pearson correlation between the fully hedged returns of the equally weighted economy portfolios and the excess currency returns in the medium investable universe (top 50 stocks per economy). The US dollar is excluded from the currency axis, since it serves as the base currency and therefore its excess return is zero.

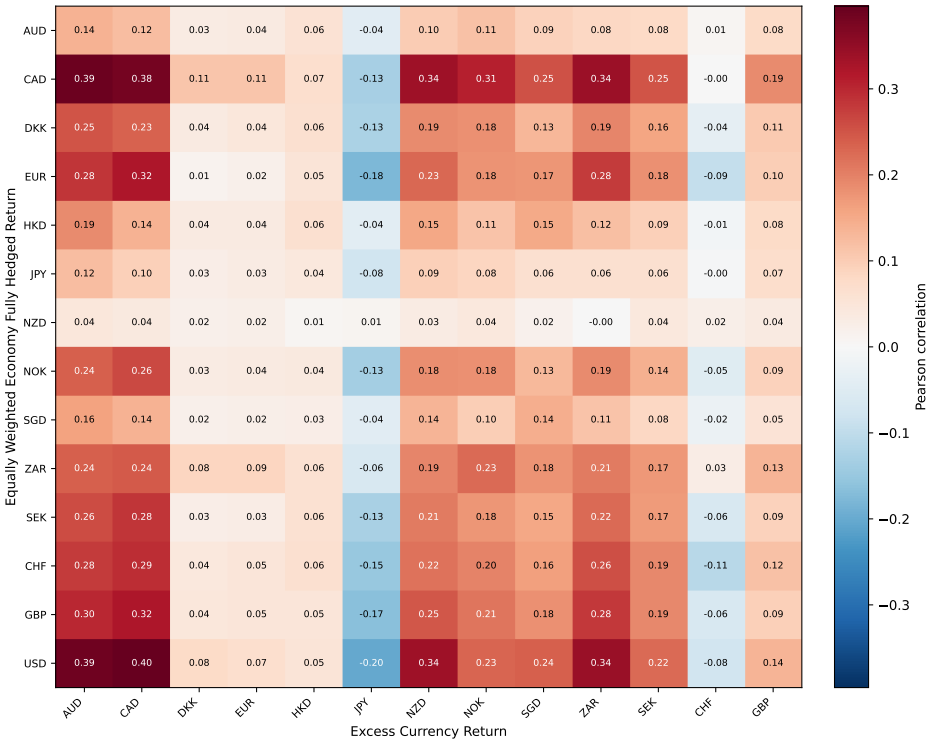


Figure 4.3: Pearson correlation between the fully hedged returns of the equally weighted economy portfolios and the excess currency returns in the large investable universe (top 100 stocks per economy). The US dollar is excluded from the currency axis, since it serves as the base currency and therefore its excess return is zero.

4.2 Backtest Implementation

4.2.1 Backtesting Framework

In Chapter 5 we evaluate a range of optimization strategies under different risk measures (variance, semi-variance, mean absolute deviation, Expected Shortfall, and second-order Expected Shortfall), their treatment of currency exposure (overlay or joint), and the size of the investable universe. To keep these comparisons fair, the backtesting framework separates between the orchestration of the backtest and the optimization logic it evaluates.[2] The main backtesting loop that ingests data, rolls the estimation window forward, selects hyperparameters, and records performance is fixed, while the optimizer that produces portfolio weights is a swappable component. Any optimizer that conforms to a common interface is therefore evaluated under the same conditions. The framework is composed of three main components, a data handler, the backtester, and a family of optimizers under a common abstract base class, whose interaction is summarized in Figure 4.4.

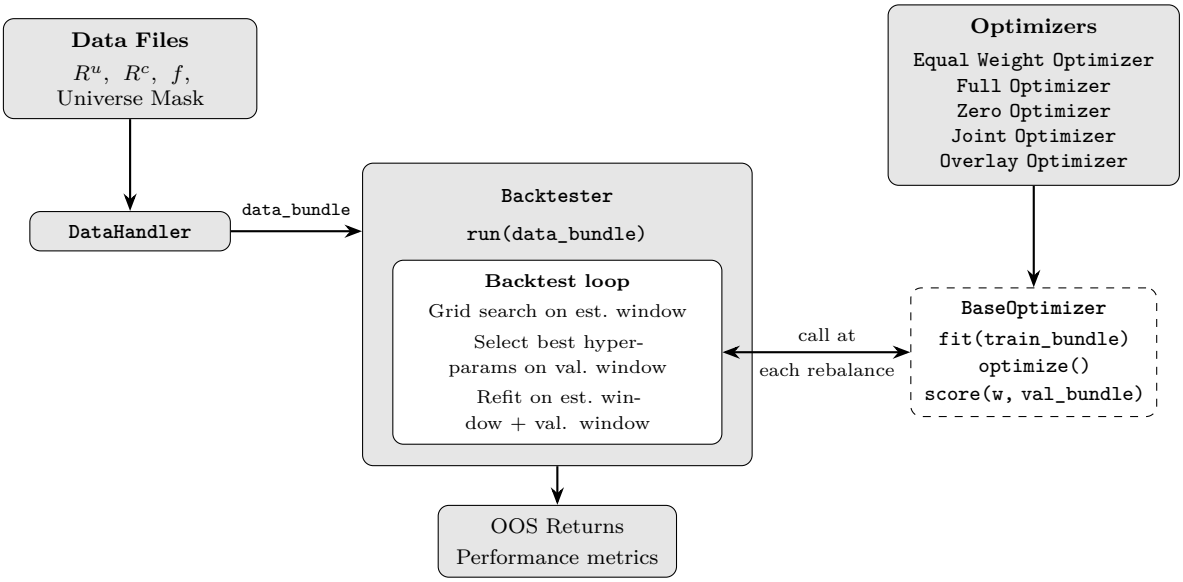


Figure 4.4: Schematic of the backtesting framework.

Data ingestion and alignment are implemented in the `DataHandler`. It loads the input data, the local unhedged asset returns R^u, the currency returns R^c, the forward premia f, and the universe mask that identifies which assets constitute the investable universe at each date. It then aligns them into a `data_bundle`, which is passed to the backtester. The optimizers operate only on the bundled series, and the construction of returns from these inputs follows Section 3.2.

All implemented optimizers inherit from an abstract base class, `BaseOptimizer`, that defines the three-method interface through which the backtester runs them. The `fit` method receives a training data bundle and estimates and stores the quantities the optimizer requires for its objective (e.g., the covariance matrix for the variance optimizers), so that estimation is performed entirely within each optimizer. The `optimize` method solves the specific optimization problem. The `score` method takes a set of weights and a validation bundle and returns the validation performance used for hyperparameter selection. More details on the estimators and

[2]The full codebase with the backtest implementation is publicly available on GitHub: https://github.com/anejrozman/Multi-Currency-Portfolio-Optimization-with-General-Risk-Measures.

solver internals invoked inside the methods `fit` and `optimize` can be found in Chapter 3 and Appendix E.2, respectively.

The backtest itself is implemented in the `Backtester.run` method, which advances through the data sample on a set rebalancing frequency (daily, weekly, monthly, quarterly, or annual). At each rebalancing date, the optimizer is calibrated and deployed using a fixed rolling lookback window, partitioned into an estimation window and a validation window. Hyperparameters are selected by a grid search over this partition. Each configuration is fitted on the estimation window via `fit`, used to produce weights via `optimize`, and evaluated on the validation window via `score`, and the configuration with the best validation score is retrained on both the estimation and validation windows and finally used to produce the portfolio weights for the next out-of-sample period. These weights are held until the next rebalancing date. The window then rolls forward and the procedure repeats.

Each invocation of the backtester for a given optimizer produces a self-contained run directory. A new `run_<n>` folder is created in `results/runs/<optimizer_name>/` and collects the full record of that run: the out-of-sample return series `oos_returns.csv`, the performance metrics `metrics.json`, a cumulative-return plot `cumulative_returns_plot.png`, the hyperparameters selected at each rebalancing date `selected_params.csv`, the asset and currency weights `weights_assets.csv` and `weights_currencies.csv`, and a configuration file `config.json` recording the run specification. Storing the configuration alongside its outputs makes each result reproducible and attributable to the settings that generated it.

4.2.2 Backtest Configuration

We now specify the configuration under which the considered strategies in the thesis are backtested. We first list the backtested strategies, then describe the chosen backtest specification, and finally the execution of the currency positions and the transaction costs.

We backtest the joint and the sequential currency overlay optimizers under the five risk measures considered in the thesis, variance, semi-variance, mean absolute deviation, Expected Shortfall, and second-order Expected Shortfall, each in an unregularized, a regularized, and a mean-risk formulation, as introduced in Chapter 3. As benchmarks we use the equally weighted portfolio and two regularized asset-only optimizers, Full and Zero, under the same five risk measures. EW and Full are evaluated on fully hedged returns, and the joint and overlay optimizers on fully hedged returns plus the excess currency returns of their currency positions. Zero optimizes the same objective as Full but is evaluated on unhedged returns, so it keeps the full currency exposure of its asset positions. The following portfolio strategies are considered:

- **EW**: The fully hedged equally weighted portfolio.
- **Full**: The fully hedged regularized asset portfolio with both L^1 and L^2 penalties.
- **Zero**: The unhedged regularized asset portfolio with both L^1 and L^2 penalties.
- **Overlay**: The unregularized sequential currency overlay portfolio.
- **Regularized Overlay**: The overlay portfolio with both L^1 and L^2 penalties on the asset and currency weights.
- **Mean-risk Overlay**: The unregularized overlay portfolio with an added mean term.
- **Joint**: The unregularized joint optimization portfolio.
- **Regularized Joint**: The joint optimization portfolio with both L^1 and L^2 penalties on the asset and currency weights.

- **Mean-risk Joint**: The unregularized joint optimization portfolio with an added mean term.

Portfolios are rebalanced on the last trading day of each quarter. The first portfolio is formed on 30 June 2000, and the last on 30 December 2022, which gives 91 rebalancing dates. The lookback window covers the ten years before each rebalancing date, approximately 2,500 daily observations. All strategies invest in the constituents of the current year's universe that have a complete return history over the lookback window.

The penalty weights of the regularized optimizers are selected at every rebalancing date by a grid search, for which the lookback window is partitioned into a nine-year estimation window and a one-year validation window. Each configuration is fitted on the estimation window, and the resulting weights are scored by the risk measure they realize on the validation window. The configuration with the lowest validation risk is then re-fitted on both the estimation and validation windows, so that the final weights also use the most recent year of data. More details on the grid search and the hyperparameter range analysis are given in Section G.1.

The mean-risk optimizers also require expected asset and currency returns, which are obtained at each rebalancing date from the lookback window. Since individual stock means are hard to estimate accurately, we combine them by economy, as derived in Chapter 3. That is, the means of the equally weighted economy returns are shrunk toward zero with a James–Stein estimator, and every asset inherits the shrunk mean of its economy. The expected excess currency returns are not estimated from returns but set to the negative forward premium at the end of the lookback window. Due to computational complexity considerations, the risk aversion coefficient γ is not selected on the validation window. Instead, we run each mean-risk optimizer over a grid of γ values on the first lookback window between July 1990 and June 2000 and fix one value for each risk measure. More details on the risk aversion parameter selection can be found in Section G.3.

The currency positions of the hedged strategies are implemented with three-month forward contracts. Since each contract expires at the next rebalancing date, the full forward position is traded at every rebalancing date rather than only its change, which is why the currency turnover (4.1) in Section 4.2.3 equals the notional of the forward positions. All returns are reported net of transaction costs. We assume constant half-spreads of 10 basis points for equities and 1 basis point for currencies, which gives transaction costs of $\tau^{asset} \approx 0.002$ and $\tau^{curr} \approx 0.0002$ per unit traded. We derive the transaction costs in detail in Appendix F.

4.2.3 Performance Evaluation Metrics

In Chapter 5 we evaluate the performance of all considered portfolio optimization strategies on a common set of performance metrics computed from their out-of-sample return series, each capturing a different view of the portfolio's performance. How well the joint and overlay programs reduce the risk measure they minimize is analyzed separately in Chapter 6. Throughout this section, let T be the number of trading days in the out-of-sample period, and let $R_{\mathcal{P},t}$ denote the realized daily out-of-sample portfolio return, $\bar{R}_{\mathcal{P}}$ its sample mean, $\sigma_{\mathcal{P}}$ its annualized volatility, $\sigma_{\mathcal{P},\text{down}}$ its annualized downside volatility, $R_{f,t}$ the daily risk-free rate, and $\bar{R}_f$ the average daily risk-free rate. Daily figures are annualized using the standard 252 trading days per year assumption. The metrics we report are the annualized excess return, annualized volatility, maximum drawdown, the Sharpe ratio, the Sortino ratio, and average asset and currency turnover.

The annualized excess return is the compound annual growth rate of the portfolio in excess of

that of the risk-free rate

$$\mathrm{ER}_{\mathcal{P}} = \left(\prod_{t=1}^{T}(1+R_{\mathcal{P},t})\right)^{252/T} - \left(\prod_{t=1}^{T}(1+R_{f,t})\right)^{252/T}.$$

It is the annual return an investor holding the portfolio over the full sample would have earned on top of the risk-free rate. Since the risk-minimizing optimizers do not target returns, inspecting the excess return shows how much return they give up in exchange for lower risk.

Annualized volatility is the standard deviation of portfolio returns

$$\sigma_{\mathcal{P}} = \sqrt{252}\sqrt{\frac{1}{T-1}\sum_{t=1}^{T}\left(R_{\mathcal{P},t}-\bar{R}_{\mathcal{P}}\right)^2}.$$

It is the most common measure of risk used to evaluate portfolio performance and is also used in the calculation of the Sharpe ratio to measure risk-adjusted performance. Because a portfolio can achieve low volatility while still suffering severe cumulative losses over time, it is most informative when interpreted together with the maximum drawdown.

Maximum drawdown is the largest decline of the cumulative return path of the portfolio. Unlike volatility, which summarizes a one-period return distribution, drawdown is a property of the realized wealth path, capturing the cumulative and sequential experience of losses that an investor goes through. Such path-dependent losses matter in practice. For example, deep drawdowns can breach margin requirements. Importantly, none of the optimizers considered in the thesis targets drawdown directly, so it serves as an out-of-sample robustness check. Let $V_{\mathcal{P},t} = \prod_{s\le t}(1+R_{\mathcal{P},s})$ denote the cumulative wealth of the portfolio, with $V_{\mathcal{P},0} = 1$, and $P_{\mathcal{P},t} = \max_{s\le t} V_{\mathcal{P},s}$ its running maximum. The maximum drawdown is defined as

$$\mathrm{MD}_{\mathcal{P}} = \max_{t}\frac{P_{\mathcal{P},t}-V_{\mathcal{P},t}}{P_{\mathcal{P},t}}.$$

The Sharpe ratio is the annualized ratio of the average excess portfolio return relative to its total volatility. It is the standard measure of return per unit of total risk. The Sharpe ratio is given by

$$\mathrm{SR}_{\mathcal{P}} = 252\,\frac{\bar{R}_{\mathcal{P}}-\bar{R}_f}{\sigma_{\mathcal{P}}}.$$

Since the risk-minimizing optimizers run at a much lower volatility than the equally weighted portfolio, their returns alone cannot be compared fairly. The Sharpe ratio corrects for this, and a strategy with a higher Sharpe ratio earns a higher return at any level of risk an investor chooses. Like volatility, however, it penalizes favorable and adverse deviations equally.

The Sortino ratio replaces volatility in the Sharpe ratio with downside volatility, so that a strategy is not penalized for upside dispersion. Read against the Sharpe ratio, it indicates whether a strategy's risk-adjusted performance stems from downside protection or from lower overall variability. The Sortino ratio is given by

$$\mathrm{SOR}_{\mathcal{P}} = 252\,\frac{\bar{R}_{\mathcal{P}}-\bar{R}_f}{\sigma_{\mathcal{P},\mathrm{down}}},$$

where the annualized downside volatility

$$\sigma_{\mathcal{P},\mathrm{down}} = \sqrt{252}\sqrt{\frac{1}{T}\sum_{t=1}^{T}\min\left(R_{\mathcal{P},t},\,0\right)^2}$$

only considers negative returns.

Finally, turnover proxies the trading activity and the implementation costs of each strategy. Although the reported returns are already net of transaction costs, turnover remains informative because it measures how much the portfolio must trade to maintain its positions, its capacity to scale, and its exposure to frictions such as slippage and market impact that our analysis does not capture. A strategy whose performance depends on heavy rebalancing can be hard to implement, so low and stable turnover is a desirable property. Turnover is reported separately for the asset and currency legs, since the two are traded through different instruments with different cost levels. Asset turnover is the average rebalancing volume over the out-of-sample period

$$\text{Turnover}^a = \frac{1}{D}\sum_{d=1}^{D}\sum_{i=1}^{N}|x_{i,d} - x_{i,d}^{-}|,$$

where D is the number of out-of-sample rebalancing dates, $x_{i,d}^{-}$ the weight of asset i just before the rebalance at date d, and $x_{i,d}$ the weight set at that rebalance. Currency turnover is the average absolute notional of the forward positions

$$\text{Turnover}^c = \frac{1}{D}\sum_{d=1}^{D}\sum_{c=1}^{M}|\phi_{c,d}|. \tag{4.1}$$

Chapter 5

Empirical Analysis

In this chapter, we analyze the out-of-sample performance of the joint and sequential overlay programs across the five risk measures and three universe sizes introduced in Chapters 3 and 4. In Section 5.1 we report the baseline backtests of the unregularized joint and overlay programs, together with the equally weighted benchmark. In Section 5.2 we repeat the analysis for the regularized joint and overlay programs and compare them against two regularized asset-only benchmarks introduced in Section 4.2.2. In Section 5.3 we add a mean term to the objectives of the unregularized programs and analyze how it changes their performance.

5.1 Baseline Backtest Results

We first report the baseline backtests of the unregularized joint and overlay optimizers across the five risk measures and three universe sizes considered in the thesis. We compare them against the equally weighted portfolio, which serves as the no-optimization benchmark. For Expected Shortfall and second-order Expected Shortfall, we use the 80% confidence level throughout the chapter.

For the small, medium, and large universes, we report in Table 5.1 the out-of-sample annualized excess return over the risk-free rate, annualized volatility, maximum drawdown, Sharpe ratio, Sortino ratio, average asset turnover, and average currency turnover of the overlay and joint optimizers and of the equally weighted portfolio. All quantities are computed on the out-of-sample excess return series produced by the backtest configuration presented in Section 4.2.2. We also report in Figures 5.1 to 5.3 the cumulative out-of-sample returns of the joint optimizers together with the equally weighted benchmark, and in the lower panel of each figure the cumulative difference between the joint and overlay optimizers for each risk measure. The latter helps us visualize the realized return difference between the two optimization approaches over time. In Section G.5, we also report the realized semi-deviation, mean absolute deviation, Expected Shortfall, and second-order Expected Shortfall of all optimizers in this chapter.

Across all five risk measures and universe sizes, the overlay and joint optimizers achieve substantially lower volatility than the equally weighted portfolio. In the small universe, their volatility ranges from 7.71% to 9.17%, against 13.96% for the equally weighted portfolio, a reduction of 34% to 45%. In the medium and large universes, the reduction is even larger, by up to 59% and 64%, respectively. The other four risk measures in Section G.5 fall by a similar amount. The equally weighted portfolio ignores the dependence between asset returns. Since it already holds a broad cross-section of stocks, most idiosyncratic risk is diversified away, and further adding stocks does little to reduce the remaining market risk, which can only be addressed by taking the correlation structure of the assets and currencies

Panel A: Small Investable Universe (top 10 stocks per economy)											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	1.77	1.12	1.25	0.67	0.61	1.26	1.39	0.61	0.62	0.69	0.19
Vol (%)	13.96	7.97	7.73	8.15	7.84	7.87	7.71	8.18	7.97	9.17	8.72
MD (%)	54.77	28.23	27.92	32.17	32.63	27.67	27.31	29.31	28.31	34.43	35.38
SR	0.13	0.14	0.16	0.08	0.08	0.16	0.18	0.08	0.08	0.07	0.02
SOR	0.17	0.19	0.22	0.11	0.11	0.22	0.25	0.10	0.11	0.10	0.03
Turnover^a	0.30	0.87	0.82	0.99	0.97	1.10	1.06	1.30	1.27	1.38	1.47
Turnover^c	0.92	1.70	1.88	2.13	2.38	1.51	1.84	2.39	2.98	2.66	3.46
Panel B: Medium Investable Universe (top 50 stocks per economy)											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	4.09	2.41	2.53	2.37	2.54	2.75	2.65	3.16	2.95	2.39	2.70
Vol (%)	12.50	5.17	5.09	5.27	5.16	5.73	5.56	5.68	5.53	5.86	5.64
MD (%)	55.75	35.85	34.90	37.88	36.71	35.93	36.94	38.59	38.84	35.05	33.09
SR	0.33	0.47	0.50	0.45	0.49	0.48	0.48	0.56	0.53	0.41	0.48
SOR	0.44	0.62	0.67	0.60	0.66	0.69	0.68	0.75	0.73	0.55	0.66
Turnover^a	0.27	1.05	1.05	1.25	1.26	1.66	1.68	1.78	1.79	2.13	2.23
Turnover^c	0.92	1.16	1.37	1.20	1.36	0.98	1.25	0.97	1.67	1.02	1.85
Panel C: Large Investable Universe (top 100 stocks per economy)											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	4.26	2.74	2.79	2.73	2.76	2.44	2.40	3.28	3.44	2.39	2.46
Vol (%)	11.66	4.24	4.19	4.26	4.20	4.63	4.57	4.57	4.55	4.67	4.66
MD (%)	56.26	26.72	26.37	28.74	28.29	31.25	31.42	31.32	30.46	31.29	30.31
SR	0.37	0.65	0.66	0.64	0.66	0.53	0.52	0.72	0.76	0.51	0.53
SOR	0.49	0.85	0.88	0.84	0.87	0.70	0.70	0.95	1.01	0.68	0.70
Turnover^a	0.28	1.09	1.09	1.34	1.36	1.76	1.77	1.89	1.91	2.15	2.18
Turnover^c	0.91	1.06	1.24	1.07	1.21	0.97	1.16	0.97	1.43	0.99	1.62

Table 5.1: Out-of-sample performance of the unregularized overlay (Overlay) and joint (Joint) optimizers and the equally weighted portfolio (EW) in the small (Panel A), medium (Panel B), and large (Panel C) investable universes. Columns are grouped by the minimized risk measure: variance, semi-variance, mean absolute deviation (MAD), Expected Shortfall (ES), and second-order Expected Shortfall (SOES). The metrics are the annualized excess return over the risk-free rate (ER), volatility (Vol), maximum drawdown (MD), Sharpe ratio (SR), Sortino ratio (SOR), and average asset (Turnover^a) and currency (Turnover^c) turnover. Portfolios are rebalanced quarterly, returns are net of transaction costs, and the out-of-sample period runs from July 2000 to April 2023. Additionally, the realized semi-deviation, MAD, ES, and SOES for the optimizers are reported in Table G.3.

into account. The optimizers instead tilt the portfolio toward assets with low risk, and a larger universe offers more such assets to choose from. As a result, maximum drawdowns are also substantially reduced, by up to 27 percentage points in the small, 23 in the medium, and 30 in the large universe, relative to drawdowns of 54.77% to 56.26% for the equally weighted portfolio. Large drawdowns occur in market crashes, when the correlations between assets rise and diversification offers less protection. The optimizers reduce the portfolio's exposure to these crashes but cannot completely mitigate them, since in a crash even low-risk equities tend to fall together, which is why the drawdowns remain sizeable for all strategies. Section G.4 shows the drawdowns over the whole out-of-sample period and compares the optimizers in the main market crashes of the sample.

For excess returns, we observe the opposite effect. The overlay and joint optimizers deliver lower excess returns than the equally weighted portfolio for every risk measure and universe size. This is partially due to the fact that the optimizers are minimizing risk without any

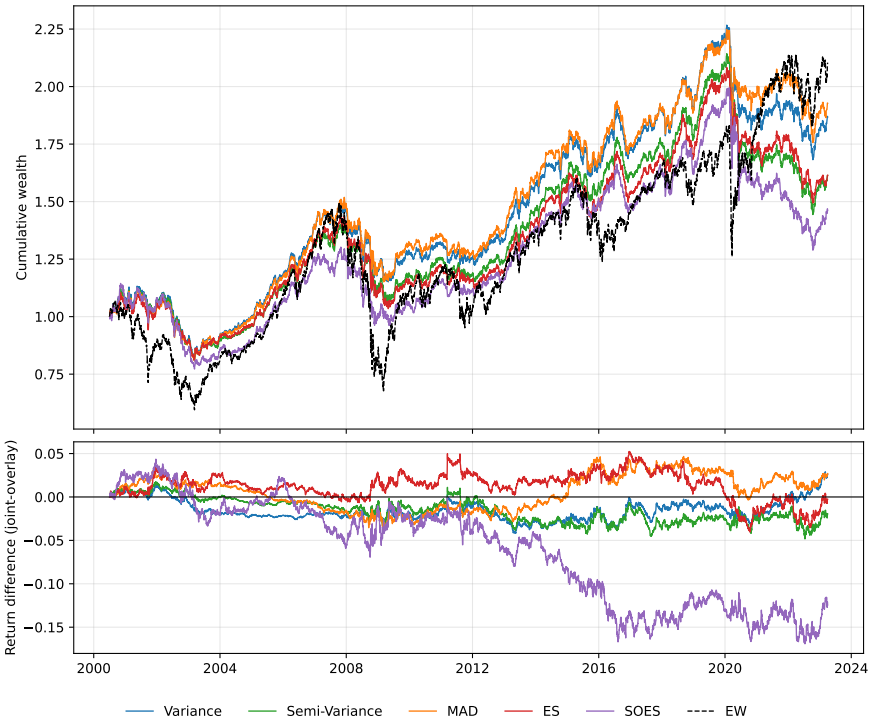


Figure 5.1: Cumulative performance of the unregularized joint optimizers and their return difference to the overlay optimizers in the small investable universe (top 10 stocks per economy). The top panel shows the cumulative wealth of the joint optimizer of each risk measure and of the equally weighted portfolio (EW), and the bottom panel the cumulative sum of the daily return differences between the joint and the overlay optimizers. Portfolios are rebalanced quarterly, and returns are net of transaction costs. The performance metrics are reported in Panel A of Table 5.1.

explicit return term in the objective function. We address this in more detail in Section 5.3. Secondly, much of the volatility reduction achieved by the risk-minimizing portfolios may come from a lower exposure to the common movement of the equity markets, in which case the optimizers also earn a smaller share of the equity risk premium. But excess returns alone say little about performance when the portfolios carry different levels of risk, so we compare them on a risk-adjusted basis.

In the small universe, the Sharpe ratios of the optimizers range from 0.02 to 0.18, against 0.13 for the equally weighted portfolio, and only the variance and mean absolute deviation optimizers improve on the benchmark. In the medium and large universes, every optimizer outperforms the equally weighted portfolio, with Sharpe ratios between 0.41 and 0.56 against 0.33, and between 0.51 and 0.76 against 0.37, respectively. In the small universe, the optimizers lose more excess return than they reduce volatility, while in larger universes they can diversify across more low-risk stocks, and the reduction in volatility outweighs the lost return. Sortino ratios follow the same pattern, so accounting only for downside risk does not change the comparison.

Both the overlay and joint optimizers have higher asset and currency turnover than the equally weighted portfolio. This is expected since the equally weighted portfolio has fixed target weights and only trades to undo the price drift. Comparing the asset turnover of the joint and overlay optimizers, we see that it is very similar, differing at most by 7%. This suggests that the asset weights are determined primarily by the risk measure being optimized and not by the optimization approach itself. The difference between the two optimizers is in currency turnover, where that of the joint optimizer is higher than that of the overlay optimizer in every measure-universe combination, by between 11% and 82%. Since returns are reported net

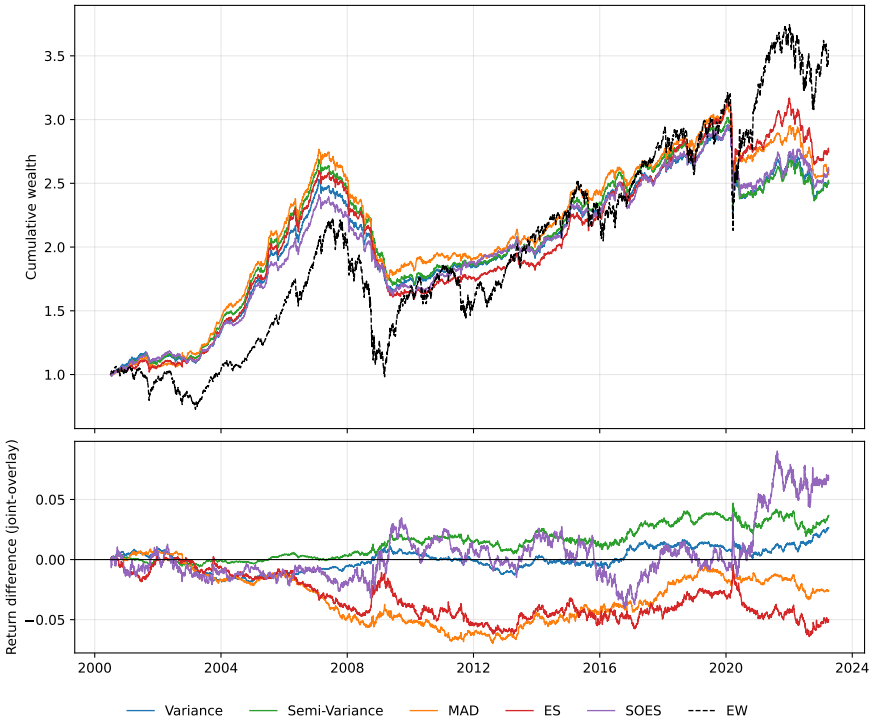


Figure 5.2: Cumulative performance of the unregularized joint optimizers and their return difference to the overlay optimizers in the medium investable universe (top 50 stocks per economy). Panels are as in Figure 5.1. The performance metrics are reported in Panel B of Table 5.1.

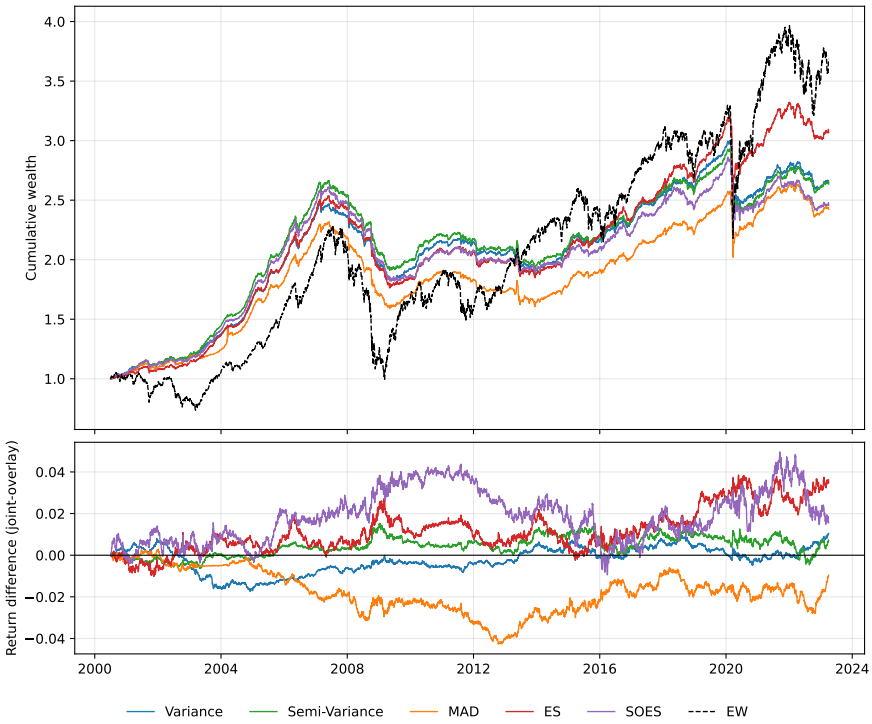


Figure 5.3: Cumulative performance of the unregularized joint optimizers and their return difference to the overlay optimizers in the large investable universe (top 100 stocks per economy). Panels are as in Figure 5.1. The performance metrics are reported in Panel C of Table 5.1.

of transaction costs, the comparisons above already incorporate these differences. The extra currency turnover does not reduce the risk-adjusted returns of the joint optimizer relative to the overlay optimizer, and the higher turnover of both optimizers does not change their advantage over the equally weighted portfolio.

Finally, comparing the performance of the unregularized joint and overlay optimizers under the same risk measures, across all reported metrics, we see that they perform very similarly. The joint optimizer does consistently achieve a lower volatility, but the differences shrink as we increase the universe size, from up to a 5% decrease in the small universe to essentially no difference in the large universe. This is in line with the joint program being able to exploit the asset and currency dependence structure when the number of assets compared to the number of currencies is smaller. In Chapter 6, we study in more detail how much the joint program reduces the minimized risk measure relative to the overlay program, and why. The other metrics differ by small amounts, with excess returns differing by at most 0.5 percentage points, maximum drawdowns by at most 1.96 percentage points, and Sharpe ratios by at most 0.07. However, the sign of these differences changes across risk measures within every universe, so one optimization approach does not consistently outperform the other.

Figures 5.1 to 5.3 present the cumulative performance of the overlay and joint optimizers alongside the equally weighted benchmark. In each universe, the top panel shows that the joint optimizers perform better than the equally weighted portfolio during the first major market crashes in the sample, the dot-com crash and the global financial crisis. This advantage is particularly evident in the medium and large universes during the global financial crisis, where the optimizer strategies exhibit a much smaller drawdown compared to the equally weighted portfolio. The equally weighted portfolio recovers quickly after the COVID-19 market shock and ultimately attains the highest terminal wealth across all three universes, albeit at considerably higher volatility and with deeper drawdowns throughout the sample. In contrast, the overlay and joint optimizers follow more stable growth paths, consistent with their lower realized volatility reported in Table 5.1.

The bottom panels show the cumulative daily return difference between the joint and overlay optimizers for each risk measure, and confirm that the two optimization approaches track each other closely throughout the sample in terms of returns. The cumulative differences oscillate around zero in every universe size and for every measure, with the exception of second-order Expected Shortfall in the small universe, where the joint optimizer drifts persistently below the overlay optimizer from the early 2010s onward.

Overall, we see a consistent pattern across all the considered risk measures. The overlay and joint optimizers reduce volatility and drawdowns substantially relative to the equally weighted portfolio. The Sharpe comparison depends on the universe size. In the small universe, the joint and overlay optimizers do not consistently outperform the equally weighted portfolio, while in the medium and large universes they do. The joint and overlay optimizers produce very similar out-of-sample metrics across all risk measures and universe sizes, with the joint optimizer consistently delivering lower volatility and higher currency turnover but no consistent advantage on risk-adjusted returns.

5.2 Regularization

We now repeat the analysis of Section 5.1 for the regularized versions of the joint and overlay optimizers. We add L^1 and L^2 penalties on the asset weights and, separately, on the currency weights to the objective, following the regularization scheme of Chapter 3. The penalty weights are selected at every rebalancing date by a grid search over the validation window described in Section 4.2.2. All other aspects of the backtests remain unchanged. We also compare

	Overlay			Joint		
Risk Measure	**Small**	**Medium**	**Large**	**Small**	**Medium**	**Large**
Panel A: Absolute Change						
Variance	+0.02	−0.01	+0.12	0.00	−0.04	+0.09
Semi-Variance	+0.05	+0.02	+0.19	+0.03	−0.03	+0.18
MAD	+0.04	−0.02	+0.02	+0.03	+0.01	+0.04
ES	+0.06	−0.01	+0.15	+0.09	+0.02	+0.13
SOES	+0.04	+0.12	+0.38	−0.03	+0.08	+0.38
Average	+0.04	+0.02	+0.17	+0.02	+0.01	+0.17
Panel B: Relative Change (%)						
Variance	+17	−3	+19	−2	−9	+14
Semi-Variance	+60	+4	+30	+39	−6	+28
MAD	+28	−5	+4	+17	+2	+7
ES	+78	−1	+21	+113	+3	+18
SOES	+53	+29	+73	−117	+17	+72
Average	+41	+4	+28	+24	+1	+26

Table 5.2: Change in the out-of-sample Sharpe ratio of the overlay and joint optimizers from regularization in the small, medium, and large investable universes. Panel A reports the absolute change, the Sharpe ratio of each regularized optimizer in Tables 5.3 to 5.5 minus that of its unregularized version in Table 5.1, and Panel B the relative change in percent of the unregularized Sharpe ratio. Rows correspond to the minimized risk measure, and the last row of each panel reports the change of the average Sharpe ratio across the five risk measures.

the joint and overlay optimizers against the regularized Assets-Full (Full) and Assets-Zero (Zero) optimizers, which serve as benchmarks for optimizing only the asset weights under the same risk measures, with the currency weights either fully hedged or set to zero, respectively. Table 5.2 reports the change in the Sharpe ratio between the unregularized and regularized optimizers, and Tables 5.3 to 5.5 report performance metrics of the regularized optimizers and the three benchmarks.

We see in Table 5.2 that regularization improves the Sharpe ratios of the overlay and joint optimizers for most risk measures and universe sizes. In absolute terms, the improvement is the largest in the large universe, where the Sharpe ratio rises by 0.17 on average, while the effect is modest in the small universe and mixed in the medium universe. Relative to the low unregularized Sharpe ratios of the small universe, the gain there is substantial as well, 41% on average for the overlay optimizers and 24% for the joint optimizers, against 28% and 26% in the large universe and at most 4% in the medium universe. The penalties stabilize the weights by limiting the extreme positions that the unregularized programs fit to noise in the lookback window, which becomes more important when the number of assets increases relative to the length of the lookback window. The unregularized second-order Expected Shortfall optimizers are the least stable, since the measure is driven by the squared losses of the few most extreme observations in the lookback window. Accordingly, regularization benefits them the most, raising their Sharpe ratios by 0.38 in the large universe.

Comparing the performance of the Full optimizer with the overlay and joint optimizers in Tables 5.3 to 5.5, we see that actively trading the currencies reduces risk relative to fully hedging the currency exposure, without giving up risk-adjusted returns. Full solves the same asset-only problem as the first step of the overlay program, and the regularized overlay and joint optimizers realize a lower value of the risk measure they minimize than Full for every

Panel A1: Regularized Overlay and Joint Optimizers										
	Variance		Semi-Variance		MAD		ES		SOES	
Metric	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	1.29	1.24	1.05	0.84	1.60	1.64	1.07	1.31	0.98	0.03
Vol (%)	7.82	7.76	7.91	7.72	7.81	7.75	7.99	7.85	8.52	8.64
MD (%)	27.81	27.80	30.63	30.43	27.83	26.01	29.96	28.89	36.02	36.22
SR	0.16	0.16	0.13	0.11	0.20	0.21	0.13	0.17	0.11	0.00
SOR	0.22	0.22	0.18	0.15	0.28	0.29	0.18	0.23	0.16	0.00
Turnover^a	0.93	0.96	1.09	1.01	1.04	1.06	1.07	1.17	1.40	1.59
Turnover^c	1.43	1.47	1.72	1.77	1.36	1.44	1.70	1.77	2.04	2.12

Panel B1: Benchmarks											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**
ER (%)	1.77	1.05	2.64	1.08	2.65	1.66	3.13	0.85	2.44	0.55	1.93
Vol (%)	13.96	8.39	11.80	8.41	11.72	8.36	11.70	8.51	11.84	9.12	12.35
MD (%)	54.77	28.00	42.18	29.21	41.12	28.59	40.88	30.81	44.28	33.06	44.84
SR	0.13	0.12	0.22	0.13	0.23	0.20	0.27	0.10	0.21	0.06	0.16
SOR	0.17	0.17	0.30	0.17	0.31	0.27	0.36	0.13	0.28	0.08	0.21
Turnover^a	0.30	1.02	1.02	1.16	1.17	1.08	1.08	1.27	1.23	1.66	1.67
Turnover^c	0.92	0.93	0	0.93	0	0.92	0	0.93	0	0.94	0

Table 5.3: Out-of-sample performance of the regularized overlay (Overlay) and joint (Joint) optimizers (Panel A1) and of the benchmarks (Panel B1) in the small investable universe (top 10 stocks per economy). The benchmarks are the equally weighted portfolio (EW) and the regularized asset-only optimizers Assets-Full (Full), which fully hedges the currency exposure of its assets, and Assets-Zero (Zero), which leaves it unhedged. Columns are grouped by the minimized risk measure: variance, semi-variance, mean absolute deviation (MAD), Expected Shortfall (ES), and second-order Expected Shortfall (SOES). The metrics are the annualized excess return over the risk-free rate (ER), volatility (Vol), maximum drawdown (MD), Sharpe ratio (SR), Sortino ratio (SOR), and average asset (Turnover^a) and currency (Turnover^c) turnover. Portfolios are rebalanced quarterly, returns are net of transaction costs, and the out-of-sample period runs from July 2000 to April 2023. Additionally, the realized semi-deviation, MAD, ES, and SOES for the optimizers are reported in Table G.4.

Panel A2: Regularized Overlay and Joint Optimizers										
	Variance		Semi-Variance		MAD		ES		SOES	
Metric	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	2.38	2.35	2.41	2.33	2.83	2.95	3.16	3.08	2.91	3.10
Vol (%)	5.26	5.19	5.16	5.08	6.16	6.08	5.74	5.58	5.52	5.52
MD (%)	35.03	34.25	36.39	34.57	35.93	35.52	38.57	36.38	30.41	30.20
SR	0.45	0.45	0.47	0.46	0.46	0.49	0.55	0.55	0.53	0.56
SOR	0.59	0.59	0.62	0.61	0.75	0.79	0.77	0.76	0.71	0.76
Turnover^a	1.14	1.13	1.24	1.27	1.31	1.32	1.53	1.57	1.56	1.97
Turnover^c	1.11	1.11	1.18	1.17	1.09	1.06	1.26	1.30	1.39	1.32

Panel B2: Benchmarks											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**
ER (%)	4.09	2.52	3.86	2.55	3.92	2.94	3.84	3.26	4.54	3.02	4.48
Vol (%)	12.50	5.31	9.58	5.22	9.43	6.19	9.97	5.84	9.87	5.60	9.78
MD (%)	55.75	35.05	42.05	36.53	41.31	35.90	43.81	38.54	42.37	29.76	36.74
SR	0.33	0.48	0.40	0.49	0.42	0.48	0.38	0.56	0.46	0.54	0.46
SOR	0.44	0.62	0.55	0.65	0.57	0.77	0.57	0.78	0.64	0.72	0.63
Turnover^a	0.27	1.07	1.07	1.34	1.35	1.30	1.30	1.60	1.60	1.81	1.81
Turnover^c	0.92	0.95	0	0.95	0	0.96	0	0.95	0	0.95	0

Table 5.4: Out-of-sample performance of the regularized optimizers (Panel A2) and the benchmarks (Panel B2) in the medium investable universe (top 50 stocks per economy). The layout and reported metrics are as in Table 5.3.

Panel A3: Regularized Overlay and Joint Optimizers										
	Variance		Semi-Variance		MAD		ES		SOES	
Metric	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ER (%)	3.33	3.25	3.56	3.52	2.72	2.73	3.88	3.91	4.05	4.09
Vol (%)	4.35	4.29	4.27	4.19	4.97	4.85	4.45	4.40	4.57	4.52
MD (%)	26.27	26.40	27.33	27.29	31.28	31.63	30.61	30.11	25.84	24.70
SR	0.77	0.76	0.83	0.84	0.55	0.56	0.87	0.89	0.89	0.91
SOR	0.99	0.98	1.09	1.10	0.73	0.75	1.14	1.17	1.17	1.20
Turnover^a	1.14	1.10	1.40	1.33	1.60	1.56	1.42	1.64	1.65	1.87
Turnover^c	1.06	1.03	1.06	1.06	1.01	1.03	1.15	1.23	1.23	1.26

Panel B3: Benchmarks											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**
ER (%)	4.26	3.16	4.40	3.56	4.94	2.83	4.13	4.00	5.28	3.75	4.92
Vol (%)	11.66	4.37	8.99	4.28	8.92	4.94	9.36	4.46	9.21	4.57	9.28
MD (%)	56.26	26.60	33.06	27.84	34.27	31.28	36.99	31.60	35.12	30.41	35.50
SR	0.37	0.72	0.49	0.83	0.55	0.57	0.44	0.90	0.57	0.82	0.53
SOR	0.49	0.93	0.68	1.09	0.77	0.77	0.62	1.18	0.80	1.09	0.74
Turnover^a	0.28	1.10	1.10	1.35	1.35	1.59	1.59	1.61	1.62	1.84	1.84
Turnover^c	0.91	0.96	0	0.97	0	0.97	0	0.96	0	0.97	0

Table 5.5: Out-of-sample performance of the regularized optimizers (Panel A3) and the benchmarks (Panel B3) in the large investable universe (top 100 stocks per economy). The layout and reported metrics are as in Table 5.3.

risk measure and universe size, and a lower volatility in almost every case (see also Table G.4). The reduction is substantial in the small universe, up to 13% for the minimized risk measure and 8% for volatility, but becomes smaller as the universe grows and stays below 3% in the large universe. At the same time, the Sharpe ratios of the overlay and joint optimizers are on par with those of Full in every universe, slightly higher for some risk measures and slightly lower for others. A likely reason why the lower risk does not also raise the risk-adjusted returns is that the objective contains no mean term, so the currency weights are chosen only to minimize the risk measure and not to earn the excess currency returns. Looking at the currency turnover, we see that active currency trading is the mechanism through which the risk reduction is achieved. In the medium and large universes, the currency turnover of the overlay and joint optimizers is only 4% to 46% higher than that of Full, so their currency positions deviate only moderately from a full hedge, whereas in the small universe it is 1.5 to 2.3 times as high. On the other hand, the Zero optimizer keeps the full currency exposure and with it the excess currency returns that the risk-minimizing optimizers give up, and earns a higher excess return at a much higher volatility. In the small universe this gives Zero the higher Sharpe ratio for every risk measure, while in the medium and large universes the overlay and joint optimizers outperform it for every risk measure, by up to 70%. This performance advantage of the risk-minimizing optimizers over Zero shows the benefits of active currency management in larger universes, where equity risk is reduced further and currency risk makes up a larger share of the portfolio's total risk.

After regularization, the performance of the joint and overlay optimizers remains very similar. Their Sharpe ratios in Tables 5.3 to 5.5 differ only marginally for every risk measure and universe size, with the sign of the difference changing across risk measures within each universe. The joint optimizers again achieve a lower volatility in every measure-universe combination, the only exception being second-order Expected Shortfall in the small universe, where the overlay optimizer beats the joint optimizer on every metric. The bottom panel of Figure 5.4 shows that this happens almost entirely after the onset of the COVID-19 crisis in 2020, so it is

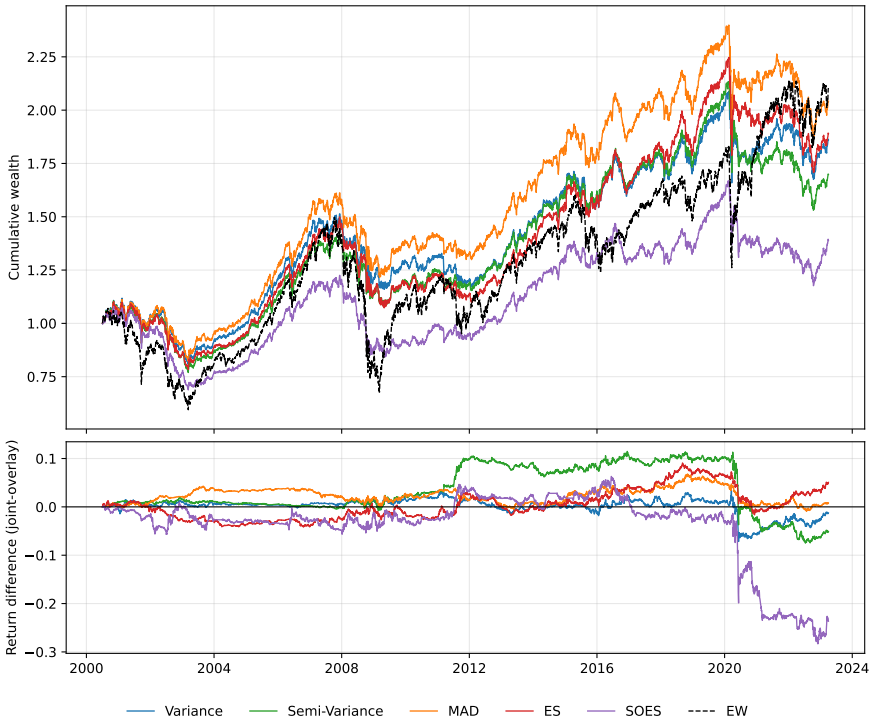


Figure 5.4: Cumulative performance of the regularized joint optimizers and their return difference to the overlay optimizers in the small investable universe (top 10 stocks per economy). The top panel shows the cumulative wealth of the joint optimizer of each risk measure and of the equally weighted portfolio (EW), and the bottom panel the cumulative sum of the daily return differences between the joint and the overlay optimizers. Portfolios are rebalanced quarterly, and returns are net of transaction costs. The performance metrics are reported in Table 5.3.

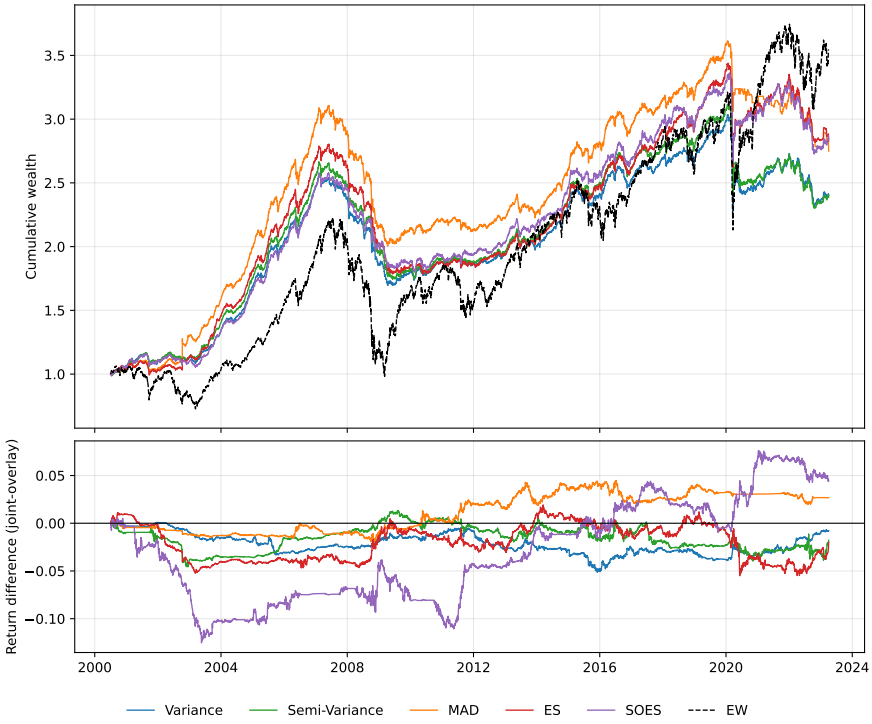


Figure 5.5: Cumulative performance of the regularized joint optimizers and their return difference to the overlay optimizers in the medium investable universe (top 50 stocks per economy). Panels are as in Figure 5.4. The performance metrics are reported in Table 5.4.

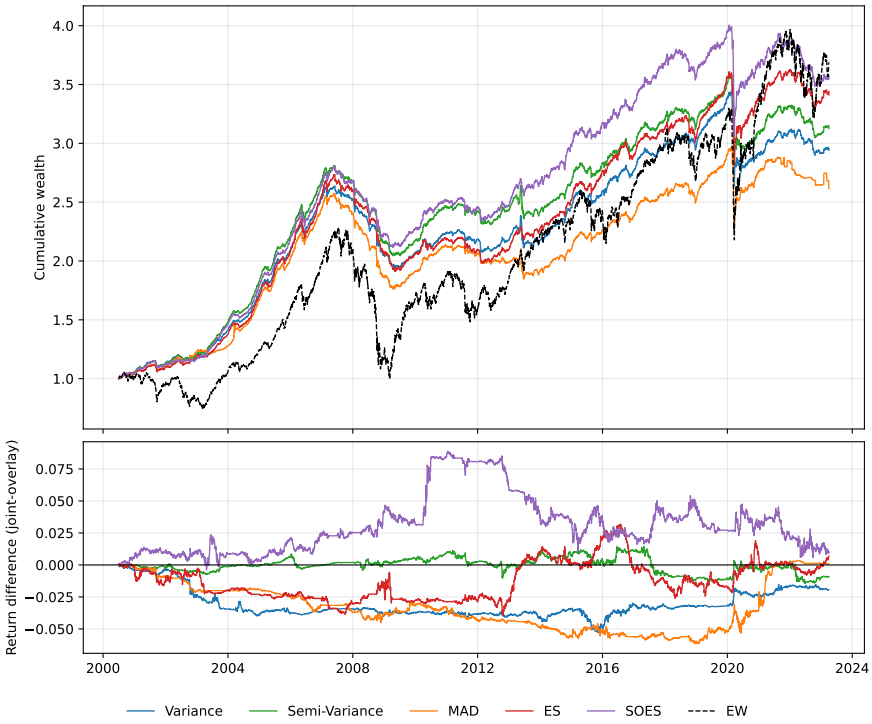


Figure 5.6: Cumulative performance of the regularized joint optimizers and their return difference to the overlay optimizers in the large investable universe (top 100 stocks per economy). Panels are as in Figure 5.4. The performance metrics are reported in Table 5.5.

reflective of this period rather than a persistent difference. The bottom panels of Figures 5.5 and 5.6 show that the cumulative return differences in the medium and large universes remain small for all risk measures throughout the sample. The main effect of regularization on the comparison is on currency turnover. Without regularization, the currency turnover of the joint optimizer exceeds that of the overlay optimizer by 11% to 82%, while with regularization the two differ by at most 7%. The penalties on the currency weights thus shrink the larger currency positions of the joint program to about the size of those of the overlay program. This does not have an effect on their risk-adjusted returns, which remain very similar, with no optimizer consistently outperforming the other.

The results so far share a common feature. All optimizers minimize a risk measure without any return term in the objective, and while this delivers substantially lower volatility and drawdowns than the equally weighted portfolio, it also gives up potential excess asset and currency returns. A natural extension is therefore to add a mean term to the objective, which lets the optimizers trade off expected return against risk. A risk-return objective is also where joint optimization could be more beneficial, as the joint program can account for the dependence structure between asset and currency returns when choosing positions that raise the expected return.

5.3 Mean-Risk Optimization

In this section, we add a mean term to the objective of the unregularized joint and overlay programs of Section 5.1. Each program now minimizes the risk measure with an added expected portfolio return scaled by $1/\gamma$, where γ is a risk aversion coefficient, as introduced in Section 3.4. In the overlay program, the asset step includes the expected asset returns and the currency step the expected excess currency returns. The expected asset returns are averaged

	Overlay			Joint		
Risk Measure	**Small**	**Medium**	**Large**	**Small**	**Medium**	**Large**
Panel A: Absolute Change						
Variance	+0.23	+0.25	+0.23	+0.21	+0.24	+0.25
Semi-Variance	+0.30	+0.16	+0.12	+0.30	+0.16	+0.18
MAD	+0.18	+0.15	+0.14	+0.19	+0.21	+0.17
ES	+0.04	0.00	0.00	+0.05	+0.05	+0.05
SOES	+0.09	+0.04	+0.02	+0.10	+0.14	+0.09
Deviation measures	+0.24	+0.19	+0.16	+0.24	+0.20	+0.20
Tail measures	+0.07	+0.02	+0.01	+0.07	+0.10	+0.07
Panel B: Relative Change (%)						
Variance	+167	+53	+35	+130	+48	+37
Semi-Variance	+361	+36	+18	+389	+33	+28
MAD	+111	+32	+27	+108	+44	+33
ES	+55	0	0	+63	+10	+6
SOES	+120	+11	+4	+448	+30	+17
Deviation measures	+185	+40	+27	+168	+42	+33
Tail measures	+88	+4	+2	+146	+19	+10

Table 5.6: Change in the out-of-sample Sharpe ratio of the overlay and joint optimizers from adding a mean term in the small, medium, and large investable universes. Panel A reports the absolute change, the Sharpe ratio of each mean-risk optimizer in Table 5.7 minus that of the same unregularized optimizer without a mean term in Table 5.1, and Panel B the relative change in percent of the latter. Rows correspond to the risk measure in the objective, and the last two rows of each panel report the change of the average Sharpe ratio across the deviation measures (variance, semi-variance, and MAD) and across the tail measures (ES and SOES). The relative changes in the small universe are large because several Sharpe ratios without a mean term are close to zero.

across the stocks of each economy and shrunk toward zero, while the expected excess currency returns are set to the negative forward premium, that is, the difference between the foreign and the US interest rate. The choice of γ for each risk measure is described in Section G.3. In both programs, we also introduce a currency leverage constraint. Table 5.6 reports the change in the Sharpe ratio from adding the mean term, and Table 5.7 reports the performance metrics.

Looking at Table 5.6, we see that adding a mean term to the objective raises the Sharpe ratios of the overlay and joint optimizers in 28 of the 30 combinations of risk measure and universe size. Unlike regularization, which has the highest absolute gain in the large universe, the addition of a mean term improves the Sharpe ratios by a similar absolute amount in all three universes. Minimizing risk alone ignores expected returns, and the mean term recovers part of the excess return the optimizers otherwise give up, though we can see that the improvement is very uneven across risk measures. The Sharpe ratios of the deviation measures rise by 0.16 to 0.24 on average, while those of the tail measures rise by at most 0.10.

This split is in line with how the risk measures handle the mean of the portfolio return. Expected Shortfall and second-order Expected Shortfall are translation equivariant, so minimizing them already rewards portfolios with a higher mean. The two tail measures therefore carry an implicit mean term, which the explicit one only strengthens. The deviation measures are invariant to a shift in the portfolio return, so for them the addition of a mean term introduces new information in the optimization problem. The average currency turnover is consistent with this interpretation. In the medium and large universes, the mean term raises the currency

Panel A: Small Investable Universe (top 10 stocks per economy)											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric	EW	Overlay	Joint	Overlay	Joint	Overlay	Joint	Overlay	Joint	Overlay	Joint
ER (%)	1.77	3.21	3.13	3.75	3.75	2.81	3.05	0.96	1.01	1.51	1.03
Vol (%)	13.96	8.56	8.40	9.85	9.81	8.31	8.13	8.20	7.95	9.19	8.75
MD (%)	54.77	26.41	25.91	30.52	30.17	26.52	25.58	29.58	27.57	33.77	37.06
SR	0.13	0.37	0.37	0.38	0.38	0.34	0.37	0.12	0.13	0.16	0.12
SOR	0.17	0.51	0.51	0.53	0.53	0.46	0.51	0.16	0.17	0.23	0.16
Turnover^a	0.30	0.94	0.92	1.17	1.22	1.13	1.13	1.30	1.28	1.39	1.48
Turnover^c	0.92	2.36	2.41	2.65	2.61	2.14	2.25	2.15	2.35	2.25	2.42
Panel B: Medium Investable Universe (top 50 stocks per economy)											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric	EW	Overlay	Joint	Overlay	Joint	Overlay	Joint	Overlay	Joint	Overlay	Joint
ER (%)	4.09	4.54	4.66	4.78	5.21	3.59	3.82	3.15	3.26	2.67	3.54
Vol (%)	12.50	6.37	6.33	7.79	7.95	5.68	5.57	5.68	5.58	5.89	5.70
MD (%)	55.75	37.66	36.85	40.93	41.50	36.19	37.92	39.52	38.75	34.11	32.47
SR	0.33	0.71	0.74	0.61	0.66	0.63	0.69	0.55	0.58	0.45	0.62
SOR	0.44	0.96	1.00	0.83	0.90	0.87	0.95	0.75	0.79	0.61	0.86
Turnover^a	0.27	1.17	1.21	1.43	1.52	1.64	1.69	1.78	1.80	2.13	2.25
Turnover^c	0.92	2.62	2.71	3.15	3.10	1.60	2.01	0.99	1.72	1.03	1.92
Panel C: Large Investable Universe (top 100 stocks per economy)											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric	EW	Overlay	Joint	Overlay	Joint	Overlay	Joint	Overlay	Joint	Overlay	Joint
ER (%)	4.26	5.02	5.26	5.57	6.39	3.20	3.25	3.30	3.66	2.51	2.90
Vol (%)	11.66	5.73	5.76	7.36	7.59	4.79	4.68	4.59	4.57	4.74	4.72
MD (%)	56.26	31.32	30.62	37.45	37.04	31.43	33.29	31.65	30.47	31.65	29.57
SR	0.37	0.87	0.91	0.76	0.84	0.67	0.70	0.72	0.80	0.53	0.61
SOR	0.49	1.17	1.23	1.03	1.16	0.89	0.91	0.95	1.07	0.70	0.82
Turnover^a	0.28	1.24	1.29	1.56	1.65	1.74	1.85	1.89	1.92	2.14	2.20
Turnover^c	0.91	2.88	2.95	3.40	3.36	1.37	1.94	0.97	1.49	1.00	1.80

Table 5.7: Out-of-sample performance of the mean-risk overlay (Overlay) and joint (Joint) optimizers and the equally weighted portfolio (EW) in the small (Panel A), medium (Panel B), and large (Panel C) investable universes. The mean-risk optimizers are the unregularized optimizers of Table 5.1 with a mean term added to the objective. Columns are grouped by the risk measure in the objective: variance, semi-variance, mean absolute deviation (MAD), Expected Shortfall (ES), and second-order Expected Shortfall (SOES). The metrics are the annualized excess return over the risk-free rate (ER), volatility (Vol), maximum drawdown (MD), Sharpe ratio (SR), Sortino ratio (SOR), and average asset (Turnover^a) and currency (Turnover^c) turnover. Portfolios are rebalanced quarterly, returns are net of transaction costs, and the out-of-sample period runs from July 2000 to April 2023. Additionally, the realized semi-deviation, MAD, ES, and SOES for the optimizers are reported in Table G.5.

turnover of the variance and semi-variance optimizers to two to three times its previous level, while for the tail-measure optimizers it remains stable around its previous level. We analyze the implicit risk-return tradeoff of the tail measures in more detail in Section 6.2.

As we can see in Table 5.7, the higher Sharpe ratios come from substantially higher excess returns achieved by the mean-risk optimizers. The excess returns are increased by 1.9 to 3.6 percentage points for the variance and semi-variance optimizers and by 0.8 to 1.7 percentage points for the mean absolute deviation optimizers, but by at most 0.8 percentage points for the tail measures. This comes at the cost of an increase in volatility, which rises by 0.6 to 3.4 percentage points for the variance and semi-variance optimizers, while for mean absolute deviation and the tail measures it changes by less than half a percentage point. Even with the increase, the volatility of the variance and semi-variance optimizers remains at 50% to 70% of that of the equally weighted portfolio, and the maximum drawdowns of all mean-risk

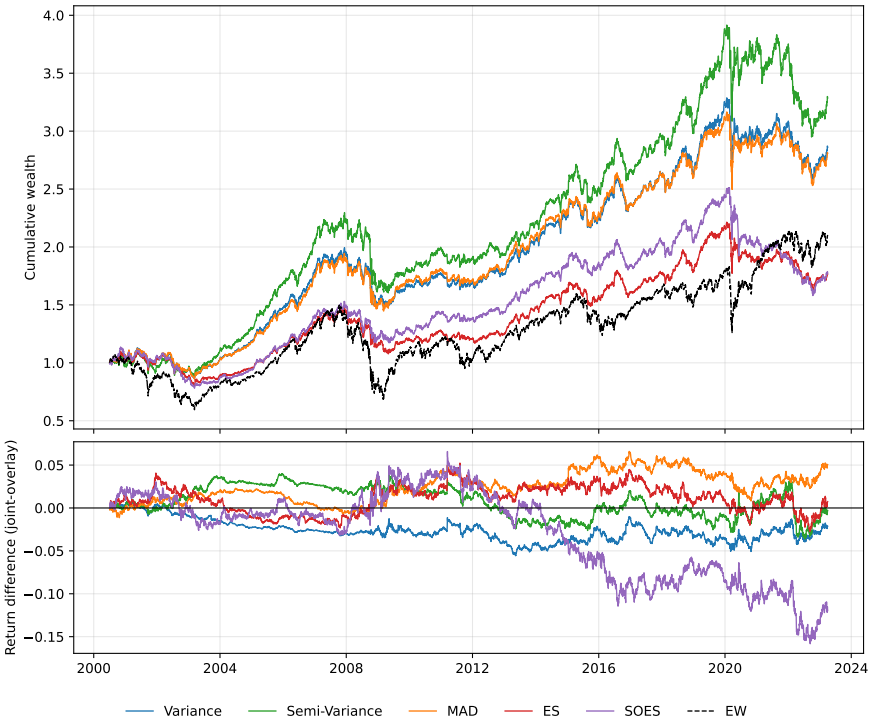


Figure 5.7: Cumulative performance of the mean-risk joint optimizers and their return difference to the overlay optimizers in the small investable universe (top 10 stocks per economy). The top panel shows the cumulative wealth of the joint optimizer of each risk measure and of the equally weighted portfolio (EW), and the bottom panel the cumulative sum of the daily return differences between the joint and the overlay optimizers. Portfolios are rebalanced quarterly, and returns are net of transaction costs. The performance metrics are reported in Panel A of Table 5.7.

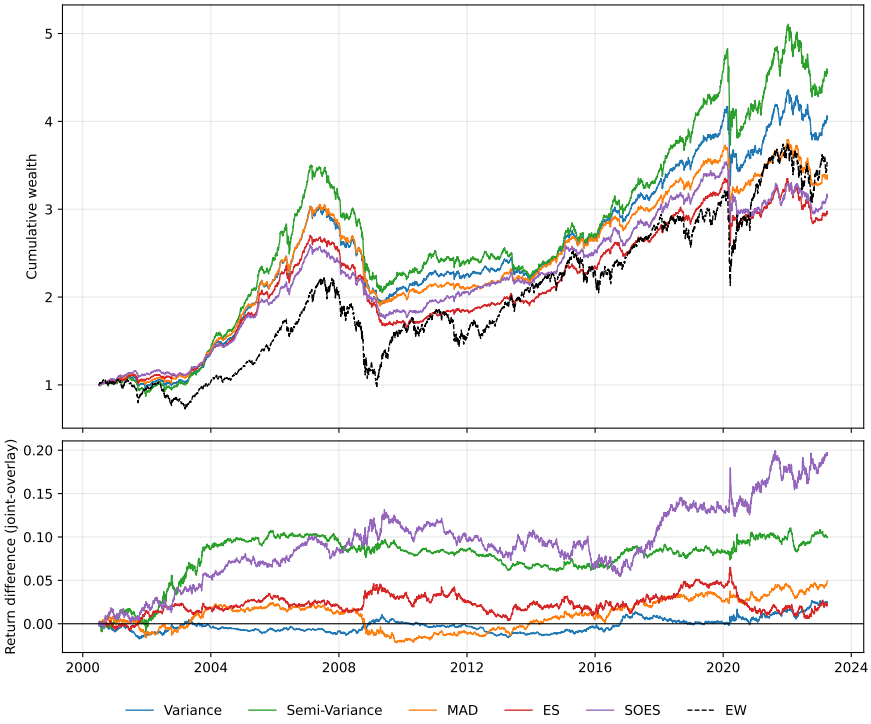


Figure 5.8: Cumulative performance of the mean-risk joint optimizers and their return difference to the overlay optimizers in the medium investable universe (top 50 stocks per economy). Panels are as in Figure 5.7. The performance metrics are reported in Panel B of Table 5.7.

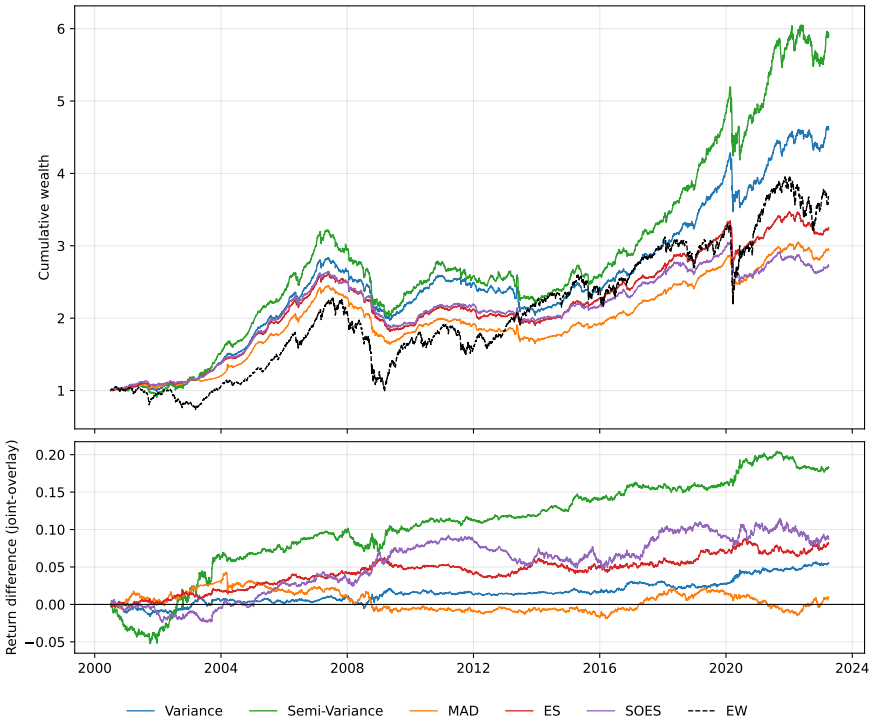


Figure 5.9: Cumulative performance of the mean-risk joint optimizers and their return difference to the overlay optimizers in the large investable universe (top 100 stocks per economy). Panels are as in Figure 5.7. The performance metrics are reported in Panel C of Table 5.7.

optimizers stay below 75% of that of the equally weighted portfolio. The mean term thus raises returns without giving up the lower volatility and drawdowns that distinguish the optimizers from the equally weighted portfolio. As a result, every mean-risk optimizer outperforms the equally weighted portfolio on a risk-adjusted basis in the medium and large universes. In the small universe, the deviation measures reach Sharpe ratios of 0.34 to 0.38 against 0.13 for the equally weighted portfolio, while the tail measures perform similarly at 0.12 to 0.16. For the deviation measures, the underperformance of the pure risk-minimizing optimizers in the small universe is therefore not a failure to reduce volatility but a consequence of ignoring expected returns. The Sortino ratios show the same pattern, so a comparison focused on downside risk does not change the conclusion.

The addition of a mean term also changes the comparison between the joint and overlay optimizers. In the previous sections, neither optimization approach consistently outperformed the other. Now the joint optimizer attains a higher Sharpe ratio than the overlay optimizer for every risk measure in the medium and large universes, by 0.03 to 0.17 and by about 0.07 on average. The Sortino ratios confirm this, with the joint optimizer performing better, by 0.02 to 0.25. The bottom panels of Figures 5.8 and 5.9 show that this advantage builds up gradually over the sample rather than in a single period, with the cumulative return difference reaching about 0.18 for semi-variance in the large universe and 0.19 for second-order Expected Shortfall in the medium universe by the end of the sample. At the same time, the joint optimizer attains a lower volatility than the overlay optimizer for all risk measures, by up to 0.44 percentage points. The lower volatility of the joint optimizers is a persistent finding across all backtests in Sections 5.1 and 5.2. Intuitively, the joint program can choose assets and currencies that offset each other's risk, while the overlay program picks its assets before considering the currencies. Without a mean term, this flexibility is used only to minimize the risk measure, which lowers volatility but leaves returns largely unchanged. With a mean term, the joint program can

also weigh the expected returns of assets and currencies against each other, keeping currency exposure where it is expected to pay and hedging it where it is not.

5.4 Summary

All optimizers considered in this chapter achieve a much lower volatility and smaller drawdowns than the equally weighted portfolio. Without a mean term, the excess return of the joint and overlay optimizers is also lower. They minimize a risk measure without any return term in the objective and possibly have a lower exposure to the common movement of the equity markets, which would also give them a smaller share of the equity risk premium. In the medium and large universes, the optimizers achieve a higher Sharpe ratio than the equally weighted portfolio, but in the small universe they do not outperform it consistently. With fewer stocks to choose from, the optimizers lose more excess return than they reduce volatility, while a larger universe offers more low-risk stocks, and the reduction in volatility outweighs the lost return.

Regularization increases the Sharpe ratios, especially in the large universe. The penalties limit the extreme positions that the unregularized programs fit to noise in the lookback window, which matters more when the number of assets is large relative to the length of the window. Compared with the Full optimizer, which keeps the currency exposure fully hedged, the regularized joint and overlay optimizers realize a lower value of the risk measure they minimize, at a similar Sharpe ratio. Without a mean term in the objective, actively trading the currencies therefore reduces risk relative to full hedging without giving up risk-adjusted returns but does not raise them either, likely because the currency weights are then chosen only to minimize the risk measure and not to earn the excess currency returns. Compared with the Zero optimizer, which keeps the full currency exposure, managing the currencies pays off in the medium and large universes, where equity risk is reduced further and currency risk makes up a larger share of the portfolio's total risk.

Adding a mean term to the objective raises the Sharpe ratios in almost every case and by a similar amount in all three universes. The gain comes from higher excess returns, while the volatility and drawdowns stay well below those of the equally weighted portfolio. The most significant improvement is for the deviation measures. Since they are invariant to a shift in the portfolio return, minimizing them ignores expected returns entirely, and the mean term adds information the optimizers did not use before. Expected Shortfall and second-order Expected Shortfall are translation equivariant and already reward a higher mean, so the explicit mean term changes them much less.

The joint and overlay optimizers perform very similarly when they only minimize a risk measure. The joint optimizer achieves a lower volatility and, without regularization, a higher currency turnover. With regularization, the currency turnover decreases to about the size of that of the overlay optimizer. In both cases, neither approach achieves consistently higher risk-adjusted returns. This changes when a mean term is added to the unregularized programs. The joint optimizer then has higher Sharpe and Sortino ratios than the overlay optimizer for every risk measure in the medium and large universes, and this advantage is consistent over the sample. Once expected returns enter the objective, the joint program can optimize the expected returns of assets and currencies against each other, which the overlay program cannot do since it fixes the asset weights first. This is the setting in which active currency management adds to the risk-adjusted returns, whereas without a mean term it mainly reduces risk.

The lower volatility of the joint optimizer is the most consistent difference between the two approaches. We would expect a difference in this direction, since the joint program can always

choose the same portfolio as the overlay program. Intuitively, the joint program can choose assets and currencies that offset each other's risk, while the overlay program picks its assets before considering the currencies, and this matters most in the small universe, where the number of assets is small relative to the number of currencies. In the next chapter, we therefore focus on comparing the optimizers in terms of reducing the risk measure they are optimizing for.

Chapter 6

Joint versus Sequential Optimization

Chapter 5 shows that the joint and overlay programs achieve similar returns, and that the most consistent difference between them is the lower realized volatility of the joint optimizers. In this chapter, we compare how well the two programs minimize their target risk measure, in sample and out of sample. In Section 6.1 we show why the joint program has an in-sample advantage over the overlay program by construction, define the in-sample and out-of-sample gaps between the two programs, and estimate them for the five risk measures and three universe sizes. In Section 6.2 we study the tail measures in more detail. We repeat the gap analysis across confidence levels in Section 6.2.1 and for the de-meaned tail measures in Section 6.2.2. In Section 6.3 we analyze the deviation measures and compare them with the tail measures in terms of how much of the in-sample advantage is realized out of sample.

6.1 Advantage of Joint Optimization for Risk Minimization

In sample, joint optimization provides a benefit over sequential optimization by construction when it comes to minimizing the target risk measure. We write the weight vector as $\boldsymbol{\theta} = (\boldsymbol{x}, \boldsymbol{\psi})$ and let $\Theta = \mathcal{X} \times \mathbb{R}^M$ denote the feasible set defined by the budget and leverage constraints as in Chapter 3. For a rebalancing period $t \in \{1, \ldots, T\}$, let ρ_t^{IS} be the in-sample risk over the lookback window and ρ_t^{OOS} the out-of-sample realized risk over the next holding period. Let $\hat{\boldsymbol{\theta}}_t$ denote the weight vector an optimizer returns on the lookback window, so that $\rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_t)$ and $\rho_t^{\mathrm{OOS}}(\hat{\boldsymbol{\theta}}_t)$ are the in-sample and out-of-sample risks it realizes. The two optimization programs are written as

$$\hat{\boldsymbol{\theta}}_{J,t} \in \operatorname*{arg\,min}_{\boldsymbol{\theta} \in \Theta} \rho_t^{\mathrm{IS}}(\boldsymbol{\theta}),$$

and

$$\hat{\boldsymbol{x}}_{O,t} \in \operatorname*{arg\,min}_{\boldsymbol{x} \in \mathcal{X}} \rho_{A,t}^{\mathrm{IS}}(\boldsymbol{x}), \quad \hat{\boldsymbol{\psi}}_{O,t} \in \operatorname*{arg\,min}_{\boldsymbol{\psi} \in \mathbb{R}^M} \rho_t^{\mathrm{IS}}(\hat{\boldsymbol{x}}_{O,t}, \boldsymbol{\psi}),$$

where $\rho_{A,t}^{\mathrm{IS}}(\boldsymbol{x}) \coloneqq \rho_t^{\mathrm{IS}}(\boldsymbol{x}, \mathbf{0})$ is the in-sample risk of the fully hedged asset portfolio, and $\hat{\boldsymbol{\theta}}_{i,t} = (\hat{\boldsymbol{x}}_{i,t}, \hat{\boldsymbol{\psi}}_{i,t})$, $i \in \{J, O\}$. The subscript J denotes the joint optimizer and O the sequential overlay optimizer. The overlay optimizer is a restricted minimizer. It first minimizes over $\boldsymbol{x}$ against a different objective, then over $\boldsymbol{\psi}$ with $\hat{\boldsymbol{x}}_{O,t}$ held fixed.

The overlay optimizer's output satisfies the joint program's constraints, so $\hat{\boldsymbol{\theta}}_{O,t} \in \Theta$, and its second step is a partial minimization of the same ρ_t^{IS}. Hence

$$\rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{J,t}) = \min_{\boldsymbol{\theta} \in \Theta} \rho_t^{\mathrm{IS}}(\boldsymbol{\theta}) \;\leq\; \min_{\boldsymbol{\psi} \in \mathbb{R}^M} \rho_t^{\mathrm{IS}}(\hat{\boldsymbol{x}}_{O,t}, \boldsymbol{\psi}) = \rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{O,t}).$$

This inequality holds for exact minimizers. In practice, both programs are solved numerically, so it holds only up to the convergence tolerance of the solver, and we assume that the solver

converges to a solution within a sufficiently small numerical tolerance. The solver configuration is given in Section E.2. We define the in-sample gap between the two optimizers as

$$g_t^{\mathrm{IS}} \coloneqq \rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{O,t}) - \rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{J,t}) \;\geq\; 0.$$

The in-sample gap is non-negative and measures the advantage of joint optimization over sequential optimization in the lookback window. Naturally, the question we are interested in is whether this advantage is realized out of sample.

We define the out-of-sample error of a fitted $\hat{\boldsymbol{\theta}}_t$ as

$$\epsilon_t(\hat{\boldsymbol{\theta}}_t) \coloneqq \rho_t^{\mathrm{OOS}}(\hat{\boldsymbol{\theta}}_t) - \rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_t),$$

the amount by which realized risk in the holding quarter exceeds the risk the optimizer achieved in sample.[1] Then the out-of-sample gap is defined as

$$\begin{aligned} g_t^{\mathrm{OOS}} &\coloneqq \rho_t^{\mathrm{OOS}}(\hat{\boldsymbol{\theta}}_{O,t}) - \rho_t^{\mathrm{OOS}}(\hat{\boldsymbol{\theta}}_{J,t}) \\ &= \underbrace{\left[\rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{O,t}) - \rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{J,t})\right]}_{g_t^{\mathrm{IS}} \geq 0} \;-\; \underbrace{\left[\epsilon_t(\hat{\boldsymbol{\theta}}_{J,t}) - \epsilon_t(\hat{\boldsymbol{\theta}}_{O,t})\right]}_{\Delta\epsilon_t}. \end{aligned}$$

The difference between the out-of-sample errors $\Delta\epsilon_t$ can be either positive or negative. For the variance, the in-sample risk of an optimized portfolio is known to understate its true risk, and the more weights are fitted relative to the sample size, the larger this understatement becomes [17]. Since the joint program fits all of its weights to the same in-sample objective, we expect it to adapt more closely to the noise in the lookback window than the overlay program, and hence $\Delta\epsilon_t$ to be typically positive. If $\Delta\epsilon_t$ exceeds g_t^{IS}, the overlay program could in some cases realize lower risk than the joint program out of sample. We also expect $\Delta\epsilon_t$ to be measure-dependent. Tail measures, in our case Expected Shortfall and second-order Expected Shortfall, are estimated from a smaller fraction of the sample, whereas the deviation measures variance and mean absolute deviation use every observation, and semi-variance those below the mean. For a given sample size, we therefore expect $\Delta\epsilon_t$ to be larger and more volatile for the tail measures.

We perform the following gap analysis to investigate the in-sample and out-of-sample performance of the joint and overlay programs with respect to the risk measure they are minimizing. The in-sample gap is guaranteed to be non-negative only when each program minimizes the risk measure it is scored on, so we consider the unregularized formulation from Chapter 3. At each of the $T = 91$ quarterly rebalancing dates the joint and the overlay optimizers, for a given risk measure, are solved on the same ten-year lookback window and under the same budget and leverage constraints. The resulting weights are then held fixed and scored twice, once on the lookback window itself and once on the following holding quarter, which gives ρ_t^{IS} and ρ_t^{OOS} for each of the two programs. Both gaps are reported relative to the risk of the overlay optimizer

$$\tilde{g}_t^{\mathrm{IS}} \coloneqq \frac{g_t^{\mathrm{IS}}}{\rho_t^{\mathrm{IS}}(\hat{\boldsymbol{\theta}}_{O,t})} \qquad \text{and} \qquad \tilde{g}_t^{\mathrm{OOS}} \coloneqq \frac{g_t^{\mathrm{OOS}}}{\rho_t^{\mathrm{OOS}}(\hat{\boldsymbol{\theta}}_{O,t})},$$

so they are to be interpreted as the risk reduction achieved by the joint optimizer over the overlay optimizer, expressed as a percentage of the latter's risk. For each risk measure and universe size we report the mean of the relative gaps across the rebalancing dates together with a 95% bootstrap confidence interval based on 10,000 resamples.

[1] This quantity is the portfolio analogue of the optimism of the training error, see, for example, Section 7.4 in [22].

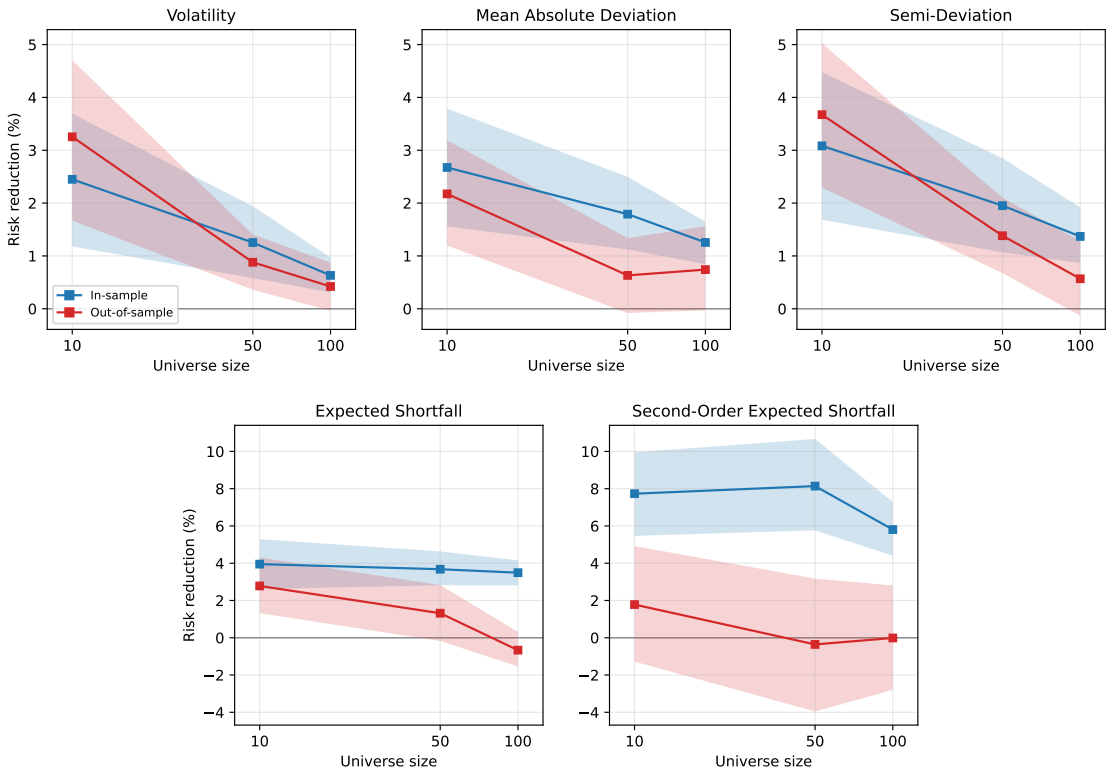


Figure 6.1: In-sample and out-of-sample relative gaps between the unregularized joint and overlay optimizers, by risk measure and universe size. Each panel shows one risk measure, with the size of the investable universe on the horizontal axis and the risk reduction of the joint optimizer over the overlay optimizer on the vertical axis. Squares mark the mean across the $T = 91$ quarterly rebalancing dates, in blue for the in-sample gap $\tilde{g}_t^{\mathsf{IS}}$ and in red for the out-of-sample gap $\tilde{g}_t^{\mathsf{OOS}}$, and shaded areas mark the 95% bootstrap confidence intervals.

We see in Figure 6.1 that the in-sample gap of joint optimization is positive for all risk measures across all universe sizes and that it is largest for the two considered tail measures. The out-of-sample gap is smaller than the in-sample gap for almost all risk measures and universe sizes, the only exception being the variance and semi-variance measures in the small universe. Out of sample, we see a negative dependence on the size of the investable universe. In the small universe the realized out-of-sample advantage of joint optimization is in the range between 1.78% and 3.67%, with second-order Expected Shortfall having the smallest benefit and semi-variance having the largest. In the medium universe, the advantage drops to zero for second-order Expected Shortfall and remains around 1% for the other measures, and in the large universe the advantage for Expected Shortfall drops slightly below zero while the remaining measures stay below 1%.

The in-sample series separates the two groups of risk measures the same way as in Chapter 5, that is, whether they are deviation or tail measures. The in-sample advantage falls as the universe grows, from 2.45% to 0.63% for variance, from 2.67% to 1.26% for mean absolute deviation, and from 3.08% to 1.37% for semi-variance, whereas for the two tail measures it remains larger and more stable, from 3.95% to 3.49% for Expected Shortfall and from 7.73% to 5.81% for second-order Expected Shortfall. For the deviation measures the in-sample and out-of-sample series fall together, whereas for the tail measures, the out-of-sample series falls to zero, and the in-sample series remains large and positive. We visualize the difference between the two series in Figure 6.2.

The first thing we notice in Figure 6.2 is that the bootstrapped confidence intervals for the two tail measures are wider than those for the three deviation measures, averaging 4.37 against

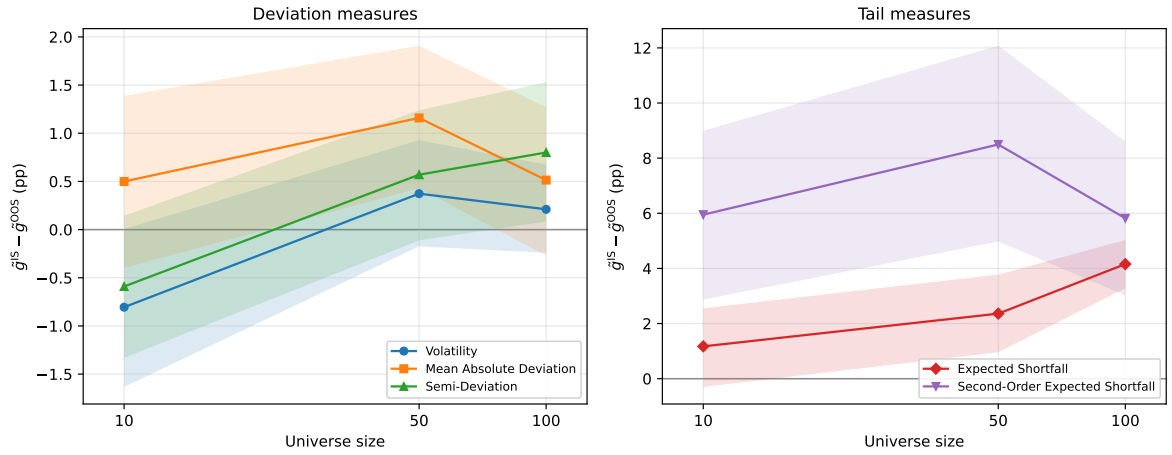


Figure 6.2: Unrealized in-sample advantage of unregularized joint optimization, $\tilde{g}_t^{\mathsf{IS}} - \tilde{g}_t^{\mathsf{OOS}}$, by risk measure and universe size. The reported quantity is the percentage advantage the joint optimizer has in sample minus the percentage advantage it realizes out of sample, measured in percentage points. Markers show the mean across the 91 quarterly rebalancing dates, and shaded areas the 95% bootstrap confidence intervals as in Figure 6.1. The deviation and tail measures are shown on separate panels with independent vertical scales, since the latter are an order of magnitude larger.

1.41 percentage points. This is expected, since the tail measures are estimated on a smaller fraction of the sample. The second thing is the size of the unrealized advantage itself, which is between 1.17 and 8.50 percentage points for the two tail measures while the three deviation measures are between −0.81 and 1.16 percentage points. This, together with the in-sample series of Figure 6.1 (which for the tail measures stays between 3.49% and 8.14% at every universe size), shows that the tail measures have a larger in-sample advantage coming from joint optimization that is not realized out of sample, whereas for the deviation measures the advantage that is achieved in sample is largely realized out of sample but drops with universe size. The two groups produce different results, and we analyze the mechanism behind them separately in Sections 6.2 and 6.3.

6.2 Tail Measures

6.2.1 Level Analysis

To investigate the mechanism behind the in-sample advantage of joint optimization for the two tail measures, Expected Shortfall and second-order Expected Shortfall, we perform the same gap analysis as in Section 6.1 across a range of confidence levels. We report the results in Figures 6.3 and 6.4.

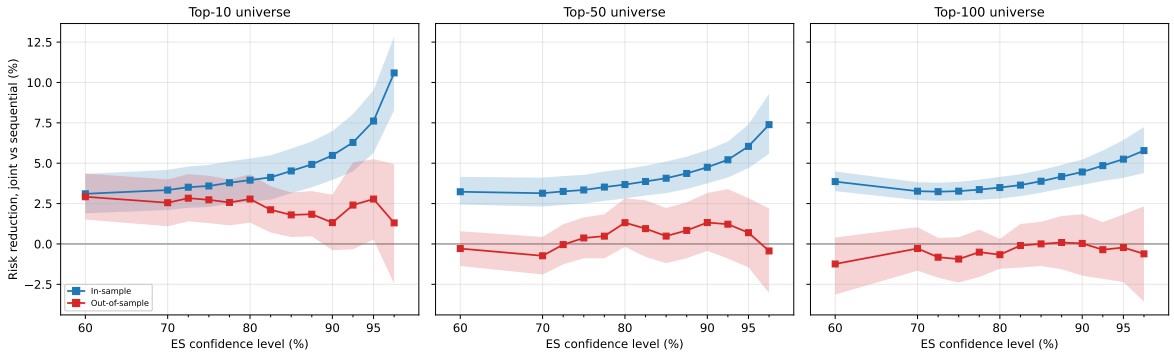


Figure 6.3: In-sample and out-of-sample relative gaps between the unregularized joint and overlay Expected Shortfall optimizers, by confidence level and universe size. Each panel shows one universe size, with thirteen confidence levels from 60% to 97.5% on the horizontal axis. Squares and shaded areas denote the same quantities as in Figure 6.1.

We see in Figure 6.3 that the in-sample advantage of joint optimization for minimizing Expected Shortfall grows with the confidence level and the growth is more pronounced in the smaller universe. The out-of-sample advantage is much smaller and does not rise with the confidence level at any universe size. In the small universe, the in-sample advantage rises from 3.11% at confidence level 60% to 10.59% at level 97.5%, while the out-of-sample advantage stays in the 1.30 to 2.92% range. In the larger universes the in-sample advantage grows at a lower rate, going from 3.23% to 7.39% in the medium and from 3.86% to 5.78% in the large universe, while the realized advantage never exceeds 1.32% and 0.09%, respectively.

Expected Shortfall at confidence level α is the average loss over the worst $1-\alpha$ fraction of the lookback window, which is roughly 1,000 observations at 60% and 65 at 97.5%. Which observations fall in the tail depends largely on the portfolio weights. The joint program chooses the asset and currency weights together, so it can select assets whose worst days the currency positions can offset. The overlay program fixes the asset weights of the fully hedged portfolio, and its currency step can adjust M weights against a largely fixed set of the worst-performing days. As we increase the confidence level, the number of observations in the tail shrinks, so the joint program's currency weights can cover a larger share of the worst-performing days, and the in-sample advantage grows. The growth is largest in the small universe, where the currency weights represent a larger share of all the decision variables. In larger universes, the overlay program's asset step has more weights to diversify, so the joint program's advantage decreases. Since the weights are fitted to specific worst-performing days of the lookback window that do not necessarily occur in the next period, the mechanism generalizes poorly out of sample.

In Figure 6.4 we see the same pattern for second-order Expected Shortfall. The in-sample advantage of joint optimization grows with the confidence level and the growth is faster in the smaller universe. We also observe that above certain confidence levels, the in-sample advantage becomes constant. The out-of-sample advantage is much smaller and does not rise with the confidence level at any universe size. In the small universe, the in-sample advantage

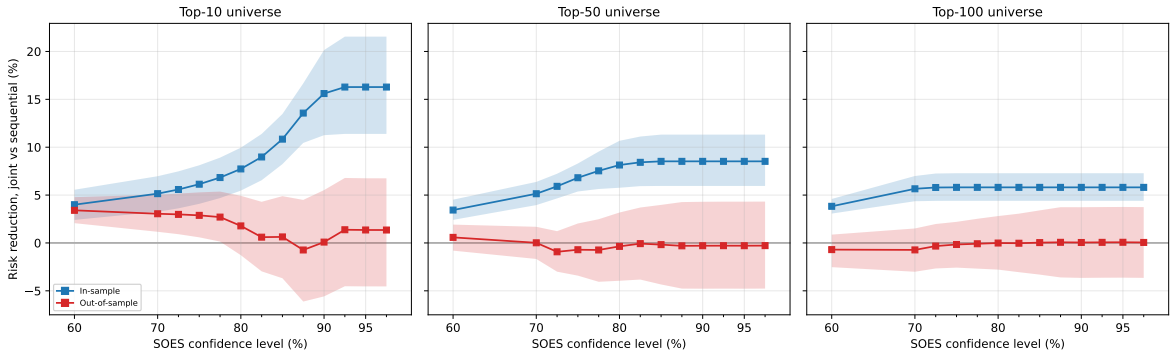


Figure 6.4: In-sample and out-of-sample relative gaps between the unregularized joint and overlay second-order Expected Shortfall optimizers, by confidence level and universe size. Squares and shaded areas denote the same quantities as in Figure 6.3.

rises from 4.00% at level 60% to 16.28% at level 92.5% and from there remains constant. The out-of-sample advantage stays in the -0.45 to 3.88% range. In the larger universes the in-sample advantage grows at a lower rate and becomes constant at lower confidence levels, going from 3.44% to 8.52% in the medium and from 3.83% to 5.81% in the large universe, while the out-of-sample advantage fluctuates around zero.

Second-order Expected Shortfall squares the tail observations, so more weight is placed on the most extreme observations. At level α it therefore behaves like Expected Shortfall at a higher confidence level, which is why the in-sample advantage is larger at every confidence level and every universe size. As we increase the confidence level, the squared weighting concentrates on a smaller and more stable set of the worst-performing days, and above a certain level, raising the confidence level further no longer changes which observations dominate the objective, and the in-sample series becomes constant. In larger universes this happens at lower confidence levels, since the overlay program's asset step has more weights to fit those days. Out of sample, the same reasoning as for Expected Shortfall applies, and the realized advantage does not rise with the confidence level at any universe size and is only present in the small universe.

6.2.2 De-meaning the Tail Measures

As discussed in Section 3.1.6, Expected Shortfall and second-order Expected Shortfall are coherent (strictly expectation bounded) risk measures rather than deviation measures. Unlike variance, mean absolute deviation, and semi-variance, they are not invariant to translation, so a portfolio with a higher mean return is scored more favorably than one with a lower mean return, everything else being equal. The objectives therefore carry an implicit risk-return tradeoff that the three deviation measures do not. We can transform a coherent strictly expectation bounded risk measure $\mathcal{R}$ into a deviation measure $\mathcal{D}$ via a simple de-meaning operation. More precisely

Theorem D.3 *Let $\mathcal{R}$ be a coherent strictly expectation bounded risk measure, and let $\mathcal{D}$ be the functional defined by*

$$\mathcal{D}(X) \coloneqq \mathcal{R}(X - \mathbb{E}[X]).$$

Then $\mathcal{D}$ is a deviation measure.

The proof of Theorem D.3 and further details can be found in Appendix D. We perform the same gap analysis as in Section 6.1 on the de-meaned Expected Shortfall and second-order Expected Shortfall optimization programs, and report the results in Figure 6.5.

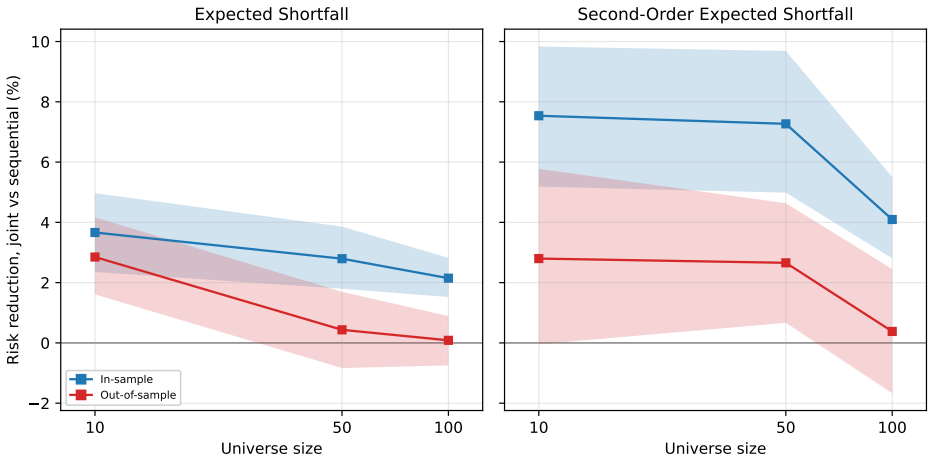


Figure 6.5: In-sample and out-of-sample relative gaps between the de-meaned unregularized joint and overlay Expected Shortfall and second-order Expected Shortfall optimizers, by universe size. Squares and shaded areas denote the same quantities as in Figure 6.1.

We see in Figure 6.5 that de-meaning the optimizers steepens the decline of the in-sample relative gaps. For Expected Shortfall the in-sample gap falls from 3.66% to 2.15% across the three universes, against 3.95% to 3.49% for the original tail measure, and for second-order Expected Shortfall from 7.54% to 4.09% against 7.73% to 5.81%. The drop in the gap across universes is 41.3% as opposed to 11.6% for Expected Shortfall and 45.7% as opposed to 24.8% for second-order Expected Shortfall. This indicates that the implicit risk-return tradeoff is one of the driving factors for the tail measures' in-sample advantage as the universe grows. With it removed, they begin to decay in the way the deviation measures do. In the large universe the de-meaned Expected Shortfall gap is still 2.15% against, for example, 0.63% for variance, so the mean term accounts for only part of the in-sample advantage.

6.3 Deviation Measures

For the three considered deviation measures the advantage of joint optimization is smaller in sample than for the tail measures, and most of it survives out of sample. If we compare the two series from Figure 6.1 measure by measure, the realized gap $\tilde{g}^{\mathrm{OOS}}$ as a fraction of the in-sample gap $\tilde{g}^{\mathrm{IS}}$ is 133%, 70%, and 67% for variance across the small, medium, and large universes, 81%, 35%, and 59% for mean absolute deviation, and 119%, 71%, and 42% for semi-variance. For the tail measures the same ratio is 70%, 36%, and −19% for Expected Shortfall and 23%, −4%, and 0% for second-order Expected Shortfall.

Variance, semi-variance, and mean absolute deviation are estimated from all observations in the window (half for semi-variance), so the objective is estimated more accurately and the out-of-sample gap follows the in-sample gap more closely. Expected Shortfall and its second-order variant are determined by a small fraction of the observations at the confidence levels considered, so the same gap is a noisier quantity. Figures 6.3 and 6.4 make this explicit. As the tail is made thinner, the in-sample gap grows while the realized gap stays flat or decays. The deviation measures are also translation invariant, so they carry no implicit risk-return tradeoff, and the joint program cannot produce an in-sample advantage by tilting the weights toward a high realized mean. The tail measures can, and Figure 6.5 shows that removing the mean term accounts for part of the in-sample advantage across universes. So deviation measures are left with a small advantage that shrinks both in sample and out of sample as the investable universe grows.

6.4 Summary

Even though joint optimization does not consistently outperform in terms of returns, it has an advantage over sequential optimization by construction when it comes to minimizing the targeted risk measure. In sample, the advantage for the deviation measures falls with universe size, while for the two considered tail measures it stays stable. Out of sample, the advantage falls with universe size for all five considered risk measures.

The tail measures' in-sample advantage is driven by two mechanisms. The first is a tail-fitting mechanism, which lets the joint program align the currency weights with the specific worst-performing days of the lookback window, and which does not generalize out of sample. The second is an implicit risk-return tradeoff, which lets the joint program produce an in-sample advantage by tilting the weights toward a high realized mean. Once the tail measures are de-meaned, their in-sample advantage begins to fall with universe size, in the same way as the deviation measures. Neither of the two mechanisms is available to the deviation measures, which is why their in-sample advantage was smaller to begin with and a larger fraction of it survives out of sample.

The out-of-sample advantage of joint optimization is present only in the small universe, where the currency weights represent a larger share of the decision variables. In larger universes the overlay program's asset step has enough weights to fit the objective on its own, and the joint program's advantage drops to zero. Joint optimization is therefore beneficial for risk reduction when the number of assets is small relative to the number of currencies. These results agree with the findings in Chapter 5, where the joint program's lower realized volatility was largest in the small universe.

Chapter 7

Conclusion

In this thesis, we studied how risk-based portfolio optimization performs for an investor who holds assets denominated in several currencies and whether optimizing the asset and currency weights jointly improves on the industry-standard sequential currency overlay. In Section 7.1 we summarize the findings of Chapters 5 and 6, and in Section 7.2 we discuss the limitations of our analysis and directions for future research.

7.1 Findings

Out of sample, the unregularized risk-minimizing optimizers reduce volatility by 34% to 64% relative to the equally weighted portfolio and their maximum drawdown by up to 30 percentage points. Without a return term in the objective, they earn lower excess returns, possibly also because a lower exposure to the common movement of the equity markets gives them a smaller share of the equity risk premium. On a risk-adjusted basis, they outperform the equally weighted portfolio, with Sharpe ratios of up to 70% higher in the medium universe and up to 105% higher in the large universe, but do not consistently outperform in the small universe. The optimizers tilt the portfolio toward low-risk stocks, and a larger universe offers more of them, while in the small universe they lose more excess return than they reduce volatility.

Regularizing the optimizers further improves their risk-adjusted performance, most in the large universe, where the Sharpe ratios increase by 0.17 on average. The penalties limit the extreme positions that the unregularized programs fit to noise in the lookback window. Compared with the Full optimizer, which keeps the currency exposure fully hedged, the regularized joint and overlay optimizers realize a lower value of the risk measure they minimize at a similar Sharpe ratio, so actively trading the currencies reduces risk relative to full hedging without giving up risk-adjusted returns. The Zero optimizer, which leaves the currency exposure unhedged, outperforms them only in the small universe. In the medium and large universes, where currency risk makes up a larger share of the portfolio's total risk, managing it actively is more effective than leaving it unhedged, and the regularized joint and overlay optimizers achieve better risk-adjusted performance than the Zero optimizer.

Adding a mean term to the optimizers transforms the problem into a risk-return optimization problem. It has a large effect on risk-adjusted performance, increasing the Sharpe ratios by 0.16 to 0.24 on average for the deviation measures and by 0.01 to 0.10 for the tail measures. This is not surprising, as deviation measures are invariant to a shift in the portfolio return, so minimizing them ignores expected returns entirely, whereas Expected Shortfall and second-order Expected Shortfall already reward a higher mean. A possible further source of the gain is currency carry. Since the expected excess currency returns are set to the interest rate

differential, the mean term rewards exposure to higher-yielding currencies, so the currency positions may earn part of the carry premium that pure risk minimization gives up. The optimizers earn higher risk-adjusted returns than the equally weighted portfolio in the medium and large universes, and the deviation measures also in the small universe, at about 40% to 75% of its volatility and maximum drawdown.

Comparing all backtests in Tables 5.1, 5.3 to 5.5, and 5.7, the mean-variance joint optimizer performs best out of sample, and in the small universe the mean-risk semi-variance optimizers perform similarly. It more than doubles the Sharpe ratio of the equally weighted portfolio in every universe. Although several of the risk-minimizing optimizers have smaller drawdowns, the mean-variance joint optimizer still outperforms them on a risk-adjusted basis. The optimizer benefits from two sources of improvement over its other formulations. Variance is a deviation measure, which gains the most from the mean term, and the joint program can weigh the expected returns of assets and currencies against each other, unlike the overlay program.

The benefit of optimizing assets and currencies jointly rather than in two steps depends on whether expected returns are part of the objective. When the optimizers only minimize a risk measure, the joint and overlay optimizers perform very similarly, with or without regularization, with joint optimization consistently achieving lower volatility, although predominantly in the small universe. Once a mean term is added to the unregularized programs, the joint optimizers achieve higher Sharpe and Sortino ratios than the overlay optimizers for every risk measure in the medium and large universes.

Since the lower volatility of the joint optimizers is the most consistent difference between the two approaches, we compare them in Chapter 6 on the risk measure they minimize. In sample, the joint program has an advantage by construction, since the solution of the overlay program is feasible for it. Out of sample, this advantage is realized only in the small universe, where the currency weights represent a larger share of the decision variables. In the medium and large universes it drops to zero, as the asset step of the overlay program has enough weights to fit the objective on its own. Since the dependence between asset and currency returns barely changes across universe sizes, this decline is driven by the number of assets relative to the number of currencies and not by a change in the dependence structure.

For the deviation measures, most of the small in-sample advantage in risk reduction is realized out of sample, while the larger in-sample advantage of the tail measures is not. We identify two mechanisms that drive this discrepancy, a tail-fitting mechanism, in which the joint program aligns the currency weights with the specific worst days of the lookback window, and an implicit risk-return tradeoff, through which the tail measures reward a high realized mean. Both mechanisms fail to generalize out of sample.

Our results agree with the finding of [37] that the benefit of joint optimization is most pronounced in small universes, which we find for risk minimization, and with the finding of [8] that joint optimization outperforms sequential optimization in a mean-variance setting, which we find in the medium and large universes once a mean term is added to the unregularized programs.

7.2 Limitations and Future Research

Our analysis covers one base currency (the US dollar), 14 economies, and one out-of-sample period from 2000 to 2023, so the results reflect the dependence between asset and currency returns for a US investor over this period. For an investor based in another currency, such as the euro, the Swiss franc, or the Japanese yen, the excess currency returns and their dependence with the asset returns differ, and repeating the analysis for other base currencies would show how much the results depend on this structure.

The backtests are also based on simplified trading assumptions, namely constant transaction costs, no market impact, and quarterly rebalancing, among others. Since the unregularized and mean-risk joint optimizers have a much higher currency turnover than the overlay optimizers in most cases, higher trading costs, more frequent rebalancing, and market impact would likely reduce the advantage of the joint optimizers over the overlay optimizers. A more realistic analysis could include a dynamic transaction cost model and multiple rebalancing frequencies, and test the sensitivity of the results to these assumptions.

Two further extensions follow from the results. First, the tail measures are estimated from the empirical distribution of the lookback window and are sensitive to estimation error. Estimators or formulations that stabilize the tail, such as the extreme value approach proposed in [42], could improve the out-of-sample performance. Second, the explanations given for the optimizers' performance, a lower market exposure, a tilt toward low-risk stocks, and currency carry, could be further investigated by regressing their returns on market, low-volatility, carry, and other factors.

Appendices

Appendix A

Optimization Derivation: Expected Shortfall and Second-Order Expected Shortfall

In this appendix, we derive the portfolio optimization problems used in the main text, building up from the definitions of Expected Shortfall and second-order Expected Shortfall to the programs solved in the empirical analysis. In Section A.1 we derive the auxiliary formulation of Expected Shortfall and introduce second-order Expected Shortfall. In Section A.2 we write the portfolio optimization problem as a linear program for Expected Shortfall and as a second-order cone program for second-order Expected Shortfall. In Section A.3 we regularize the programs by adding an L^1 penalty, an L^2 penalty, and a leverage constraint. We then extend the programs to the multi-currency setting, for sequential optimization in Section A.4 and for joint optimization in Section A.5. Finally, in Section A.6 we add the expected portfolio return to the objectives of the sequential and joint problems.

A.1 Auxiliary Formulations of Expected Shortfall and Second-Order Expected Shortfall

We derive the equivalence between the direct minimization of Expected Shortfall and the auxiliary formulation proposed by Rockafellar and Uryasev [54], and introduce second-order Expected Shortfall, which Krokhmal [33] defines directly through an auxiliary formulation of the same type. The two formulations allow for writing the portfolio optimization problem as a linear program for Expected Shortfall and as a second-order cone program for second-order Expected Shortfall in Section A.2.

A.1.1 Expected Shortfall

We assume that the loss distribution is continuous and has a strictly increasing cumulative distribution function and additionally that its probability density function exists. Under these assumptions, the Value-at-Risk (VaR) defined below is the unique point at which the cumulative distribution function equals the confidence level [43]. The assumptions simplify the derivation but are not needed for the result. Rockafellar and Uryasev [55] show that the equivalence derived below holds for general loss distributions, including the empirical distribution used in Section A.2, when Expected Shortfall is defined as the average of the quantiles of the loss distribution over all levels above the confidence level [1].

Define the loss random variable at time $t+1$, given portfolio weights $\boldsymbol{x}_t \in \mathbb{R}^N$, as $L(\boldsymbol{x}_t) = -\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}$, where $\boldsymbol{R}_{t+1} \in \mathbb{R}^N$ is the asset return vector. Let $f(l)$ denote the probability density function of the loss $L(\boldsymbol{x}_t)$. The VaR at confidence level β is defined as the β-quantile of the loss distribution

$$\mathrm{VaR}_\beta(\boldsymbol{x}_t) = \min\{\alpha \mid \mathbb{P}_t(L(\boldsymbol{x}_t) \leq \alpha) \geq \beta\}.$$

The Expected Shortfall at confidence level β is defined as the conditional expectation of losses exceeding VaR_β

$$\mathrm{ES}_\beta(\boldsymbol{x}_t) = \mathbb{E}_t[L(\boldsymbol{x}_t) \mid L(\boldsymbol{x}_t) \geq \mathrm{VaR}_\beta(\boldsymbol{x}_t)] = \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} l f(l)\, dl.$$

We now consider the auxiliary function F_β defined as

$$\begin{aligned} F_\beta(\boldsymbol{x}_t, \alpha) &= \alpha + \frac{1}{1-\beta}\mathbb{E}_t[(L(\boldsymbol{x}_t) - \alpha)^+] \\ &= \alpha + \frac{1}{1-\beta}\int_\alpha^\infty (l-\alpha) f(l)\, dl. \end{aligned} \tag{A.1}$$

Before minimizing (A.1) with respect to α, we verify convexity. Using Leibniz's rule, the first and second derivatives of F_β with respect to α are

$$\begin{aligned} \frac{\partial F_\beta}{\partial \alpha} &= 1 + \frac{1}{1-\beta}\frac{\partial}{\partial \alpha}\int_\alpha^\infty (l-\alpha) f(l)\, dl \\ &= 1 + \frac{1}{1-\beta}\left[-f(\alpha)\cdot 0 + \int_\alpha^\infty (-1) f(l)\, dl\right] \\ &= 1 - \frac{1}{1-\beta}\int_\alpha^\infty f(l)\, dl, \\ \frac{\partial^2 F_\beta}{\partial \alpha^2} &= \frac{1}{1-\beta} f(\alpha). \end{aligned}$$

Since the probability density function $f(\alpha) \geq 0$ and $\beta < 1$, the second derivative is non-negative. Thus, (A.1) is convex in α, ensuring that the first-order condition is a necessary and sufficient condition for a global minimum.

Setting the derivative to zero gives the optimal α^*

$$\begin{aligned} \frac{\partial F_\beta}{\partial \alpha} &= 0 \\ 1 - \frac{1}{1-\beta}\int_{\alpha^*}^\infty f(l)\, dl &= 0 \\ 1 - \frac{1}{1-\beta}\mathbb{P}_t(L(\boldsymbol{x}_t) \geq \alpha^*) &= 0 \\ \mathbb{P}_t(L(\boldsymbol{x}_t) \geq \alpha^*) &= 1-\beta \\ \mathbb{P}_t(L(\boldsymbol{x}_t) \leq \alpha^*) &= \beta. \end{aligned}$$

We see that the optimal α^* that minimizes the auxiliary function for a fixed $\boldsymbol{x}_t$ is exactly the Value-at-Risk at level β.

Substituting $\alpha^* = \mathrm{VaR}_\beta(\boldsymbol{x}_t)$ back into (A.1) we obtain

$$\begin{aligned} F_\beta(\boldsymbol{x}_t, \mathrm{VaR}_\beta(\boldsymbol{x}_t)) &= \mathrm{VaR}_\beta(\boldsymbol{x}_t) + \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^\infty (l - \mathrm{VaR}_\beta(\boldsymbol{x}_t)) f(l)\, dl \\ &= \mathrm{VaR}_\beta(\boldsymbol{x}_t) + \frac{1}{1-\beta}\left(\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^\infty l f(l)\, dl - \mathrm{VaR}_\beta(\boldsymbol{x}_t)\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^\infty f(l)\, dl\right). \end{aligned}$$

Recognizing that $\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} f(l)\,dl = 1-\beta$, we have

$$\begin{aligned} F_\beta(\boldsymbol{x}_t, \mathrm{VaR}_\beta(\boldsymbol{x}_t)) &= \mathrm{VaR}_\beta(\boldsymbol{x}_t) + \frac{1}{1-\beta}\int_{\mathrm{VaR}_\beta(\boldsymbol{x}_t)}^{\infty} l f(l)\,dl - \frac{1}{1-\beta}\mathrm{VaR}_\beta(\boldsymbol{x}_t)(1-\beta) \\ &= \mathrm{VaR}_\beta(\boldsymbol{x}_t) + \mathrm{ES}_\beta(\boldsymbol{x}_t) - \mathrm{VaR}_\beta(\boldsymbol{x}_t) \\ &= \mathrm{ES}_\beta(\boldsymbol{x}_t). \end{aligned}$$

Thus, minimizing $\mathrm{ES}_\beta(\boldsymbol{x}_t)$ with respect to $\boldsymbol{x}_t$ is equivalent to minimizing $F_\beta(\boldsymbol{x}_t, \alpha)$ jointly with respect to $\boldsymbol{x}_t$ and α

$$\min_{\boldsymbol{x}_t} \mathrm{ES}_\beta(\boldsymbol{x}_t) = \min_{\boldsymbol{x}_t, \alpha} \left(\alpha + \frac{1}{1-\beta}\mathbb{E}_t[(-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+] \right). \tag{A.2}$$

Finally, we show that $F_\beta(\boldsymbol{x}_t, \alpha)$ is jointly convex in $(\boldsymbol{x}_t, \alpha)$. The inner map $(\boldsymbol{x}_t, \alpha) \mapsto (-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)$ is affine and the function $z \mapsto z^+$ is convex and nondecreasing. The composition of a convex function with an affine map is convex [6, Section 3.2.2]. Thus, $(-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+$ is jointly convex in $(\boldsymbol{x}_t, \alpha)$. Expectation is a linear operator which preserves convexity, and the sum of convex functions is convex.

A.1.2 Second-Order Expected Shortfall

Second-order Expected Shortfall belongs to the family of higher-moment risk measures introduced by Krokhmal [33]. It is obtained from the auxiliary function F_β by replacing the expectation of the excess loss $(L(\boldsymbol{x}_t) - \alpha)^+$ with its L^2 norm.[1] This gives the auxiliary function

$$G_\beta(\boldsymbol{x}_t, \alpha) = \alpha + \frac{1}{1-\beta}\left\|(L(\boldsymbol{x}_t) - \alpha)^+\right\|_{L^2},$$

and second-order Expected Shortfall at confidence level β is defined as its minimum over α

$$\mathrm{SOES}_\beta(\boldsymbol{x}_t) = \min_{\alpha} G_\beta(\boldsymbol{x}_t, \alpha). \tag{A.3}$$

The L^2 norm squares the excess losses, so large losses receive more weight than in Expected Shortfall. Second-order Expected Shortfall is a coherent risk measure, which we prove in Proposition D.5.

Unlike Expected Shortfall, second-order Expected Shortfall is defined directly through the minimization over α, so there is no equivalence to derive. Instead, we only have to verify that G_β is jointly convex in $(\boldsymbol{x}_t, \alpha)$.

We argued for the case of F_β that the map $(\boldsymbol{x}_t, \alpha) \mapsto (-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+$ is convex. The L^2 norm is a convex function [6, Section 3.1.5] and nondecreasing for non-negative random variables. The composition of a convex nondecreasing function with a convex function is again convex [6, Section 3.2.4]. Hence $\left\|(L(\boldsymbol{x}_t) - \alpha)^+\right\|_{L^2}$ is jointly convex in $(\boldsymbol{x}_t, \alpha)$, and the sum of convex functions is convex.

By definition (A.3), minimizing $\mathrm{SOES}_\beta(\boldsymbol{x}_t)$ with respect to $\boldsymbol{x}_t$ is equivalent to minimizing $G_\beta(\boldsymbol{x}_t, \alpha)$ jointly with respect to $\boldsymbol{x}_t$ and α

$$\min_{\boldsymbol{x}_t} \mathrm{SOES}_\beta(\boldsymbol{x}_t) = \min_{\boldsymbol{x}_t, \alpha} \left(\alpha + \frac{1}{1-\beta}\left\|(-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+\right\|_{L^2} \right). \tag{A.4}$$

Since both F_β and G_β are convex, any local minimum is global, ensuring that both optimization problems can be solved reliably with standard convex optimization methods.

[1] We define the L^2 norm in the standard way as $\|Z\|_{L^2} = (\mathbb{E}_t[Z^2])^{1/2}$ for a random variable Z.

A.2 Representation for Portfolio Optimization

Let ρ_β denote either of the two risk measures, written as a function of the portfolio return $\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}$. The portfolio optimization problem is

$$\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \rho_\beta(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}), \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1, \tag{A.5}$$

where the budget constraint enforces full investment. By (A.2) and (A.4), problem (A.5) is equivalent to minimizing $F_\beta(\boldsymbol{x}_t, \alpha)$ or $G_\beta(\boldsymbol{x}_t, \alpha)$ jointly with respect to $\boldsymbol{x}_t$ and α. Each program in this appendix defines its own decision vector, vector of constants, and penalty and constraint matrices, so the same symbols denote slightly different objects in different programs, and we refer to a specific program by its equation number when discussing them.

A.2.1 Expected Shortfall

Let $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_q \in \mathbb{R}^N$ denote q historical realizations of the asset return vector $\boldsymbol{R}_{t+1}$. We approximate the conditional expectation in F_β by the empirical average over this sample

$$\mathbb{E}_t[(-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+] \approx \frac{1}{q} \sum_{k=1}^{q} (-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+,$$

so that the portfolio optimization problem under Expected Shortfall becomes

$$\boldsymbol{x}_t^*, \alpha^* = \arg\min_{\boldsymbol{x}_t, \alpha} \left\{ \alpha + \frac{1}{q(1-\beta)} \sum_{k=1}^{q} (-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+ \right\}, \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1. \tag{A.6}$$

The summand $(-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+$ is convex but non-linear. We linearize it by introducing, for each observation k, an auxiliary variable $u_k \geq 0$ subject to

$$u_k \geq -\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha \qquad \Longleftrightarrow \qquad \boldsymbol{x}_t^\top \boldsymbol{y}_k + \alpha + u_k \geq 0. \tag{A.7}$$

Replacing each summand by u_k, the objective becomes $\alpha + \frac{1}{q(1-\beta)} \sum_{k=1}^{q} u_k$. It is strictly increasing in each u_k and minimized, so every u_k is driven to its lower bound at the optimum, $u_k^* = (-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+$.

By collecting the decision variables in the vector

$$\tilde{\boldsymbol{x}}_t = \Big[x_{1,t}, \ldots, x_{N,t}, \quad \alpha, \quad u_1, \ldots, u_q\Big]^\top = \Big[\boldsymbol{x}_t^\top, \quad \alpha, \quad \boldsymbol{u}^\top\Big]^\top \in \mathbb{R}^{N+q+1},$$

and the constants in the vector

$$\boldsymbol{c} = \left[\mathbf{0}_N^\top, \quad 1, \quad \frac{1}{q(1-\beta)} \mathbf{1}_q^\top\right]^\top \in \mathbb{R}^{N+q+1},$$

the portfolio optimization problem (A.6) can be written as a linear program

$$\begin{aligned} \tilde{\boldsymbol{x}}_t^* &= \arg\min_{\tilde{\boldsymbol{x}}_t} \quad \boldsymbol{c}^\top \tilde{\boldsymbol{x}}_t \\ \text{s.t.} \quad \boldsymbol{D}\tilde{\boldsymbol{x}}_t &= \Big[\mathbf{1}_N^\top \quad \mathbf{0}_{q+1}^\top\Big] \tilde{\boldsymbol{x}}_t = E = 1, \\ \boldsymbol{A}\tilde{\boldsymbol{x}}_t &= \Big[-\boldsymbol{R}^{\text{data}} \quad -\mathbf{1}_q \quad -\boldsymbol{I}_q\Big] \tilde{\boldsymbol{x}}_t \leq \boldsymbol{b} = \mathbf{0}_q, \\ u_k &\geq 0 \quad \text{for } k = 1, \ldots, q. \end{aligned} \tag{A.8}$$

Here $\boldsymbol{D} \in \mathbb{R}^{1\times(N+q+1)}$ encodes the budget constraint, while $\boldsymbol{A} \in \mathbb{R}^{q\times(N+q+1)}$ and $\boldsymbol{b} \in \mathbb{R}^q$ encode the q inequality constraints arising from the linearization (A.7). The matrix $\boldsymbol{R}^{\text{data}} \in \mathbb{R}^{q\times N}$ is composed of the sampled return vectors $\boldsymbol{y}_k^\top$ row by row, $\boldsymbol{0}_q$ is the q-dimensional zero vector, and $\boldsymbol{I}_q$ the q-dimensional identity matrix.

Problem (A.8) has a linear objective subject to one linear equality and q linear inequalities, and is thus a linear program solvable with standard methods. This formulation serves as the base unregularized portfolio optimization problem under Expected Shortfall.

A.2.2 Second-Order Expected Shortfall

Using the same historical realizations $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_q$ of Section A.2.1, we approximate the L^2 norm in G_β by its empirical average over the sample

$$\left\|(-\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1} - \alpha)^+\right\|_{L^2} \approx \left(\frac{1}{q}\sum_{k=1}^{q}\left((-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+\right)^2\right)^{1/2},$$

so that the portfolio optimization problem under second-order Expected Shortfall becomes

$$\begin{gathered}\boldsymbol{x}_t^*, \alpha^* = \underset{\boldsymbol{x}_t,\alpha}{\arg\min}\left\{\alpha + \frac{1}{1-\beta}\left(\frac{1}{q}\sum_{k=1}^{q}\left((-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+\right)^2\right)^{1/2}\right\}, \\ \text{s.t. } \boldsymbol{x}_t^\top \boldsymbol{1}_N = 1.\end{gathered} \tag{A.9}$$

As in Section A.2.1, we replace each term $(-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+$ by u_k subject to (A.7). The empirical norm then becomes $\left(\frac{1}{q}\sum_{k=1}^{q} u_k^2\right)^{1/2} = \frac{1}{\sqrt{q}}\|\boldsymbol{u}\|_2$, and the objective becomes $\alpha + \frac{1}{(1-\beta)\sqrt{q}}\|\boldsymbol{u}\|_2$. The Euclidean norm is strictly increasing in every positive u_k, so, as for Expected Shortfall, every u_k is driven to its lower bound at the optimum, $u_k^* = (-\boldsymbol{x}_t^\top \boldsymbol{y}_k - \alpha)^+$.

The objective is still not linear, so we introduce one more auxiliary variable s subject to the constraint $\|\boldsymbol{u}\|_2 \le s$ and minimize $\alpha + \frac{1}{(1-\beta)\sqrt{q}}s$ instead. Since $\frac{1}{(1-\beta)\sqrt{q}} > 0$, $s^* = \|\boldsymbol{u}^*\|_2$ at the optimum, and the objective is linear in the decision variables. The constraint $\|\boldsymbol{u}\|_2 \le s$ requires the vector $(\boldsymbol{u}, s) \in \mathbb{R}^{q+1}$ to lie in the second-order cone, the set of vectors whose last entry is at least the Euclidean norm of the remaining entries [6, Section 2.2.3]. This set is a cone, since it contains every non-negative multiple of its elements, and it is convex, so the constraint keeps the optimization problem convex. An optimization problem with a linear objective, linear constraints, and second-order cone constraints is a second-order cone program [6, Section 4.4.2].

By collecting the decision variables in the vector

$$\tilde{\boldsymbol{x}}_t = \left[\boldsymbol{x}_t^\top, \quad \alpha, \quad \boldsymbol{u}^\top, \quad s\right]^\top \in \mathbb{R}^{N+q+2},$$

and the constants in the vector

$$\boldsymbol{c} = \left[\boldsymbol{0}_N^\top, \quad 1, \quad \boldsymbol{0}_q^\top, \quad \frac{1}{(1-\beta)\sqrt{q}}\right]^\top \in \mathbb{R}^{N+q+2},$$

the portfolio optimization problem (A.9) can be written as a second-order cone program

$$\begin{aligned} \tilde{\boldsymbol{x}}_t^* = \arg\min_{\tilde{\boldsymbol{x}}_t} \quad & \boldsymbol{c}^\top \tilde{\boldsymbol{x}}_t \\ \text{s.t.} \quad & \boldsymbol{D}\tilde{\boldsymbol{x}}_t = \begin{bmatrix} \mathbf{1}_N^\top & \mathbf{0}_{q+2}^\top \end{bmatrix} \tilde{\boldsymbol{x}}_t = E = 1, \\ & \boldsymbol{A}\tilde{\boldsymbol{x}}_t = \begin{bmatrix} -\boldsymbol{R}^{\text{data}} & -\mathbf{1}_q & -\boldsymbol{I}_q & \mathbf{0}_q \end{bmatrix} \tilde{\boldsymbol{x}}_t \leq \boldsymbol{b} = \mathbf{0}_q, \\ & \|\boldsymbol{u}\|_2 \leq s, \\ & u_k \geq 0 \quad \text{for } k = 1, \ldots, q. \end{aligned} \tag{A.10}$$

Here $\boldsymbol{D} \in \mathbb{R}^{1\times(N+q+2)}$ and $\boldsymbol{A} \in \mathbb{R}^{q\times(N+q+2)}$ are the constraint matrices of (A.8) with a zero column added for the variable s. This formulation serves as the base unregularized portfolio optimization problem under second-order Expected Shortfall.

A.3 Regularization

The base problems (A.8) and (A.10) are known to be numerically unstable in high-dimensional settings, producing extreme and concentrated weights [12, 28]. We stabilize them with three regularization techniques, an L^1-norm penalty that promotes sparsity, an L^2-norm penalty that promotes shrinkage, and an asset leverage constraint that bounds gross exposure. Each is appended to the base problems while keeping them convex.

A.3.1 L^1 Regularization

The L^1 penalty $\lambda_1\|\boldsymbol{x}_t\|_1 = \lambda_1 \sum_{i=1}^N |x_{i,t}|$, with $\lambda_1 \geq 0$, is non-linear through the absolute value. Following [57], we linearize it by splitting each weight into its positive and negative parts via the auxiliary variables $\Delta_{i,t}^+ \geq 0$ and $\Delta_{i,t}^- \geq 0$

$$x_{i,t} = \Delta_{i,t}^+ - \Delta_{i,t}^-, \qquad |x_{i,t}| = \Delta_{i,t}^+ + \Delta_{i,t}^-. \tag{A.11}$$

The second identity holds at the optimum. Minimizing the L^1 penalty drives the $\Delta_{i,t}^+$ and $\Delta_{i,t}^-$ to their smallest feasible values, so at most one is non-zero, giving $\Delta_{i,t}^+ = \max(x_{i,t}, 0)$ and $\Delta_{i,t}^- = \max(-x_{i,t}, 0)$. Writing the positive and negative parts as vectors $\boldsymbol{\Delta}_t^+ = [\Delta_{1,t}^+, \ldots, \Delta_{N,t}^+]^\top$ and $\boldsymbol{\Delta}_t^- = [\Delta_{1,t}^-, \ldots, \Delta_{N,t}^-]^\top$, the L^1 penalty is written as

$$\lambda_1\|\boldsymbol{x}_t\|_1 = \lambda_1 \mathbf{1}_N^\top(\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-).$$

We append $\boldsymbol{\Delta}_t^-$ and $\boldsymbol{\Delta}_t^+$ to the decision vector, with the penalty coefficient λ_1 on each of their entries in the vector of constants. The decomposition (A.11) is enforced as N additional linear equality constraints, $\boldsymbol{x}_t + \boldsymbol{\Delta}_t^- - \boldsymbol{\Delta}_t^+ = \mathbf{0}_N$, alongside the original budget constraint.

A.3.2 L^2 Regularization

The L^2 penalty $\lambda_2\|\boldsymbol{x}_t\|_2^2 = \lambda_2 \boldsymbol{x}_t^\top \boldsymbol{x}_t$, with $\lambda_2 \geq 0$, is quadratic in the weights and is written as a quadratic form

$$\lambda_2\|\boldsymbol{x}_t\|_2^2 = \boldsymbol{x}_t^\top \boldsymbol{Q}\, \boldsymbol{x}_t, \qquad \boldsymbol{Q} = \lambda_2 \boldsymbol{I}_N.$$

Embedding this in the decision vector $\tilde{\boldsymbol{x}}_t$, the penalty becomes $\tilde{\boldsymbol{x}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{x}}_t$, where $\tilde{\boldsymbol{Q}}$ places the block $\lambda_2 \boldsymbol{I}_N$ on the asset weights and zeros for the rest of the variables. Adding this term to the linear objective of the base problems makes the objective quadratic, and since $\tilde{\boldsymbol{Q}}$ is positive semi-definite, it remains convex.

A.3.3 Leverage Constraint

The leverage constraint $\|\boldsymbol{x}_t\|_1 \leq \ell^a$, with $\ell^a \geq 1$, bounds the portfolio's gross exposure and guarantees a bounded feasible region even when $\lambda_1 = 0$. Reusing the decomposition (A.11), it is expressed as a single linear inequality

$$\|\boldsymbol{x}_t\|_1 = \mathbf{1}_N^\top(\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-) \leq \ell^a. \tag{A.12}$$

The choice $\ell^a = 1$ rules out leverage entirely and is equivalent to a hard L^1 constraint.

A.3.4 Expected Shortfall

Adding all three regularization terms to the base problem (A.8) extends the decision vector to

$$\tilde{\boldsymbol{x}}_t = \left[\boldsymbol{x}_t^\top, \quad \alpha, \quad \boldsymbol{u}^\top, \quad (\boldsymbol{\Delta}_t^-)^\top, \quad (\boldsymbol{\Delta}_t^+)^\top\right]^\top \in \mathbb{R}^{3N+q+1},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \left[\mathbf{0}_N^\top, \quad 1, \quad \tfrac{1}{q(1-\beta)}\mathbf{1}_q^\top, \quad \lambda_1\mathbf{1}_N^\top, \quad \lambda_1\mathbf{1}_N^\top\right]^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \lambda_2\boldsymbol{I}_N & \mathbf{0}_{N\times(2N+q+1)} \\ \mathbf{0}_{(2N+q+1)\times N} & \mathbf{0}_{(2N+q+1)\times(2N+q+1)} \end{bmatrix}.$$

The portfolio optimization problem under Expected Shortfall then becomes a quadratic program with linear constraints

$$\begin{aligned} \tilde{\boldsymbol{x}}_t^* = \arg\min_{\tilde{\boldsymbol{x}}_t} \quad & \tilde{\boldsymbol{c}}^\top\tilde{\boldsymbol{x}}_t + \tilde{\boldsymbol{x}}_t^\top\tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{x}}_t, \\ \text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{x}}_t = & \begin{bmatrix} \mathbf{1}_N^\top & \mathbf{0}_{q+1}^\top & \mathbf{0}_N^\top & \mathbf{0}_N^\top \\ \boldsymbol{I}_N & \mathbf{0}_{N\times(q+1)} & \boldsymbol{I}_N & -\boldsymbol{I}_N \end{bmatrix}\tilde{\boldsymbol{x}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_N \end{bmatrix}, \\ \tilde{\boldsymbol{A}}\tilde{\boldsymbol{x}}_t = & \begin{bmatrix} -\boldsymbol{R}^{\text{data}} & -\mathbf{1}_q & -\boldsymbol{I}_q & \mathbf{0}_{q\times N} & \mathbf{0}_{q\times N} \\ \mathbf{0}_N^\top & 0 & \mathbf{0}_q^\top & \mathbf{1}_N^\top & \mathbf{1}_N^\top \end{bmatrix}\tilde{\boldsymbol{x}}_t \leq \tilde{\boldsymbol{b}} = \begin{bmatrix} \mathbf{0}_q \\ \ell^a \end{bmatrix}, \\ & u_k \geq 0, \quad \Delta_{i,t}^- \geq 0, \quad \Delta_{i,t}^+ \geq 0 \quad \text{for } i = 1,\ldots,N,\ k = 1,\ldots,q. \end{aligned} \tag{A.13}$$

The first row of $\tilde{\boldsymbol{D}}$ enforces the budget constraint and the remaining N rows enforce the weight decomposition (A.11). The first q rows of $\tilde{\boldsymbol{A}}$ encode the inequality constraints from the linearization (A.7), extended with zeros for the new variables, and the final row of $\tilde{\boldsymbol{A}}$ imposes the leverage constraint (A.12).

A.3.5 Second-Order Expected Shortfall

Adding the same three regularization terms to the base problem (A.10) extends the decision vector to

$$\tilde{\boldsymbol{x}}_t = \left[\boldsymbol{x}_t^\top, \quad \alpha, \quad \boldsymbol{u}^\top, \quad (\boldsymbol{\Delta}_t^-)^\top, \quad (\boldsymbol{\Delta}_t^+)^\top, \quad s\right]^\top \in \mathbb{R}^{3N+q+2},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \left[\mathbf{0}_N^\top, \quad 1, \quad \mathbf{0}_q^\top, \quad \lambda_1\mathbf{1}_N^\top, \quad \lambda_1\mathbf{1}_N^\top, \quad \tfrac{1}{(1-\beta)\sqrt{q}}\right]^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \lambda_2 \boldsymbol{I}_N & \boldsymbol{0}_{N\times(2N+q+2)} \\ \boldsymbol{0}_{(2N+q+2)\times N} & \boldsymbol{0}_{(2N+q+2)\times(2N+q+2)} \end{bmatrix}.$$

The portfolio optimization problem under second-order Expected Shortfall then becomes a second-order cone program with a convex quadratic objective

$$\begin{aligned} \tilde{\boldsymbol{x}}_t^* = \arg\min_{\tilde{\boldsymbol{x}}_t} \quad & \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{x}}_t + \tilde{\boldsymbol{x}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{x}}_t, \\ \text{s.t.} \quad & \tilde{\boldsymbol{D}}\tilde{\boldsymbol{x}}_t = \begin{bmatrix} \boldsymbol{1}_N^\top & \boldsymbol{0}_{q+1}^\top & \boldsymbol{0}_N^\top & \boldsymbol{0}_N^\top & 0 \\ \boldsymbol{I}_N & \boldsymbol{0}_{N\times(q+1)} & \boldsymbol{I}_N & -\boldsymbol{I}_N & \boldsymbol{0}_N \end{bmatrix} \tilde{\boldsymbol{x}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \boldsymbol{0}_N \end{bmatrix}, \\ & \tilde{\boldsymbol{A}}\tilde{\boldsymbol{x}}_t = \begin{bmatrix} -\boldsymbol{R}^{\text{data}} & -\boldsymbol{1}_q & -\boldsymbol{I}_q & \boldsymbol{0}_{q\times N} & \boldsymbol{0}_{q\times N} & \boldsymbol{0}_q \\ \boldsymbol{0}_N^\top & 0 & \boldsymbol{0}_q^\top & \boldsymbol{1}_N^\top & \boldsymbol{1}_N^\top & 0 \end{bmatrix} \tilde{\boldsymbol{x}}_t \le \tilde{\boldsymbol{b}} = \begin{bmatrix} \boldsymbol{0}_q \\ \ell^a \end{bmatrix}, \\ & \|\boldsymbol{u}\|_2 \le s, \\ & \qquad u_k \ge 0, \quad \Delta_{i,t}^- \ge 0, \quad \Delta_{i,t}^+ \ge 0 \quad \text{for } i = 1,\dots,N,\ k = 1,\dots,q. \end{aligned} \tag{A.14}$$

Here $\tilde{\boldsymbol{D}}$ and $\tilde{\boldsymbol{A}}$ are the constraint matrices of (A.13) with a zero column added for the variable s, and the cone constraint is the same as in (A.10).

A.4 Sequential Optimization

So far, we have formulated the portfolio optimization problem over the asset weights only. An international portfolio, however, holds assets and currency positions as well, so constructing it also requires selecting currency exposures. In the sequential portfolio optimization approach, the asset weights and currency exposures are optimized in two steps. First, the asset weights $\boldsymbol{x}_t$ are optimized on the fully hedged returns. Then, conditional on the optimal $\boldsymbol{x}_t^*$, the currency exposures $\boldsymbol{\psi}_t \in \mathbb{R}^M$ are optimized. The two optimization steps are given by

$$\begin{aligned} \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} & \left\{ \rho_\beta\left(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}^{fh} \right) + \lambda_1^a \|\boldsymbol{x}_t\|_1 + \lambda_2^a \|\boldsymbol{x}_t\|_2^2 \right\} \\ & \text{s.t. } \boldsymbol{x}_t^\top \boldsymbol{1}_N = 1,\ \|\boldsymbol{x}_t\|_1 \le \ell^a, \end{aligned} \tag{A.15}$$

$$\boldsymbol{\psi}_t^* \mid \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{\psi}_t} \left\{ \rho_\beta\left(\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t) \right) + \lambda_1^c \|\boldsymbol{\psi}_t\|_1 + \lambda_2^c \|\boldsymbol{\psi}_t\|_2^2 \right\}. \tag{A.16}$$

The regularization parameters are kept separate across assets $(\lambda_1^a, \lambda_2^a)$ and currencies $(\lambda_1^c, \lambda_2^c)$.

The fully hedged return vector $\boldsymbol{R}_{t+1}^{fh}$ (defined in Section 3.2.3) takes the place of the asset return vector $\boldsymbol{R}_{t+1}$ of Sections A.1–A.3, so the asset step (A.15) is identical to the regularized programs (A.13) for Expected Shortfall and (A.14) for second-order Expected Shortfall, and is obtained from them by substituting the matrix of historically sampled fully hedged returns $\boldsymbol{R}^{fh} \in \mathbb{R}^{q\times N}$ for $\boldsymbol{R}^{\text{data}}$ and the asset-specific penalties λ_1^a, λ_2^a for λ_1, λ_2.

In the currency step (A.16), the asset weights $\boldsymbol{x}_t^*$ are fixed and the currency exposures $\boldsymbol{\psi}_t$ are optimized. The currency exposure $\psi_{c,t}$ is the net exposure of the portfolio to the foreign currency c as defined in Section 3.2.3, and each unit of net exposure earns the excess currency return $e_{c,t+1} - f_{c,t}$. Using the same auxiliary representation for the risk measures and the empirical approximation as in Section A.2, we introduce α_ψ and auxiliary variables $u_{k,\psi} \ge 0$, and the slack constraint (A.7) becomes

$$u_{k,\psi} \ge -\boldsymbol{\psi}_t^\top (\boldsymbol{e}_k - \boldsymbol{f}_k) - \boldsymbol{x}_t^{*\top} \boldsymbol{R}_k^{fh} - \alpha_\psi \quad \Longleftrightarrow \quad -\boldsymbol{\psi}_t^\top (\boldsymbol{e}_k - \boldsymbol{f}_k) - \alpha_\psi - u_{k,\psi} \le \boldsymbol{x}_t^{*\top} \boldsymbol{R}_k^{fh},$$

where $\boldsymbol{R}_k^{fh}$ is the k-th sampled fully hedged return vector, and $\boldsymbol{e}_k$ and $\boldsymbol{f}_k$ are the k-th sampled currency return and forward premium vectors. We stack the sampled excess currency returns $(\boldsymbol{e}_k - \boldsymbol{f}_k)^\top$ row by row into the matrix $\boldsymbol{R}^c \in \mathbb{R}^{q \times M}$ and collect the fixed asset-portfolio returns in $\boldsymbol{R}_{\mathcal{P}}^{fh} = \boldsymbol{R}^{fh}\boldsymbol{x}_t^* \in \mathbb{R}^q$.

The currency step has no budget constraint, since the currency overlay is a zero-investment portfolio whose domestic exposure $\psi_{0,t} = -\mathbf{1}_M^\top \boldsymbol{\psi}_t$ is inferred residually, and no leverage constraint is imposed on the currency exposures. We linearize the currency L^1 penalty with the decomposition $\psi_{c,t} = \Delta_{\psi,c,t}^+ - \Delta_{\psi,c,t}^-$ as in Section A.3.1 and write the currency L^2 penalty as a quadratic form with the block $\lambda_2^c \boldsymbol{I}_M$ as in Section A.3.2.

A.4.1 Currency Overlay under Expected Shortfall

For Expected Shortfall, the decision vector of the currency step is

$$\tilde{\boldsymbol{\psi}}_t = \left[\boldsymbol{\psi}_t^\top, \quad \alpha_\psi, \quad \boldsymbol{u}_\psi^\top, \quad (\boldsymbol{\Delta}_{\psi,t}^-)^\top, \quad (\boldsymbol{\Delta}_{\psi,t}^+)^\top\right]^\top \in \mathbb{R}^{3M+q+1},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \left[\mathbf{0}_M^\top, \quad 1, \quad \tfrac{1}{q(1-\beta)}\mathbf{1}_q^\top, \quad \lambda_1^c \mathbf{1}_M^\top, \quad \lambda_1^c \mathbf{1}_M^\top\right]^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \lambda_2^c \boldsymbol{I}_M & \mathbf{0}_{M \times (2M+q+1)} \\ \mathbf{0}_{(2M+q+1)\times M} & \mathbf{0}_{(2M+q+1)\times(2M+q+1)} \end{bmatrix}.$$

The currency overlay optimization under Expected Shortfall then becomes a quadratic program with linear constraints

$$\begin{aligned} \tilde{\boldsymbol{\psi}}_t^* &= \arg\min_{\tilde{\boldsymbol{\psi}}_t} \quad \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{\psi}}_t + \tilde{\boldsymbol{\psi}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{\psi}}_t, \\ \text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\psi}}_t &= \left[\boldsymbol{I}_M \quad \mathbf{0}_{M\times(q+1)} \quad \boldsymbol{I}_M \quad -\boldsymbol{I}_M\right] \tilde{\boldsymbol{\psi}}_t = \tilde{\boldsymbol{E}} = \mathbf{0}_M, \\ \tilde{\boldsymbol{A}}\tilde{\boldsymbol{\psi}}_t &= \left[-\boldsymbol{R}^c \quad -\mathbf{1}_q \quad -\boldsymbol{I}_q \quad \mathbf{0}_{q\times M} \quad \mathbf{0}_{q\times M}\right] \tilde{\boldsymbol{\psi}}_t \le \tilde{\boldsymbol{b}} = \boldsymbol{R}_{\mathcal{P}}^{fh}, \\ & u_{k,\psi} \ge 0, \quad \Delta_{\psi,c,t}^- \ge 0, \quad \Delta_{\psi,c,t}^+ \ge 0 \quad \text{for } c = 1,\dots,M,\ k = 1,\dots,q. \end{aligned} \tag{A.17}$$

The M rows of $\tilde{\boldsymbol{D}}$ enforce the weight decomposition (A.11) for the currency exposures, and the q rows of $\tilde{\boldsymbol{A}}$ encode the slack constraints of the currency step, with the fixed asset-portfolio returns $\boldsymbol{R}_{\mathcal{P}}^{fh}$ on the right-hand side.

A.4.2 Currency Overlay under Second-Order Expected Shortfall

For second-order Expected Shortfall, we introduce, as in Section A.2.2, one more auxiliary variable s_ψ subject to the cone constraint $\|\boldsymbol{u}_\psi\|_2 \le s_\psi$, which extends the decision vector to

$$\tilde{\boldsymbol{\psi}}_t = \left[\boldsymbol{\psi}_t^\top, \quad \alpha_\psi, \quad \boldsymbol{u}_\psi^\top, \quad (\boldsymbol{\Delta}_{\psi,t}^-)^\top, \quad (\boldsymbol{\Delta}_{\psi,t}^+)^\top, \quad s_\psi\right]^\top \in \mathbb{R}^{3M+q+2},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \left[\mathbf{0}_M^\top, \quad 1, \quad \mathbf{0}_q^\top, \quad \lambda_1^c \mathbf{1}_M^\top, \quad \lambda_1^c \mathbf{1}_M^\top, \quad \tfrac{1}{(1-\beta)\sqrt{q}}\right]^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \lambda_2^c \boldsymbol{I}_M & \mathbf{0}_{M \times (2M+q+2)} \\ \mathbf{0}_{(2M+q+2)\times M} & \mathbf{0}_{(2M+q+2)\times(2M+q+2)} \end{bmatrix}.$$

The currency overlay optimization under second-order Expected Shortfall then becomes a second-order cone program with a convex quadratic objective

$$\begin{aligned} \tilde{\boldsymbol{\psi}}_t^* &= \arg\min_{\tilde{\boldsymbol{\psi}}_t} \quad \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{\psi}}_t + \tilde{\boldsymbol{\psi}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{\psi}}_t, \\ \text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\psi}}_t &= \begin{bmatrix} \boldsymbol{I}_M & \boldsymbol{0}_{M\times(q+1)} & \boldsymbol{I}_M & -\boldsymbol{I}_M & \boldsymbol{0}_M \end{bmatrix} \tilde{\boldsymbol{\psi}}_t = \tilde{\boldsymbol{E}} = \boldsymbol{0}_M, \\ \tilde{\boldsymbol{A}}\tilde{\boldsymbol{\psi}}_t &= \begin{bmatrix} -\boldsymbol{R}^c & -\boldsymbol{1}_q & -\boldsymbol{I}_q & \boldsymbol{0}_{q\times M} & \boldsymbol{0}_{q\times M} & \boldsymbol{0}_q \end{bmatrix} \tilde{\boldsymbol{\psi}}_t \le \tilde{\boldsymbol{b}} = \boldsymbol{R}_{\mathcal{P}}^{fh}, \\ \|\boldsymbol{u}_\psi\|_2 &\le s_\psi, \\ & u_{k,\psi} \ge 0, \quad \Delta^-_{\psi,c,t} \ge 0, \quad \Delta^+_{\psi,c,t} \ge 0 \quad \text{for } c = 1,\dots,M,\ k = 1,\dots,q. \end{aligned} \tag{A.18}$$

Here $\tilde{\boldsymbol{D}}$ and $\tilde{\boldsymbol{A}}$ are the constraint matrices of (A.17) with a zero column added for the variable s_ψ.

A.5 Joint Optimization

In the joint portfolio optimization approach, the asset weights and currency exposures are optimized simultaneously rather than sequentially. We stack them into the combined decision vector $\boldsymbol{\theta}_t = [\boldsymbol{x}_t^\top, \boldsymbol{\psi}_t^\top]^\top \in \mathbb{R}^{N+M}$ and the corresponding returns into the combined return vector $\boldsymbol{R}_{t+1}^{\text{all}} = [\boldsymbol{R}_{t+1}^{fh\top}, (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)^\top]^\top \in \mathbb{R}^{N+M}$, so that $\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}} = \boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)$ is the hedged portfolio return of Section A.4. The joint optimization problem is

$$\begin{aligned} \boldsymbol{\theta}_t^* = \arg\min_{\boldsymbol{\theta}_t} & \left\{ \rho_\beta\left(\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}}\right) + \|\boldsymbol{S}_1\boldsymbol{\theta}_t\|_1 + \|\boldsymbol{S}_2\boldsymbol{\theta}_t\|_2^2 \right\} \\ & \text{s.t. } \boldsymbol{\theta}_t^\top \boldsymbol{q}_{N,M} = 1, \quad \|\boldsymbol{L}^a\boldsymbol{\theta}_t\|_1 \le \ell^a. \end{aligned} \tag{A.19}$$

The following matrices and vector adapt the problem to the joint optimization setting

$$\boldsymbol{q}_{N,M} = \begin{bmatrix} \boldsymbol{1}_N \\ \boldsymbol{0}_M \end{bmatrix}, \quad \boldsymbol{L}^a = \begin{bmatrix} \boldsymbol{I}_N & \boldsymbol{0}_{N\times M} \\ \boldsymbol{0}_{M\times N} & \boldsymbol{0}_{M\times M} \end{bmatrix},$$

$$\boldsymbol{S}_1 = \begin{bmatrix} \lambda_1^a \boldsymbol{I}_N & \boldsymbol{0}_{N\times M} \\ \boldsymbol{0}_{M\times N} & \lambda_1^c \boldsymbol{I}_M \end{bmatrix}, \quad \boldsymbol{S}_2 = \begin{bmatrix} \sqrt{\lambda_2^a}\, \boldsymbol{I}_N & \boldsymbol{0}_{N\times M} \\ \boldsymbol{0}_{M\times N} & \sqrt{\lambda_2^c}\, \boldsymbol{I}_M \end{bmatrix}.$$

The vector $\boldsymbol{q}_{N,M}$ applies the budget constraint to the asset weights only, so that $\boldsymbol{\theta}_t^\top \boldsymbol{q}_{N,M} = \boldsymbol{x}_t^\top \boldsymbol{1}_N = 1$, leaving the currency exposures as a zero-investment overlay. The matrix $\boldsymbol{L}^a$ applies the leverage constraint to the asset weights only, giving $\|\boldsymbol{L}^a\boldsymbol{\theta}_t\|_1 = \|\boldsymbol{x}_t\|_1$. The diagonal regularization matrices $\boldsymbol{S}_1$ and $\boldsymbol{S}_2$ allow the asset and currency penalties to differ within a single objective

$$\|\boldsymbol{S}_1\boldsymbol{\theta}_t\|_1 = \lambda_1^a\|\boldsymbol{x}_t\|_1 + \lambda_1^c\|\boldsymbol{\psi}_t\|_1, \qquad \|\boldsymbol{S}_2\boldsymbol{\theta}_t\|_2^2 = \lambda_2^a\|\boldsymbol{x}_t\|_2^2 + \lambda_2^c\|\boldsymbol{\psi}_t\|_2^2.$$

Let $\boldsymbol{R}^{\text{all}} = [\, \boldsymbol{R}^{fh} \;\; \boldsymbol{R}^c \,] \in \mathbb{R}^{q\times(N+M)}$ denote the matrix of jointly sampled asset and currency returns, whose k-th row $(\boldsymbol{R}_k^{\text{all}})^\top = [(\boldsymbol{R}_k^{fh})^\top, (\boldsymbol{e}_k - \boldsymbol{f}_k)^\top]$ is the k-th historical realization of $\boldsymbol{R}_{t+1}^{\text{all}}$. We replace the expectation in F_β and the L^2 norm in G_β by their empirical counterparts as in Section A.2, with the joint return $\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}}$ in place of $\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}$. Introducing α and the auxiliary variables $u_k \ge 0$ for $k = 1,\dots,q$ as in Section A.2, the slack constraint (A.7) becomes

$$u_k \ge -\boldsymbol{\theta}_t^\top \boldsymbol{R}_k^{\text{all}} - \alpha \qquad \Longleftrightarrow \qquad \boldsymbol{\theta}_t^\top \boldsymbol{R}_k^{\text{all}} + \alpha + u_k \ge 0.$$

To linearize the L^1 penalty, we decompose all $N+M$ components of $\boldsymbol{\theta}_t$ into positive and negative parts as in Section A.3.1, $\boldsymbol{\theta}_t = \boldsymbol{\Delta}_t^+ - \boldsymbol{\Delta}_t^-$ with $\boldsymbol{\Delta}_t^\pm = [(\boldsymbol{\Delta}_{x,t}^\pm)^\top, (\boldsymbol{\Delta}_{\psi,t}^\pm)^\top]^\top$, so that at the optimum

$$\begin{aligned}\|\boldsymbol{S}_1\boldsymbol{\theta}_t\|_1 &= \lambda_1^a \mathbf{1}_N^\top(\boldsymbol{\Delta}_{x,t}^+ + \boldsymbol{\Delta}_{x,t}^-) + \lambda_1^c \mathbf{1}_M^\top(\boldsymbol{\Delta}_{\psi,t}^+ + \boldsymbol{\Delta}_{\psi,t}^-),\\ \|\boldsymbol{L}^a\boldsymbol{\theta}_t\|_1 &= \boldsymbol{q}_{N,M}^\top(\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-) \le \ell^a.\end{aligned}$$

The L^2 penalty is the quadratic form $\|\boldsymbol{S}_2\boldsymbol{\theta}_t\|_2^2 = \boldsymbol{\theta}_t^\top \boldsymbol{S}_2^2\,\boldsymbol{\theta}_t$, where $\boldsymbol{S}_2^2$ is diagonal with the blocks $\lambda_2^a \boldsymbol{I}_N$ and $\lambda_2^c \boldsymbol{I}_M$, which enter the penalty matrix $\tilde{\boldsymbol{Q}}$ of the two programs below.

A.5.1 Joint Optimization under Expected Shortfall

For Expected Shortfall, the decision vector is

$$\tilde{\boldsymbol{\theta}}_t = \Big[\, \boldsymbol{x}_t^\top,\ \ \boldsymbol{\psi}_t^\top,\ \ \alpha,\ \ \boldsymbol{u}^\top,\ \ (\boldsymbol{\Delta}_{x,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{x,t}^+)^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^+)^\top \,\Big]^\top \in \mathbb{R}^{3N+3M+q+1},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \Big[\, \mathbf{0}_{N+M}^\top,\ \ 1,\ \ \tfrac{1}{q(1-\beta)}\mathbf{1}_q^\top,\ \ \lambda_1^a \mathbf{1}_N^\top,\ \ \lambda_1^c \mathbf{1}_M^\top,\ \ \lambda_1^a \mathbf{1}_N^\top,\ \ \lambda_1^c \mathbf{1}_M^\top \,\Big]^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \lambda_2^a \boldsymbol{I}_N & \mathbf{0}_{N\times M} & \mathbf{0}_{N\times(2N+2M+q+1)} \\ \mathbf{0}_{M\times N} & \lambda_2^c \boldsymbol{I}_M & \mathbf{0}_{M\times(2N+2M+q+1)} \\ \mathbf{0}_{(2N+2M+q+1)\times N} & \mathbf{0}_{(2N+2M+q+1)\times M} & \mathbf{0}_{(2N+2M+q+1)\times(2N+2M+q+1)} \end{bmatrix}.$$

The joint portfolio optimization problem under Expected Shortfall then becomes a quadratic program with linear constraints

$$\begin{aligned} \tilde{\boldsymbol{\theta}}_t^* &= \arg\min_{\tilde{\boldsymbol{\theta}}_t} \quad \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{\theta}}_t + \tilde{\boldsymbol{\theta}}_t^\top \tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{\theta}}_t, \\ \text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\theta}}_t &= \begin{bmatrix} \boldsymbol{q}_{N,M}^\top & \mathbf{0}_{q+1}^\top & \mathbf{0}_{N+M}^\top & \mathbf{0}_{N+M}^\top \\ \boldsymbol{I}_{N+M} & \mathbf{0}_{(N+M)\times(q+1)} & \boldsymbol{I}_{N+M} & -\boldsymbol{I}_{N+M} \end{bmatrix} \tilde{\boldsymbol{\theta}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_{N+M} \end{bmatrix}, \\ \tilde{\boldsymbol{A}}\tilde{\boldsymbol{\theta}}_t &= \begin{bmatrix} -\boldsymbol{R}^{\text{all}} & -\mathbf{1}_q & -\boldsymbol{I}_q & \mathbf{0}_{q\times(N+M)} & \mathbf{0}_{q\times(N+M)} \\ \mathbf{0}_{N+M}^\top & 0 & \mathbf{0}_q^\top & \boldsymbol{q}_{N,M}^\top & \boldsymbol{q}_{N,M}^\top \end{bmatrix} \tilde{\boldsymbol{\theta}}_t \le \tilde{\boldsymbol{b}} = \begin{bmatrix} \mathbf{0}_q \\ \ell^a \end{bmatrix}, \\ & u_k \ge 0 \quad \text{for } k = 1,\dots,q, \\ & \Delta_{x,i,t}^\pm \ge 0, \quad \Delta_{\psi,c,t}^\pm \ge 0 \quad \text{for } i = 1,\dots,N,\ c = 1,\dots,M. \end{aligned} \tag{A.20}$$

In $\tilde{\boldsymbol{D}}$, the first row enforces the asset budget constraint $\boldsymbol{x}_t^\top \mathbf{1}_N = 1$, and the remaining $N+M$ rows enforce the weight decomposition (A.11) for all components of $\boldsymbol{\theta}_t$. In $\tilde{\boldsymbol{A}}$, the first q rows encode the linearization inequalities (A.7) applied to the joint return matrix $\boldsymbol{R}^{\text{all}}$. The final row imposes the asset-only leverage constraint (A.12), giving $\mathbf{1}_N^\top(\boldsymbol{\Delta}_{x,t}^+ + \boldsymbol{\Delta}_{x,t}^-) = \|\boldsymbol{x}_t\|_1 \le \ell^a$.

A.5.2 Joint Optimization under Second-Order Expected Shortfall

For second-order Expected Shortfall, we introduce, as in Section A.2.2, one more auxiliary variable s subject to the cone constraint $\|\boldsymbol{u}\|_2 \le s$, which extends the decision vector to

$$\tilde{\boldsymbol{\theta}}_t = \Big[\, \boldsymbol{x}_t^\top,\ \ \boldsymbol{\psi}_t^\top,\ \ \alpha,\ \ \boldsymbol{u}^\top,\ \ (\boldsymbol{\Delta}_{x,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{x,t}^+)^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^+)^\top,\ \ s \,\Big]^\top \in \mathbb{R}^{3N+3M+q+2},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \Big[\, \mathbf{0}_{N+M}^\top, \quad 1, \quad \mathbf{0}_q^\top, \quad \lambda_1^a \mathbf{1}_N^\top, \quad \lambda_1^c \mathbf{1}_M^\top, \quad \lambda_1^a \mathbf{1}_N^\top, \quad \lambda_1^c \mathbf{1}_M^\top, \quad \tfrac{1}{(1-\beta)\sqrt{q}} \,\Big]^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \lambda_2^a \boldsymbol{I}_N & \mathbf{0}_{N\times M} & \mathbf{0}_{N\times(2N+2M+q+2)} \\ \mathbf{0}_{M\times N} & \lambda_2^c \boldsymbol{I}_M & \mathbf{0}_{M\times(2N+2M+q+2)} \\ \mathbf{0}_{(2N+2M+q+2)\times N} & \mathbf{0}_{(2N+2M+q+2)\times M} & \mathbf{0}_{(2N+2M+q+2)\times(2N+2M+q+2)} \end{bmatrix}.$$

The joint portfolio optimization problem under second-order Expected Shortfall then becomes a second-order cone program with a convex quadratic objective

$$\begin{aligned} \tilde{\boldsymbol{\theta}}_t^* &= \arg\min_{\tilde{\boldsymbol{\theta}}_t} \quad \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{\theta}}_t + \tilde{\boldsymbol{\theta}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{\theta}}_t, \\ \text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\theta}}_t &= \begin{bmatrix} \boldsymbol{q}_{N,M}^\top & \mathbf{0}_{q+1}^\top & \mathbf{0}_{N+M}^\top & \mathbf{0}_{N+M}^\top & 0 \\ \boldsymbol{I}_{N+M} & \mathbf{0}_{(N+M)\times(q+1)} & \boldsymbol{I}_{N+M} & -\boldsymbol{I}_{N+M} & \mathbf{0}_{N+M} \end{bmatrix} \tilde{\boldsymbol{\theta}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_{N+M} \end{bmatrix}, \\ \tilde{\boldsymbol{A}}\tilde{\boldsymbol{\theta}}_t &= \begin{bmatrix} -\boldsymbol{R}^{\text{all}} & -\mathbf{1}_q & -\boldsymbol{I}_q & \mathbf{0}_{q\times(N+M)} & \mathbf{0}_{q\times(N+M)} & \mathbf{0}_q \\ \mathbf{0}_{N+M}^\top & 0 & \mathbf{0}_q^\top & \boldsymbol{q}_{N,M}^\top & \boldsymbol{q}_{N,M}^\top & 0 \end{bmatrix} \tilde{\boldsymbol{\theta}}_t \le \tilde{\boldsymbol{b}} = \begin{bmatrix} \mathbf{0}_q \\ \ell^a \end{bmatrix}, \qquad \text{(A.21)} \\ \|\boldsymbol{u}\|_2 &\le s, \\ & u_k \ge 0 \quad \text{for } k = 1, \dots, q, \\ & \Delta_{x,i,t}^{\pm} \ge 0, \quad \Delta_{\psi,c,t}^{\pm} \ge 0 \quad \text{for } i = 1, \dots, N,\ c = 1, \dots, M. \end{aligned}$$

Here $\tilde{\boldsymbol{D}}$ and $\tilde{\boldsymbol{A}}$ are the constraint matrices of (A.20) with a zero column added for the variable s.

A.6 Mean-Risk Optimization

The programs of Sections A.2–A.5 minimize the risk measure only. A mean-risk program subtracts the expected portfolio return, scaled by $1/\gamma$, from the risk measure, where $\gamma > 0$ is the risk aversion coefficient. For problem (A.5), this gives

$$\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \left\{ \rho_\beta(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}) - \frac{1}{\gamma}\, \boldsymbol{\mu}_t^\top \boldsymbol{x}_t \right\}, \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1, \qquad \text{(A.22)}$$

where $\boldsymbol{\mu}_t = \mathbb{E}_t[\boldsymbol{R}_{t+1}]$ is the expected asset return.

Let $\boldsymbol{\mu}_t^{fh} = \mathbb{E}_t[\boldsymbol{R}_{t+1}^{fh}] \in \mathbb{R}^N$ and $\boldsymbol{\mu}_t^c = \mathbb{E}_t[\boldsymbol{e}_{t+1} - \boldsymbol{f}_t] \in \mathbb{R}^M$ denote the expected fully hedged asset returns and the expected excess currency returns, and $\boldsymbol{\mu}_t^{\text{all}} = [(\boldsymbol{\mu}_t^{fh})^\top, (\boldsymbol{\mu}_t^c)^\top]^\top \in \mathbb{R}^{N+M}$. The asset means are estimated from the equally weighted return of each economy, whose mean is shrunk toward zero with a James–Stein estimator, and the currency means are set to the negative forward premium $-\boldsymbol{f}_t$. Both estimators are defined and further discussed in Chapter 3.

In the sequential approach, the asset step and the currency step become

$$\begin{aligned} \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \Big\{ \rho_\beta\Big(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}^{fh}\Big) - \frac{1}{\gamma}\, (\boldsymbol{\mu}_t^{fh})^\top \boldsymbol{x}_t + \lambda_1^a \|\boldsymbol{x}_t\|_1 + \lambda_2^a \|\boldsymbol{x}_t\|_2^2 \Big\} \\ \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1,\ \|\boldsymbol{x}_t\|_1 \le \ell^a, \end{aligned} \qquad \text{(A.23)}$$

$$\begin{aligned}\boldsymbol{\psi}_t^* \mid \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{\psi}_t} \Big\{ & \rho_\beta\Big(\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)\Big) - \frac{1}{\gamma} (\boldsymbol{\mu}_t^c)^\top \boldsymbol{\psi}_t \\ & + \lambda_1^c \|\boldsymbol{\psi}_t\|_1 + \lambda_2^c \|\boldsymbol{\psi}_t\|_2^2 \Big\} \\ & \text{s.t. } \|\boldsymbol{\psi}_t\|_1 \le \ell^c.\end{aligned} \tag{A.24}$$

In the joint approach, problem (A.19) becomes

$$\begin{aligned}\boldsymbol{\theta}_t^* = \arg\min_{\boldsymbol{\theta}_t} & \left\{ \rho_\beta\Big(\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}}\Big) - \frac{1}{\gamma} (\boldsymbol{\mu}_t^{\text{all}})^\top \boldsymbol{\theta}_t + \|\boldsymbol{S}_1 \boldsymbol{\theta}_t\|_1 + \|\boldsymbol{S}_2 \boldsymbol{\theta}_t\|_2^2 \right\} \\ & \text{s.t. } \boldsymbol{\theta}_t^\top \boldsymbol{q}_{N,M} = 1, \quad \|\boldsymbol{L}^a \boldsymbol{\theta}_t\|_1 \le \ell^a, \quad \|\boldsymbol{L}^c \boldsymbol{\theta}_t\|_1 \le \ell^c,\end{aligned} \tag{A.25}$$

where

$$\boldsymbol{L}^c = \begin{bmatrix} \mathbf{0}_{N\times N} & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \boldsymbol{I}_M \end{bmatrix}$$

applies the leverage constraint to the currency exposures only. We add the currency leverage constraint to both the currency step and the joint program, since without it, the mean term can push the currency exposures toward very large positions.

Appendix B

Optimization Derivation: Variance and Semi-Variance

In this appendix, we derive the portfolio optimization problems used in the main text, building up from the definitions of variance and semi-variance to the programs solved in the empirical analysis. In Section B.1 we define both risk measures and show that they are convex in the portfolio weights. In Section B.2 we write the portfolio optimization problem as a quadratic program for both risk measures. In Section B.3 we regularize the programs by adding an L^1 penalty, an L^2 penalty, and a leverage constraint. We then extend the programs to the multi-currency setting, for sequential optimization in Section B.4 and for joint optimization in Section B.5. Finally, in Section B.6 we add the expected portfolio return to the objectives of the sequential and joint problems.

B.1 Definitions and Convexity

We define the variance of the portfolio return, which Markowitz [39] uses as the measure of portfolio risk, and the semi-variance, which Markowitz et al. [41] consider as a downside alternative to the variance, and show that both are convex in the portfolio weights. The two definitions allow for writing the portfolio optimization problem as a quadratic program for both risk measures in Section B.2.

Let $\boldsymbol{x}_t \in \mathbb{R}^N$ denote the portfolio weights, $\boldsymbol{R}_{t+1} \in \mathbb{R}^N$ the asset return vector, and $\boldsymbol{\mu}_t = \mathbb{E}_t[\boldsymbol{R}_{t+1}] \in \mathbb{R}^N$ the expected asset return. The variance of the portfolio return is

$$\mathrm{Var}_t(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}) = \mathbb{E}_t\big[(\boldsymbol{x}_t^\top \boldsymbol{\mu}_t - \boldsymbol{x}_t^\top \boldsymbol{R}_{t+1})^2\big] = \boldsymbol{x}_t^\top \boldsymbol{\Sigma}_t \boldsymbol{x}_t, \tag{B.1}$$

where $\boldsymbol{\Sigma}_t = \mathbb{E}_t\big[(\boldsymbol{R}_{t+1} - \boldsymbol{\mu}_t)(\boldsymbol{R}_{t+1} - \boldsymbol{\mu}_t)^\top\big] \in \mathbb{R}^{N \times N}$ is the conditional covariance matrix of the asset returns. Since the variance (B.1) is non-negative for every $\boldsymbol{x}_t$, the matrix $\boldsymbol{\Sigma}_t$ is positive semi-definite, and the quadratic form is convex in $\boldsymbol{x}_t$ [6, Section 3.1.4].

The variance penalizes returns above the expected return as much as returns below it, although an investor is typically concerned with underperformance rather than outperformance [41]. This motivates us to consider the semi-variance of the portfolio return as a downside risk measure. The semi-variance is defined as

$$\mathrm{SVar}_t(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}) = \mathbb{E}_t\Big[\big((\boldsymbol{x}_t^\top \boldsymbol{\mu}_t - \boldsymbol{x}_t^\top \boldsymbol{R}_{t+1})^+\big)^2\Big]. \tag{B.2}$$

The inner map $\boldsymbol{x}_t \mapsto \boldsymbol{x}_t^\top(\boldsymbol{\mu}_t - \boldsymbol{R}_{t+1})$ is linear and the function $z \mapsto (z^+)^2$ is convex. The composition of a convex function with an affine map is convex [6, Section 3.2.2]. Expectation is a linear operator which preserves convexity, so the semi-variance is convex in $\boldsymbol{x}_t$.

Since both the variance and the semi-variance are convex, any local minimum is global, ensuring that the portfolio optimization problems under both risk measures can be solved reliably with standard convex optimization methods.

B.2 Representation for Portfolio Optimization

Let ρ denote either of the two risk measures, written as a function of the portfolio return $\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}$. The portfolio optimization problem is

$$\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \rho(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}), \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1, \tag{B.3}$$

where the budget constraint enforces full investment. Each program in this appendix defines its own decision vector, vector of constants, and penalty and constraint matrices, so the same symbols denote slightly different objects in different programs, and we refer to a specific program by its equation number when discussing them.

B.2.1 Variance

By (B.1), the objective of problem (B.3) under variance is the quadratic form $\boldsymbol{x}_t^\top \boldsymbol{\Sigma}_t \boldsymbol{x}_t$. In the programs, $\boldsymbol{\Sigma}_t$ is the Ledoit–Wolf estimator of the covariance matrix, denoted $\hat{\boldsymbol{\Sigma}}_t$ in Section 3.1.1, which is positive semi-definite, so the convexity shown in Section B.1 is preserved. With the decision vector $\boldsymbol{x}_t \in \mathbb{R}^N$ and the penalty matrix $\boldsymbol{Q} = \boldsymbol{\Sigma}_t \in \mathbb{R}^{N\times N}$, the portfolio optimization problem under variance can be written as a quadratic program

$$\begin{aligned} \boldsymbol{x}_t^* &= \arg\min_{\boldsymbol{x}_t} \quad \boldsymbol{x}_t^\top \boldsymbol{Q}\, \boldsymbol{x}_t \\ \text{s.t.} \quad \boldsymbol{D}\boldsymbol{x}_t &= \mathbf{1}_N^\top \boldsymbol{x}_t = E = 1. \end{aligned} \tag{B.4}$$

Here $\boldsymbol{D} = \mathbf{1}_N^\top \in \mathbb{R}^{1\times N}$ encodes the budget constraint. This formulation serves as the base unregularized portfolio optimization problem under variance.

B.2.2 Semi-Variance

Let $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_q \in \mathbb{R}^N$ denote q historical realizations of the asset return vector $\boldsymbol{R}_{t+1}$, as in Section A.2.1. We approximate the expected asset return $\boldsymbol{\mu}_t$ by the sample mean $\bar{\boldsymbol{y}} = \frac{1}{q}\sum_{k=1}^q \boldsymbol{y}_k$ and the conditional expectation in (B.2) by the empirical average over this sample

$$\mathrm{SVar}_t(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}) \approx \frac{1}{q}\sum_{k=1}^q \big((\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k)^+\big)^2,$$

so that the portfolio optimization problem under semi-variance becomes

$$\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \frac{1}{q}\sum_{k=1}^q \big((\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k)^+\big)^2, \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1. \tag{B.5}$$

Each summand is only piecewise quadratic in $\boldsymbol{x}_t$. Following [41], and as for Expected Shortfall in Section A.2.1, we modify it by introducing, for each observation k, an auxiliary variable $u_k \geq 0$ subject to

$$u_k \geq \boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k \qquad \Longleftrightarrow \qquad \boldsymbol{x}_t^\top (\boldsymbol{y}_k - \bar{\boldsymbol{y}}) + u_k \geq 0. \tag{B.6}$$

The objective becomes $\frac{1}{q}\sum_{k=1}^q u_k^2 = \frac{1}{q}\|\boldsymbol{u}\|_2^2$. It is strictly increasing in each $u_k \geq 0$ and minimized, so every u_k is driven to its lower bound at the optimum, $u_k^* = (\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k)^+$.

By collecting the decision variables in the vector

$$\tilde{\boldsymbol{x}}_t = \begin{bmatrix} x_{1,t}, \dots, x_{N,t}, & u_1, \dots, u_q \end{bmatrix}^\top = \begin{bmatrix} \boldsymbol{x}_t^\top, & \boldsymbol{u}^\top \end{bmatrix}^\top \in \mathbb{R}^{N+q},$$

and the penalty matrix

$$\boldsymbol{Q} = \begin{bmatrix} \mathbf{0}_{N\times N} & \mathbf{0}_{N\times q} \\ \mathbf{0}_{q\times N} & \frac{1}{q}\boldsymbol{I}_q \end{bmatrix} \in \mathbb{R}^{(N+q)\times(N+q)},$$

the portfolio optimization problem (B.5) can be written as a quadratic program

$$\begin{aligned} \tilde{\boldsymbol{x}}_t^* = \arg\min_{\tilde{\boldsymbol{x}}_t} \quad & \tilde{\boldsymbol{x}}_t^\top \boldsymbol{Q}\, \tilde{\boldsymbol{x}}_t \\ \text{s.t.} \quad & \boldsymbol{D}\tilde{\boldsymbol{x}}_t = \begin{bmatrix} \mathbf{1}_N^\top & \mathbf{0}_q^\top \end{bmatrix} \tilde{\boldsymbol{x}}_t = E = 1, \\ & \boldsymbol{A}\tilde{\boldsymbol{x}}_t = \begin{bmatrix} -\mathring{\boldsymbol{R}}^{\text{data}} & -\boldsymbol{I}_q \end{bmatrix} \tilde{\boldsymbol{x}}_t \le \boldsymbol{b} = \mathbf{0}_q, \\ & u_k \ge 0 \quad \text{for } k = 1, \dots, q. \end{aligned} \tag{B.7}$$

Here $\boldsymbol{D} \in \mathbb{R}^{1\times(N+q)}$ encodes the budget constraint, while $\boldsymbol{A} \in \mathbb{R}^{q\times(N+q)}$ and $\boldsymbol{b} \in \mathbb{R}^q$ encode the q inequality constraints (B.6). The matrix $\mathring{\boldsymbol{R}}^{\text{data}} = \boldsymbol{R}^{\text{data}} - \mathbf{1}_q \bar{\boldsymbol{y}}^\top \in \mathbb{R}^{q\times N}$ is the demeaned sample matrix, where $\boldsymbol{R}^{\text{data}}$ is composed of the sampled return vectors $\boldsymbol{y}_k^\top$ row by row, so that the k-th row of $\mathring{\boldsymbol{R}}^{\text{data}}$ is $(\boldsymbol{y}_k - \bar{\boldsymbol{y}})^\top$. This formulation serves as the base unregularized portfolio optimization problem under semi-variance.

B.3 Regularization

The base problems (B.4) and (B.7) are known to be unstable in high-dimensional settings, where the programs are prone to producing extreme and concentrated weights [47, 28]. As for Expected Shortfall, we stabilize them with the L^1 penalty, the L^2 penalty, and the asset leverage constraint of Sections A.3.1–A.3.3, using the decomposition (A.11) of the weights into positive and negative parts and the leverage constraint (A.12).

B.3.1 Variance

Adding the three regularization terms to the base problem (B.4) extends the decision vector to

$$\tilde{\boldsymbol{x}}_t = \begin{bmatrix} \boldsymbol{x}_t^\top, & (\boldsymbol{\Delta}_t^-)^\top, & (\boldsymbol{\Delta}_t^+)^\top \end{bmatrix}^\top \in \mathbb{R}^{3N},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \begin{bmatrix} \mathbf{0}_N^\top, & \lambda_1 \mathbf{1}_N^\top, & \lambda_1 \mathbf{1}_N^\top \end{bmatrix}^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \boldsymbol{\Sigma}_t + \lambda_2 \boldsymbol{I}_N & \mathbf{0}_{N\times 2N} \\ \mathbf{0}_{2N\times N} & \mathbf{0}_{2N\times 2N} \end{bmatrix}.$$

Since $\boldsymbol{x}_t^\top \boldsymbol{\Sigma}_t \boldsymbol{x}_t + \lambda_2 \boldsymbol{x}_t^\top \boldsymbol{x}_t = \boldsymbol{x}_t^\top (\boldsymbol{\Sigma}_t + \lambda_2 \boldsymbol{I}_N) \boldsymbol{x}_t$, the L^2 penalty shrinks the covariance estimate toward a scaled identity matrix [36]. The portfolio optimization problem under variance then

becomes a quadratic program with linear constraints

$$\begin{aligned} \tilde{\boldsymbol{x}}_t^* &= \arg\min_{\tilde{\boldsymbol{x}}_t} \quad \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{x}}_t + \tilde{\boldsymbol{x}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{x}}_t, \\ \text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{x}}_t &= \begin{bmatrix} \mathbf{1}_N^\top & \mathbf{0}_N^\top & \mathbf{0}_N^\top \\ \boldsymbol{I}_N & \boldsymbol{I}_N & -\boldsymbol{I}_N \end{bmatrix} \tilde{\boldsymbol{x}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_N \end{bmatrix}, \\ \tilde{\boldsymbol{A}}\tilde{\boldsymbol{x}}_t &= \begin{bmatrix} \mathbf{0}_N^\top & \mathbf{1}_N^\top & \mathbf{1}_N^\top \end{bmatrix} \tilde{\boldsymbol{x}}_t \le \tilde{b} = \ell^a, \\ \Delta_{i,t}^- \ge 0, &\quad \Delta_{i,t}^+ \ge 0 \quad \text{for } i = 1, \dots, N. \end{aligned} \tag{B.8}$$

The first row of $\tilde{\boldsymbol{D}}$ enforces the budget constraint and the remaining N rows enforce the weight decomposition (A.11), while the single row of $\tilde{\boldsymbol{A}}$ imposes the leverage constraint (A.12).

B.3.2 Semi-Variance

Adding the same three regularization terms to the base problem (B.7) extends the decision vector to

$$\tilde{\boldsymbol{x}}_t = \begin{bmatrix} \boldsymbol{x}_t^\top, & \boldsymbol{u}^\top, & (\boldsymbol{\Delta}_t^-)^\top, & (\boldsymbol{\Delta}_t^+)^\top \end{bmatrix}^\top \in \mathbb{R}^{3N+q},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \begin{bmatrix} \mathbf{0}_N^\top, & \mathbf{0}_q^\top, & \lambda_1 \mathbf{1}_N^\top, & \lambda_1 \mathbf{1}_N^\top \end{bmatrix}^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \lambda_2 \boldsymbol{I}_N & \mathbf{0}_{N\times q} & \mathbf{0}_{N\times 2N} \\ \mathbf{0}_{q\times N} & \frac{1}{q}\boldsymbol{I}_q & \mathbf{0}_{q\times 2N} \\ \mathbf{0}_{2N\times N} & \mathbf{0}_{2N\times q} & \mathbf{0}_{2N\times 2N} \end{bmatrix}.$$

The sample semi-variance in (B.5) equals $\boldsymbol{x}_t^\top \boldsymbol{\Sigma}^-(\boldsymbol{x}_t)\, \boldsymbol{x}_t$, where

$$\boldsymbol{\Sigma}^-(\boldsymbol{x}_t) = \frac{1}{q} \sum_{k\,:\,\boldsymbol{x}_t^\top(\boldsymbol{y}_k - \bar{\boldsymbol{y}}) < 0} (\boldsymbol{y}_k - \bar{\boldsymbol{y}})(\boldsymbol{y}_k - \bar{\boldsymbol{y}})^\top$$

is the sample semi-covariance matrix of the observations below the portfolio mean, so the semi-variance and the L^2 penalty together equal $\boldsymbol{x}_t^\top(\boldsymbol{\Sigma}^-(\boldsymbol{x}_t) + \lambda_2 \boldsymbol{I}_N)\boldsymbol{x}_t$. The penalty shrinks this semi-covariance matrix toward a scaled identity matrix in the same way as it does for the variance. The portfolio optimization problem under semi-variance then becomes a quadratic program with linear constraints

$$\begin{aligned} \tilde{\boldsymbol{x}}_t^* &= \arg\min_{\tilde{\boldsymbol{x}}_t} \quad \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{x}}_t + \tilde{\boldsymbol{x}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{x}}_t, \\ \text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{x}}_t &= \begin{bmatrix} \mathbf{1}_N^\top & \mathbf{0}_q^\top & \mathbf{0}_N^\top & \mathbf{0}_N^\top \\ \boldsymbol{I}_N & \mathbf{0}_{N\times q} & \boldsymbol{I}_N & -\boldsymbol{I}_N \end{bmatrix} \tilde{\boldsymbol{x}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_N \end{bmatrix}, \\ \tilde{\boldsymbol{A}}\tilde{\boldsymbol{x}}_t &= \begin{bmatrix} -\mathring{\boldsymbol{R}}^{\text{data}} & -\boldsymbol{I}_q & \mathbf{0}_{q\times N} & \mathbf{0}_{q\times N} \\ \mathbf{0}_N^\top & \mathbf{0}_q^\top & \mathbf{1}_N^\top & \mathbf{1}_N^\top \end{bmatrix} \tilde{\boldsymbol{x}}_t \le \tilde{\boldsymbol{b}} = \begin{bmatrix} \mathbf{0}_q \\ \ell^a \end{bmatrix}, \\ u_k \ge 0, &\quad \Delta_{i,t}^- \ge 0, \quad \Delta_{i,t}^+ \ge 0 \quad \text{for } i = 1, \dots, N,\ k = 1, \dots, q. \end{aligned} \tag{B.9}$$

The first row of $\tilde{\boldsymbol{D}}$ enforces the budget constraint and the remaining N rows enforce the weight decomposition (A.11). The first q rows of $\tilde{\boldsymbol{A}}$ encode the inequality constraints (B.6), extended with zeros for the new variables, and the final row of $\tilde{\boldsymbol{A}}$ imposes the leverage constraint (A.12).

B.4 Sequential Optimization

As we have considered for Expected Shortfall in Section A.4, constructing an international portfolio also requires selecting currency exposures. In the sequential portfolio optimization approach, the asset weights and currency exposures are optimized in two steps. First, the asset weights $\boldsymbol{x}_t$ are optimized on the fully hedged returns. Then, conditional on the optimal $\boldsymbol{x}_t^*$, the currency exposures $\boldsymbol{\psi}_t \in \mathbb{R}^M$ are optimized. The two optimization steps are given by

$$\begin{aligned} \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \left\{ \rho\Big(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}^{fh} \Big) + \lambda_1^a \|\boldsymbol{x}_t\|_1 + \lambda_2^a \|\boldsymbol{x}_t\|_2^2 \right\} \\ \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1,\ \|\boldsymbol{x}_t\|_1 \le \ell^a, \end{aligned} \tag{B.10}$$

$$\boldsymbol{\psi}_t^* \mid \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{\psi}_t} \left\{ \rho\Big(\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t) \Big) + \lambda_1^c \|\boldsymbol{\psi}_t\|_1 + \lambda_2^c \|\boldsymbol{\psi}_t\|_2^2 \right\}. \tag{B.11}$$

The regularization parameters are kept separate across assets $(\lambda_1^a, \lambda_2^a)$ and currencies $(\lambda_1^c, \lambda_2^c)$.

The fully hedged return vector $\boldsymbol{R}_{t+1}^{fh}$, defined in Section 3.2.3, replaces the asset return vector $\boldsymbol{R}_{t+1}$ of Sections B.1–B.3, with expected value $\boldsymbol{\mu}_t^{fh} = \mathbb{E}_t[\boldsymbol{R}_{t+1}^{fh}] \in \mathbb{R}^N$ and conditional covariance matrix

$$\boldsymbol{\Sigma}_t^a = \mathbb{E}_t\big[(\boldsymbol{R}_{t+1}^{fh} - \boldsymbol{\mu}_t^{fh})(\boldsymbol{R}_{t+1}^{fh} - \boldsymbol{\mu}_t^{fh})^\top\big] \in \mathbb{R}^{N\times N},$$

which in the programs is the Ledoit–Wolf estimator of Chapter 3. The asset step (B.10) is therefore identical to the regularized programs (B.8) for variance and (B.9) for semi-variance, and is obtained from them by substituting $\boldsymbol{\Sigma}_t^a$ for $\boldsymbol{\Sigma}_t$ in (B.8) and the demeaned matrix $\mathring{\boldsymbol{R}}^{fh}$ of the historically sampled fully hedged returns $\boldsymbol{R}^{fh} \in \mathbb{R}^{q\times N}$ for $\mathring{\boldsymbol{R}}^{\text{data}}$ in (B.9).

In the currency step (B.11), the asset weights $\boldsymbol{x}_t^*$ are fixed and the currency exposures $\boldsymbol{\psi}_t$ are optimized. The currency exposure $\psi_{c,t}$ is the net exposure of the portfolio to the foreign currency c as defined in Section 3.2.3, and each unit of net exposure earns the excess currency return $e_{c,t+1} - f_{c,t}$. The currency step has no budget constraint, since the currency overlay is a zero-investment portfolio whose domestic exposure $\psi_{0,t} = -\mathbf{1}_M^\top \boldsymbol{\psi}_t$ is inferred residually, and no leverage constraint is imposed on the currency exposures. We linearize the currency L^1 penalty with the decomposition $\psi_{c,t} = \Delta_{\psi,c,t}^+ - \Delta_{\psi,c,t}^-$ as in Section A.3.1 and write the currency L^2 penalty as a quadratic form as in Section A.3.2.

B.4.1 Currency Overlay under Variance

Let $\boldsymbol{\mu}_t^c = \mathbb{E}_t[\boldsymbol{e}_{t+1} - \boldsymbol{f}_t] \in \mathbb{R}^M$ denote the expected excess currency return, and

$$\begin{aligned} \boldsymbol{\Sigma}_t^c &= \mathbb{E}_t\big[(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t - \boldsymbol{\mu}_t^c)(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t - \boldsymbol{\mu}_t^c)^\top\big] \in \mathbb{R}^{M\times M}, \\ \boldsymbol{\Sigma}_t^{ac} &= \mathbb{E}_t\big[(\boldsymbol{R}_{t+1}^{fh} - \boldsymbol{\mu}_t^{fh})(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t - \boldsymbol{\mu}_t^c)^\top\big] \in \mathbb{R}^{N\times M} \end{aligned}$$

the conditional covariance matrix of the excess currency returns and their conditional cross-covariance matrix with the fully hedged returns. In the programs, $\boldsymbol{\Sigma}_t^c$ is the Ledoit–Wolf estimator and $\boldsymbol{\Sigma}_t^{ac}$ the sample cross-covariance matrix. The variance of the hedged portfolio return decomposes as

$$\operatorname{Var}_t(\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)) = \boldsymbol{x}_t^{*\top} \boldsymbol{\Sigma}_t^a \boldsymbol{x}_t^* + 2\, \boldsymbol{x}_t^{*\top} \boldsymbol{\Sigma}_t^{ac} \boldsymbol{\psi}_t + \boldsymbol{\psi}_t^\top \boldsymbol{\Sigma}_t^c \boldsymbol{\psi}_t,$$

where the first term does not depend on $\boldsymbol{\psi}_t$ and is subsequently dropped from the currency step. The decision vector of the currency step is

$$\tilde{\boldsymbol{\psi}}_t = \Big[\boldsymbol{\psi}_t^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^-)^\top,\ \ (\boldsymbol{\Delta}_{\psi,t}^+)^\top\Big]^\top \in \mathbb{R}^{3M},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \left[2\,\boldsymbol{x}_t^{*\top}\boldsymbol{\Sigma}_t^{ac},\quad \lambda_1^c\mathbf{1}_M^\top,\quad \lambda_1^c\mathbf{1}_M^\top\right]^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \boldsymbol{\Sigma}_t^c + \lambda_2^c\boldsymbol{I}_M & \mathbf{0}_{M\times 2M} \\ \mathbf{0}_{2M\times M} & \mathbf{0}_{2M\times 2M}\end{bmatrix}.$$

The currency overlay optimization under variance then becomes a quadratic program with linear constraints

$$\begin{aligned} \tilde{\boldsymbol{\psi}}_t^* &= \arg\min_{\tilde{\boldsymbol{\psi}}_t} \quad \tilde{\boldsymbol{c}}^\top\tilde{\boldsymbol{\psi}}_t + \tilde{\boldsymbol{\psi}}_t^\top\tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{\psi}}_t, \\ \text{s.t.}\quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\psi}}_t &= \left[\boldsymbol{I}_M \quad \boldsymbol{I}_M \quad -\boldsymbol{I}_M\right]\tilde{\boldsymbol{\psi}}_t = \tilde{\boldsymbol{E}} = \mathbf{0}_M, \\ \Delta^-_{\psi,c,t} \geq 0,&\quad \Delta^+_{\psi,c,t} \geq 0 \quad \text{for } c = 1,\ldots,M. \end{aligned} \tag{B.12}$$

The M rows of $\tilde{\boldsymbol{D}}$ enforce the weight decomposition (A.11) for the currency exposures, and the fixed asset weights are present through the linear term $2\,(\boldsymbol{\Sigma}_t^{ac})^\top\boldsymbol{x}_t^*$ in $\tilde{\boldsymbol{c}}$.

B.4.2 Currency Overlay under Semi-Variance

For semi-variance, we use the empirical approximation of Section B.2.2 for the hedged portfolio return. As in Section A.4, let $\boldsymbol{R}_k^{fh}$ denote the k-th sampled fully hedged return vector, and $\boldsymbol{e}_k$ and $\boldsymbol{f}_k$ the k-th sampled currency return and forward premium vectors. We stack the sampled excess currency returns $(\boldsymbol{e}_k - \boldsymbol{f}_k)^\top$ row by row into the matrix $\boldsymbol{R}^c \in \mathbb{R}^{q\times M}$ and collect the fixed asset-portfolio returns in $\boldsymbol{R}_{\mathcal{P}}^{fh} = \boldsymbol{R}^{fh}\boldsymbol{x}_t^* \in \mathbb{R}^q$. Since the sample mean of the hedged portfolio returns $\boldsymbol{R}_{\mathcal{P}}^{fh} + \boldsymbol{R}^c\boldsymbol{\psi}_t$ is linear in $\boldsymbol{\psi}_t$, their demeaned formulation is simply $\mathring{\boldsymbol{R}}_{\mathcal{P}}^{fh} + \mathring{\boldsymbol{R}}^c\boldsymbol{\psi}_t$, where $\mathring{\boldsymbol{R}}_{\mathcal{P}}^{fh} = \mathring{\boldsymbol{R}}^{fh}\boldsymbol{x}_t^*$. Introducing the auxiliary variables $\boldsymbol{u}_\psi \geq \mathbf{0}_q$, the slack constraint (B.6) becomes

$$\boldsymbol{u}_\psi \geq -\mathring{\boldsymbol{R}}^c\boldsymbol{\psi}_t - \mathring{\boldsymbol{R}}_{\mathcal{P}}^{fh} \qquad\Longleftrightarrow\qquad -\mathring{\boldsymbol{R}}^c\boldsymbol{\psi}_t - \boldsymbol{u}_\psi \leq \mathring{\boldsymbol{R}}_{\mathcal{P}}^{fh}.$$

The decision vector of the currency step is

$$\tilde{\boldsymbol{\psi}}_t = \left[\boldsymbol{\psi}_t^\top,\quad \boldsymbol{u}_\psi^\top,\quad (\boldsymbol{\Delta}^-_{\psi,t})^\top,\quad (\boldsymbol{\Delta}^+_{\psi,t})^\top\right]^\top \in \mathbb{R}^{3M+q},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \left[\mathbf{0}_M^\top,\quad \mathbf{0}_q^\top,\quad \lambda_1^c\mathbf{1}_M^\top,\quad \lambda_1^c\mathbf{1}_M^\top\right]^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \lambda_2^c\boldsymbol{I}_M & \mathbf{0}_{M\times q} & \mathbf{0}_{M\times 2M} \\ \mathbf{0}_{q\times M} & \frac{1}{q}\boldsymbol{I}_q & \mathbf{0}_{q\times 2M} \\ \mathbf{0}_{2M\times M} & \mathbf{0}_{2M\times q} & \mathbf{0}_{2M\times 2M}\end{bmatrix}.$$

The currency overlay optimization under semi-variance then becomes a quadratic program with linear constraints

$$\begin{aligned} \tilde{\boldsymbol{\psi}}_t^* &= \arg\min_{\tilde{\boldsymbol{\psi}}_t} \quad \tilde{\boldsymbol{c}}^\top\tilde{\boldsymbol{\psi}}_t + \tilde{\boldsymbol{\psi}}_t^\top\tilde{\boldsymbol{Q}}\,\tilde{\boldsymbol{\psi}}_t, \\ \text{s.t.}\quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\psi}}_t &= \left[\boldsymbol{I}_M \quad \mathbf{0}_{M\times q} \quad \boldsymbol{I}_M \quad -\boldsymbol{I}_M\right]\tilde{\boldsymbol{\psi}}_t = \tilde{\boldsymbol{E}} = \mathbf{0}_M, \\ \tilde{\boldsymbol{A}}\tilde{\boldsymbol{\psi}}_t &= \left[-\mathring{\boldsymbol{R}}^c \quad -\boldsymbol{I}_q \quad \mathbf{0}_{q\times M} \quad \mathbf{0}_{q\times M}\right]\tilde{\boldsymbol{\psi}}_t \leq \tilde{\boldsymbol{b}} = \mathring{\boldsymbol{R}}_{\mathcal{P}}^{fh}, \\ u_{k,\psi} \geq 0,&\quad \Delta^-_{\psi,c,t} \geq 0, \quad \Delta^+_{\psi,c,t} \geq 0 \quad \text{for } c = 1,\ldots,M,\ k = 1,\ldots,q. \end{aligned} \tag{B.13}$$

The M rows of $\tilde{\boldsymbol{D}}$ enforce the weight decomposition (A.11) for the currency exposures, and the q rows of $\tilde{\boldsymbol{A}}$ encode the slack constraints of the currency step.

B.5 Joint Optimization

In the joint portfolio optimization approach, the asset weights and currency exposures are optimized simultaneously rather than sequentially. As for Expected Shortfall in Section A.5, we stack the weights into the combined decision vector $\boldsymbol{\theta}_t = [\boldsymbol{x}_t^\top, \boldsymbol{\psi}_t^\top]^\top \in \mathbb{R}^{N+M}$ and the corresponding returns into the combined return vector $\boldsymbol{R}_{t+1}^{\text{all}} = [\boldsymbol{R}_{t+1}^{fh\top}, (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)^\top]^\top \in \mathbb{R}^{N+M}$, so that $\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}} = \boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)$ is the hedged portfolio return of Section B.4. The joint optimization problem is

$$\begin{aligned} \boldsymbol{\theta}_t^* = \arg\min_{\boldsymbol{\theta}_t} \left\{ \rho\Big(\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}}\Big) + \|\boldsymbol{S}_1 \boldsymbol{\theta}_t\|_1 + \|\boldsymbol{S}_2 \boldsymbol{\theta}_t\|_2^2 \right\} \\ \text{s.t. } \boldsymbol{\theta}_t^\top \boldsymbol{q}_{N,M} = 1, \quad \|\boldsymbol{L}^a \boldsymbol{\theta}_t\|_1 \le \ell^a, \end{aligned} \tag{B.14}$$

with

$$\boldsymbol{q}_{N,M} = \begin{bmatrix} \mathbf{1}_N \\ \mathbf{0}_M \end{bmatrix}, \quad \boldsymbol{L}^a = \begin{bmatrix} \boldsymbol{I}_N & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \mathbf{0}_{M\times M} \end{bmatrix},$$

$$\boldsymbol{S}_1 = \begin{bmatrix} \lambda_1^a \boldsymbol{I}_N & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \lambda_1^c \boldsymbol{I}_M \end{bmatrix}, \quad \boldsymbol{S}_2 = \begin{bmatrix} \sqrt{\lambda_2^a}\, \boldsymbol{I}_N & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \sqrt{\lambda_2^c}\, \boldsymbol{I}_M \end{bmatrix}.$$

The vector $\boldsymbol{q}_{N,M}$ and the matrix $\boldsymbol{L}^a$ apply the budget and leverage constraints to the asset weights only, and $\boldsymbol{S}_1$ and $\boldsymbol{S}_2$ encode the asset and currency penalties. As in Section A.5, we decompose $\boldsymbol{\theta}_t = \boldsymbol{\Delta}_t^+ - \boldsymbol{\Delta}_t^-$ with $\boldsymbol{\Delta}_t^\pm = [(\boldsymbol{\Delta}_{x,t}^\pm)^\top, (\boldsymbol{\Delta}_{\psi,t}^\pm)^\top]^\top$, which linearizes $\|\boldsymbol{S}_1 \boldsymbol{\theta}_t\|_1$ and the leverage constraint $\|\boldsymbol{L}^a \boldsymbol{\theta}_t\|_1 = \boldsymbol{q}_{N,M}^\top (\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-) \le \ell^a$, and write the L^2 penalty as $\boldsymbol{\theta}_t^\top \boldsymbol{S}_2^2\, \boldsymbol{\theta}_t$.

B.5.1 Joint Optimization under Variance

Let $\boldsymbol{\mu}_t^{\text{all}} = [(\boldsymbol{\mu}_t^{fh})^\top, (\boldsymbol{\mu}_t^c)^\top]^\top \in \mathbb{R}^{N+M}$ denote the expected combined return. The variance of the joint return is $\text{Var}_t(\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}}) = \boldsymbol{\theta}_t^\top \boldsymbol{\Sigma}_t^{\text{all}} \boldsymbol{\theta}_t$ with the conditional covariance matrix

$$\boldsymbol{\Sigma}_t^{\text{all}} = \mathbb{E}_t[(\boldsymbol{R}_{t+1}^{\text{all}} - \boldsymbol{\mu}_t^{\text{all}})(\boldsymbol{R}_{t+1}^{\text{all}} - \boldsymbol{\mu}_t^{\text{all}})^\top] = \begin{bmatrix} \boldsymbol{\Sigma}_t^a & \boldsymbol{\Sigma}_t^{ac} \\ (\boldsymbol{\Sigma}_t^{ac})^\top & \boldsymbol{\Sigma}_t^c \end{bmatrix} \in \mathbb{R}^{(N+M)\times(N+M)}.$$

In the programs, $\boldsymbol{\Sigma}_t^{\text{all}}$ is the Ledoit–Wolf estimator, so its blocks differ from the separately estimated blocks of Section B.4. The decision vector is

$$\tilde{\boldsymbol{\theta}}_t = \Big[\, \boldsymbol{x}_t^\top, \;\; \boldsymbol{\psi}_t^\top, \;\; (\boldsymbol{\Delta}_{x,t}^-)^\top, \;\; (\boldsymbol{\Delta}_{\psi,t}^-)^\top, \;\; (\boldsymbol{\Delta}_{x,t}^+)^\top, \;\; (\boldsymbol{\Delta}_{\psi,t}^+)^\top \,\Big]^\top \in \mathbb{R}^{3N+3M},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \Big[\, \mathbf{0}_{N+M}^\top, \;\; \lambda_1^a \mathbf{1}_N^\top, \;\; \lambda_1^c \mathbf{1}_M^\top, \;\; \lambda_1^a \mathbf{1}_N^\top, \;\; \lambda_1^c \mathbf{1}_M^\top \,\Big]^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \boldsymbol{\Sigma}_t^a + \lambda_2^a \boldsymbol{I}_N & \boldsymbol{\Sigma}_t^{ac} & \mathbf{0}_{N\times(2N+2M)} \\ (\boldsymbol{\Sigma}_t^{ac})^\top & \boldsymbol{\Sigma}_t^c + \lambda_2^c \boldsymbol{I}_M & \mathbf{0}_{M\times(2N+2M)} \\ \mathbf{0}_{(2N+2M)\times N} & \mathbf{0}_{(2N+2M)\times M} & \mathbf{0}_{(2N+2M)\times(2N+2M)} \end{bmatrix}.$$

The joint portfolio optimization problem under variance then becomes a quadratic program with linear constraints

$$\begin{aligned}
\tilde{\boldsymbol{\theta}}_t^* &= \arg\min_{\tilde{\boldsymbol{\theta}}_t} \quad \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{\theta}}_t + \tilde{\boldsymbol{\theta}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{\theta}}_t, \\
\text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\theta}}_t &= \begin{bmatrix} \boldsymbol{q}_{N,M}^\top & \mathbf{0}_{N+M}^\top & \mathbf{0}_{N+M}^\top \\ \boldsymbol{I}_{N+M} & \boldsymbol{I}_{N+M} & -\boldsymbol{I}_{N+M} \end{bmatrix} \tilde{\boldsymbol{\theta}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_{N+M} \end{bmatrix}, \\
\tilde{\boldsymbol{A}}\tilde{\boldsymbol{\theta}}_t &= \begin{bmatrix} \mathbf{0}_{N+M}^\top & \boldsymbol{q}_{N,M}^\top & \boldsymbol{q}_{N,M}^\top \end{bmatrix} \tilde{\boldsymbol{\theta}}_t \le \tilde{b} = \ell^a, \\
&\Delta_{x,i,t}^{\pm} \ge 0, \quad \Delta_{\psi,c,t}^{\pm} \ge 0 \quad \text{for } i = 1, \dots, N,\ c = 1, \dots, M.
\end{aligned} \tag{B.15}$$

In $\tilde{\boldsymbol{D}}$, the first row enforces the asset budget constraint and the remaining $N + M$ rows the weight decomposition (A.11) for all components of $\boldsymbol{\theta}_t$. The single row of $\tilde{\boldsymbol{A}}$ imposes the asset-only leverage constraint (A.12).

B.5.2 Joint Optimization under Semi-Variance

As in Section A.5, let $\boldsymbol{R}^{\text{all}} = [\, \boldsymbol{R}^{fh} \;\; \boldsymbol{R}^{c} \,] \in \mathbb{R}^{q \times (N+M)}$ denote the matrix of jointly sampled asset and currency returns. Since the sample mean of the joint returns $\boldsymbol{R}^{\text{all}}\boldsymbol{\theta}_t$ is linear in $\boldsymbol{\theta}_t$, the demeaned formulation is simply $\mathring{\boldsymbol{R}}^{\text{all}}\boldsymbol{\theta}_t$. Introducing the auxiliary variables $\boldsymbol{u} \ge \mathbf{0}_q$, the slack constraint (B.6) becomes

$$\boldsymbol{u} \ge -\mathring{\boldsymbol{R}}^{\text{all}}\boldsymbol{\theta}_t \qquad \Longleftrightarrow \qquad \mathring{\boldsymbol{R}}^{\text{all}}\boldsymbol{\theta}_t + \boldsymbol{u} \ge \mathbf{0}_q.$$

The decision vector is

$$\tilde{\boldsymbol{\theta}}_t = \Big[\, \boldsymbol{x}_t^\top, \;\; \boldsymbol{\psi}_t^\top, \;\; \boldsymbol{u}^\top, \;\; (\boldsymbol{\Delta}_{x,t}^{-})^\top, \;\; (\boldsymbol{\Delta}_{\psi,t}^{-})^\top, \;\; (\boldsymbol{\Delta}_{x,t}^{+})^\top, \;\; (\boldsymbol{\Delta}_{\psi,t}^{+})^\top \,\Big]^\top \in \mathbb{R}^{3N+3M+q},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \Big[\, \mathbf{0}_{N+M}^\top, \;\; \mathbf{0}_q^\top, \;\; \lambda_1^a \mathbf{1}_N^\top, \;\; \lambda_1^c \mathbf{1}_M^\top, \;\; \lambda_1^a \mathbf{1}_N^\top, \;\; \lambda_1^c \mathbf{1}_M^\top \,\Big]^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \lambda_2^a \boldsymbol{I}_N & \mathbf{0}_{N\times M} & \mathbf{0}_{N\times q} & \mathbf{0}_{N\times(2N+2M)} \\ \mathbf{0}_{M\times N} & \lambda_2^c \boldsymbol{I}_M & \mathbf{0}_{M\times q} & \mathbf{0}_{M\times(2N+2M)} \\ \mathbf{0}_{q\times N} & \mathbf{0}_{q\times M} & \frac{1}{q}\boldsymbol{I}_q & \mathbf{0}_{q\times(2N+2M)} \\ \mathbf{0}_{(2N+2M)\times N} & \mathbf{0}_{(2N+2M)\times M} & \mathbf{0}_{(2N+2M)\times q} & \mathbf{0}_{(2N+2M)\times(2N+2M)} \end{bmatrix}.$$

The joint portfolio optimization problem under semi-variance then becomes a quadratic program with linear constraints

$$\begin{aligned}
\tilde{\boldsymbol{\theta}}_t^* &= \arg\min_{\tilde{\boldsymbol{\theta}}_t} \quad \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{\theta}}_t + \tilde{\boldsymbol{\theta}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{\theta}}_t, \\
\text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\theta}}_t &= \begin{bmatrix} \boldsymbol{q}_{N,M}^\top & \mathbf{0}_q^\top & \mathbf{0}_{N+M}^\top & \mathbf{0}_{N+M}^\top \\ \boldsymbol{I}_{N+M} & \mathbf{0}_{(N+M)\times q} & \boldsymbol{I}_{N+M} & -\boldsymbol{I}_{N+M} \end{bmatrix} \tilde{\boldsymbol{\theta}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_{N+M} \end{bmatrix}, \\
\tilde{\boldsymbol{A}}\tilde{\boldsymbol{\theta}}_t &= \begin{bmatrix} -\mathring{\boldsymbol{R}}^{\text{all}} & -\boldsymbol{I}_q & \mathbf{0}_{q\times(N+M)} & \mathbf{0}_{q\times(N+M)} \\ \mathbf{0}_{N+M}^\top & \mathbf{0}_q^\top & \boldsymbol{q}_{N,M}^\top & \boldsymbol{q}_{N,M}^\top \end{bmatrix} \tilde{\boldsymbol{\theta}}_t \le \tilde{\boldsymbol{b}} = \begin{bmatrix} \mathbf{0}_q \\ \ell^a \end{bmatrix}, \\
&u_k \ge 0 \quad \text{for } k = 1, \dots, q, \\
&\Delta_{x,i,t}^{\pm} \ge 0, \quad \Delta_{\psi,c,t}^{\pm} \ge 0 \quad \text{for } i = 1, \dots, N,\ c = 1, \dots, M.
\end{aligned} \tag{B.16}$$

Here $\tilde{\boldsymbol{D}}$ and the final row of $\tilde{\boldsymbol{A}}$ are the constraint matrices of (B.15) with a zero block added for $\boldsymbol{u}$, and the first q rows of $\tilde{\boldsymbol{A}}$ encode the slack constraints.

B.6 Mean-Risk Optimization

The programs of Sections B.2–B.5 minimize the risk measure only. As for Expected Shortfall in Section A.6, a mean-risk program subtracts the expected portfolio return, scaled by $1/\gamma$, from the risk measure, where $\gamma > 0$ is the risk aversion coefficient. For problem (B.3), this gives

$$\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \left\{ \rho(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}) - \frac{1}{\gamma}\, \boldsymbol{\mu}_t^\top \boldsymbol{x}_t \right\}, \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1, \tag{B.17}$$

where $\boldsymbol{\mu}_t$ is the expected asset return of Section B.1.

For the expected returns $\boldsymbol{\mu}_t^{fh}$, $\boldsymbol{\mu}_t^{c}$, and $\boldsymbol{\mu}_t^{\text{all}}$ of Sections B.4 and B.5, the asset means are estimated from the equally weighted return of each economy, whose mean is shrunk toward zero with a James–Stein estimator, and the currency means are set to the negative forward premium $-\boldsymbol{f}_t$. Both estimators are defined and further discussed in Chapter 3.

In the sequential approach, the asset step and the currency step become

$$\begin{aligned} \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \left\{ \rho\Big(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}^{fh}\Big) - \frac{1}{\gamma}\, (\boldsymbol{\mu}_t^{fh})^\top \boldsymbol{x}_t + \lambda_1^a \|\boldsymbol{x}_t\|_1 + \lambda_2^a \|\boldsymbol{x}_t\|_2^2 \right\} \\ \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1,\ \|\boldsymbol{x}_t\|_1 \le \ell^a, \end{aligned} \tag{B.18}$$

$$\begin{aligned} \boldsymbol{\psi}_t^* \mid \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{\psi}_t} \Big\{ \rho\Big(\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)\Big) - \frac{1}{\gamma}\, (\boldsymbol{\mu}_t^{c})^\top \boldsymbol{\psi}_t \\ + \lambda_1^c \|\boldsymbol{\psi}_t\|_1 + \lambda_2^c \|\boldsymbol{\psi}_t\|_2^2 \Big\} \\ \text{s.t. } \|\boldsymbol{\psi}_t\|_1 \le \ell^c. \end{aligned} \tag{B.19}$$

In the joint approach, problem (B.14) becomes

$$\begin{aligned} \boldsymbol{\theta}_t^* = \arg\min_{\boldsymbol{\theta}_t} \left\{ \rho\Big(\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}}\Big) - \frac{1}{\gamma}\, (\boldsymbol{\mu}_t^{\text{all}})^\top \boldsymbol{\theta}_t + \|\boldsymbol{S}_1 \boldsymbol{\theta}_t\|_1 + \|\boldsymbol{S}_2 \boldsymbol{\theta}_t\|_2^2 \right\} \\ \text{s.t. } \boldsymbol{\theta}_t^\top \boldsymbol{q}_{N,M} = 1, \quad \|\boldsymbol{L}^a \boldsymbol{\theta}_t\|_1 \le \ell^a, \quad \|\boldsymbol{L}^c \boldsymbol{\theta}_t\|_1 \le \ell^c, \end{aligned} \tag{B.20}$$

where

$$\boldsymbol{L}^c = \begin{bmatrix} \mathbf{0}_{N\times N} & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \boldsymbol{I}_M \end{bmatrix}$$

applies the leverage constraint to the currency exposures only. We add the currency leverage constraint to both the currency step and the joint program, since without it, the mean term can push the currency exposures toward very large positions.

Appendix C

Optimization Derivation: Mean Absolute Deviation

In this appendix, we derive the portfolio optimization problems used in the main text, building up from the definition of the mean absolute deviation to the programs solved in the empirical analysis. In Section C.1 we define the mean absolute deviation, show that it is convex in the portfolio weights, and write the portfolio optimization problem as a linear program. In Section C.2 we regularize the program by adding an L^1 penalty, an L^2 penalty, and a leverage constraint. We then extend the program to the multi-currency setting, for sequential optimization in Section C.3 and for joint optimization in Section C.4. Finally, in Section C.5 we add the expected portfolio return to the objectives of the sequential and joint problems.

C.1 Definition and Representation for Portfolio Optimization

We define the mean absolute deviation of the portfolio return, which Konno and Yamazaki [32] propose as an alternative to variance for portfolio optimization. Unlike variance, mean absolute deviation requires no estimate of the covariance matrix and can be implemented efficiently as a linear program instead of a quadratic program.

Let $\boldsymbol{x}_t \in \mathbb{R}^N$ denote the portfolio weights, $\boldsymbol{R}_{t+1} \in \mathbb{R}^N$ the asset return vector, and $\boldsymbol{\mu}_t = \mathbb{E}_t[\boldsymbol{R}_{t+1}] \in \mathbb{R}^N$ the expected asset return. The mean absolute deviation of the portfolio return is

$$\mathrm{MAD}_t(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}) = \mathbb{E}_t\big[|\boldsymbol{x}_t^\top \boldsymbol{\mu}_t - \boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}|\big]. \tag{C.1}$$

The inner map $\boldsymbol{x}_t \mapsto \boldsymbol{x}_t^\top(\boldsymbol{\mu}_t - \boldsymbol{R}_{t+1})$ is linear and the function $z \mapsto |z|$ is convex. The composition of a convex function with an affine map is convex [6, Section 3.2.2]. Expectation is a linear operator which preserves convexity, so the mean absolute deviation is convex in $\boldsymbol{x}_t$. Hence any local minimum is global, ensuring that the portfolio optimization problem under mean absolute deviation can be solved reliably with standard convex optimization methods.

The portfolio optimization problem is

$$\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \mathrm{MAD}_t(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}), \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1, \tag{C.2}$$

where the budget constraint enforces full investment. Let $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_q \in \mathbb{R}^N$ denote q historical realizations of the asset return vector $\boldsymbol{R}_{t+1}$. As for semi-variance in Section B.2.2, we approximate the expected asset return $\boldsymbol{\mu}_t$ by the sample mean $\bar{\boldsymbol{y}} = \frac{1}{q}\sum_{k=1}^{q} \boldsymbol{y}_k$ and the conditional expectation in (C.1) by the empirical average over this sample, so that the portfolio

optimization problem under mean absolute deviation becomes

$$\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \frac{1}{q}\sum_{k=1}^{q} |\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k|, \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1. \tag{C.3}$$

Each summand is convex but non-linear. Since the deviations $\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k$ sum to zero over the sample, the sum of their positive parts equals the sum of their negative parts, and the absolute value of each deviation is the sum of its two parts. The empirical mean absolute deviation is therefore twice the average shortfall of the portfolio return below its sample mean [59, 49]

$$\frac{1}{q}\sum_{k=1}^{q} |\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k| = \frac{2}{q}\sum_{k=1}^{q} (\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k)^+. \tag{C.4}$$

We linearize the right-hand side by introducing, for each observation k, an auxiliary variable $u_k \geq 0$ subject to

$$u_k \geq \boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k \qquad \Longleftrightarrow \qquad \boldsymbol{x}_t^\top (\boldsymbol{y}_k - \bar{\boldsymbol{y}}) + u_k \geq 0. \tag{C.5}$$

The objective becomes $\frac{2}{q}\sum_{k=1}^{q} u_k = \frac{2}{q}\mathbf{1}_q^\top \boldsymbol{u}$. It is strictly increasing in each u_k and minimized, so every u_k is driven to its lower bound at the optimum, $u_k^* = (\boldsymbol{x}_t^\top \bar{\boldsymbol{y}} - \boldsymbol{x}_t^\top \boldsymbol{y}_k)^+$. The portfolio optimization problem therefore has the constraints of the semi-variance program (B.7), with the auxiliary variables entering the objective linearly.

By writing the decision variables in the vector $\tilde{\boldsymbol{x}}_t = [\boldsymbol{x}_t^\top, \boldsymbol{u}^\top]^\top \in \mathbb{R}^{N+q}$ and the constants in the vector

$$\boldsymbol{c} = \left[\mathbf{0}_N^\top, \quad \frac{2}{q}\mathbf{1}_q^\top\right]^\top \in \mathbb{R}^{N+q},$$

the portfolio optimization problem (C.3) can be written as a linear program

$$\begin{aligned} \tilde{\boldsymbol{x}}_t^* = \arg\min_{\tilde{\boldsymbol{x}}_t} \quad & \boldsymbol{c}^\top \tilde{\boldsymbol{x}}_t \\ \text{s.t.} \quad & \boldsymbol{D}\tilde{\boldsymbol{x}}_t = \begin{bmatrix}\mathbf{1}_N^\top & \mathbf{0}_q^\top\end{bmatrix} \tilde{\boldsymbol{x}}_t = E = 1, \\ & \boldsymbol{A}\tilde{\boldsymbol{x}}_t = \begin{bmatrix}-\mathring{\boldsymbol{R}}^{\text{data}} & -\boldsymbol{I}_q\end{bmatrix} \tilde{\boldsymbol{x}}_t \leq \boldsymbol{b} = \mathbf{0}_q, \\ & u_k \geq 0 \quad \text{for } k = 1, \ldots, q. \end{aligned} \tag{C.6}$$

Here $\boldsymbol{D} \in \mathbb{R}^{1\times(N+q)}$ encodes the budget constraint, while $\boldsymbol{A} \in \mathbb{R}^{q\times(N+q)}$ and $\boldsymbol{b} \in \mathbb{R}^q$ encode the q inequality constraints (C.5). The matrix $\mathring{\boldsymbol{R}}^{\text{data}} = \boldsymbol{R}^{\text{data}} - \mathbf{1}_q \bar{\boldsymbol{y}}^\top \in \mathbb{R}^{q\times N}$ is the demeaned sample matrix introduced in Section B.2.2, where $\boldsymbol{R}^{\text{data}}$ is composed of the sampled return vectors $\boldsymbol{y}_k^\top$ row by row, so that the k-th row of $\mathring{\boldsymbol{R}}^{\text{data}}$ is $(\boldsymbol{y}_k - \bar{\boldsymbol{y}})^\top$. This formulation serves as the base unregularized portfolio optimization problem under mean absolute deviation.

C.2 Regularization

The base problem (C.6) is known to be unstable in high-dimensional settings, where the portfolio weights can fluctuate strongly from sample to sample [28]. As for the other risk measures in Appendices A and B, we stabilize it with the three regularization techniques of Sections A.3.1–A.3.3, an L^1 penalty $\lambda_1\|\boldsymbol{x}_t\|_1$, an L^2 penalty $\lambda_2\|\boldsymbol{x}_t\|_2^2$, and an asset leverage constraint $\|\boldsymbol{x}_t\|_1 \leq \ell^a$, with $\lambda_1, \lambda_2 \geq 0$ and $\ell^a \geq 1$. Splitting the weights into their positive and negative parts gives

$$\boldsymbol{x}_t = \boldsymbol{\Delta}_t^+ - \boldsymbol{\Delta}_t^-, \qquad \|\boldsymbol{x}_t\|_1 = \mathbf{1}_N^\top(\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-) \leq \ell^a, \qquad \boldsymbol{\Delta}_t^+, \boldsymbol{\Delta}_t^- \geq \mathbf{0}_N, \tag{C.7}$$

where the identity for $\|\boldsymbol{x}_t\|_1$ holds at the optimum, so that the L^1 penalty becomes $\lambda_1 \mathbf{1}_N^\top(\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-)$ and the L^2 penalty is the quadratic form $\boldsymbol{x}_t^\top(\lambda_2 \boldsymbol{I}_N)\boldsymbol{x}_t$.

Adding the three regularization terms to the base problem (C.6) extends the decision vector to

$$\tilde{\boldsymbol{x}}_t = \Big[\boldsymbol{x}_t^\top, \quad \boldsymbol{u}^\top, \quad (\boldsymbol{\Delta}_t^-)^\top, \quad (\boldsymbol{\Delta}_t^+)^\top\Big]^\top \in \mathbb{R}^{3N+q},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \Big[\mathbf{0}_N^\top, \quad \tfrac{2}{q}\mathbf{1}_q^\top, \quad \lambda_1 \mathbf{1}_N^\top, \quad \lambda_1 \mathbf{1}_N^\top\Big]^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \lambda_2 \boldsymbol{I}_N & \mathbf{0}_{N\times(2N+q)} \\ \mathbf{0}_{(2N+q)\times N} & \mathbf{0}_{(2N+q)\times(2N+q)} \end{bmatrix}.$$

The portfolio optimization problem under mean absolute deviation then becomes a quadratic program with linear constraints

$$\begin{aligned} \tilde{\boldsymbol{x}}_t^* &= \arg\min_{\tilde{\boldsymbol{x}}_t} \quad \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{x}}_t + \tilde{\boldsymbol{x}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{x}}_t, \\ \text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{x}}_t &= \begin{bmatrix} \mathbf{1}_N^\top & \mathbf{0}_q^\top & \mathbf{0}_N^\top & \mathbf{0}_N^\top \\ \boldsymbol{I}_N & \mathbf{0}_{N\times q} & \boldsymbol{I}_N & -\boldsymbol{I}_N \end{bmatrix} \tilde{\boldsymbol{x}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_N \end{bmatrix}, \\ \tilde{\boldsymbol{A}}\tilde{\boldsymbol{x}}_t &= \begin{bmatrix} -\mathring{\boldsymbol{R}}^{\text{data}} & -\boldsymbol{I}_q & \mathbf{0}_{q\times N} & \mathbf{0}_{q\times N} \\ \mathbf{0}_N^\top & \mathbf{0}_q^\top & \mathbf{1}_N^\top & \mathbf{1}_N^\top \end{bmatrix} \tilde{\boldsymbol{x}}_t \le \tilde{\boldsymbol{b}} = \begin{bmatrix} \mathbf{0}_q \\ \ell^a \end{bmatrix}, \\ u_k \ge 0, \quad \Delta_{i,t}^- \ge 0, \quad \Delta_{i,t}^+ \ge 0 \quad &\text{for } i = 1,\dots,N,\ k = 1,\dots,q. \end{aligned} \tag{C.8}$$

The first row of $\tilde{\boldsymbol{D}}$ enforces the budget constraint and the remaining N rows enforce the weight decomposition (C.7). The first q rows of $\tilde{\boldsymbol{A}}$ encode the inequality constraints (C.5), extended with zeros for the new variables, and the final row of $\tilde{\boldsymbol{A}}$ imposes the leverage constraint (C.7).

C.3 Sequential Optimization

As for the other risk measures in Sections A.4 and B.4, we consider an international portfolio optimization problem that also requires selecting currency exposures. In the sequential portfolio optimization approach, the asset weights and currency exposures are optimized in two steps. First, the asset weights $\boldsymbol{x}_t$ are optimized on the fully hedged returns. Then, conditional on the optimal $\boldsymbol{x}_t^*$, the currency exposures $\boldsymbol{\psi}_t \in \mathbb{R}^M$ are optimized. The two optimization steps are given by

$$\begin{aligned} \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} &\left\{ \text{MAD}_t\Big(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}^{fh}\Big) + \lambda_1^a \|\boldsymbol{x}_t\|_1 + \lambda_2^a \|\boldsymbol{x}_t\|_2^2 \right\} \\ &\text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1,\ \|\boldsymbol{x}_t\|_1 \le \ell^a, \end{aligned} \tag{C.9}$$

$$\boldsymbol{\psi}_t^* \mid \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{\psi}_t} \left\{ \text{MAD}_t\Big(\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)\Big) + \lambda_1^c \|\boldsymbol{\psi}_t\|_1 + \lambda_2^c \|\boldsymbol{\psi}_t\|_2^2 \right\}. \tag{C.10}$$

The regularization parameters are kept separate across assets $(\lambda_1^a, \lambda_2^a)$ and currencies $(\lambda_1^c, \lambda_2^c)$.

The fully hedged return vector $\boldsymbol{R}_{t+1}^{fh}$, defined in Section 3.2.3, takes the place of the asset return vector $\boldsymbol{R}_{t+1}$ of Sections C.1–C.2, so the asset step (C.9) is the regularized program (C.8) with the demeaned matrix $\mathring{\boldsymbol{R}}^{fh}$ of the historically sampled fully hedged returns $\boldsymbol{R}^{fh} \in \mathbb{R}^{q\times N}$ in place of $\mathring{\boldsymbol{R}}^{\text{data}}$.

In the currency step (C.10), the asset weights $\boldsymbol{x}_t^*$ are fixed and the currency exposures $\boldsymbol{\psi}_t$ are optimized. The currency step has no budget constraint, since the currency overlay is a zero-investment portfolio whose domestic exposure $\psi_{0,t} = -\mathbf{1}_M^\top \boldsymbol{\psi}_t$ is inferred residually, and no leverage constraint is imposed on the currency exposures. We split the currency exposures as $\boldsymbol{\psi}_t = \boldsymbol{\Delta}_{\psi,t}^+ - \boldsymbol{\Delta}_{\psi,t}^-$ with $\boldsymbol{\Delta}_{\psi,t}^+, \boldsymbol{\Delta}_{\psi,t}^- \geq \mathbf{0}_M$ as in (C.7), which linearizes the currency L^1 penalty, and write the currency L^2 penalty as the quadratic form $\boldsymbol{\psi}_t^\top (\lambda_2^c \boldsymbol{I}_M) \boldsymbol{\psi}_t$.

For the empirical approximation of the currency step, let $\boldsymbol{R}_k^{fh}$ denote the k-th sampled fully hedged return vector, and $\boldsymbol{e}_k$ and $\boldsymbol{f}_k$ the k-th sampled currency return and forward premium vectors. We stack the sampled excess currency returns $(\boldsymbol{e}_k - \boldsymbol{f}_k)^\top$ row by row into the matrix $\boldsymbol{R}^c \in \mathbb{R}^{q \times M}$ and collect the fixed asset-portfolio returns in $\boldsymbol{R}_{\mathcal{P}}^{fh} = \boldsymbol{R}^{fh} \boldsymbol{x}_t^* \in \mathbb{R}^q$. Since the sample mean is linear in the returns, the demeaned hedged portfolio returns are $\mathring{\boldsymbol{R}}_{\mathcal{P}}^{fh} + \mathring{\boldsymbol{R}}^c \boldsymbol{\psi}_t$, as for semi-variance in Section B.4.2, where $\mathring{\boldsymbol{R}}_{\mathcal{P}}^{fh} = \mathring{\boldsymbol{R}}^{fh} \boldsymbol{x}_t^*$ and $\mathring{\boldsymbol{R}}^c$ is the demeaned matrix of the sampled excess currency returns. These demeaned returns also sum to zero, so the identity (C.4) applies, and introducing the auxiliary variables $\boldsymbol{u}_\psi \geq \mathbf{0}_q$, the slack constraint (C.5) becomes

$$\boldsymbol{u}_\psi \geq -\mathring{\boldsymbol{R}}^c \boldsymbol{\psi}_t - \mathring{\boldsymbol{R}}_{\mathcal{P}}^{fh} \qquad \Longleftrightarrow \qquad -\mathring{\boldsymbol{R}}^c \boldsymbol{\psi}_t - \boldsymbol{u}_\psi \leq \mathring{\boldsymbol{R}}_{\mathcal{P}}^{fh},$$

with the objective $\frac{2}{q} \mathbf{1}_q^\top \boldsymbol{u}_\psi$.

The decision vector of the currency step is

$$\tilde{\boldsymbol{\psi}}_t = \left[\boldsymbol{\psi}_t^\top, \quad \boldsymbol{u}_\psi^\top, \quad (\boldsymbol{\Delta}_{\psi,t}^-)^\top, \quad (\boldsymbol{\Delta}_{\psi,t}^+)^\top \right]^\top \in \mathbb{R}^{3M+q},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \left[\mathbf{0}_M^\top, \quad \tfrac{2}{q} \mathbf{1}_q^\top, \quad \lambda_1^c \mathbf{1}_M^\top, \quad \lambda_1^c \mathbf{1}_M^\top \right]^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \lambda_2^c \boldsymbol{I}_M & \mathbf{0}_{M \times (2M+q)} \\ \mathbf{0}_{(2M+q) \times M} & \mathbf{0}_{(2M+q) \times (2M+q)} \end{bmatrix}.$$

The currency overlay optimization under mean absolute deviation then becomes a quadratic program with linear constraints

$$\begin{aligned} \tilde{\boldsymbol{\psi}}_t^* = \arg\min_{\tilde{\boldsymbol{\psi}}_t} \quad & \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{\psi}}_t + \tilde{\boldsymbol{\psi}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{\psi}}_t, \\ \text{s.t.} \quad \tilde{\boldsymbol{D}} \tilde{\boldsymbol{\psi}}_t &= \begin{bmatrix} \boldsymbol{I}_M & \mathbf{0}_{M \times q} & \boldsymbol{I}_M & -\boldsymbol{I}_M \end{bmatrix} \tilde{\boldsymbol{\psi}}_t = \tilde{\boldsymbol{E}} = \mathbf{0}_M, \\ \tilde{\boldsymbol{A}} \tilde{\boldsymbol{\psi}}_t &= \begin{bmatrix} -\mathring{\boldsymbol{R}}^c & -\boldsymbol{I}_q & \mathbf{0}_{q \times M} & \mathbf{0}_{q \times M} \end{bmatrix} \tilde{\boldsymbol{\psi}}_t \leq \tilde{\boldsymbol{b}} = \mathring{\boldsymbol{R}}_{\mathcal{P}}^{fh}, \\ u_{k,\psi} \geq 0, \quad & \Delta_{\psi,c,t}^- \geq 0, \quad \Delta_{\psi,c,t}^+ \geq 0 \quad \text{for } c = 1, \ldots, M,\ k = 1, \ldots, q. \end{aligned} \tag{C.11}$$

The M rows of $\tilde{\boldsymbol{D}}$ enforce the weight decomposition (C.7) for the currency exposures, and the q rows of $\tilde{\boldsymbol{A}}$ encode the slack constraints of the currency step.

C.4 Joint Optimization

In the joint portfolio optimization approach, the asset weights and currency exposures are optimized simultaneously rather than sequentially. As for the other risk measures in Sections A.5 and B.5, we stack the weights into the combined decision vector $\boldsymbol{\theta}_t = [\boldsymbol{x}_t^\top, \boldsymbol{\psi}_t^\top]^\top \in \mathbb{R}^{N+M}$ and

the corresponding returns into the combined return vector $\boldsymbol{R}_{t+1}^{\text{all}} = [\boldsymbol{R}_{t+1}^{fh\top}, (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)^\top]^\top \in \mathbb{R}^{N+M}$, so that $\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}} = \boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top(\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)$ is the hedged portfolio return of Section C.3. The joint optimization problem is

$$\begin{aligned} \boldsymbol{\theta}_t^* = \arg\min_{\boldsymbol{\theta}_t} \left\{ \text{MAD}_t\left(\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}}\right) + \|\boldsymbol{S}_1\boldsymbol{\theta}_t\|_1 + \|\boldsymbol{S}_2\boldsymbol{\theta}_t\|_2^2 \right\} \\ \text{s.t. } \boldsymbol{\theta}_t^\top \boldsymbol{q}_{N,M} = 1, \quad \|\boldsymbol{L}^a\boldsymbol{\theta}_t\|_1 \le \ell^a, \end{aligned} \tag{C.12}$$

with

$$\boldsymbol{q}_{N,M} = \begin{bmatrix} \mathbf{1}_N \\ \mathbf{0}_M \end{bmatrix}, \quad \boldsymbol{L}^a = \begin{bmatrix} \boldsymbol{I}_N & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \mathbf{0}_{M\times M} \end{bmatrix},$$

$$\boldsymbol{S}_1 = \begin{bmatrix} \lambda_1^a \boldsymbol{I}_N & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \lambda_1^c \boldsymbol{I}_M \end{bmatrix}, \quad \boldsymbol{S}_2 = \begin{bmatrix} \sqrt{\lambda_2^a}\, \boldsymbol{I}_N & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \sqrt{\lambda_2^c}\, \boldsymbol{I}_M \end{bmatrix}.$$

The vector $\boldsymbol{q}_{N,M}$ and the matrix $\boldsymbol{L}^a$ apply the budget and leverage constraints to the asset weights only, and $\boldsymbol{S}_1$ and $\boldsymbol{S}_2$ encode the asset and currency penalties. We decompose $\boldsymbol{\theta}_t = \boldsymbol{\Delta}_t^+ - \boldsymbol{\Delta}_t^-$ as in (C.7), with $\boldsymbol{\Delta}_t^\pm = [(\boldsymbol{\Delta}_{x,t}^\pm)^\top, (\boldsymbol{\Delta}_{\psi,t}^\pm)^\top]^\top \ge \mathbf{0}_{N+M}$, which linearizes $\|\boldsymbol{S}_1\boldsymbol{\theta}_t\|_1$ and the leverage constraint $\|\boldsymbol{L}^a\boldsymbol{\theta}_t\|_1 = \boldsymbol{q}_{N,M}^\top(\boldsymbol{\Delta}_t^+ + \boldsymbol{\Delta}_t^-) \le \ell^a$, and write the L^2 penalty as the quadratic form $\boldsymbol{\theta}_t^\top \boldsymbol{S}_2^2\, \boldsymbol{\theta}_t$.

For the empirical approximation, let $\boldsymbol{R}^{\text{all}} = [\, \boldsymbol{R}^{fh} \;\; \boldsymbol{R}^c \,] \in \mathbb{R}^{q\times(N+M)}$ denote the matrix of jointly sampled asset and currency returns, with $\boldsymbol{R}^{fh}$ and $\boldsymbol{R}^c$ of Section C.3. Since the sample mean is linear in the returns, the demeaned joint returns are $\mathring{\boldsymbol{R}}^{\text{all}}\boldsymbol{\theta}_t$, as for semi-variance in Section B.5.2, where $\mathring{\boldsymbol{R}}^{\text{all}}$ is the demeaned matrix of the jointly sampled returns. These demeaned returns sum to zero, so the identity (C.4) applies, and introducing the auxiliary variables $\boldsymbol{u} \ge \mathbf{0}_q$, the slack constraint (C.5) becomes

$$\boldsymbol{u} \ge -\mathring{\boldsymbol{R}}^{\text{all}}\boldsymbol{\theta}_t \qquad \Longleftrightarrow \qquad \mathring{\boldsymbol{R}}^{\text{all}}\boldsymbol{\theta}_t + \boldsymbol{u} \ge \mathbf{0}_q,$$

with the objective $\frac{2}{q}\mathbf{1}_q^\top \boldsymbol{u}$.

The decision vector is

$$\tilde{\boldsymbol{\theta}}_t = \left[\, \boldsymbol{x}_t^\top, \;\; \boldsymbol{\psi}_t^\top, \;\; \boldsymbol{u}^\top, \;\; (\boldsymbol{\Delta}_{x,t}^-)^\top, \;\; (\boldsymbol{\Delta}_{\psi,t}^-)^\top, \;\; (\boldsymbol{\Delta}_{x,t}^+)^\top, \;\; (\boldsymbol{\Delta}_{\psi,t}^+)^\top \,\right]^\top \in \mathbb{R}^{3N+3M+q},$$

with the constants in the vector

$$\tilde{\boldsymbol{c}} = \left[\, \mathbf{0}_{N+M}^\top, \;\; \tfrac{2}{q}\mathbf{1}_q^\top, \;\; \lambda_1^a\mathbf{1}_N^\top, \;\; \lambda_1^c\mathbf{1}_M^\top, \;\; \lambda_1^a\mathbf{1}_N^\top, \;\; \lambda_1^c\mathbf{1}_M^\top \,\right]^\top$$

and the penalty matrix

$$\tilde{\boldsymbol{Q}} = \begin{bmatrix} \lambda_2^a \boldsymbol{I}_N & \mathbf{0}_{N\times M} & \mathbf{0}_{N\times(2N+2M+q)} \\ \mathbf{0}_{M\times N} & \lambda_2^c \boldsymbol{I}_M & \mathbf{0}_{M\times(2N+2M+q)} \\ \mathbf{0}_{(2N+2M+q)\times N} & \mathbf{0}_{(2N+2M+q)\times M} & \mathbf{0}_{(2N+2M+q)\times(2N+2M+q)} \end{bmatrix}.$$

The joint portfolio optimization problem under mean absolute deviation then becomes a

quadratic program with linear constraints

$$\begin{aligned}
\tilde{\boldsymbol{\theta}}_t^* &= \arg\min_{\tilde{\boldsymbol{\theta}}_t} \quad \tilde{\boldsymbol{c}}^\top \tilde{\boldsymbol{\theta}}_t + \tilde{\boldsymbol{\theta}}_t^\top \tilde{\boldsymbol{Q}}\, \tilde{\boldsymbol{\theta}}_t, \\
\text{s.t.} \quad \tilde{\boldsymbol{D}}\tilde{\boldsymbol{\theta}}_t &= \begin{bmatrix} \boldsymbol{q}_{N,M}^\top & \mathbf{0}_q^\top & \mathbf{0}_{N+M}^\top & \mathbf{0}_{N+M}^\top \\ \boldsymbol{I}_{N+M} & \mathbf{0}_{(N+M)\times q} & \boldsymbol{I}_{N+M} & -\boldsymbol{I}_{N+M} \end{bmatrix} \tilde{\boldsymbol{\theta}}_t = \tilde{\boldsymbol{E}} = \begin{bmatrix} 1 \\ \mathbf{0}_{N+M} \end{bmatrix}, \\
\tilde{\boldsymbol{A}}\tilde{\boldsymbol{\theta}}_t &= \begin{bmatrix} -\mathring{\boldsymbol{R}}^{\text{all}} & -\boldsymbol{I}_q & \mathbf{0}_{q\times(N+M)} & \mathbf{0}_{q\times(N+M)} \\ \mathbf{0}_{N+M}^\top & \mathbf{0}_q^\top & \boldsymbol{q}_{N,M}^\top & \boldsymbol{q}_{N,M}^\top \end{bmatrix} \tilde{\boldsymbol{\theta}}_t \le \tilde{\boldsymbol{b}} = \begin{bmatrix} \mathbf{0}_q \\ \ell^a \end{bmatrix}, \\
& u_k \ge 0 \quad \text{for } k = 1, \dots, q, \\
& \Delta_{x,i,t}^{\pm} \ge 0, \quad \Delta_{\psi,c,t}^{\pm} \ge 0 \quad \text{for } i = 1, \dots, N,\ c = 1, \dots, M.
\end{aligned} \tag{C.13}$$

In $\tilde{\boldsymbol{D}}$, the first row enforces the asset budget constraint and the remaining $N + M$ rows the weight decomposition (C.7) for all components of $\boldsymbol{\theta}_t$. In $\tilde{\boldsymbol{A}}$, the first q rows encode the slack constraints, and the final row imposes the asset-only leverage constraint (C.7).

C.5 Mean-Risk Optimization

The programs of Sections C.1–C.4 minimize the risk measure only. As for the other risk measures in Sections A.6 and B.6, a mean-risk program subtracts the expected portfolio return, scaled by $1/\gamma$, from the risk measure, where $\gamma > 0$ is the risk aversion coefficient. For problem (C.2), this gives

$$\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \left\{ \text{MAD}_t(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}) - \frac{1}{\gamma}\, \boldsymbol{\mu}_t^\top \boldsymbol{x}_t \right\}, \quad \text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1. \tag{C.14}$$

Let $\boldsymbol{\mu}_t^{fh} = \mathbb{E}_t[\boldsymbol{R}_{t+1}^{fh}] \in \mathbb{R}^N$ and $\boldsymbol{\mu}_t^c = \mathbb{E}_t[\boldsymbol{e}_{t+1} - \boldsymbol{f}_t] \in \mathbb{R}^M$ denote the expected fully hedged asset returns and the expected excess currency returns, and $\boldsymbol{\mu}_t^{\text{all}} = [(\boldsymbol{\mu}_t^{fh})^\top, (\boldsymbol{\mu}_t^c)^\top]^\top \in \mathbb{R}^{N+M}$. The asset means are estimated from the equally weighted return of each economy, whose mean is shrunk toward zero with a James–Stein estimator, and the currency means are set to the negative forward premium $-\boldsymbol{f}_t$. Both estimators are defined and further discussed in Chapter 3.

In the sequential approach, the asset step and the currency step become

$$\begin{aligned}
\boldsymbol{x}_t^* = \arg\min_{\boldsymbol{x}_t} \Big\{ &\text{MAD}_t\Big(\boldsymbol{x}_t^\top \boldsymbol{R}_{t+1}^{fh}\Big) - \frac{1}{\gamma}\, (\boldsymbol{\mu}_t^{fh})^\top \boldsymbol{x}_t + \lambda_1^a \|\boldsymbol{x}_t\|_1 + \lambda_2^a \|\boldsymbol{x}_t\|_2^2 \Big\} \\
&\text{s.t. } \boldsymbol{x}_t^\top \mathbf{1}_N = 1,\ \|\boldsymbol{x}_t\|_1 \le \ell^a,
\end{aligned} \tag{C.15}$$

$$\begin{aligned}
\boldsymbol{\psi}_t^* \mid \boldsymbol{x}_t^* = \arg\min_{\boldsymbol{\psi}_t} \Big\{ &\text{MAD}_t\Big(\boldsymbol{x}_t^{*\top} \boldsymbol{R}_{t+1}^{fh} + \boldsymbol{\psi}_t^\top (\boldsymbol{e}_{t+1} - \boldsymbol{f}_t)\Big) - \frac{1}{\gamma}\, (\boldsymbol{\mu}_t^c)^\top \boldsymbol{\psi}_t \\
&+ \lambda_1^c \|\boldsymbol{\psi}_t\|_1 + \lambda_2^c \|\boldsymbol{\psi}_t\|_2^2 \Big\} \\
&\text{s.t. } \|\boldsymbol{\psi}_t\|_1 \le \ell^c.
\end{aligned} \tag{C.16}$$

In the joint approach, problem (C.12) becomes

$$\begin{aligned}
\boldsymbol{\theta}_t^* = \arg\min_{\boldsymbol{\theta}_t} \Big\{ &\text{MAD}_t\Big(\boldsymbol{\theta}_t^\top \boldsymbol{R}_{t+1}^{\text{all}}\Big) - \frac{1}{\gamma}\, (\boldsymbol{\mu}_t^{\text{all}})^\top \boldsymbol{\theta}_t + \|\boldsymbol{S}_1 \boldsymbol{\theta}_t\|_1 + \|\boldsymbol{S}_2 \boldsymbol{\theta}_t\|_2^2 \Big\} \\
&\text{s.t. } \boldsymbol{\theta}_t^\top \boldsymbol{q}_{N,M} = 1, \quad \|\boldsymbol{L}^a \boldsymbol{\theta}_t\|_1 \le \ell^a, \quad \|\boldsymbol{L}^c \boldsymbol{\theta}_t\|_1 \le \ell^c,
\end{aligned} \tag{C.17}$$

where

$$\boldsymbol{L}^c = \begin{bmatrix} \mathbf{0}_{N\times N} & \mathbf{0}_{N\times M} \\ \mathbf{0}_{M\times N} & \boldsymbol{I}_M \end{bmatrix}$$

applies the leverage constraint to the currency exposures only. We add the currency leverage constraint to both the currency step and the joint program, since without it, the mean term can push the currency exposures toward very large positions.

Appendix D

Coherent Risk Measures and Deviation Measures

In this appendix, we define the notion of a deviation measure and a coherent risk measure, and prove that a strictly expectation bounded coherent risk measure can be transformed into a deviation measure by a simple de-meaning operation. This result is used in Section 6.2 to de-mean the Expected Shortfall and second-order Expected Shortfall risk measures and transform them into deviation measures. Throughout this appendix, let $X \in L^2(\Omega)$ denote a square-integrable random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition D.1 (Deviation measure) *A functional $\mathcal{D} : L^2(\Omega) \to [0, \infty]$ is a* deviation measure *if for all $X, X' \in L^2(\Omega)$, every constant C, and every $\lambda > 0$,*

(D1)	$\mathcal{D}(X + C) = \mathcal{D}(X)$,
(D2)	$\mathcal{D}(0) = 0 \quad \textit{and} \quad \mathcal{D}(\lambda X) = \lambda \mathcal{D}(X)$,
(D3)	$\mathcal{D}(X + X') \leq \mathcal{D}(X) + \mathcal{D}(X')$,
(D4)	$\mathcal{D}(X) \geq 0$, *with* $\mathcal{D}(X) > 0$ *for nonconstant* X.

The property *(D1)* is the main one that distinguishes deviation measures from coherent risk measures. It is a form of translation invariance.

Definition D.2 (Coherent and strictly expectation bounded risk measure) *A functional $\mathcal{R} : L^2(\Omega) \to (-\infty, \infty]$ is a* coherent risk measure *if for all $X, X' \in L^2(\Omega)$ and every constant C,*

(R1)	$\mathcal{R}(X + C) = \mathcal{R}(X) - C$,
(R2)	$\mathcal{R}(0) = 0 \quad \textit{and} \quad \mathcal{R}(\lambda X) = \lambda \mathcal{R}(X)$ *for* $\lambda > 0$,
(R3)	$\mathcal{R}(X + X') \leq \mathcal{R}(X) + \mathcal{R}(X')$,
(R4)	$\mathcal{R}(X) \leq \mathcal{R}(X')$ *whenever* $X \geq X'$,

and it is strictly expectation bounded *if in addition*

(R5) $\mathcal{R}(X) > \mathbb{E}[-X] \quad$ *for every nonconstant* X, $\quad$ *and* $\quad \mathcal{R}(X) = \mathbb{E}[-X]$ *for constant* X.

Properties *(R1)* and *(D1)* are incompatible, so no non-degenerate functional can be a risk measure and a deviation measure at once. The de-meaning transformation is what moves between the two classes. We state and prove only the operation for transforming a coherent and strictly expectation bounded risk measure into a deviation measure. This is one direction of the more general result of Theorem 2 in [56].

Theorem D.3 *Let $\mathcal{R}$ be a strictly expectation bounded coherent risk measure. Then*

$$\mathcal{D}(X) \coloneqq \mathcal{R}(X - \mathbb{E}[X]) \tag{D.1}$$

is a deviation measure.

Proof The proof is a straightforward verification of the four properties in Definition D.1. Write $\bar{X} = X - \mathbb{E}[X]$. Then $\mathcal{D}(X) = \mathcal{R}(\bar{X})$.

(D1): For any constant C we have $(X+C)-\mathbb{E}[X+C] = \bar{X}$. Hence, $\mathcal{D}(X+C) = \mathcal{R}(\bar{X}) = \mathcal{D}(X)$.

(D2): Taking $X = 0$ gives $\mathcal{D}(0) = \mathcal{R}(0) = 0$ by *(R2)*. For $\lambda > 0$, $\lambda X - \mathbb{E}[\lambda X] = \lambda\bar{X}$, so again, *(R2)* gives $\mathcal{D}(\lambda X) = \mathcal{R}(\lambda\bar{X}) = \lambda\mathcal{R}(\bar{X}) = \lambda\mathcal{D}(X)$.

(D3): Since $(X + X') - \mathbb{E}[X + X'] = \bar{X} + \bar{X}'$, *(R3)* gives $\mathcal{D}(X + X') = \mathcal{R}(\bar{X} + \bar{X}') \leq \mathcal{R}(\bar{X}) + \mathcal{R}(\bar{X}') = \mathcal{D}(X) + \mathcal{D}(X')$.

(D4): If X is constant then $\bar{X} = 0$ and $\mathcal{D}(X) = 0$. If X is nonconstant then so is $\bar{X}$, and *(R5)* applied to $\bar{X}$ gives $\mathcal{D}(X) = \mathcal{R}(\bar{X}) > \mathbb{E}[-\bar{X}] = 0$. □

The proof uses the properties *(R2)*, *(R3)*, and *(R5)* only. Monotonicity *(R4)* is never used, which is the reason the original theorem in [56] is stated more generally for strictly expectation bounded risk measures rather than coherent risk measures.

We now show that both Expected Shortfall and second-order Expected Shortfall are strictly expectation bounded coherent risk measures, and we can therefore apply Theorem D.3 to transform them into deviation measures. Write $z^+ = \max\{z, 0\}$ and $z^- = \max\{-z, 0\}$, fix a confidence level $\beta \in (0, 1)$, and set $\kappa \coloneqq (1 - \beta)^{-1} > 1$. In the Rockafellar–Uryasev form [55] the Expected Shortfall is

$$\mathrm{ES}_\beta(X) = \min_{\alpha\in\mathbb{R}} F_\beta(X, \alpha), \tag{D.2}$$

and its second-order analogue, obtained by replacing the expectation in F_β with the L^2 norm [33], is

$$\mathrm{SOES}_\beta(X) = \min_{\alpha\in\mathbb{R}} G_\beta(X, \alpha), \tag{D.3}$$

where

$$F_\beta(X, \alpha) = \alpha + \kappa\,\mathbb{E}[(-X - \alpha)^+], \qquad G_\beta(X, \alpha) = \alpha + \kappa\,\big\|(-X - \alpha)^+\big\|_{L^2}$$

are the auxiliary functions of Section A.1. Both $F_\beta(X, \cdot)$ and $G_\beta(X, \cdot)$ are convex, and both diverge as $|\alpha| \to \infty$, so the minima in (D.2) and (D.3) are attained. It is well known that ES_β is a coherent risk measure, see, for example, Proposition 3.1 and Corollary 4.3 in [1]. We show it is strictly expectation bounded.

Proposition D.4 (Expected Shortfall) ES_β *is strictly expectation bounded.*

Proof Fix α, write $\mu = \mathbb{E}[X]$, and put $Y = -X - \alpha$, so that $\mathbb{E}[Y] = -\mu - \alpha$ and hence $\alpha + \mu = -\mathbb{E}[Y]$. Using $\mathbb{E}[Y] = \mathbb{E}[Y^+] - \mathbb{E}[Y^-]$ we write

$$\begin{aligned} F_\beta(X, \alpha) - \mathbb{E}[-X] &= \alpha + \mu + \kappa\,\mathbb{E}[Y^+] \\ &= -\mathbb{E}[Y] + \kappa\,\mathbb{E}[Y^+] \\ &= \mathbb{E}[Y^-] + (\kappa - 1)\,\mathbb{E}[Y^+]. \end{aligned}$$

Both terms on the right are non-negative, the second because $\kappa > 1$, so $F_\beta(X, \alpha) \geq \mathbb{E}[-X]$ for every α, with equality if and only if $Y = 0$ almost surely, that is, $X \equiv -\alpha$. If X is nonconstant the inequality is therefore strict at every α, and since the minimum is attained, $\mathrm{ES}_\beta(X) > \mathbb{E}[-X]$. If $X \equiv C$, then $\alpha = -C$ attains equality, so $\mathrm{ES}_\beta(X) = \mathbb{E}[-X]$. □

Proposition D.5 (Second-order Expected Shortfall) SOES_β *is a coherent, strictly expectation bounded risk measure.*

Proof *(R1):* Substituting $\alpha' = \alpha + C$ gives $-X - C - \alpha = -X - \alpha'$ and $G_\beta(X + C, \alpha) = \alpha' - C + \kappa\|(-X - \alpha')^+\|_{L^2} = G_\beta(X, \alpha') - C$. As α ranges over $\mathbb{R}$, so does α', so minimizing both sides gives $\mathrm{SOES}_\beta(X + C) = \mathrm{SOES}_\beta(X) - C$.

(R2): For $X = 0$ we have $G_\beta(0, \alpha) = \alpha + \kappa\,\alpha^-$, which equals $\alpha \geq 0$ for $\alpha \geq 0$ and $(1 - \kappa)\alpha > 0$ for $\alpha < 0$. The minimum is 0, attained at $\alpha = 0$. For $\lambda > 0$, substituting $\alpha = \lambda\alpha'$ and using positive homogeneity of $z \mapsto z^+$ and of the norm,

$$G_\beta(\lambda X, \lambda\alpha') = \lambda\alpha' + \kappa\|\lambda(-X - \alpha')^+\|_{L^2} = \lambda\left[\alpha' + \kappa\|(-X - \alpha')^+\|_{L^2}\right] = \lambda\, G_\beta(X, \alpha'),$$

and minimizing over α' gives $\mathrm{SOES}_\beta(\lambda X) = \lambda\,\mathrm{SOES}_\beta(X)$.

(R3): Let α and α' attain the minima for X and X'. Since $(a + b)^+ \leq a^+ + b^+$ pointwise, and since $\|\cdot\|_{L^2}$ is monotone on non-negative random variables and satisfies the triangle inequality,

$$\begin{aligned}\|(-(X + X') - (\alpha + \alpha'))^+\|_{L^2} &\leq \|(-X - \alpha)^+ + (-X' - \alpha')^+\|_{L^2} \\ &\leq \|(-X - \alpha)^+\|_{L^2} + \|(-X' - \alpha')^+\|_{L^2}.\end{aligned}$$

Adding $\alpha + \alpha'$ and multiplying the norms by κ, $\mathrm{SOES}_\beta(X + X') \leq G_\beta(X + X', \alpha + \alpha') \leq G_\beta(X, \alpha) + G_\beta(X', \alpha') = \mathrm{SOES}_\beta(X) + \mathrm{SOES}_\beta(X')$.

(R4): If $X \geq X'$ then $-X - \alpha \leq -X' - \alpha$, so $(-X - \alpha)^+ \leq (-X' - \alpha)^+$ pointwise and therefore $G_\beta(X, \alpha) \leq G_\beta(X', \alpha)$ for every α. Taking the minimum gives $\mathrm{SOES}_\beta(X) \leq \mathrm{SOES}_\beta(X')$.

(R5): On a probability space $\|Z\|_{L^2} \geq \|Z\|_{L^1}$, so for $Z = (-X - \alpha)^+ \geq 0$ we have $\|(-X - \alpha)^+\|_{L^2} \geq \mathbb{E}[(-X - \alpha)^+]$ and hence $G_\beta(X, \alpha) \geq F_\beta(X, \alpha)$ for every α. Minimizing over α gives $\mathrm{SOES}_\beta(X) \geq \mathrm{ES}_\beta(X)$, and Proposition D.4 then gives $\mathrm{SOES}_\beta(X) > \mathbb{E}[-X]$ for nonconstant X, with equality for constant X by *(R1)* and *(R2)*. □

Appendix E

Numerical Implementation and Analysis of the Optimization Problems

In this appendix, we describe how the programs derived in Appendices A–C are implemented and solved numerically. In Section E.1 we introduce the CVXPY library. In Section E.2 we show the implementation in CVXPY for the example of the regularized joint program for Expected Shortfall and give the configuration of the Gurobi solver. Finally, in Section E.3 we discuss how the size of the programs grows with the investable universe and analyze their running time.

E.1 The CVXPY Library

All optimization problems in this thesis are implemented with CVXPY, an open-source, Python-embedded modeling language for convex optimization problems [16]. It lets us express a problem in a natural way that follows the mathematics, rather than in the restrictive standard form required by solvers, and it automatically transforms the problem into standard form, calls a solver, and returns the results in a readable format [13]. CVXPY acts as an interface between the problem written in Python and an external numerical solver, which in this thesis is the Gurobi solver. In this section, we summarize the parts of the CVXPY library needed to understand the implementation of the programs. Further details can be found in the CVXPY documentation [13].

When defining a program in CVXPY, variables, created with `cp.Variable`, are the decision variables of the problem and can be scalars, vectors, or matrices. Parameters, created with `cp.Parameter`, are symbolic representations of constants, whose purpose is to change the value of a constant in a problem without reconstructing the entire problem. Variables, parameters, and numerical constants are combined into expressions through the usual arithmetic operations and a library of functions called atoms, such as `cp.sum`, `cp.norm`, and `cp.sum_squares`. An objective, `cp.Minimize` or `cp.Maximize` of a scalar expression, and a list of constraints then define a problem `cp.Problem`. Calling its method `solve` returns the optimal value and stores the optimal solution in the attribute `value` of every variable.

Listing E.1, adapted from the CVXPY documentation, illustrates these objects on the least-squares problem with the constraint that the solution lies in the unit box

$$\min_{\boldsymbol{x}} \ \|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2, \quad \text{s.t. } \boldsymbol{x} \in [0,1]^n,$$

for a matrix $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ and a vector $\boldsymbol{b} \in \mathbb{R}^m$, in this example drawn at random with $m = 30$ and $n = 20$. Lines 5–8 of the code generate the problem data and line 11 declares the

variable $\boldsymbol{x} \in \mathbb{R}^n$. Line 12 states the objective with the atom `cp.sum_squares`, line 13 the two elementwise constraints, and line 14 combines them into a problem. Line 17 solves the problem with Gurobi set as the solver, and line 18 prints the optimal solution stored in `x.value` as a NumPy array.

```
1  import cvxpy as cp
2  import numpy as np
3
4  # Problem data.
5  m = 30
6  n = 20
7  A = np.random.randn(m, n)
8  b = np.random.randn(m)
9
10 # Construct the problem.
11 x = cp.Variable(n)
12 objective = cp.Minimize(cp.sum_squares(A @ x - b))
13 constraints = [0 <= x, x <= 1]
14 prob = cp.Problem(objective, constraints)
15
16 # Solve the problem.
17 result = prob.solve(solver=cp.GUROBI)
18 print(x.value)
```

Listing E.1: Least-squares problem with box constraints in CVXPY, adapted from [13].

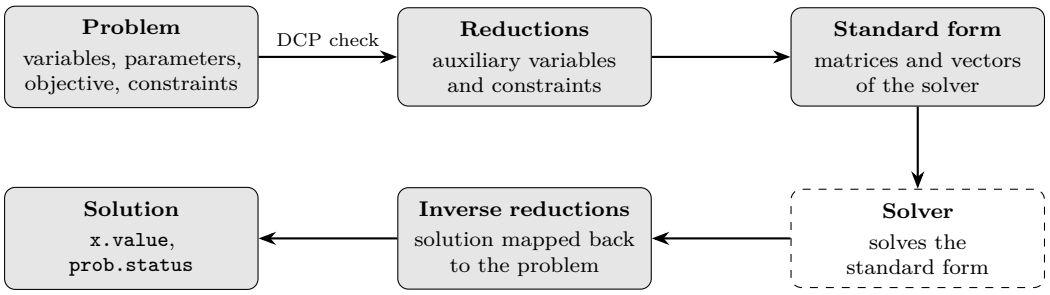


Figure E.1: Compilation of a problem in CVXPY. The problem is checked against the DCP rules and rewritten by a chain of reductions into the standard form of the solver, whose solution is mapped back through the inverted reductions to the variables of the original problem.

When `solve` is called, CVXPY compiles the problem in the steps shown in Figure E.1. It first checks that the problem is convex using the rules of disciplined convex programming (DCP). Each variable and function has a curvature label (affine, convex, or concave), and simple rules combine these labels as the expression is built. For example, a sum of convex terms is convex. If the objective and constraints do not end up with labels that make the problem convex, CVXPY rejects it before any solver is called. The problem is then passed through a chain of reductions. A reduction is a transformation from one problem to an equivalent one. The reductions replace each atom that the solver does not accept directly by auxiliary variables and constraints. For example, each absolute value $|x_i|$ in the atom `cp.norm(x, 1)` is replaced by variables $\Delta_i^+, \Delta_i^- \geq 0$ and the constraints $\Delta_i^+ - \Delta_i^- = x_i$. The final reductions save the coefficients of the resulting problem into the matrices and vectors of the standard form accepted by the solver. The solver returns a solution of this standard-form problem, which CVXPY maps back to the original problem by inverting the reductions. In Section E.2, we go through the implementation of the regularized joint optimization problem for Expected Shortfall (A.20) and what these steps generate for it.

E.2 Numerical Implementation

The programs derived in Appendices A–C are written in CVXPY close to their mathematical formulations, and the standard form passed to the solver is generated as described in Section E.1. We showcase this for the regularized joint optimization problem for Expected Shortfall (A.20). In the code, this program is written as

$$\begin{aligned} \boldsymbol{x}_t^*, \boldsymbol{\psi}_t^*, \alpha^*, \boldsymbol{u}^* = \underset{\boldsymbol{x}_t, \boldsymbol{\psi}_t, \alpha, \boldsymbol{u}}{\arg\min} \quad & \alpha + \frac{1}{q(1-\beta)} \mathbf{1}_q^\top \boldsymbol{u} + \lambda_1^a \|\boldsymbol{x}_t\|_1 + \lambda_1^c \|\boldsymbol{\psi}_t\|_1 \\ & + \lambda_2^a \|\boldsymbol{x}_t\|_2^2 + \lambda_2^c \|\boldsymbol{\psi}_t\|_2^2, \\ \text{s.t.} \quad & \boldsymbol{u} \geq -\boldsymbol{R}^{fh} \boldsymbol{x}_t - \boldsymbol{R}^c \boldsymbol{\psi}_t - \alpha \mathbf{1}_q, \quad \boldsymbol{u} \geq \mathbf{0}_q, \\ & \boldsymbol{x}_t^\top \mathbf{1}_N = 1, \quad \|\boldsymbol{x}_t\|_1 \leq \ell^a. \end{aligned} \tag{E.1}$$

As in (A.20), the Expected Shortfall is linearized with the auxiliary variables $\boldsymbol{u}$ of (A.7), but the penalties and the leverage constraint are kept as norms instead of being split as in (A.11).

```python
import cvxpy as cp

# Data and configuration.
beta, ell_a = 0.8, 2.0
q, N = R_fh.shape
M = R_c.shape[1]

# Variables.
x = cp.Variable(N)
psi = cp.Variable(M)
alpha = cp.Variable()
u = cp.Variable(q)

# Penalties as parameters.
lambda1_a, lambda2_a = cp.Parameter(nonneg=True), cp.Parameter(nonneg=True)
lambda1_c, lambda2_c = cp.Parameter(nonneg=True), cp.Parameter(nonneg=True)

# Objective and constraints.
es = alpha + cp.sum(u) / (q * (1 - beta))
penalty = (lambda1_a * cp.norm(x, 1) + lambda1_c * cp.norm(psi, 1)
           + lambda2_a * cp.sum_squares(x) + lambda2_c * cp.sum_squares(psi))
constraints = [
    u >= -R_fh @ x - R_c @ psi - alpha,
    u >= 0,
    cp.sum(x) == 1,
    cp.norm(x, 1) <= ell_a,
]
problem = cp.Problem(cp.Minimize(es + penalty), constraints)

# One setting of the hyperparameter grid is selected.
lambda1_a.value, lambda2_a.value = 5e-4, 0.03
lambda1_c.value, lambda2_c.value = 5e-5, 5e-5

# Solve.
problem.solve(solver=cp.GUROBI)
x_opt, psi_opt = x.value, psi.value
```

Listing E.2: Regularized joint Expected Shortfall program implementation in CVXPY.

Lines 5 and 6 of Listing E.2 read the dimensions q, N, and M from the return matrices $\boldsymbol{R}^{fh}$ and $\boldsymbol{R}^{c}$, and lines 9–12 declare the variables $\boldsymbol{x}_t$, $\boldsymbol{\psi}_t$, α, and $\boldsymbol{u}$. Lines 15 and 16 declare the four penalties as parameters, whose values are set only after the problem is built. Line 19 is the Expected Shortfall term of (E.1), lines 20 and 21 are the L^1 and L^2 penalties, and lines 22–27 are the constraints, namely the linearization of the Expected Shortfall in line 23, the non-negativity of $\boldsymbol{u}$ in line 24, the budget constraint in line 25, and the leverage constraint in line 26. Line 28 combines the objective and the constraints into a problem. Lines 31 and 32 assign the penalties of one point of the hyperparameter grid. Finally, line 35 solves the problem with Gurobi, configured as in Listing E.3, and line 36 reads out the solution.

None of the matrices of (A.20) are written in the code. When `problem.solve` is called, CVXPY carries out the steps of Figure E.1. The norms in lines 20 and 26 are replaced by the split of the weights into positive and negative parts (A.11), the variables are combined into one vector, which corresponds to the decision vector $\tilde{\boldsymbol{\theta}}_t$ of (A.20), and the coefficients of the objective and the constraints are combined into the vector of constants, the penalty matrix, and the constraint matrices, which take the form of $\tilde{\boldsymbol{c}}$, $\tilde{\boldsymbol{Q}}$, $\tilde{\boldsymbol{D}}$, $\tilde{\boldsymbol{E}}$, $\tilde{\boldsymbol{A}}$, and $\tilde{\boldsymbol{b}}$ in (A.20).

In the backtest, this program is solved at every rebalancing date for all 256 points of the hyperparameter grid (further details on hyperparameter selection can be found in Appendix G). Since the grid points differ only in the penalties, the problem of Listing E.2 does not have to be rebuilt for each grid point. To speed up the runtime, we keep the problem of each window cached and, for every further grid point, only assign the new penalties before calling `solve`. As a result, CVXPY compiles the problem only once per window. The cache holds two problems at a time, one for the estimation window, used by all grid points, and one for the estimation and validation windows together, used for the final refit.

All programs are solved with Gurobi, a state-of-the-art solver for linear, quadratic, and second-order cone programs. The solver is configured to use the barrier method with tight convergence and accuracy tolerances. The exact configuration is given in Listing E.3.

```
problem.solve(
    solver=cp.GUROBI,
    verbose=False,
    BarConvTol=1e-8,
    Method=2,
    ScaleFlag=0,
    AggFill=0,
    PreDual=1,
    Presolve=1,
    Threads=3,
    FeasibilityTol=1e-9,
    OptimalityTol=1e-9,
)
```

Listing E.3: Configuration of Gurobi for all programs.

The configuration was kindly provided by Dr. Patrick Lucescu, who calibrated it on the same dataset as used in this thesis.

E.3 Computational Complexity and Performance

The size of the programs derived in Appendices A–C grows with the investable universe in two ways. In the regularized programs, every asset weight and currency exposure is split into positive and negative parts, so the number of variables always grows linearly with the number of assets and currencies. For all risk measures except variance, every observation of

the lookback window adds one more auxiliary variable and one constraint to the program. The second-order Expected Shortfall programs additionally have a second-order cone constraint, which makes them harder to solve than the linear or quadratic programs of the other measures. The joint programs are the largest, since they optimize all asset weights and currency exposures at once, whereas sequential optimization splits the problem into an asset program and a small currency program. Regularization hardly changes the size of a program, but it multiplies the number of programs solved at every rebalancing date, since the regularized programs are solved for every point of the hyperparameter grid.

The backtests were run on the Euler cluster of ETH Zürich, each as one job with four CPU cores and Gurobi using three threads, as described in Listing E.3. The backtester implementation of Section 4.2.1 logs a timestamp at the start of every rebalancing date. For each program, we average these step times over all logged backtests.

Figures E.2 to E.4 show the step times at each rebalancing date for the small, medium, and large universes. The times grow with the size of the universe, from about a minute per rebalancing date for the regularized joint Expected Shortfall program in the small universe to about half an hour in the large universe, and within each universe they rise over the backtest period as the number of assets in the investable universe grows. The regularized programs are considerably slower than the unregularized ones, and the regularized joint programs are the slowest of all programs. Across the risk measures, the second-order Expected Shortfall programs are computationally the most expensive, followed by the Expected Shortfall programs. The variance programs are the cheapest in the small and medium universes, but their cost grows faster because the size of the covariance matrix grows quadratically with the number of assets. In the large universe they are solved as fast as the Expected Shortfall programs but slower than the semi-variance and mean absolute deviation programs.

Without regularization, the joint and the overlay programs are similar in terms of computational cost. With regularization, the joint programs are slower than the overlay programs, about twice as slow in the small universe and up to 14 times slower in the large universe.



Figure E.2: Time per rebalancing date of the backtests in the small investable universe (top 10 stocks per economy), by risk measure and program, on a logarithmic scale, averaged over all logged backtests of each program. Times below the logging resolution of one second are drawn at one second.



Figure E.3: Time per rebalancing date of the backtests in the medium investable universe (top 50 stocks per economy), by risk measure and program, on a logarithmic scale, averaged over all logged backtests of each program.



Figure E.4: Time per rebalancing date of the backtests in the large investable universe (top 100 stocks per economy), by risk measure and program, on a logarithmic scale, averaged over all logged backtests of each program.

Appendix F

Transaction Costs

In this appendix, we detail the modeling of transaction costs in the context of international portfolio optimization. Following [8, 37, 36], and keeping the notation introduced in Chapter 3, we model transaction costs as proportional to the volume of traded assets and currencies, ignoring any fixed trading costs. The return of the hedged portfolio is expressed as the sum of unhedged asset returns and gains from forward currency positions

$$R^h_{\mathcal{P},t+1} = \sum_{i=1}^{N} x_{i,t} R^u_{i,t+1} + \sum_{c=1}^{M} \phi_{c,t}(f_{c,t} - e_{c,t+1}). \tag{3.4}$$

Let τ^{asset}_{t+1} and τ^{curr}_{t+1} denote the total transaction costs associated with asset and currency trades, respectively. The portfolio return net of transaction costs is then given by

$$R^{h,net}_{\mathcal{P},t+1} = R^h_{\mathcal{P},t+1} - \tau^{asset}_{t+1} - \tau^{curr}_{t+1}. \tag{F.1}$$

The total transaction costs are computed as the sum of the costs incurred from trading individual assets and currencies, weighted by the trading volumes

$$\tau^{asset}_{t+1} = \sum_{i=1}^{N} \tau^{asset}_{i,t+1} \left(x_{i,t+1} - x^-_{i,t+1} \right),$$

$$\tau^{curr}_{t+1} = \sum_{c=1}^{M} \tau^{curr}_{c,t+1} \phi_{c,t+1} + \sum_{c=1}^{M} \tau^{curr}_{c,t+1} \left(w_{c,t+1} - w^-_{c,t+1} \right),$$

where $\tau^{asset}_{i,t+1}$ and $\tau^{curr}_{c,t+1}$ are the per-unit transaction costs for asset i and currency c. The terms $x^-_{i,t+1}$ and $w^-_{c,t+1}$ denote the asset weights and implied currency weights immediately prior to rebalancing at time $t+1$ (i.e., the drifted weights).

Asset transaction costs are incurred only when rebalancing individual asset holdings. Currency transaction costs, however, arise from two sources, adjustments in forward hedge positions (assuming the forward contract maturity matches the rebalancing frequency, the old contract expires and a new contract of full notional size $\phi_{c,t+1}$ must be established at each rebalancing date) and currency exchanges required to fund purchases of assets denominated in different currencies ($w_{c,t+1} - w^-_{c,t+1}$). If the portfolio is rebalanced more frequently than the forward contract maturity, existing forward contracts are kept open and only incremental adjustments are traded. In this case, the forward transaction cost term is modeled as the absolute difference, replacing $\sum_{c=1}^{M} \tau^{curr}_{c,t+1} \phi_{c,t+1}$ with $\sum_{c=1}^{M} \tau^{curr}_{c,t+1} (\phi_{c,t+1} - \phi^-_{c,t+1})$, where $\phi^-_{c,t+1}$ represents the drifted hedge weight. Following [8], these costs are aggregated at the currency level, operating under

the assumption that proceeds from selling assets denominated in currency c can fund the purchase of other assets in the same currency without incurring currency conversion fees.

To define the individual per-unit costs $\tau^{asset}_{i,t+1}$ and $\tau^{curr}_{c,t+1}$, we consider an asset i denominated in currency c_i. Let $P^{ask}_{i,t}$, $P^{bid}_{i,t}$, and $P_{i,t}$ represent the ask, bid, and mid prices of the asset, and let $S^{ask}_{c_i,t}$, $S^{bid}_{c_i,t}$, and $S_{c_i,t}$ represent the corresponding currency prices. If asset i is purchased at time t and sold at time $t+1$, its realized log return is given by

$$\begin{aligned}
\log\left(\frac{P^{bid}_{i,t+1}S^{bid}_{c_i,t+1}}{P^{ask}_{i,t}S^{ask}_{c_i,t}}\right) &= \log\left(\frac{\left(P_{i,t+1}-\frac{\delta^{asset}_{i,t+1}}{2}\right)\left(S_{c_i,t+1}-\frac{\delta^{curr}_{c_i,t+1}}{2}\right)}{\left(P_{i,t}+\frac{\delta^{asset}_{i,t}}{2}\right)\left(S_{c_i,t}+\frac{\delta^{curr}_{c_i,t}}{2}\right)}\right) \\
&= \log\left(\frac{P_{i,t+1}\left(1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}\right)S_{c_i,t+1}\left(1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}\right)}{P_{i,t}\left(1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}\right)S_{c_i,t}\left(1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}\right)}\right) \\
&= \log\left(\frac{P_{i,t+1}S_{c_i,t+1}}{P_{i,t}S_{c_i,t}}\right)+\log\left(\frac{1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}}{1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}}\right)+\log\left(\frac{1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}}{1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}}\right) \\
&= \log\left(\frac{P_{i,t+1}S_{c_i,t+1}}{P_{i,t}S_{c_i,t}}\right)-\log\left(\frac{1+\frac{\delta^{asset}_{i,t}}{2P_{i,t}}}{1-\frac{\delta^{asset}_{i,t+1}}{2P_{i,t+1}}}\right)-\log\left(\frac{1+\frac{\delta^{curr}_{c_i,t}}{2S_{c_i,t}}}{1-\frac{\delta^{curr}_{c_i,t+1}}{2S_{c_i,t+1}}}\right),
\end{aligned}$$

where δ^{asset}_i and $\delta^{curr}_{c_i}$ are the bid-ask spreads. Comparing this to equation (F.1), the transaction cost is precisely the difference between the actual realized log return and mid-price returns for the asset and currency.

Defining half of the bid-ask spreads as

$$\chi^{asset}_{i,t} = \frac{\overbrace{P^{ask}_{i,t}-P^{bid}_{i,t}}^{\delta^{asset}_i}}{2P_{i,t}} \quad \text{and} \quad \chi^{curr}_{c,t} = \frac{\overbrace{S^{ask}_{c,t}-S^{bid}_{c,t}}^{\delta^{curr}_c}}{2S_{c,t}},$$

the per-unit transaction costs for trading asset i and currency c become

$$\tau^{asset}_{i,t+1} = \begin{cases} \log\left(\frac{1+\chi^{asset}_{i,t+1}}{1-\chi^{asset}_{i,t}}\right) & \text{if } x_{i,t+1} > x^{-}_{i,t+1}, \\ \log\left(\frac{1-\chi^{asset}_{i,t}}{1+\chi^{asset}_{i,t+1}}\right) & \text{otherwise,} \end{cases}$$

and

$$\tau^{curr}_{c,t+1} = \begin{cases} \log\left(\frac{1+\chi^{curr}_{c,t+1}}{1-\chi^{curr}_{c,t}}\right) & \text{if } \phi_{c,t+1} > 0 \text{ or } w_{c,t+1} > w^{-}_{c,t+1}, \\ \log\left(\frac{1-\chi^{curr}_{c,t}}{1+\chi^{curr}_{c,t+1}}\right) & \text{otherwise.} \end{cases}$$

For our empirical analysis, we adopt the simplifying assumption that proportional bid-ask spreads remain constant both over time and across the cross-sections of both assets and currencies. We set the half-spreads to $\chi^{asset} = 0.001$ (10 bps) for all equities and $\chi^{curr} = 0.0001$ (1 bp) for all currencies, consistent with empirical findings in [25, 48, 44, 19]. Under these assumptions, the transaction costs simplify to constant values of $\tau^{asset} \approx 0.002$ and $\tau^{curr} \approx 0.0002$.

Appendix G

Additional Results

In this appendix, we report additional results that support the empirical analysis of Chapter 5. In Section G.1 we justify the bounded hyperparameter grids over which the regularization penalties are selected. In Section G.2 we show how many of the available assets and currencies the overlay and joint optimizers hold and how regularization influences the portfolio construction. In Section G.3 we describe how the risk aversion coefficients of the mean-risk optimizers are calibrated. Section G.4 analyzes the drawdowns of the optimizers, with a focus on the main market crashes of the out-of-sample period, and Section G.5 reports the realized values of the risk measures and the Expected Shortfall ratios of the optimizers, together with their turnover at each rebalancing date.

G.1 Hyperparameter Grids

The regularized overlay and joint programs have four penalties in the objective, the asset penalties λ_1^a and λ_2^a and the currency penalties λ_1^c and λ_2^c, while the asset-only optimizers, Full and Zero, only have the two asset penalties. In the backtests, the penalties are selected at every rebalancing date by the grid search described in Section 4.2.1. The grid we optimize over is the Cartesian product of the penalty values, with 256 points for the overlay and joint programs and 16 points for the Full and Zero programs. The asset leverage limit is fixed at $\ell^a = 2$ for all the programs, and the confidence level of Expected Shortfall and second-order Expected Shortfall at $\beta = 0.8$.

As the penalty sizes are increased, the optimal weights approach a limiting allocation, and beyond some threshold a further increase leaves the solution unchanged. A grid whose largest value lies past that threshold covers the whole range of regularization, and extending it further would only increase the computational cost of the grid search.

To locate this threshold, we use a similar approach to [37] and analyze each penalty $\lambda \in \{\lambda_1^a, \lambda_2^a, \lambda_1^c, \lambda_2^c\}$ on its own, holding the other three fixed at zero. We consider an increasing sequence $0 = \lambda_0 < \lambda_1 < \cdots < \lambda_K$. Let $\boldsymbol{x}_t(\lambda_n)$ and $\boldsymbol{\psi}_t(\lambda_n)$ denote the optimal asset weights and currency exposures at λ_n. The marginal effect of the step from λ_{n-1} to λ_n is measured in the supremum norm

$$
\begin{aligned}
d_n^a &= \|\boldsymbol{x}_t(\lambda_n) - \boldsymbol{x}_t(\lambda_{n-1})\|_\infty \quad \text{for } \lambda \in \{\lambda_1^a, \lambda_2^a\}, \\
d_n^c &= \|\boldsymbol{\psi}_t(\lambda_n) - \boldsymbol{\psi}_t(\lambda_{n-1})\|_\infty \quad \text{for } \lambda \in \{\lambda_1^c, \lambda_2^c\}.
\end{aligned}
$$

The two penalty types approach their limits in different ways. Under an L^1 penalty, the weights reach their limit at a finite value of the penalty, a long-only portfolio for the asset

weights and a zero overlay for the currency exposures, and do not change as the penalty is increased further. An L^2 penalty instead shrinks the weights smoothly toward their limit, the equally weighted portfolio for the asset weights and a zero overlay for the currency exposures, and reaches it only asymptotically, so its marginal effect decays gradually without vanishing [22]. For this reason, we report alongside d_n^a and d_n^c the portfolio exposure, $\|\boldsymbol{x}_t(\lambda_n)\|_1$ for the asset penalties and $\|\boldsymbol{\psi}_t(\lambda_n)\|_1$ for the currency penalties.

The analysis is carried out for the regularized overlay and joint programs of each risk measure on the first lookback window of the small universe, which spans July 1990 to June 2000 and contains 2,610 daily observations. To account for sampling error, each reported quantity is averaged over five bootstrap replications of this window, each constructed by resampling its daily observations with replacement.

We visualize the results for the regularized variance programs, whose marginal effects and exposures are shown in Figures G.1 and G.2. The overlay and joint programs behave very similarly. The difference between the two penalty types is clearest in Figure G.1. Once the weights reach their limit under an L^1 penalty, the marginal effect has a sudden drop, to zero for the currency exposures and by more than two orders of magnitude for the asset weights. The L^2 penalties do not exhibit such a drop, which is why their largest grid values are interpreted from the exposure in Figure G.2 instead.

For both asset penalties, exposure falls from slightly below the leverage limit to one. Once the portfolio becomes long-only, the L^1 penalty equals the constant λ_1^a, so the solution stops changing. We set the largest grid value to 5×10^{-5}, by which this has happened. As discussed in Section B.3, the L^2 penalty instead shrinks the covariance estimate toward a scaled identity matrix, since $\boldsymbol{x}_t^\top \boldsymbol{\Sigma}_t \boldsymbol{x}_t + \lambda_2^a \|\boldsymbol{x}_t\|_2^2 = \boldsymbol{x}_t^\top (\boldsymbol{\Sigma}_t + \lambda_2^a \boldsymbol{I}_N) \boldsymbol{x}_t$, so the weights move toward the equally weighted portfolio. We set the largest grid value to 10^{-2}, by which there are no longer any short positions.

The currency exposures carry neither a budget nor a leverage constraint, so $\boldsymbol{\psi}_t$ shrinks to zero. Under λ_1^c, the currency exposures disappear at 5×10^{-5}. Under λ_2^c, they never fully disappear, so we determine the largest grid value as the point at which the overlay becomes negligible. From 1.7×10^{-3} onward, less than one percent of the unpenalized currency exposure remains, and we set the largest grid value to 2×10^{-3}.

We perform the same analysis for the regularized overlay and joint programs of the other four risk measures. They show the same pattern as the variance programs. Both asset penalties reduce the asset exposure to one, the currency L^1 penalty removes the currency exposures at a finite value, and the currency L^2 penalty shrinks them smoothly toward zero.

On the basis of this analysis, we construct the grids reported in Table G.1. The upper bounds of the asset L^1 penalty for variance and semi-variance are set to 5×10^{-5} and 10^{-4}, while those for mean absolute deviation, Expected Shortfall, and second-order Expected Shortfall are set to 2.5×10^{-3}, 5×10^{-3}, and 10^{-1}. The asset penalty grids of the variance and semi-variance programs lie one to three orders of magnitude below those of the other risk measures. This is because they are measured in squared returns.

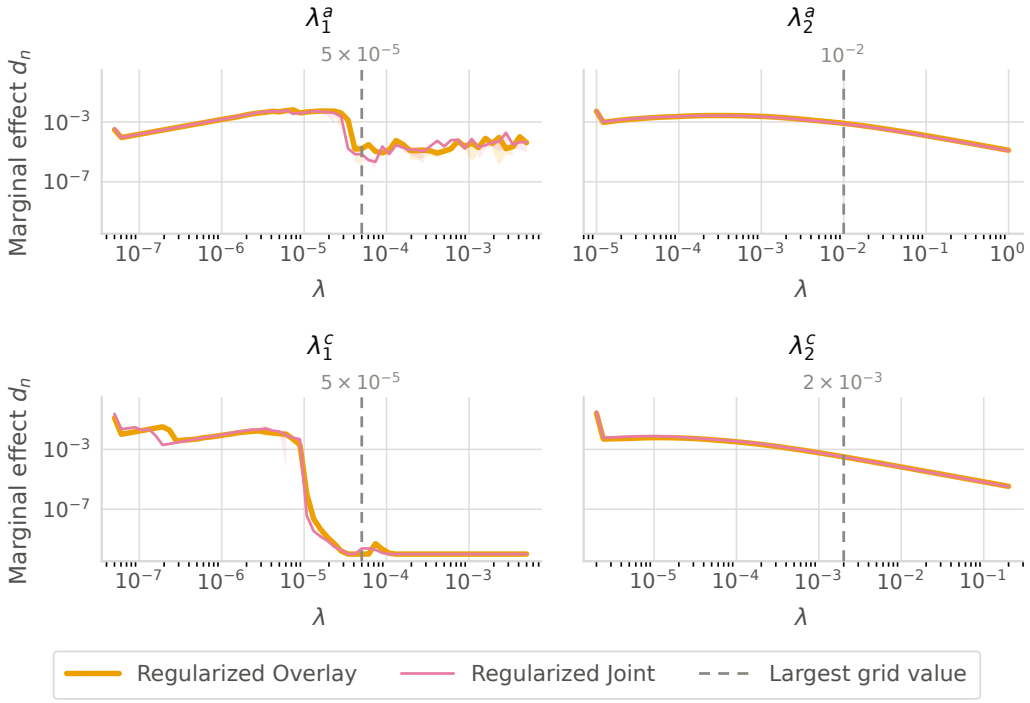


Figure G.1: Marginal effect of each penalty for the regularized variance overlay and joint programs. The upper panels report d_n^a for the asset penalties and the lower panels d_n^c for the currency penalties. The dashed lines mark the largest value of each grid in Table G.1. Each quantity is averaged over five bootstrap replications of the first lookback window of the small investable universe (top 10 stocks per economy).

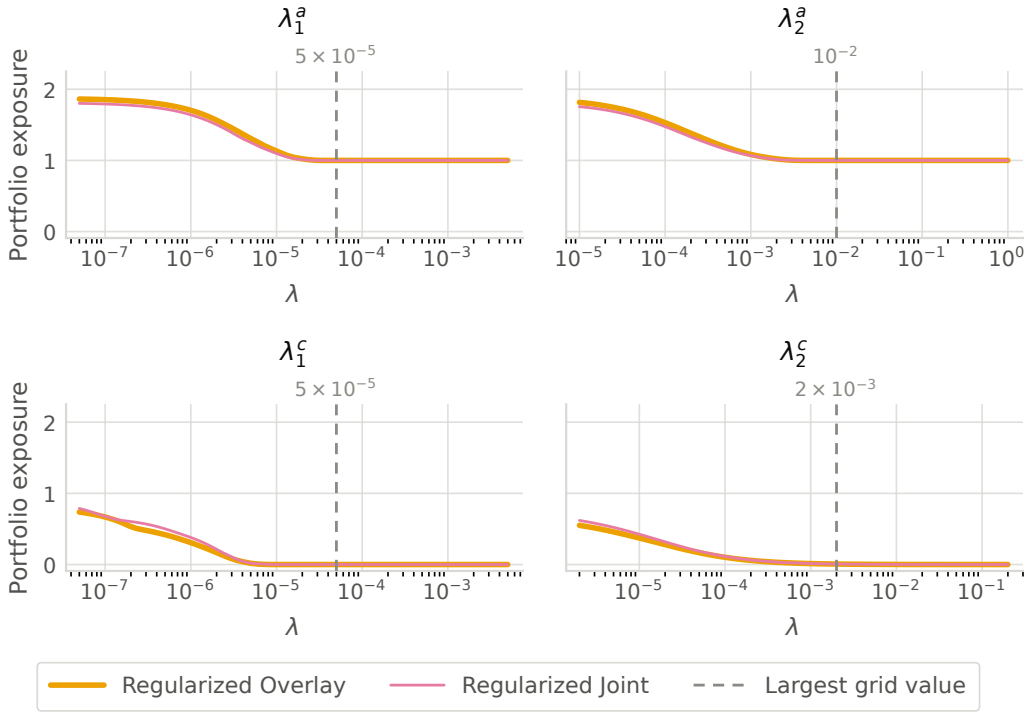


Figure G.2: Portfolio exposure as a function of each penalty for the regularized variance overlay and joint programs. The upper panels report $\|\boldsymbol{x}_t\|_1$ for the asset penalties and the lower panels $\|\boldsymbol{\psi}_t\|_1$ for the currency penalties. The dashed lines mark the largest value of each grid in Table G.1. Each quantity is averaged over five bootstrap replications of the first lookback window of the small investable universe (top 10 stocks per economy).

Risk measure	Penalty	Description	Range
Variance	λ_1^a	Asset L^1 penalty	$[0,\ 5 \times 10^{-5}]$
	λ_2^a	Asset L^2 penalty	$[0,\ 10^{-2}]$
	λ_1^c	Currency L^1 penalty	$[0,\ 5 \times 10^{-5}]$
	λ_2^c	Currency L^2 penalty	$[0,\ 2 \times 10^{-3}]$
Semi-Variance	λ_1^a	Asset L^1 penalty	$[0,\ 10^{-4}]$
	λ_2^a	Asset L^2 penalty	$[0,\ 5 \times 10^{-2}]$
	λ_1^c	Currency L^1 penalty	$[0,\ 2.5 \times 10^{-5}]$
	λ_2^c	Currency L^2 penalty	$[0,\ 10^{-3}]$
MAD	λ_1^a	Asset L^1 penalty	$[0,\ 2.5 \times 10^{-3}]$
	λ_2^a	Asset L^2 penalty	$[0,\ 1]$
	λ_1^c	Currency L^1 penalty	$[0,\ 1.5 \times 10^{-3}]$
	λ_2^c	Currency L^2 penalty	$[0,\ 2.5 \times 10^{-1}]$
ES	λ_1^a	Asset L^1 penalty	$[0,\ 5 \times 10^{-3}]$
	λ_2^a	Asset L^2 penalty	$[0,\ 3]$
	λ_1^c	Currency L^1 penalty	$[0,\ 5 \times 10^{-4}]$
	λ_2^c	Currency L^2 penalty	$[0,\ 5 \times 10^{-5}]$
SOES	λ_1^a	Asset L^1 penalty	$[0,\ 10^{-1}]$
	λ_2^a	Asset L^2 penalty	$[0,\ 15]$
	λ_1^c	Currency L^1 penalty	$[0,\ 10^{-2}]$
	λ_2^c	Currency L^2 penalty	$[0,\ 1]$

Table G.1: Ranges of the hyperparameter grids of the regularized programs.

G.2 Portfolio Sparsity Analysis

Figures G.3 and G.4 report the share of the investable universe in which the unregularized and regularized overlay and joint optimizers hold a position. Figure G.3 counts the assets whose weight exceeds 10^{-4} in absolute value, and Figure G.4 the foreign currencies whose currency exposure exceeds 10^{-4} in absolute value. On the lines of the regularized optimizers in Figure G.3, the markers indicate the rebalancing dates at which the selected asset L^1 or L^2 penalties are at the top of the range.

Regularization lowers the number of assets held in every universe size, and the share held by the regularized optimizers swings from one rebalancing date to the next due to the selection of the L^1 and L^2 penalties. An L^1-dominant selection sets weights to zero and creates a sparse, concentrated portfolio, whereas an L^2-dominant selection shrinks all weights toward the equally weighted portfolio. The swings therefore reflect the optimizers alternating between these two regimes across the rebalancing quarters, not an instability of the optimizers themselves.

The unregularized optimizers have no such degree of freedom. Besides the budget constraint, their asset weights are only restricted by the leverage limit $\|\boldsymbol{x}_t\|_1 \leq \ell^a$, which promotes sparsity in the same way as the L^1 penalty. As a result, the share of the assets they hold falls with the size of the universe. For variance, semi-variance, mean absolute deviation, and Expected Shortfall, it falls from 91–99% in the small universe to 60–73% in the large universe. And for second-order Expected Shortfall optimizers, it falls from 70–78% in the small universe to around 50% in the large universe.

For mean absolute deviation, Expected Shortfall, and second-order Expected Shortfall, the unregularized overlay optimizers hold almost no currency exposure in the medium and large universes, unlike the joint optimizers. With several hundred assets, the asset step of the overlay program likely already fits the in-sample risk so closely that no currency position can lower it further. The joint program instead chooses the asset weights and currency exposures together.

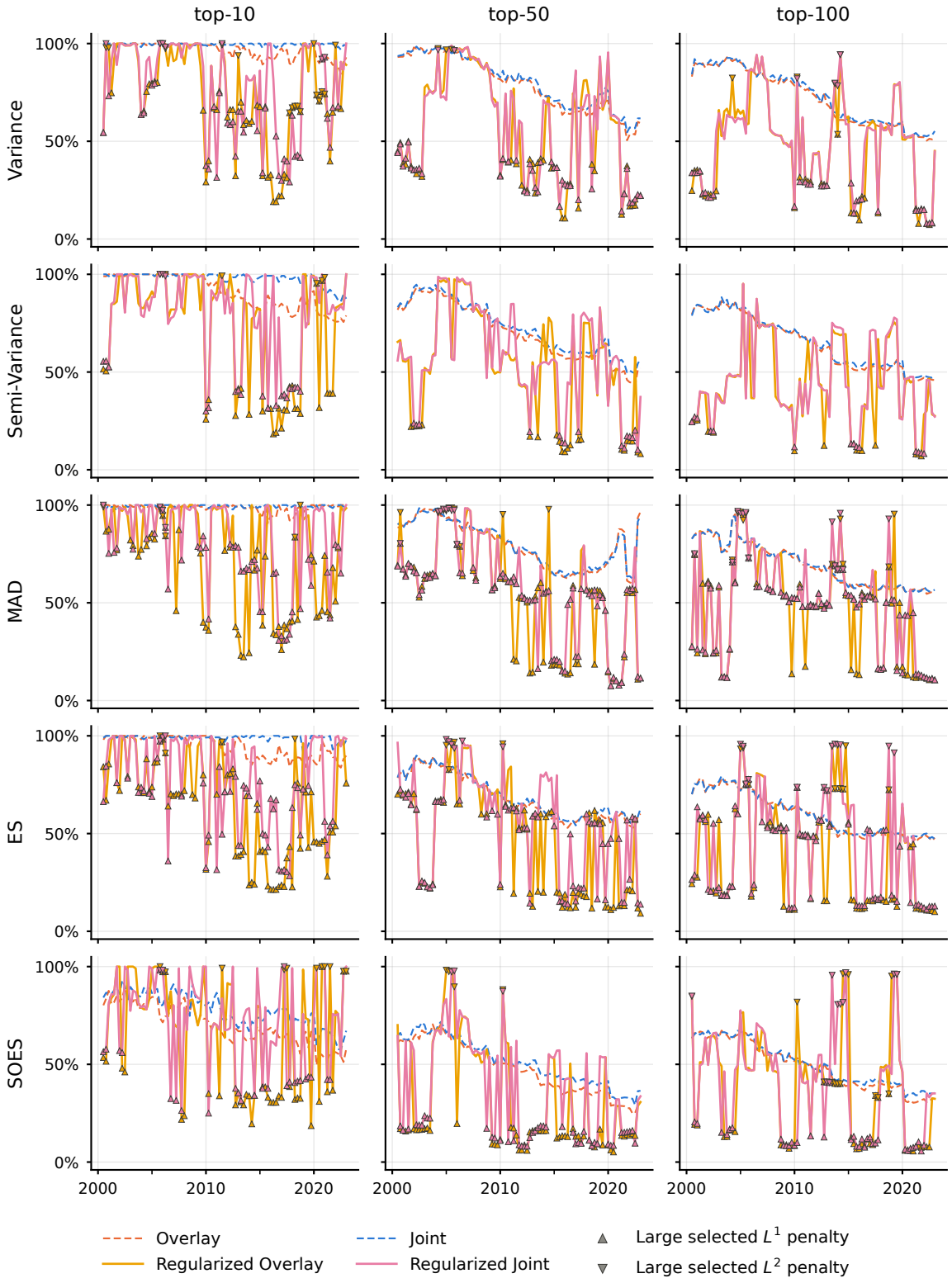


Figure G.3: Share of the assets in the investable universe held by the unregularized and regularized overlay and joint optimizers at each rebalancing date, by risk measure and universe size. An asset counts as held if the absolute value of its weight exceeds 10^{-4}. The markers on the regularized lines indicate the rebalancing dates at which the selected asset L^1 or L^2 penalty is one of the two largest values of its grid.

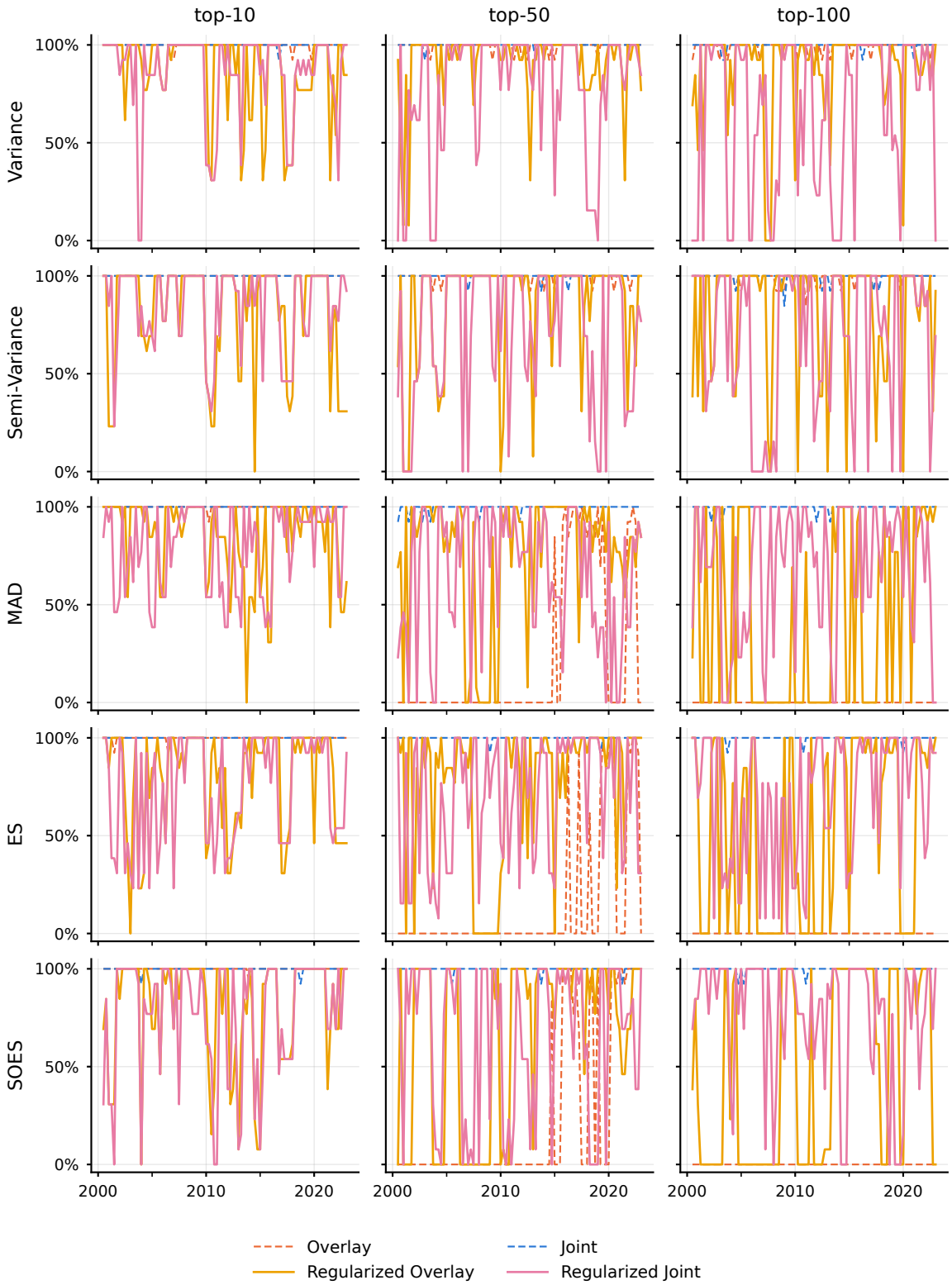


Figure G.4: Share of the foreign currencies in which the unregularized and regularized overlay and joint optimizers hold a currency exposure at each rebalancing date, by risk measure and universe size. A currency counts as held if the absolute value of its exposure exceeds 10^{-4}.

G.3 Risk Aversion Calibration

The mean-risk programs of Section 5.3 subtract the expected portfolio return, scaled by $1/\gamma$, from the risk measure, with the same risk aversion coefficient γ in the asset and currency steps of the overlay program and in the joint program. As $\gamma \to 0$, the mean term dominates, and the programs maximize the expected portfolio return subject to the budget and leverage constraints. The currency exposures then reach the currency leverage limit $\ell^c = 2$ in the currency whose interest rate differs most from the US interest rate. Without this limit, which is added to the mean-risk programs in Sections A.6, B.6, and C.5, the currency turnover reaches upwards of 20, which clearly signals instability.

The risk aversion coefficient is not selected at each rebalancing date but calibrated once on the first lookback window of the small universe, as described in Section 4.2.2. Like the lookback windows of the penalty grid search, this window is split into a nine-year estimation window and a one-year validation window. For every value of γ in a grid, the overlay and joint programs of each risk measure are fitted on the estimation window and scored on the validation window. Unlike the penalties, which are scored by the risk measure, γ is scored by the Sharpe ratio, since the mean term is meant to trade additional risk for a higher expected return. For each risk measure, we fix the value with the highest validation Sharpe ratio averaged over the overlay and joint programs and use it at every rebalancing date in all three universes. The resulting values are reported in Table G.2. Since the first lookback window ends on the first rebalancing date, 30 June 2000, the calibration only uses data that is available when the first portfolio is formed and introduces no look-ahead bias into the backtests.

Risk measure	γ
Variance	30
Semi-Variance	30
MAD	0.5
ES	2
SOES	0.5

Table G.2: Risk aversion coefficients of the mean-risk optimizers.

The values in Table G.2 are one to two orders of magnitude larger for variance and semi-variance than for mean absolute deviation, Expected Shortfall, and second-order Expected Shortfall. As for the penalty grids in Section G.1, this is because variance and semi-variance are measured in squared returns.

G.4 Drawdown Analysis

Figures G.5 and G.6 show the drawdown of the optimizers at each trading day of the out-of-sample period, that is, the decline of their cumulative wealth from its running maximum, whose largest value is the maximum drawdown as defined in Section 4.2.3. Figure G.5 reports the unregularized and regularized overlay and joint optimizers and the Zero optimizer, and Figure G.6 the mean-risk optimizers, each against the equally weighted portfolio.

To compare the optimizers in the main market crashes of the sample, we also report their crisis drawdowns, which include the dot-com crash, from the start of the out-of-sample period in July 2000 to June 2003, the global financial crisis, from 2007 to 2009, and the COVID-19 crash, from January to June 2020. In addition, we report the recovery time of the maximum drawdown, the time from the peak preceding it until the cumulative wealth regains that peak.

In the medium and large universes, all optimizers reach their maximum drawdown in the global financial crisis. The overlay and joint optimizers lose between 24.7% and 38.6%, and the mean-risk optimizers between 29.6% and 41.5%, against 55.7% and 56.3% for the equally weighted portfolio. However, they need 7.2 to 12.1 years to regain their pre-crisis peak, against 6.7 to 6.9 years for the equally weighted portfolio. Their smaller losses require smaller gains to recover, between 33% and 71% against 126% and 129%, but in the two years after each portfolio's cumulative wealth reaches its lowest point in the crisis, they gain only 7% to 32%, while the equally weighted portfolio gains 86% and 89%. The low-risk assets that limit their losses in the crisis therefore also limit their gains in the recovery. In the dot-com and COVID-19 crashes, they lose 4.7% to 26.3%, against 30.2% to 33.9% for the equally weighted portfolio.

The small-universe backtests differ in two respects. First, the overlay and joint optimizers lose as much in the dot-com crash as in the global financial crisis, between 26.0% and 36.2% against 23.3% to 34.1%, whereas in the medium and large universes they lose at most 10.2% in the dot-com crash. With ten stocks per economy, there are fewer low-risk assets for the optimizers to move into, as discussed in Section 5.1. Second, six of the ten unregularized overlay and joint optimizers and both second-order Expected Shortfall mean-risk optimizers reach their maximum drawdown, between 28.2% and 37.1%, in October 2022, and have not regained their early-2020 peak by the end of the sample in April 2023. The equally weighted portfolio had already regained its pre-COVID-19 peak in January 2021.

Zero leaves the currency exposure of its asset positions unhedged and loses between 33.1% and 44.8% in the global financial crisis, more than the overlay and joint optimizers of the same risk measure in every universe. Between July 2008 and March 2009, the foreign currencies lost 16.6% against the US dollar on average, and only the Japanese yen and the Hong Kong dollar appreciated, so the unhedged portfolio lost on its currency exposure in addition to its equities. The same exposure helps Zero in the recovery, as 12 of the 13 foreign currencies appreciated over the following two years, and in the medium and large universes Zero regains its pre-crisis peak sooner than the overlay and joint optimizers.

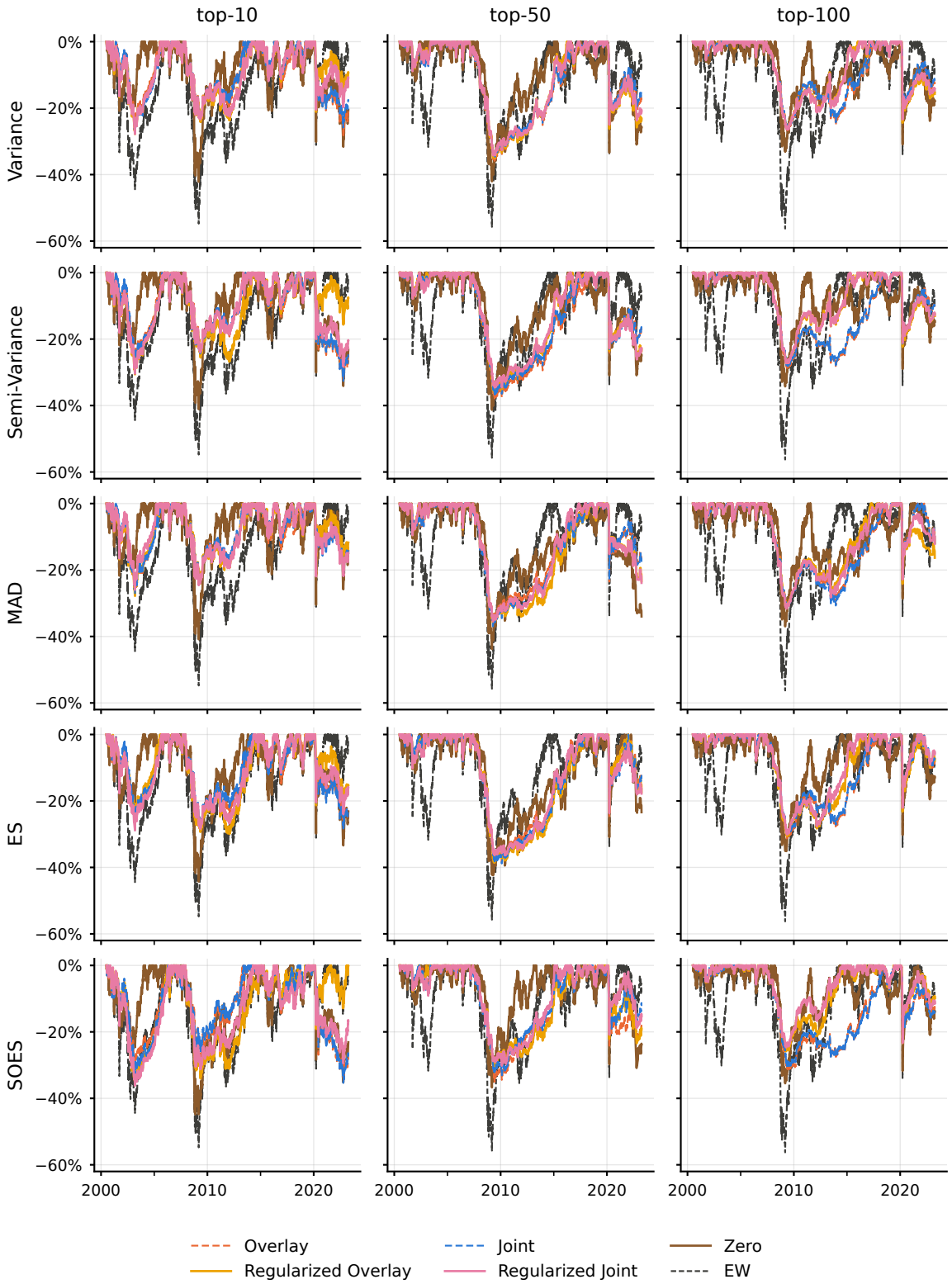


Figure G.5: Drawdown of the unregularized and regularized overlay and joint optimizers, the Zero optimizer, and the equally weighted portfolio in the out-of-sample period, by risk measure and universe size. The out-of-sample period runs from July 2000 to April 2023, and returns are net of transaction costs.

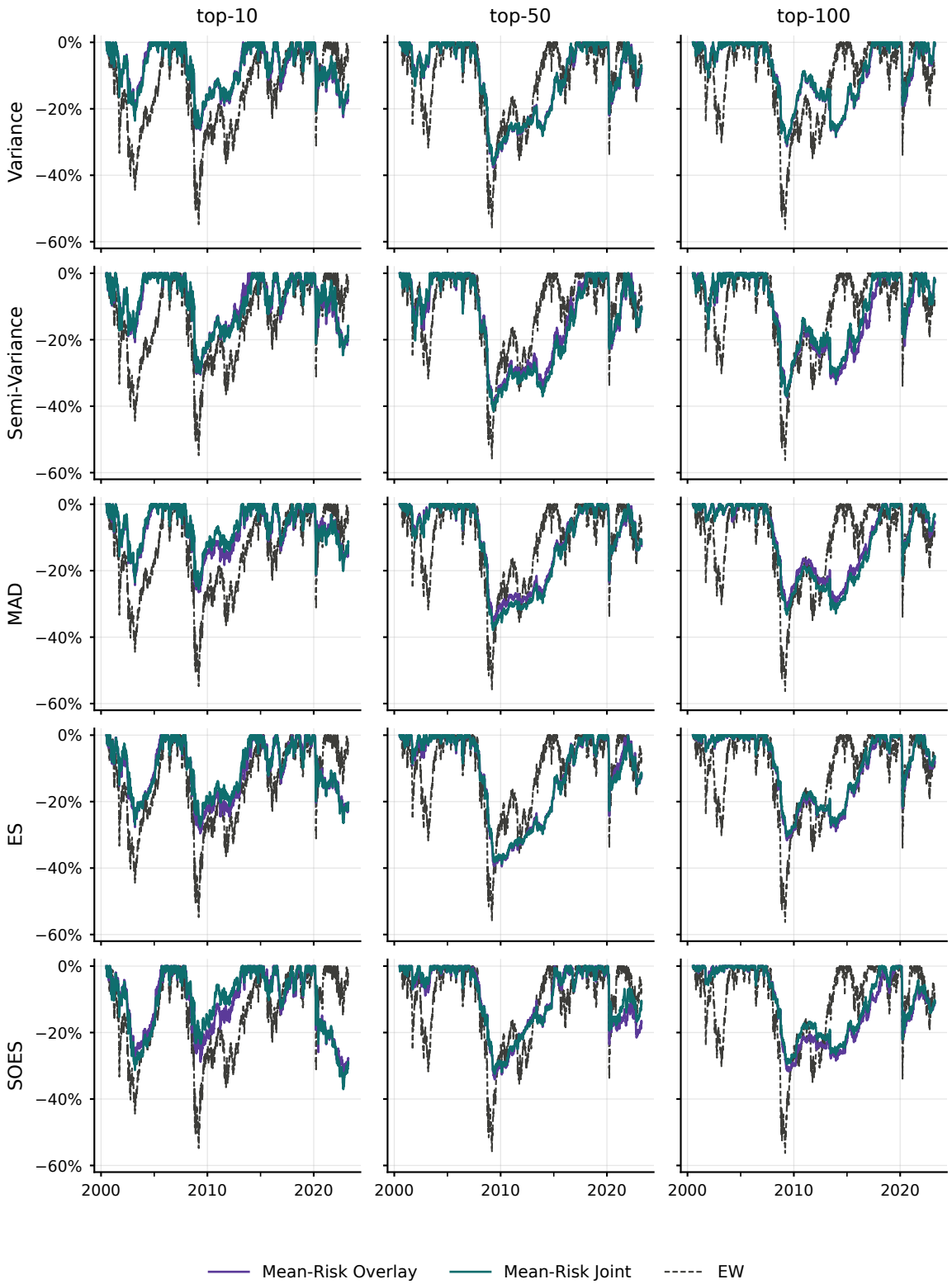


Figure G.6: Drawdown of the mean-risk overlay and joint optimizers and the equally weighted portfolio in the out-of-sample period, by risk measure and universe size. The out-of-sample period runs from July 2000 to April 2023, and returns are net of transaction costs.

G.5 Additional Backtest Performance Results

In this section, we report additional out-of-sample performance metrics for the optimizers of Chapter 5. Tables G.3 to G.5 report, for the baseline, regularized, and mean-risk optimizers, the realized values of the risk measures the optimizers minimize and the Expected Shortfall ratio. Figures G.7 and G.8 show the asset and currency turnover of the overlay and joint optimizers at each rebalancing date.

The risk measures are evaluated on the empirical distribution of the realized daily out-of-sample returns, and we report them as daily values, unlike the annualized metrics of Chapter 5. When defining the sample estimates of the risk measures, we use the notation of Section 4.2.3. The realized Expected Shortfall is the average loss on the days with a loss of at least the realized Value-at-Risk

$$\mathrm{ES}_{\beta,\mathcal{P}} = -\frac{1}{|\mathcal{T}_{\mathcal{P}}|}\sum_{t\in\mathcal{T}_{\mathcal{P}}} R_{\mathcal{P},t}, \qquad \mathcal{T}_{\mathcal{P}} = \{t \mid -R_{\mathcal{P},t} \geq \mathrm{VaR}_{\beta,\mathcal{P}}\},$$

where $\mathrm{VaR}_{\beta,\mathcal{P}}$ is the empirical β-quantile of the daily losses $-R_{\mathcal{P},t}$. The realized second-order Expected Shortfall is the sample estimate of (A.3)

$$\mathrm{SOES}_{\beta,\mathcal{P}} = \min_{\alpha}\left\{\alpha + \frac{1}{1-\beta}\left(\frac{1}{T}\sum_{t=1}^{T}\left((-R_{\mathcal{P},t}-\alpha)^{+}\right)^{2}\right)^{1/2}\right\},$$

and the realized semi-deviation and mean absolute deviation, the sample estimates of (B.2) and (C.1), are

$$\mathrm{SD}_{\mathcal{P}} = \left(\frac{1}{T}\sum_{t=1}^{T}\left(\left(\bar{R}_{\mathcal{P}} - R_{\mathcal{P},t}\right)^{+}\right)^{2}\right)^{1/2}, \qquad \mathrm{MAD}_{\mathcal{P}} = \frac{1}{T}\sum_{t=1}^{T}|R_{\mathcal{P},t} - \bar{R}_{\mathcal{P}}|.$$

The Expected Shortfall ratio replaces the volatility in the Sharpe ratio with the realized Expected Shortfall, annualized like the volatility

$$\mathrm{ESR}_{\mathcal{P}} = 252\,\frac{\bar{R}_{\mathcal{P}} - \bar{R}_{f}}{\sqrt{252}\,\mathrm{ES}_{\beta,\mathcal{P}}},$$

and measures the excess return per unit of tail risk.

As for the volatility in Section 5.1, the optimizers realize much lower values of all four risk measures than the equally weighted portfolio, and the reduction is larger in the medium and large universes than in the small universe. Without a mean term, the overlay, joint, and Full optimizers realize between 30.2% and 45.2% lower ES, SOES, SD, and MAD than the equally weighted portfolio in the small universe, between 51.4% and 61.8% in the medium universe, and between 52.4% and 67.8% in the large universe. For the mean-risk optimizers, the reduction is between 26.0% and 64.1%.

Without regularization, the joint optimizers also realize lower risk than the overlay optimizers and, as for the out-of-sample gaps in Chapter 6, their advantage falls with the size of the universe. Their realized risk is on average 3.65% lower than that of the overlay optimizers in the small universe, 2.57% in the medium, and 0.92% in the large universe. With regularization, the joint optimizers realize lower risk in almost all comparisons, while with a mean term, they do so in all comparisons in the small universe but only in about half of them in the large universe.

Leaving the currency exposure unhedged raises the risk measures by a similar factor as the volatility. The Zero optimizer optimizes the same objective as the Full optimizer but is

evaluated on unhedged returns. Its Expected Shortfall is 1.34 to 2.33 times larger than that of the Full optimizer, its semi-deviation 1.34 to 1.94 times larger, and its mean absolute deviation 1.37 to 2.40 times larger. The increase in risk when leaving the currency exposure unhedged rises proportionally with the volatility. The increase in volatility is 1.35 to 2.09 times larger for the Zero optimizer than for the Full optimizer.

The Expected Shortfall ratios lead to the same comparison with the equally weighted portfolio as the Sharpe ratios in Chapter 5. In the small universe, fewer than half of the unregularized optimizers but most of the regularized and mean-risk optimizers achieve a higher Expected Shortfall ratio than the equally weighted portfolio, and in the medium and large universes all of them do. In every case, these are the optimizers with a higher Sharpe ratio than the equally weighted portfolio.

Finally, Figure G.7 shows that the asset turnover spikes once a year, at the rebalancing date that follows the annual reconstitution of the investable universe, when the optimizers are forced to rebalance based on the new constituents. In Figure G.8, the currency turnover of the joint optimizers exceeds that of the overlay optimizers at 60% to 100% of the rebalancing dates without regularization, but only at 50% to 74% with regularization. This is in line with the analysis of the average currency turnover in Section 5.2, where the two optimizers differ by at most 7% with regularization.

Panel A: Small Investable Universe (top 10 stocks per economy)

		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ES (%)	1.15	0.67	0.64	0.68	0.66	0.65	0.64	0.69	0.67	0.77	0.73
SOES (%)	2.72	1.54	1.49	1.58	1.50	1.53	1.49	1.56	1.50	1.73	1.60
SD (%)	0.65	0.37	0.36	0.38	0.37	0.37	0.36	0.38	0.37	0.42	0.40
MAD (%)	0.59	0.35	0.34	0.36	0.35	0.35	0.34	0.37	0.36	0.41	0.40
ESR	0.10	0.11	0.12	0.06	0.06	0.12	0.14	0.06	0.06	0.06	0.02

Panel B: Medium Investable Universe (top 50 stocks per economy)

		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ES (%)	1.04	0.41	0.40	0.42	0.41	0.41	0.41	0.44	0.43	0.47	0.46
SOES (%)	2.49	1.08	1.06	1.09	1.06	1.13	1.08	1.17	1.12	1.19	1.08
SD (%)	0.59	0.25	0.24	0.25	0.25	0.26	0.25	0.27	0.26	0.28	0.26
MAD (%)	0.53	0.22	0.22	0.23	0.22	0.22	0.22	0.24	0.23	0.26	0.25
ESR	0.25	0.37	0.40	0.36	0.39	0.42	0.41	0.45	0.43	0.32	0.37

Panel C: Large Investable Universe (top 100 stocks per economy)

		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ES (%)	0.97	0.32	0.32	0.32	0.32	0.33	0.33	0.35	0.35	0.36	0.37
SOES (%)	2.37	0.93	0.91	0.93	0.91	1.02	1.01	0.99	0.97	0.99	0.98
SD (%)	0.56	0.21	0.20	0.21	0.21	0.22	0.22	0.22	0.22	0.23	0.22
MAD (%)	0.49	0.17	0.17	0.18	0.18	0.17	0.17	0.19	0.19	0.20	0.20
ESR	0.28	0.54	0.55	0.53	0.54	0.46	0.46	0.59	0.62	0.41	0.42

Table G.3: Realized risk measures and Expected Shortfall ratios of the unregularized overlay (Overlay) and joint (Joint) optimizers and the equally weighted portfolio (EW) in the small (Panel A), medium (Panel B), and large (Panel C) investable universes. Columns are grouped by the minimized risk measure. ES, SOES, SD, and MAD are daily values in percent, and ESR is the Expected Shortfall ratio. All are computed from returns net of transaction costs.

Panel A: Small Investable Universe (top 10 stocks per economy)											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric		**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ES (%)		0.66	0.65	0.67	0.65	0.65	0.64	0.67	0.66	0.72	0.71
SOES (%)		1.52	1.52	1.53	1.49	1.53	1.53	1.56	1.52	1.59	1.56
SD (%)		0.37	0.36	0.37	0.36	0.37	0.37	0.38	0.37	0.40	0.39
MAD (%)		0.35	0.34	0.35	0.34	0.34	0.34	0.35	0.35	0.39	0.38
ESR		0.12	0.12	0.10	0.08	0.16	0.16	0.10	0.13	0.09	0.00
Metric	**EW**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**
ES (%)	1.15	0.69	0.97	0.70	0.96	0.69	0.96	0.71	0.98	0.76	1.02
SOES (%)	2.72	1.70	2.33	1.69	2.29	1.69	2.31	1.70	2.32	1.79	2.38
SD (%)	0.65	0.40	0.55	0.40	0.55	0.40	0.55	0.40	0.55	0.43	0.57
MAD (%)	0.59	0.36	0.52	0.36	0.52	0.36	0.51	0.37	0.52	0.40	0.54
ESR	0.10	0.10	0.17	0.10	0.17	0.15	0.21	0.08	0.16	0.05	0.12
Panel B: Medium Investable Universe (top 50 stocks per economy)											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric		**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ES (%)		0.40	0.40	0.41	0.40	0.40	0.40	0.42	0.42	0.45	0.45
SOES (%)		1.16	1.13	1.08	1.07	1.07	1.05	1.17	1.14	1.11	1.10
SD (%)		0.26	0.25	0.25	0.25	0.24	0.24	0.26	0.26	0.26	0.26
MAD (%)		0.21	0.21	0.22	0.22	0.21	0.20	0.22	0.22	0.24	0.24
ESR		0.37	0.37	0.37	0.36	0.44	0.47	0.47	0.46	0.40	0.43
Metric	**EW**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**
ES (%)	1.04	0.41	0.79	0.41	0.78	0.40	0.79	0.43	0.80	0.46	0.81
SOES (%)	2.49	1.17	1.85	1.11	1.78	1.07	1.72	1.20	1.84	1.15	1.81
SD (%)	0.59	0.26	0.45	0.25	0.44	0.24	0.43	0.27	0.45	0.27	0.45
MAD (%)	0.53	0.21	0.43	0.22	0.43	0.21	0.41	0.22	0.43	0.24	0.45
ESR	0.25	0.39	0.31	0.39	0.32	0.46	0.31	0.48	0.36	0.42	0.35
Panel C: Large Investable Universe (top 100 stocks per economy)											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric		**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ES (%)		0.31	0.31	0.32	0.31	0.33	0.32	0.33	0.33	0.35	0.35
SOES (%)		1.00	0.98	0.95	0.93	1.13	1.10	0.99	0.96	0.98	0.96
SD (%)		0.22	0.21	0.21	0.21	0.24	0.23	0.22	0.22	0.22	0.22
MAD (%)		0.17	0.17	0.17	0.17	0.17	0.17	0.18	0.18	0.20	0.19
ESR		0.67	0.65	0.70	0.71	0.52	0.53	0.74	0.74	0.72	0.74
Metric	**EW**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**	**Full**	**Zero**
ES (%)	0.97	0.32	0.74	0.32	0.74	0.33	0.76	0.33	0.76	0.35	0.77
SOES (%)	2.37	1.01	1.68	0.96	1.63	1.11	1.78	0.99	1.69	0.98	1.70
SD (%)	0.56	0.22	0.42	0.21	0.41	0.24	0.43	0.22	0.43	0.22	0.43
MAD (%)	0.49	0.17	0.41	0.17	0.41	0.17	0.40	0.18	0.42	0.20	0.42
ESR	0.28	0.63	0.38	0.70	0.42	0.54	0.34	0.76	0.44	0.67	0.40

Table G.4: Realized risk measures and Expected Shortfall ratios of the regularized overlay (Overlay) and joint (Joint) optimizers and of the benchmarks, the equally weighted portfolio (EW) and the asset-only optimizers Full and Zero, in the small (Panel A), medium (Panel B), and large (Panel C) investable universes. Columns are grouped by the minimized risk measure. ES, SOES, SD, and MAD are daily values in percent, and ESR is the Expected Shortfall ratio. All are computed from returns net of transaction costs.

Panel A: Small Investable Universe (top 10 stocks per economy)											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ES (%)	1.15	0.71	0.70	0.82	0.81	0.69	0.67	0.69	0.67	0.77	0.74
SOES (%)	2.72	1.63	1.60	1.85	1.83	1.62	1.57	1.57	1.49	1.73	1.60
SD (%)	0.65	0.40	0.39	0.46	0.45	0.39	0.38	0.38	0.37	0.43	0.40
MAD (%)	0.59	0.38	0.37	0.43	0.43	0.36	0.36	0.37	0.36	0.41	0.40
ESR	0.10	0.28	0.28	0.29	0.29	0.26	0.29	0.09	0.10	0.12	0.09
Panel B: Medium Investable Universe (top 50 stocks per economy)											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ES (%)	1.04	0.51	0.51	0.64	0.65	0.43	0.43	0.44	0.44	0.48	0.46
SOES (%)	2.49	1.30	1.28	1.55	1.55	1.16	1.13	1.17	1.13	1.19	1.09
SD (%)	0.59	0.31	0.30	0.37	0.38	0.27	0.26	0.27	0.26	0.28	0.27
MAD (%)	0.53	0.28	0.28	0.34	0.35	0.23	0.23	0.24	0.24	0.26	0.26
ESR	0.25	0.56	0.58	0.47	0.50	0.53	0.56	0.45	0.47	0.35	0.48
Panel C: Large Investable Universe (top 100 stocks per economy)											
		Variance		Semi-Variance		MAD		ES		SOES	
Metric	**EW**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**	**Overlay**	**Joint**
ES (%)	0.97	0.45	0.46	0.59	0.61	0.35	0.35	0.35	0.35	0.37	0.37
SOES (%)	2.37	1.19	1.19	1.48	1.50	1.06	1.04	0.99	0.98	1.00	0.99
SD (%)	0.56	0.28	0.28	0.35	0.36	0.23	0.23	0.22	0.22	0.23	0.23
MAD (%)	0.49	0.25	0.25	0.32	0.33	0.18	0.19	0.19	0.19	0.20	0.20
ESR	0.28	0.69	0.73	0.60	0.66	0.58	0.58	0.59	0.66	0.43	0.49

Table G.5: Realized risk measures and Expected Shortfall ratios of the mean-risk overlay (Overlay) and joint (Joint) optimizers and the equally weighted portfolio (EW) in the small (Panel A), medium (Panel B), and large (Panel C) investable universes. Columns are grouped by the risk measure in the objective. ES, SOES, SD, and MAD are daily values in percent, and ESR is the Expected Shortfall ratio. All are computed from returns net of transaction costs.

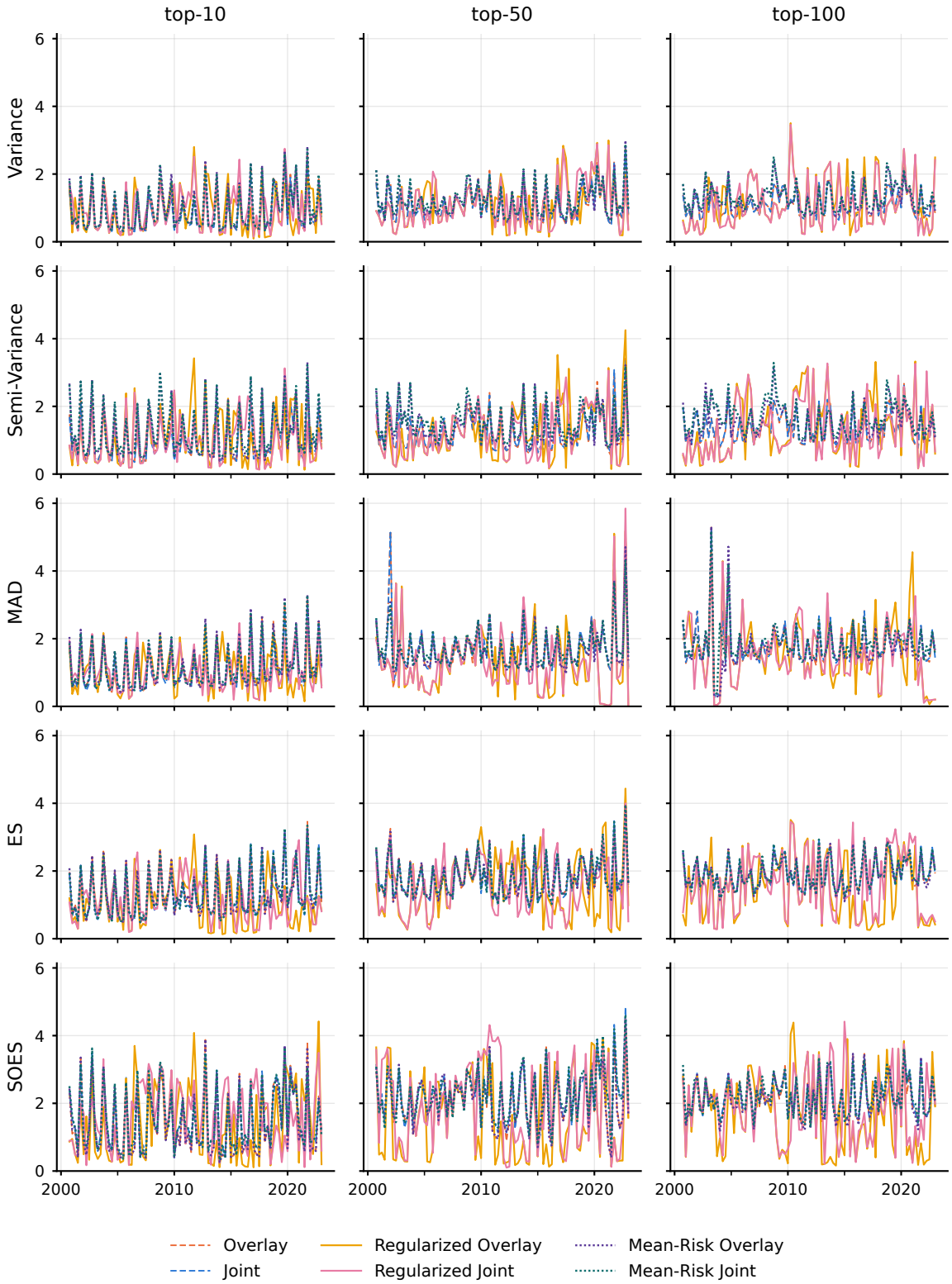


Figure G.7: Asset turnover of the unregularized, regularized, and mean-risk overlay and joint optimizers at each rebalancing date, by risk measure and universe size.

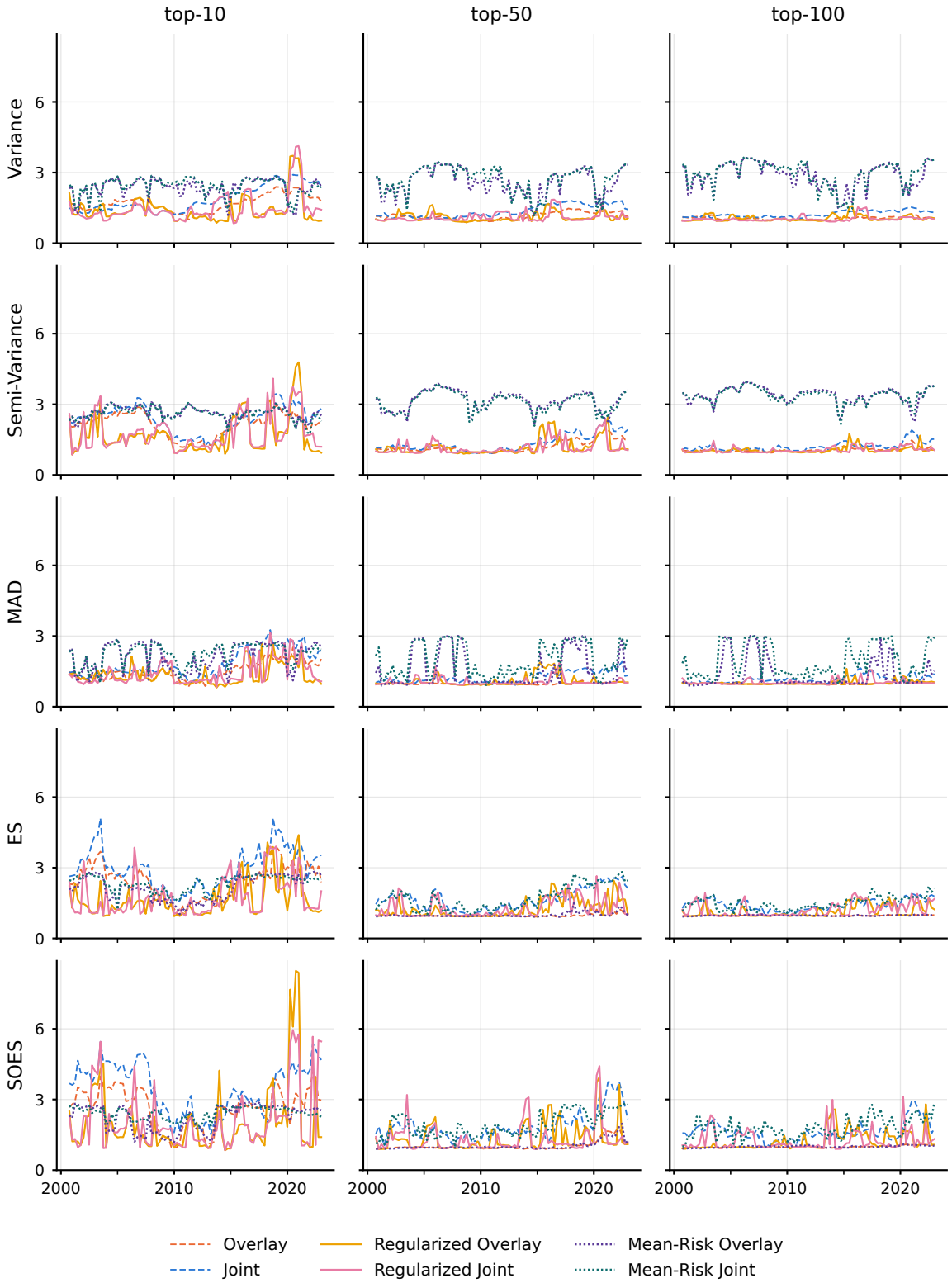


Figure G.8: Currency turnover of the unregularized, regularized, and mean-risk overlay and joint optimizers at each rebalancing date, by risk measure and universe size.

Bibliography

[1] C. Acerbi and D. Tasche. On the coherence of expected shortfall. arXiv preprint arXiv:cond-mat/0104295, 2002.

[2] M. Ammann, G. Coqueret, and J.-P. Schade. Characteristics-based portfolio choice with leverage constraints. *Journal of Banking & Finance*, 70:23–37, 2016.

[3] P. Artzner, F. Delbaen, J.-M. Eber, and D. Heath. Coherent measures of risk. *Mathematical Finance*, 9(3):203–228, 1999.

[4] Basel Committee on Banking Supervision. Minimum capital requirements for market risk. Standard, Bank for International Settlements, 2019.

[5] J. Boudoukh, M. Richardson, A. Thapar, and F. Wang. Optimal currency hedging for international equity portfolios. *Financial Analysts Journal*, 75(4):65–83, 2019.

[6] S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, Cambridge, 2004.

[7] J. Brodie, I. Daubechies, C. De Mol, D. Giannone, and I. Loris. Sparse and stable Markowitz portfolios. *Proceedings of the National Academy of Sciences*, 106(30):12267–12272, 2009.

[8] R. Burkhardt and U. Ulrych. Sparse and stable international portfolio optimization and currency risk management. *Journal of International Money and Finance*, 139:102949, 2023.

[9] F. Caccioli, I. Kondor, M. Marsili, and S. Still. Liquidity risk and instabilities in portfolio optimization. *International Journal of Theoretical and Applied Finance*, 19(5):1650035, 2016.

[10] J. Y. Campbell, K. Serfaty-de Medeiros, and L. M. Viceira. Global currency hedging. *The Journal of Finance*, 65(1):87–121, 2010.

[11] Y.-c. Chen and K. Rogoff. Commodity currencies. *Journal of International Economics*, 60(1):133–160, 2003.

[12] S. Ciliberti, I. Kondor, and M. Mézard. On the feasibility of portfolio optimization under expected shortfall. *Quantitative Finance*, 7(4):389–396, 2007.

[13] CVXPY documentation. https://www.cvxpy.org, 2026. Accessed: 2026-10-16.

[14] V. DeMiguel, L. Garlappi, F. J. Nogales, and R. Uppal. A generalized approach to portfolio optimization: Improving performance by constraining portfolio norms. *Management Science*, 55(5):798–812, 2009.

[15] V. DeMiguel, L. Garlappi, and R. Uppal. Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *The Review of Financial Studies*, 22(5):1915–1953, 2009.

[16] S. Diamond and S. Boyd. CVXPY: A Python-embedded modeling language for convex optimization. *Journal of Machine Learning Research*, 17(83):1–5, 2016.

[17] N. El Karoui. High-dimensionality effects in the Markowitz problem and other quadratic programs with linear constraints: Risk underestimation. *The Annals of Statistics*, 38(6):3487–3566, 2010.

[18] J. Estrada. Mean-semivariance optimization: A heuristic approach. *Journal of Applied Finance*, 18(1):57–72, 2008.

[19] A. Frazzini, R. Israel, and T. J. Moskowitz. Trading costs. Working paper, AQR Capital Management and Yale University, 2018.

[20] J. Glen and P. Jorion. Currency hedging for international portfolios. *The Journal of Finance*, 48(5):1865–1886, 1993.

[21] L. R. Goldberg, M. Y. Hayes, and O. Mahmoud. Minimizing shortfall. arXiv preprint arXiv:1102.0938, 2013.

[22] T. Hastie, R. Tibshirani, and J. Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer Series in Statistics. Springer, New York, second edition, 2009.

[23] R. Jagannathan and T. Ma. Risk reduction in large portfolios: Why imposing the wrong constraints helps. *The Journal of Finance*, 58(4):1651–1683, 2003.

[24] W. James and C. Stein. Estimation with quadratic loss. In *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, pages 361–379, Berkeley, CA, 1961. University of California Press.

[25] C. M. Jones. A century of stock market liquidity and trading costs. Working paper, Columbia University, 2002.

[26] P. Jorion. Bayes-Stein estimation for portfolio analysis. *Journal of Financial and Quantitative Analysis*, 21(3):279–292, 1986.

[27] P. Jorion. Mean/variance analysis of currency overlays. *Financial Analysts Journal*, 50(3):48–56, 1994.

[28] I. Kondor, S. Pafka, and G. Nagy. Noise sensitivity of portfolio selection under various risk measures. *Journal of Banking & Finance*, 31(5):1545–1573, 2007.

[29] I. Kondor and I. Varga-Haszonits. Instability of portfolio optimization under coherent risk measures. *Advances in Complex Systems*, 13(3):425–437, 2010.

[30] H. Konno and J. Li. Internationally diversified investment using an integrated portfolio model. *International Journal of Theoretical and Applied Finance*, 1(1):145–160, 1998.

[31] H. Konno and J. Li. Applications of the integrated approach to international portfolio optimization. *Asia-Pacific Financial Markets*, 7(2):121–144, 2000.

[32] H. Konno and H. Yamazaki. Mean-absolute deviation portfolio optimization model and its applications to Tokyo stock market. *Management Science*, 37(5):519–531, 1991.

[33] P. A. Krokhmal. Higher moment coherent risk measures. *Quantitative Finance*, 7(4):373–387, 2007.

[34] O. Ledoit and M. Wolf. Honey, I shrunk the sample covariance matrix. Working paper 691, Universitat Pompeu Fabra, 2003.

[35] O. Ledoit and M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

[36] J. Li. Sparse and stable portfolio selection with parameter uncertainty. *Journal of Business & Economic Statistics*, 33(3):381–392, 2015.

[37] P. Lucescu and U. Ulrych. Regularized multi-currency expected shortfall portfolios. Working paper, 2026.

[38] H. Lustig, N. Roussanov, and A. Verdelhan. Common risk factors in currency markets. *The Review of Financial Studies*, 24(11):3731–3777, 2011.

[39] H. Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.

[40] H. Markowitz. *Portfolio Selection: Efficient Diversification of Investments*. John Wiley & Sons, New York, 1959.

[41] H. Markowitz, P. Todd, G. Xu, and Y. Yamane. Computation of mean-semivariance efficient sets by the Critical Line Algorithm. *Annals of Operations Research*, 45(1):307–317, 1993.

[42] A. J. McNeil and R. Frey. Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, 7(3–4):271–300, 2000.

[43] A. J. McNeil, R. Frey, and P. Embrechts. *Quantitative Risk Management: Concepts, Techniques and Tools*. Princeton University Press, Princeton, NJ, revised edition, 2015.

[44] L. Menkhoff, L. Sarno, M. Schmeling, and A. Schrimpf. Carry trades and global foreign exchange volatility. *The Journal of Finance*, 67(2):681–718, 2012.

[45] R. C. Merton. On estimating the expected return on the market: An exploratory investigation. *Journal of Financial Economics*, 8(4):323–361, 1980.

[46] M. Mhiri and J.-L. Prigent. International portfolio optimization with higher moments. *International Journal of Economics and Finance*, 2(5):157–169, 2010.

[47] R. Michaud. The Markowitz optimization enigma: Is 'optimized' optimal? *Financial Analysts Journal*, 45(1):31–42, 1989.

[48] C. J. Neely, P. A. Weller, and J. M. Ulrich. The adaptive markets hypothesis: Evidence from the foreign exchange market. *Journal of Financial and Quantitative Analysis*, 44(2):467–488, 2009.

[49] W. Ogryczak and A. Ruszczyński. From stochastic dominance to mean-risk models: Semideviations as risk measures. *European Journal of Operational Research*, 116(1):33–50, 1999.

[50] W. Opie and S. J. Riddiough. Global currency hedging with common risk factors. *Journal of Financial Economics*, 136(3):780–805, 2020.

[51] G. Papp, I. Kondor, and F. Caccioli. Optimizing expected shortfall under an ℓ_1 constraint—an analytic approach. *Entropy*, 23(5):523, 2021.

[52] A. F. Perold and E. C. Schulman. The free lunch in currency hedging: Implications for investment policy and performance standards. *Financial Analysts Journal*, 44(3):45–50, 1988.

[53] A. Ranaldo and P. Söderlind. Safe haven currencies. *Review of Finance*, 14(3):385–407, 2010.

[54] R. T. Rockafellar and S. Uryasev. Optimization of conditional value-at-risk. *Journal of Risk*, 2(3):21–41, 2000.

[55] R. T. Rockafellar and S. Uryasev. Conditional value-at-risk for general loss distributions. *Journal of Banking & Finance*, 26(7):1443–1471, 2002.

[56] R. T. Rockafellar, S. Uryasev, and M. Zabarankin. Generalized deviations in risk analysis. *Finance and Stochastics*, 10(1):51–74, 2006.

[57] T. Roncalli. *Introduction to Risk Parity and Budgeting*. Chapman & Hall/CRC Financial Mathematics Series. Chapman & Hall/CRC, Boca Raton, FL, 2013.

[58] B. H. Solnik. An equilibrium model of the international capital market. *Journal of Economic Theory*, 8(4):500–524, 1974.

[59] M. G. Speranza. Linear programming models for portfolio optimization. *Finance*, 14:107–123, 1993.

[60] S. Still and I. Kondor. Regularizing portfolio optimization. *New Journal of Physics*, 12(7):075034, 2010.

[61] K.-I. Suzuki, H. Konno, and M. Morjiri. An international portfolio optimization model hedged with forward currency contracts. *Financial Engineering and the Japanese Markets*, 4(3):275–286, 1997.

[62] N. Topaloglou, H. Vladimirou, and S. A. Zenios. CVaR models with selective hedging for international asset allocation. *Journal of Banking & Finance*, 26(7):1535–1561, 2002.

[63] J. H. van Binsbergen, M. W. Brandt, and R. S. J. Koijen. Optimal decentralized investment management. *The Journal of Finance*, 63(4):1849–1895, 2008.

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